

Service and Business Plan for City Bus Operations - Chandigarh



Project Consultancy-7 (PC-7)

Service and Business Plan for City Bus Operations for two ESCBS Cities Chandigarh and
Mira-Bhayandar under the World Bank – GEF assisted Efficient and Sustainable City
Bus Services (ESCBS) Project

December 2020

Volume II – ANNEXURE

Submitted by:

Centre of Excellence in Urban Transport
CEPT Research and Development Foundation

CRDF **CEPT**
CEPT RESEARCH
AND DEVELOPMENT
FOUNDATION **UNIVERSITY**



Ministry of Housing
and Urban Affairs
Government of India



THE WORLD BANK

Table of Content

Annexure 1: List Existing City Bus Routes (August 2019).....	1
Annexure 2: Boarding – Alighting and section load for each existing route (one sample trip in peak hour).....	5
Annexure 3: Route-wise Operations Performance Parameters	72
Annexure 4: Alignment of Bus Routes – Business As Usual	75
Annexure 5: Alignment of Proposed Bus Routes– Alternative Scenario 2A and 2B	107
Annexure 6: Service Plan – Scenario Evaluation	139
Annexure 7: List of proposed routes & details	149
Annexure 8: Routes frequency (peak hours) and fleet requirement for fleet mix scenarios (2032) ...	151
Annexure 9: Operational cost, revenue estimation, operating ratio and load factor	153
Annexure 10: Fleet requirement and procurement plan, capital and operating cost estimates for 25% PT share (shortlisted) scenarios	155

List of Tables

Table 1 List of Existing City Bus routes (August 2019).....	1
Table 2: Details of route wise average trip length, maximum load and load factor	66
Table 3: Operations Performance parameters of existing routes (2019).....	73
Table 4: Scenario evaluation based on supply analysis	140
Table 5: Model Output - Scenario Comparison for Chandigarh and surrounding area	144
Table 6: Service Level Benchmark Indicators	147
Table 7: List of proposed routes	149
Table 8: Route frequency and fleet requirement for fleet mix scenarios	151
Table 9: Fleet procurement plan for short listed scenario – 25% PT share	155
Table 10: Total Capital Cost (in Cr.) Investment (2023 to 2032) – 25% PT share.....	155
Table 11: Total Operating Cost (in Cr.) by operational cost components (2023 to 2032) – 25% PT share	155

List of Figures

Figure 1: Peak hour Boarding & Alighting pattern along with line load of Route 3 (Up and Down)....	7
Figure 2: Peak hour Boarding & Alighting pattern along with line load of Route 11 (Up and Down) ..	8
Figure 3: Peak hour Boarding & Alighting pattern along with line load of Route 17 (Up and Down) ..	9

Figure 4: Peak hour Boarding & Alighting pattern along with line load of Route 18 (Up and Down) 10	10
Figure 5: Peak hour Boarding & Alighting pattern along with line load of Route 22 (Up and Down) 11	11
Figure 6: Peak hour Boarding & Alighting pattern along with line load of Route 28 (Up and Down) 12	12
Figure 7: Peak hour Boarding & Alighting pattern along with line load of Route 71 (Up and Down) 13	13
Figure 8: Peak hour Boarding & Alighting pattern along with line load of Route 143 (Up and Down)	14
Figure 9: Peak hour Boarding & Alighting pattern along with line load of Route 241 (Up and Down)	15
Figure 10: Peak hour Boarding & Alighting pattern along with line load of Route 1A and 1C.....	16
Figure 11: Peak hour Boarding & Alighting pattern along with line load of Route 2A and 2C.....	17
Figure 12: Peak hour Boarding & Alighting pattern along with line load of Route 2D (Up and Down)	18
Figure 13: Peak hour Boarding & Alighting pattern along with line load of Route 4A and 4C.....	19
Figure 14: Peak hour Boarding & Alighting pattern along with line load of Route 5A and 5C.....	20
Figure 15: Peak hour Boarding & Alighting pattern along with line load of Route 7A and 7C.....	21
Figure 16: Peak hour Boarding & Alighting pattern along with line load of Route 18B (Up and Down)	22
Figure 17: Peak hour Boarding & Alighting pattern along with line load of Route 23A (Up and Down)	23
Figure 18: Peak hour Boarding & Alighting pattern along with line load of Route 28A and 28C.....	24
Figure 19: Peak hour Boarding & Alighting pattern along with line load of Route 10AC (Up and Down)	25
Figure 20: Peak hour Boarding & Alighting pattern along with line load of Route 37AC (Up and Down)	26
Figure 21: Peak hour Boarding & Alighting pattern along with line load of Route 202AC (Up and Down)	27
Figure 22: Peak hour Boarding & Alighting pattern along with line load of Route 203AC (Up and Down)	28
Figure 23: Peak hour Boarding & Alighting pattern along with line load of Route 205AC (Up and Down)	29
Figure 24: Peak hour Boarding & Alighting pattern along with line load of Route 206AC (Up and Down)	30
Figure 25: Peak hour Boarding & Alighting pattern along with line load of Route 240AC (Up and Down)	31
Figure 26: Peak hour Boarding & Alighting pattern along with line load of Route 24AAC and 24CAC	32

Figure 27: Peak hour Boarding & Alighting pattern along with line load of Route 254AC (Up and Down)	33
Figure 28: Peak hour Boarding & Alighting pattern along with line load of Route 239AAC and 239CAC	34
Figure 29: Peak hour Boarding & Alighting pattern along with line load of Route 8 (Up and Down)	35
Figure 30: Peak hour Boarding & Alighting pattern along with line load of Route 25 (Up and Down)	36
Figure 31: Peak hour Boarding & Alighting pattern along with line load of Route 30 (Up and Down)	37
Figure 32: Peak hour Boarding & Alighting pattern along with line load of Route 32 (Up and Down)	38
Figure 33: Peak hour Boarding & Alighting pattern along with line load of Route 34 (Up and Down)	39
Figure 34: Peak hour Boarding & Alighting pattern along with line load of Route 35 (Up and Down)	40
Figure 35: Peak hour Boarding & Alighting pattern along with line load of Route 39 (Up and Down)	41
Figure 36: Peak hour Boarding & Alighting pattern along with line load of Route 40 (Up and Down)	42
Figure 37: Peak hour Boarding & Alighting pattern along with line load of Route 79 (Up and Down)	43
Figure 38: Peak hour Boarding & Alighting pattern along with line load of Route 80 (Up and Down)	44
Figure 39: Peak hour Boarding & Alighting pattern along with line load of Route 212 (Up and Down)	45
Figure 40: Peak hour Boarding & Alighting pattern along with line load of Route 213 (Up and Down)	46
Figure 41: Peak hour Boarding & Alighting pattern along with line load of Route 242 (Up and Down)	47
Figure 42: Peak hour Boarding & Alighting pattern along with line load of Route 2B (Up and Down)	48
Figure 43: Peak hour Boarding & Alighting pattern along with line load of Route 2F (Up and Down)	49
Figure 44: Peak hour Boarding & Alighting pattern along with line load of Route 9A and 9C.....	50
Figure 45: Peak hour Boarding & Alighting pattern along with line load of Route 20A (Up and Down)	51
Figure 46: Peak hour Boarding & Alighting pattern along with line load of Route 25A	52

Figure 47: Peak hour Boarding & Alighting pattern along with line load of Route 30A	53
Figure 48: Peak hour Boarding & Alighting pattern along with line load of Route 30B (Up and Down)	54
Figure 49: Peak hour Boarding & Alighting pattern along with line load of Route 32A (Up and Down)	55
Figure 50: Peak hour Boarding & Alighting pattern along with line load of Route 35B (Up and Down)	56
Figure 51: Peak hour Boarding & Alighting pattern along with line load of Route 123A (Up and Down)	57
Figure 52: Peak hour Boarding & Alighting pattern along with line load of Route 20AC (Up and Down)	58
Figure 53: Peak hour Boarding & Alighting pattern along with line load of Route 26AC (Up and Down)	59
Figure 54: Peak hour Boarding & Alighting pattern along with line load of Route 36AC (Up and Down)	60
Figure 55: Peak hour Boarding & Alighting pattern along with line load of Route 38AC (Up and Down)	61
Figure 56: Peak hour Boarding & Alighting pattern along with line load of Route 211AC (Up and Down)	62
Figure 57: Peak hour Boarding & Alighting pattern along with line load of Route 214AC (Up and Down)	63
Figure 58: Peak hour Boarding & Alighting pattern along with line load of Route 216AC (Up and Down)	64
Figure 59: Peak hour Boarding & Alighting pattern along with line load of Route 38AAC (Up and Down)	65
Figure 60: Existing Route3 (MAULI JAGRAN TO PGI via)	77
Figure 61: Existing Route 11 (ISBT-17 TO PB.CIVIL SECTT).....	77
Figure 62: Existing Route 17/ 17AC (ISBT-43 TO AIR PORT).....	78
Figure 63: Existing Route 18/ 18AC (ISBT-43 TO KAIMBWALA)	78
Figure 64: Existing Route 22/ 22AC (ISBT-43 TO MATA MANSA DEVI)	79
Figure 65: Existing Route 28 (MALOYA TO MATA MANSA DEVI)	79
Figure 66: Existing Route 71/ 71AC(SBT-43 TO SAKETRI (KAIMBWALA).....	80
Figure 67: Existing Route 143/ 143AC(DHANAS TO AIRPORT)	80
Figure 68: Existing Route 241/ 241AC(ISBT-43 TO Mansa Devi)	81
Figure 69: Existing Route 1A/ 1C (PGI to PGI)	81
Figure 70: Existing Route 2A/ 2AAC (MANI MAJRA TO MANI MAJRA).....	82
Figure 71: Existing Route 2C/ 2CAC (MANI MAJRA TO MANI MAJRA)	82

Figure 72: Existing Route 2D (PGI TO IT Park).....	83
Figure 73: Existing Route4A/4AAC/4C/4CAC (SBT-43 to ISBT-43)	83
Figure 74: Existing Route5A/ 5C/ 5AAC (RAM DARBAR TO RAM DARBAR).....	84
Figure 75: Existing Route7A/ 7C/ 7AAC (ISBT-17 to ISBT-17)	84
Figure 76: Existing Route 18B (ISBT-43/SESSION COURT TO H/COURT)	85
Figure 77: Existing Route 23A/ 23AAC(ISBT-43 TO KHUDA ALI SHER).....	85
Figure 78: Existing Route 28A/ 28C (ISBT-43 to ISBT-43).....	86
Figure 79: Existing Route 10/ 10AC (ISBT-43 TO MATA MANSA DEVI).....	86
Figure 80: Existing Route 37/ 37AC(ISBT 43 TO RAILWAY STATION)	87
Figure 81: Existing Route 202/ 202AC(ISBT-43 TO PB. CIVIL. SECTT. (HIGH COURT)	87
Figure 82: Existing Route 203/ 203AC(ISBT-43 TO MATA MANSA DEVI)	88
Figure 83: Existing Route 205/ 205AC(ISBT-43 to PGI)	88
Figure 84: Existing Route 206/ 206AC(ISBT-43 TO I.T.Park).....	89
Figure 85: Existing Route 240/ 240AC(ISBT-43 TO MATA MANSA DEVI).....	89
Figure 86: Existing Route 24A/ 24AAC (ISBT 43 TO PGI).....	90
Figure 87: Existing Route24C/ 24CAC(ISBT 43 to ISBT-17).....	90
Figure 88: Existing Route 254/ 254AC(ISBT 43 TO IT PARK).....	91
Figure 89: Existing Route 239A/ 239AAC/ 239C/ 239CAC(ISBT-43 to ISBT-43)	91
Figure 90: Existing Route8/ 8AC (MOHALI Ph-XI to PGI).....	92
Figure 91: Existing Route 25/ 25AC/ 25A (CHD TO KHIZRABAD).....	92
Figure 92: Existing Route 30/ 30AC(ISBT-43 TO NADA SAHIB)	93
Figure 93: Existing Route 32 (PGI TO DERABASSI).....	93
Figure 94: Existing Route 34/ 34AC(PGI TO DERABASSI)	94
Figure 95: Existing Route 35/ 35AC/ 35B/ 35BAC(ISBT-17 TO KHARAR).....	94
Figure 96: Existing Route 39/ 39AC(MOHALI PH-11 TO PGI).....	95
Figure 97: Existing Route 40 (CHD TO KHIZRABAD)	95
Figure 98: Existing Route 79/ 79AC(PGI TO CHHATTBIR ZOO)	96
Figure 99: Existing Route 80/ 80AC(ZIRAKPUR TO PGI)	96
Figure 100: Existing Route 212/ 212AC(PGI TO ZIRAKPUR (SINGHPURA BHUDA)	97
Figure 101: Existing Route 213 (LANDRAN to DERA BASSI).....	97
Figure 102: Existing Route 242/ 242AC(LANDRAN TO PGI).....	98
Figure 103: Existing Route 2B/ 2BAC (PGI TO PANCHKULA BUS STAND)	98
Figure 104: Existing Route2F/ 2FAC (PGI TO DHAKOLI BORDER)	99
Figure 105: Existing Route9A/ 9C/ 9CAC (MATA MANSA DEVI TO MATA MANSA DEVI)	99
Figure 106: Existing Route 20A/ 20AAC (CHD TO KHARAR).....	100
Figure 107: Existing Route 30A/ 30AAC(ISBT-43 TO CANTONMENT).....	100
Figure 108: Existing Route 30B/ 30BAC(ISBT-43 TO PANCHKULA SEC-4)	101

Figure 109: Existing Route 32A/ 32AAC(NEW ISBT-MOHALI TO MATA MANSA DEVI)	101
Figure 110: Existing Route 123A/ 123AAC(MOHALI R/STATION TO PGI).....	102
Figure 111: Existing Route 20/ 20AC (CHD TO KHARAR)	102
Figure 112: Existing Route 26/ 26AC (DHANAS TO DHAKOLI).....	103
Figure 113: Existing Route 36/ 36AC (PGI TO NEW AIRPORT).....	103
Figure 114: Existing Route 38/ 38AC(ISBT-17 TO NEW AIRPORT)	104
Figure 115: Existing Route 211/ 211AC (PGI TO CANTONMENT)	104
Figure 116: Existing Route 214/ 214AC(SP Barrier TO NADA SAHIB)	105
Figure 117: Existing Route 216/ 126AC(ISBT-43 TO DERABASSI)	105
Figure 118: Existing Route 38A/ 38AAC(ISBT-17 TO NEW AIRPORT).....	106
Figure 119: Proposed Route 1: (ISBT Mohali - Sukteri - ISBT 17).....	109
Figure 120: Proposed Route 2(Sohna Gurudwara - ISBT 43 - ISBT 17 - PGI - Khizrabad).....	109
Figure 121: Proposed Route 3 (ISBT 17 - ISBT 43 - Airport)	110
Figure 122: Proposed Route 4 (ISBT 17 - Railway Station – Cantonment).....	110
Figure 123: Proposed Route 5(ISBT 43 - Tribune Chowk - Derabassi).....	111
Figure 124: Proposed Route 6 (ISBT 17 - PGI - Khizrabad).....	111
Figure 125: Proposed Route 7(ISBT 17 - Mani Majra)	112
Figure 126: Proposed Route 8 (ISBT 17 - PGI - Mullanpur Barrier - ISBT 17)	112
Figure 127: Proposed Route 9 (ISBT17 -Railway Station -Dhakoli)	113
Figure 128: Proposed Route 11 (ISBT 17 - Mani Majra – Derabassi)	113
Figure 129: Proposed Route 12 (PGI - ISBT 17 – Derabassi).....	114
Figure 130: Proposed Route 12E (PGI - ISBT 17 - Zirakpur - Derabassi).....	114
Figure 131: Proposed Route 13 (Kharar - ISBT 43 - ISBT 17 – Airport)	115
Figure 132: Proposed Route 14 (ISBT 43 - Landran – Kharar).....	115
Figure 133: Proposed Route 15 (ISBT 17 - PGI – Nayagaon - ISBT 17).....	116
Figure 134: Proposed Route Route 16 (ISBT 43 - Maloya Colony).....	116
Figure 135: Proposed Route 17 (Sohna Gurudwara - ISBT Mohali - Kharar)	117
Figure 136: Proposed Route 18 (PGI - ISBT 17 - Zirakpur)	117
Figure 137: Proposed Route Route 19 (ISBT 17 - High Court - Kaimbwala).....	118
Figure 138: Proposed Route Route 20 (ISBT 17 - ISBT43 - New Airport)	118
Figure 139: Proposed Route 21(ISBT43-Landran).....	119
Figure 140: Proposed Route 22 (Sohna Gurudwara - White House - Mata Mansa Devi).....	120
Figure 141 Proposed Route 23 (ISBT17 - High Court - Kaimbwala)	120
Figure 142: Proposed Route 24 (Ramdarbar - ISBT43 -PGI - ISBT43).....	121
Figure 143: Proposed Route 25 (ISBT 43 - CTU Workshop - Mata Mansa Devi).....	121
Figure 144: Proposed Route 26 (Mata Mansa Devi - ISBT17 - Dhanas)	122
Figure 145: Proposed Route 27 (ISBT17 - SAS Railway Station)	122

Figure 146: Proposed Route 28 (ISBT43- Mohali Municipal Corporation - Phase11)	123
Figure 147: Proposed Route 29E (ISBT43- ISBT 17 - PGI)	123
Figure 148: Proposed Route 30 (ISBT Mohali-ISBT 43).....	124
Figure 149: Proposed Route 31 (PGI - ISBT Mohali – Kharar)	124
Figure 150: Proposed Route 31E (PGI - ISBT Mohali – Kharar).....	125
Figure 151: Proposed Route 33 (PGI - ISBT17 - ISBT 43 - Fortis Hospital).....	125
Figure 152: Proposed Route 34E (PGI - ISBT 17- Grain Mkt-Chandigarh Rly St- ISBT Panchkula)	126
Figure 153: Proposed Route35 (ISBT 43 -Tribune Chowk - Panchkula)	126
Figure 154: Proposed Route 36 (ISBT 17 - Railway Station - Mata Mansa Devi).....	127
Figure 155: Proposed Route 37 (Sohna Gurudwara - Sector 65 - Sohna Gurudwara)	128
Figure 156: Proposed Route 38 (ISBT 17 - PGI - ISBT 17).....	128
Figure 157: Proposed Route 39 (ISBT17 - High Court - ISBT 17)	129
Figure 158: Proposed Route 40 (Labour Chowk - Sukhna Lake - Labour Chowk)	129
Figure 159: Proposed Route 41 (Sector 21 - Purv Marg - Sector 21).....	130
Figure 160: Proposed Route 42 (Sector 15 Market - Sector 21 Market - Sector 15 Market)	130
Figure 161: Proposed Route 43 (ISBT 43 - PGI - Sector 22 Market - ISBT 43).....	131
Figure 162: Proposed Route 44 (ISBT 43 - Industrial Phase 2 - Sector 21 - ISBT 43).....	131
Figure 163: Proposed Route 45 (ISBT Mohali - Sector 36/37 - ISBT Mohali).....	132
Figure 164: Proposed Route 46 (ISBT 43 - Sector 46 - Sector 31 – ISBT 43).....	132
Figure 165: Proposed Route 47 (Sector 65 - Sector 43 Market - Sector 65)	133
Figure 166: Proposed Route 48 (Sector 43 - Sohna Gurudwara - Ptl Chowk - ISBT 43)	133
Figure 167: Proposed Route 49 (ISBT Mohali - Sector 43 - Fortis Hospital - ISBT Mohali).....	134
Figure 168: Proposed Route 50 (Daun Majra - ISBT 43 - Sector 65 - Daun Majra).....	134
Figure 169: Proposed Route 53 (Malloya - Sector 47 (Sector 40 Market Road))	135
Figure 170: Proposed Route 54 (Sector 38/39 -Industrial Area Phase 2 (Sector 36 Market Road)) ..	135
Figure 171: Proposed Route 55 (Daddu Majra Colony - Hallo Majra (Dakshin Marg)).....	136
Figure 172: Proposed Route 57 (ISBT-43 TO MATA MANSA DEVI)	136
Figure 173: Proposed Route 58 (IVY Hospital - Punjab Civil Secretariat (Udyan Path)).....	137
Figure 174: Proposed Route 59 (Sohna Gurudwara - Sukhna Lake (Himalaya Marg))	137
Figure 175: Proposed Route 60 (Sector 68/67 - Sector 26 Guru Gobind School (Sukhna Path))	138
Figure 176: Scenario Comparison - PT Network (Chandigarh and surrounding area).....	141
Figure 177: Scenario Comparison – Developed Area Coverage (Chandigarh and Mohali).....	142
Figure 178: Scenario Comparison – Routes Structure (Chandigarh and Mohali)	143
Figure 179: Scenario Comparison – Passenger Flow (Chandigarh and Mohali).....	145
Figure 180: Scenario Comparison – Bus Supply (Buses/hr.) (Chandigarh and Mohali).....	146

Annexure 1: List Existing City Bus Routes (August 2019)

Table 1 List of Existing City Bus routes (August 2019)

Sr. No.	Routes No.	EMME Code	Depot	Description	Length (km)	Headway (min)		Travel Time (min)	No of Buses	City / Suburban service	Daily ridership
						Peak	Off Peak				
1	3	C3	II	MAULI JAGRAN TO PGI	17	60	78	62	2	CITY	764
2	11	C11	II	ISBT-17 TO PB.CIVIL SECTT.	7.1	60	78	25	1	CITY	240
3	17	C17	II	ISBT-43 TO AIR PORT	15.3	20	26	64	6	CITY	2695
4	18	C18	II	ISBT-43 TO KAIMBWALA	15.5	30	39	50	4	CITY	1290
5	22	C22	IV	ISBT-43 TO MATA MANSA DEVI	19.6	15	20	73	7	CITY	4597
6	28	C28	III	MALOYA TO MATA MANSA DEVI	25.9	45	59	107	4	CITY	1094
7	71	C71	II	ISBT-43 TO SAKETRI (KAIMBWALA)	24	30	39	88	6	CITY	2755
8	143	C143	II	DHANAS TO AIRPORT	21.3	75	98	69	2	CITY	501
9	241	C241	IV	ISBT-43 TO Mansa Devi	22	20	26	70	9	CITY	3876
10	1A	C1A	III	PGI to PGI	27.4	20	26	106	5	CITY	1125
11	1C	C1C	III	PGI to PGI	27.4	20	26	100	5	CITY	1407
12	2A	C2A	II	MANI MAJRA TO MANI MAJRA	38.3	20	26	134	6	CITY	2591
13	2C	C2C	II	MANI MAJRA TO MANI MAJRA	38.3	20	26	138	6	CITY	2627
14	2D	C2D	II	PGI TO IT Park	21	35	46	71	4	CITY	992
15	4A	C4A	IV	SBT-43 to ISBT-43	31	25	33	104	4	CITY	1303
16	4C	C4C	IV	ISBT-43 to ISBT-43	31	25	33	80	4	CITY	1454
17	5A	C5A	IV	RAM DARBAR TO RAM DARBAR	33.9	15	20	126	8	CITY	5622
18	5C	C5C	IV	RAM DARBAR TO RAM DARBAR	33.9	15	20	123	8	CITY	5642
19	7A	C7A	III	ISBT-17 to ISBT-17	36	25	33	114	4	CITY	1163
20	7C	C7C	III	ISBT-17 to ISBT-17	36	25	33	109	4	CITY	819
21	18B	C18B	II	ISBT-43/SESSION COURT TO H/COURT	14	30	39	53	4	CITY	326

Sr. No.	Routes No.	EMME Code	Depot	Description	Length (km)	Headway (min)		Travel Time (min)	No of Buses	City / Suburban service	Daily ridership
						Peak	Off Peak				
22	23A	C23A	II	ISBT-43 TO KHUDA ALI SHER	16	15	20	69	7	CITY	2565
23	28A	C28A	III	ISBT-43 to ISBT-43	25.7	30	39	76	3	CITY	1054
24	28C	C28C	III	ISBT-43 to ISBT-43	25.7	30	39	89	3	CITY	1217
25	10	C10AC	II	ISBT-43 TO MATA MANSA DEVI	19.4	25	33	84	5	CITY	1083
26	37	C37AC	II	ISBT 43 TO RAILWAY STATION	12	90	117	38	1	CITY	179
27	202	C202AC	IV	ISBT-43 TO PB. CIVIL. SECTT. (HIGH COURT)	13	52	68	35	6	CITY	2144
28	203	C203AC	II	ISBT-43 TO MATA MANSA DEVI	18	35	46	80	4	CITY	653
29	205	C205AC	IV	ISBT-43 to PGI	11.1	44	57	34	6	CITY	2245
30	206	C206AC	IV	ISBT-43 TO I.T.Park	18.1	20	26	67	7	CITY	3629
31	240	C240AC	II	ISBT-43 TO MATA MANSA DEVI	20.4	30	39	92	5	CITY	1455
32	24A	C24AAC	II	ISBT 43 TO PGI	9.8	80	104	38	1	CITY	176
33	24C	C24CAC	II	ISBT 43 to ISBT-17	5.7	25	33	24	2	CITY	447
34	254	C254AC	II	ISBT 43 TO IT PARK	19.6	70	91	70	2	CITY	356
35	239A	C239A	IV	ISBT-43 to ISBT-43	23.2	25	33	97	3	CITY	722
36	239C	C239CAC	IV	ISBT-43 to ISBT-43	23.2	25	33	83	3	CITY	959
37	8	C8	II	MOHALI Ph-XI to PGI	25.6	35	46	78	4	SUB	613
38	25	C25	III	CHD TO KHIZRABAD	27.2	30	39	84	3	SUB	1037
39	30	C30	II	ISBT-43 TO NADA SAHIB	21.5	40	52	79	4	SUB	2266
40	32	C32	III	PGI TO DERABASSI	35	30	39	103	6	SUB	2577
41	34	C34	II	PGI TO DERABASSI	28	20	26	112	9	SUB	2135
42	35	C35	III	ISBT-17 TO KHARAR	18.3	20	26	76	6	SUB	3624
43	39	C39	IV	MOHALI PH-11 TO PGI	24.4	15	20	82	12	SUB	6426
44	40	C40	III	CHD TO KHIZRABAD	31	60	78	97	3	SUB	1271

Sr. No.	Routes No.	EMME Code	Depot	Description	Length (km)	Headway (min)		Travel Time (min)	No of Buses	City / Suburban service	Daily ridership
						Peak	Off Peak				
45	79	C79	II	PGI TO CHHATTBIR ZOO	27.9	45	59	95	4	SUB	1128
46	80	C80	II	ZIRAKPUR TO PGI	20	30	39	73	5	SUB	1644
47	212	C212	II	PGI TO ZIRAKPUR (SINGHPURA BHUDA)	20.4	20	26	98	6	SUB	3071
48	213	C213	II	LANDRAN to DERA BASSI	34	30	39	110	7	SUB	1181
49	242	C242	IV	LANDRAN TO PGI	24.6	20	26	77	8	SUB	4526
50	2B	C2B	II	PGI TO PANCHKULA BUS STAND	22.4	30	39	76	5	SUB	1429
51	2F	C2F	II	PGI TO DHAKOLI BORDER	27.6	30	39	95	6	SUB	2350
52	9A	C9A	II	MATA MANSA DEVI TO MATA MANSA DEVI	37	30	39	141	4	SUB	1320
53	9C	C9C	II	MATA MANSA DEVI TO MATA MANSA DEVI	37	30	39	139	4	SUB	1587
54	20A	C20A	III	CHD TO KHARAR	25.6	30	39	103	6	SUB	2145
55	25A	C25A	III	CHD TO KHIZRABAD	27	30	39	76	3	SUB	1246
56	30A	C30A	II	ISBT-43 TO CANTONMENT	21.1	30	39	80	5	SUB	2395
57	30B	C30B	II	ISBT-43 TO PANCHKULA SEC-4	20.7	25	33	66	6	SUB	2379
58	32A	C32A	III	NEW ISBT-MOHALI TO MATA MANSA DEVI	23.1	45	59	94	4	SUB	742
59	35B	C35B	III	ISBT-17 TO KHARAR	19.2	20	26	82	6	SUB	3327
60	123A	C123A	II	MOHALI R/STATION TO PGI	19.8	30	39	65	5	SUB	1855
61	20	C20AC	III	CHD TO KHARAR	25.6	30	39	89	6	SUB	3093
62	26	C26AC	II	DHANAS TO DHAKOLI	29	45	59	110	4	SUB	1445
63	36	C36AC	III	PGI TO NEW AIRPORT	30.9	45	59	91	4	SUB	622
64	38	C38AC	III	ISBT-17 TO NEW AIRPORT	21.3	45	59	73	3	SUB	757
65	211	C211AC	II	PGI TO CANTONMENT	22.3	40	52	79	4	SUB	909
66	214	C214AC	II	SP Barrier TO NADA SAHIB	25.9	40	52	106	6	SUB	721

Sr. No.	Routes No.	EMME Code	Depot	Description	Length (km)	Headway (min)		Travel Time (min)	No of Buses	City / Suburban service	Daily ridership
						Peak	Off Peak				
67	216	C216AC	IV	ISBT-43 TO DERABASSI	25.5	20	26	95	8	SUB	5158
68	38A	C38AAC	III	ISBT-17 TO NEW AIRPORT	21.1	45	59	59	3	SUB	884

Source: ETM Data from CTU/CCBSS August 2019

Annexure 2: Boarding – Alighting and section load for each existing route (one sample trip in peak hour)

Route no. 3: Mauli Jagran to PGI via Sector 18/7 Panchkula Housing Board Chowk, Kalagram, Railway Crossing, Railway Station, CTU workshop, 28/26 Grain Market, 19/7, 18/8, 22/17, 16/10 Hospital, 11/10, PEC and OPD

From figure below, it can be seen that in up direction 4 stops from total 17 stops perform boarding activity and 90% of the passengers are alighting at Sector16 and PGI. Whereas the down direction all the intermediate stops show almost same pattern of boarding which is less than 15% of the total boarding and alight at 4 stops namely ISBT17, railway station, housing board and Mauli Jagran. In up direction, the passenger load on similar line to bus capacity till 80% of the start trip, while for the rest trip, passenger load gradually decreases till PGI. In down direction, passenger volume trajectory gradually increases throughout the trip till railway station and is similar to seating capacity. The maximum load in both up and down direction is similar to bus capacity indicating optimum use of fleet during peak hours. The average trip length of passengers on this route is about 8.5 km indicating people travels 50% of the total route length.

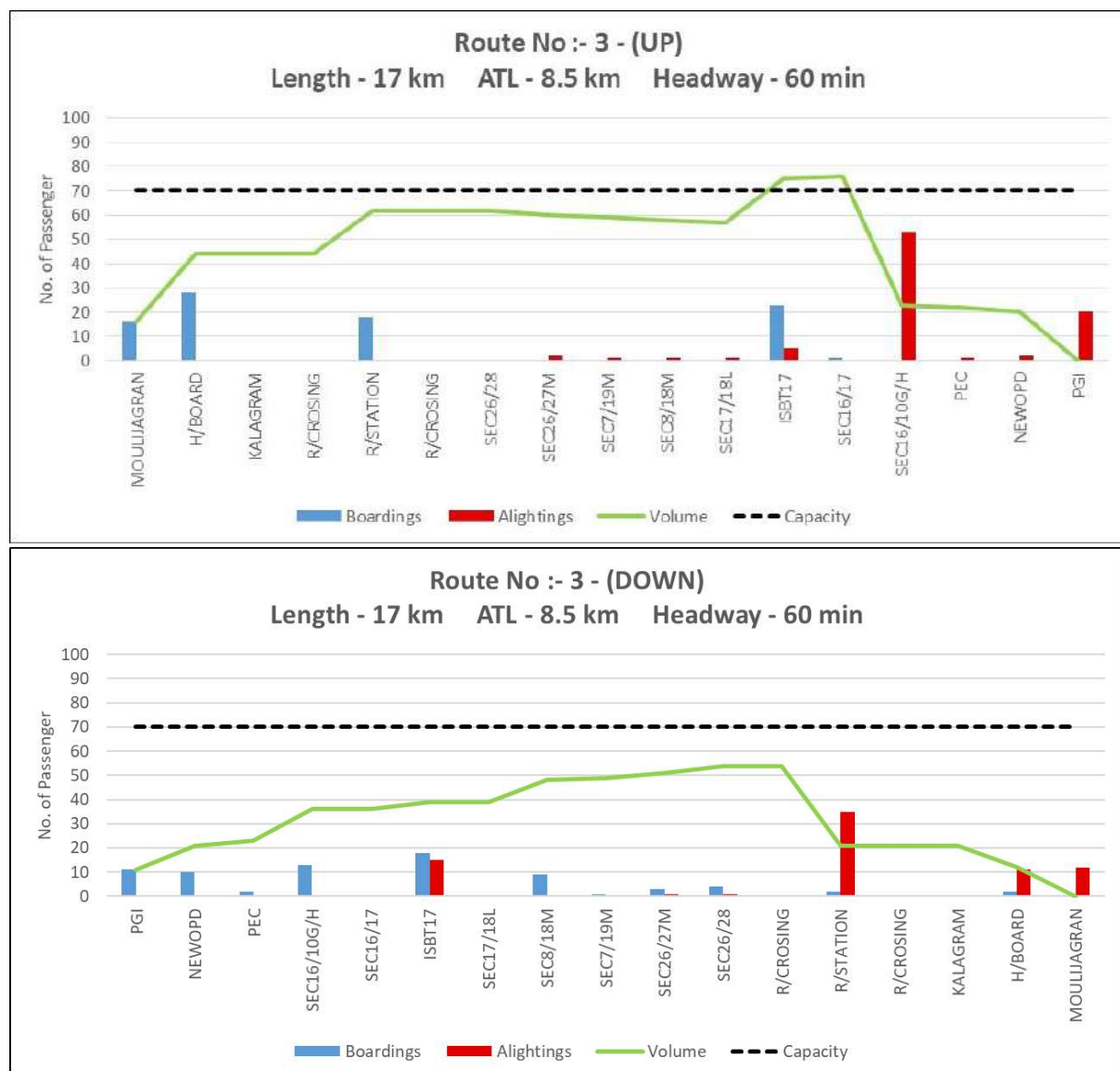


Figure 1: Peak hour Boarding & Alighting pattern along with line load of Route 3 (Up and Down)

Route no. 11: ISBT-17 TO PB. Civil Sectt. Via. Sector 17/18, 9 market, MLA Flat and High Court,

From figure below, it can be inferred that in up direction around 78% passengers are boarding from high court and all the passengers alight at ISBT17 showing no sign of alighting in the rest stops. Whereas in down direction, ISBT17 and high court perform 95% of the boarding and alighting. In both up and down direction, passenger volume in bus is similar to the seating capacity. While the maximum load in both the directions is near to the bus seating capacity average fleet utilization in the route. Also trips performed in this route are short distanced and end-to-end. The average trip length of passengers in this route is almost near to the route length i.e. 7.1 km.

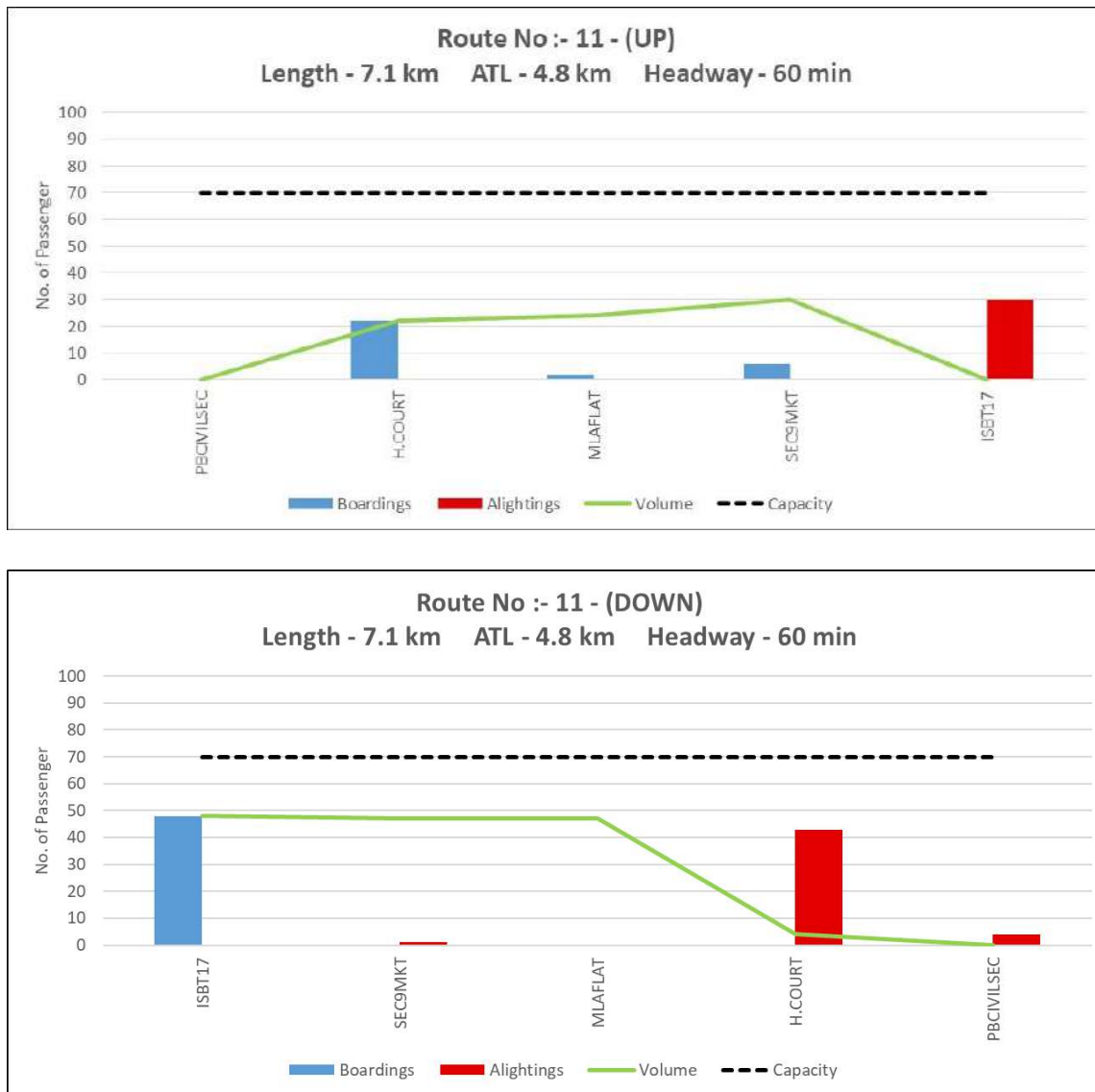


Figure 2: Peak hour Boarding & Alighting pattern along with line load of Route 11 (Up and Down)

Route 17: ISBT-43 TO Airport via Sec 43/44, 42/43, Sec- 36/35, 23/22, ISBT-17, Aroma, Sector 21, Sector 20, Sector 30, Sector 29 markets, Elante Mall, Hallo Majra and Behlana

From below figure, it can be seen that in up direction around 75% of passengers are boarding from Airport and Hallo majra and 90% of them alight at the two terminals ISBT 17 & ISBT 43. Whereas the down direction 95% of passengers board from ISBT17, ISBT 43 and Sector 29 market, while the rest of the stops have zero BA activity. All intermediate stops show almost same pattern which is less than 10% boarding. In up direction, the passenger load is equal to the bus capacity till ISBT17, while for the rest trip, passenger load is less than seating capacity even during peak hour. In down direction, passenger volume in bus is similar to the seating capacity throughout the trip. The maximum load in down direction trip is lower than the bus capacity indicating the fleet is underutilized during peak hours. The average trip length of passengers on this route is about 7.6 km indicating people travels for around 50% of the route length.

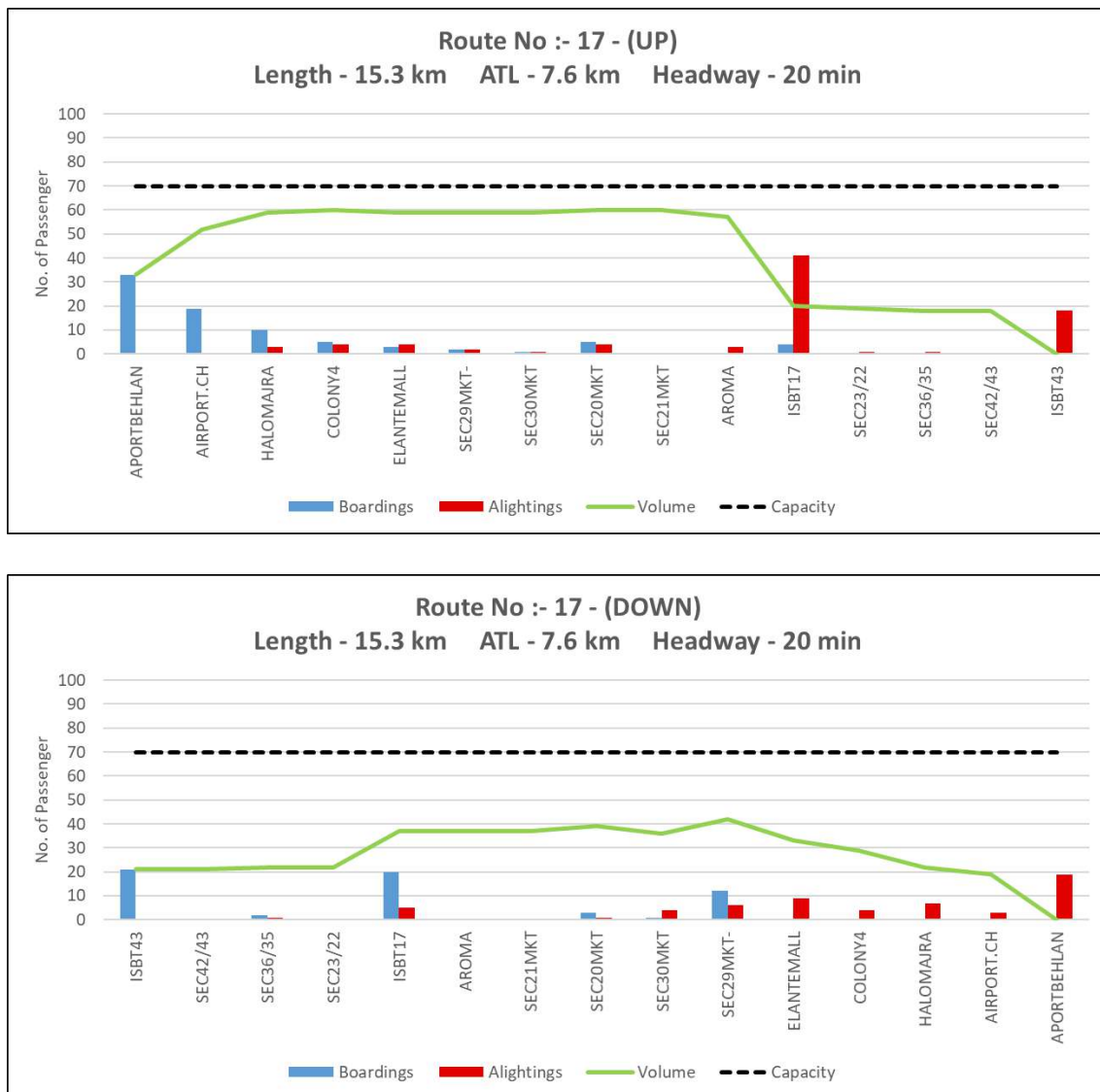


Figure 3: Peak hour Boarding & Alighting pattern along with line load of Route 17 (Up and Down)

Route no. 18: ISBT-43 to Kaimbwala via. Sector 43/44, 42/43, Sector 36/35, 23/22, ISBT-17, 17 Market., U.T., Sectt., Mini Sectt., MLA Flat, High Court and Lake

From figure below, it can be seen that in up direction 75% of the boarding and alighting are performed at highcourt, ISBT17 and ISBT43. Whereas the down direction only 2 out of 11 stops perform boarding namely ISBT17 and ISBT43 and all passengers alight at the last three stops. In up direction, the passenger load on similar line to bus seating capacity throughout the trip. While in down direction, passenger volume occupied in the bus during the trip is half of the bus seating capacity. The maximum load in up direction is similar to bus seating capacity and in down direction max load reaches only 50% of the seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 7.6 km indicating people travels 50% of the total route length.

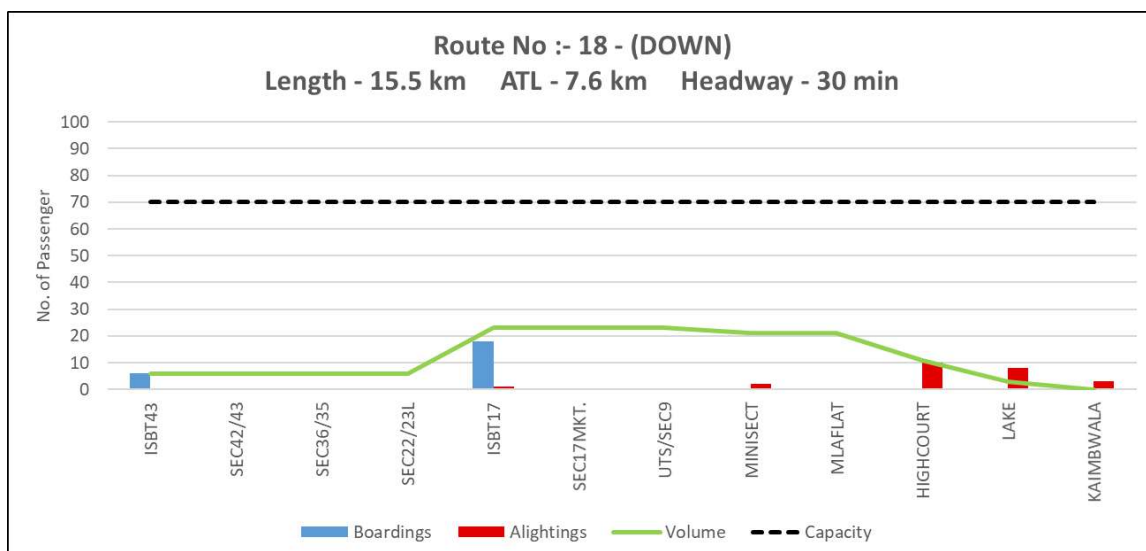
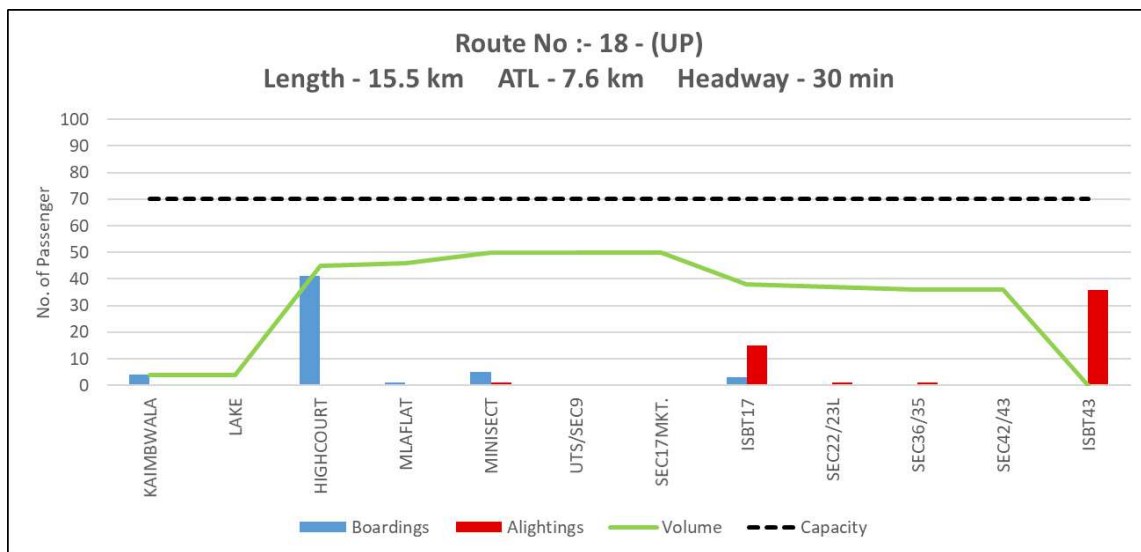


Figure 4: Peak hour Boarding & Alighting pattern along with line load of Route 18 (Up and Down)

Route no 22: ISBT-43 to Mata Mansa Devi via Sector 43/44, 42/43, 36/35, 23/22, ISBT-17, 18, 19, 27, 28 Markets, CTU workshop, Railway Station, Railway Crossing, Kalagram, Housing Board Chowk, Mani Majra, Kishangarh and I.T.Park

From figure below, it can be seen that in up direction around 85% of passenger are boarding from 5 of 22 total stops, while same pattern of alighting is seen which is less than 15% of the total boarding. Whereas the down direction 90% of passenger are boarding from 3 of 22 total stops and 75% are alighting at 3 stops namely ISBT17, railway station and housing board. In both up and down direction, the passenger load is gradually increasing and exceeds bus capacity throughout the trip. The average trip length of passengers on this route is 8.5 km indicating people travels around 45% of the total route length.

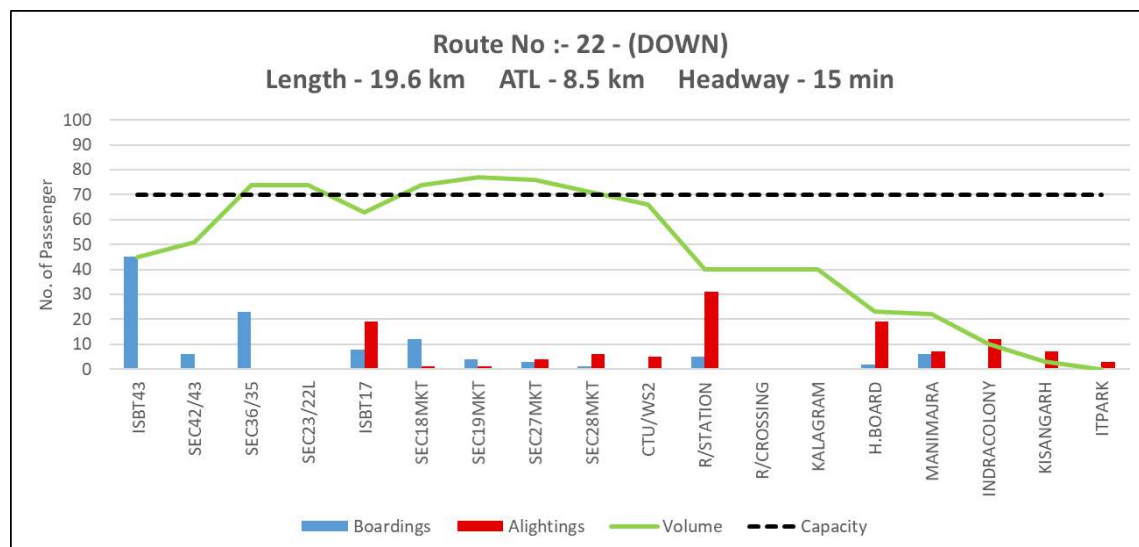
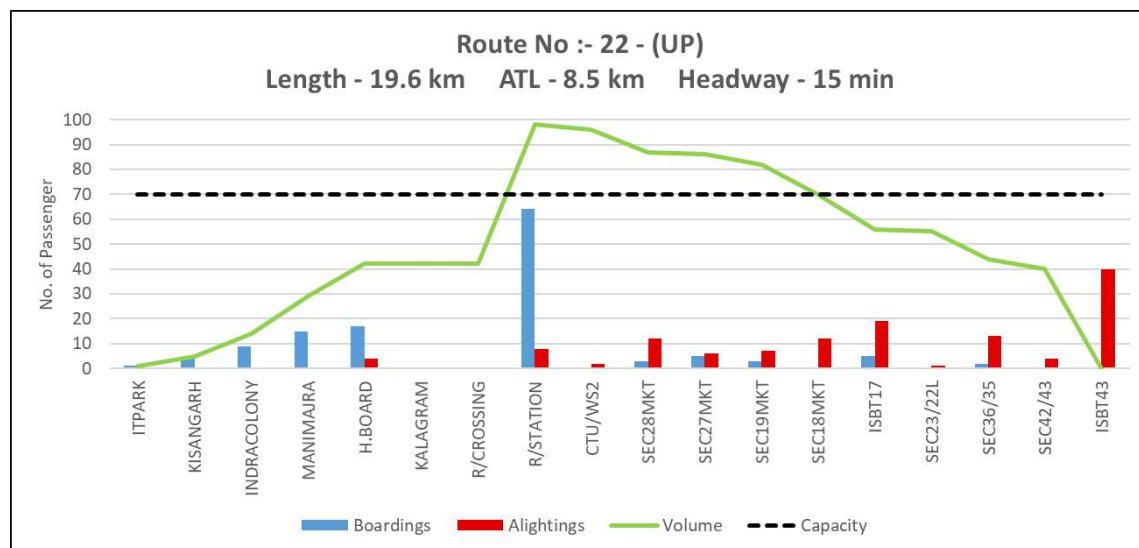


Figure 5: Peak hour Boarding & Alighting pattern along with line load of Route 22 (Up and Down)

Route no 28: Maloya to Mata Mansa Devi via DMC, 38w, 38 Market, 37 Market, 36 Market, 36/35, 23/22, ISBT-17, 8/18,7/19, 26/27, 26/28 Grain Market, CTU workshop, Railway Station, Kalagram ,Housing Borad and Mani Majra.

From figure below, it can be seen that in down direction 85% of passenger are boarding from ISBT 17 and 90% of the passengers are alighting at Railway Station, Housing board and Mansa Devi. Whereas the up direction all the intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load on gradually decreasing to half of the bus seating capacity. In down direction, passenger volume trajectory is near to bus seating capacity throughout the trip. The average trip length of passengers on this route is about 8.9 km indicating people travels 34% of the total route length.

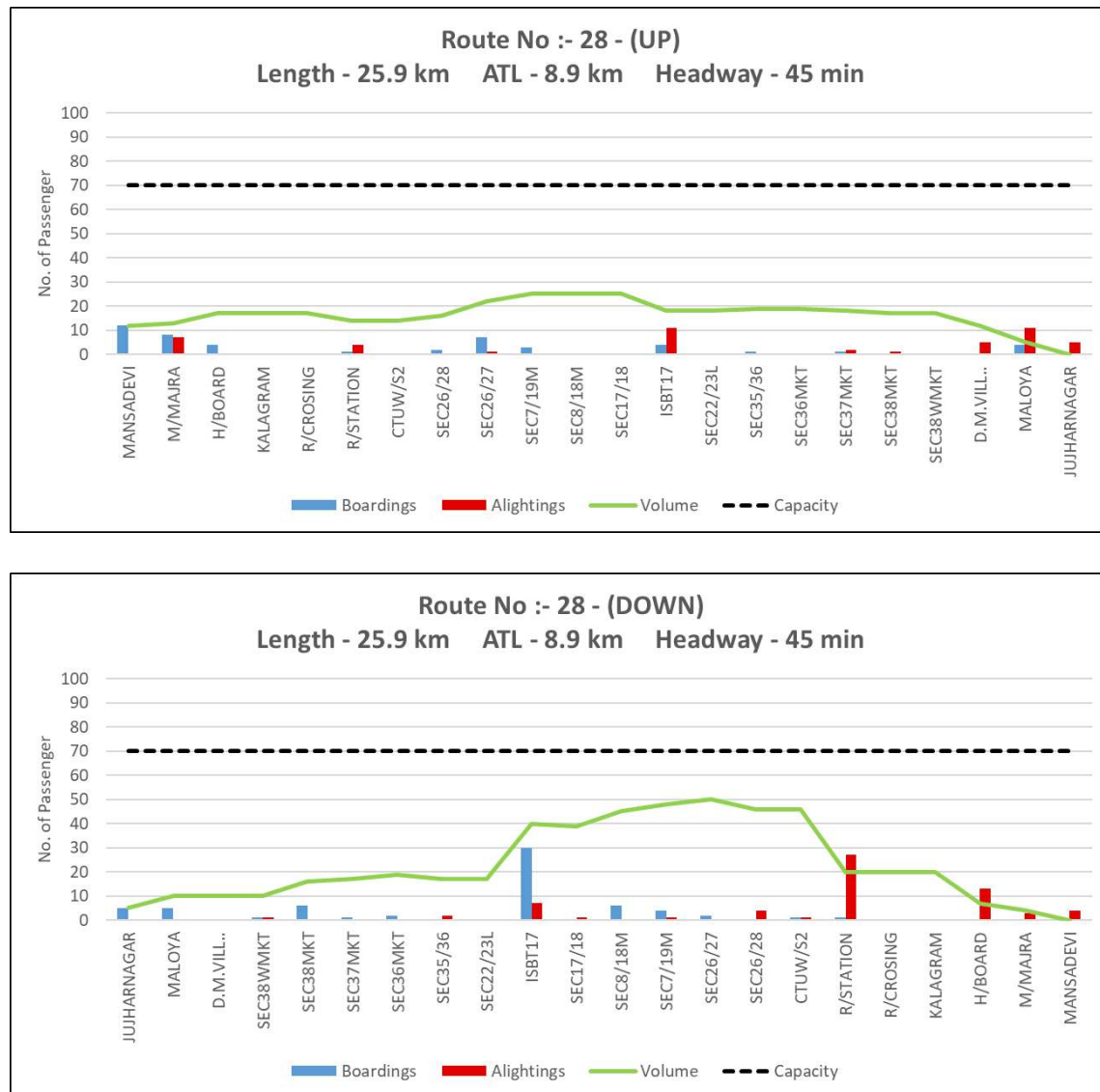


Figure 6: Peak hour Boarding & Alighting pattern along with line load of Route 28 (Up and Down)

Route no 71: ISBT-43 to Saketri (Kaimbwala) via. Sector 43/44, 35/34, Kissan Bhawan, 23/22, ISBT-17, 18, 19, 27 Markets, Grain Market, Railway Crossing, Mani majra Complex, Mani majra Town, Mani Majra, Mansa Devi and Saketri

In Up direction 90% Boarding happen at 4 stops which are Saketri, MansaDevi, M.Complex and ISBT17. Maximum alighting is at ISBT43 along with some at ISBT 17, Manimajra and Mansa Devi. Out of the total 18 stops on this route 10 stops show zero B&A activities. In down direction boarding are seen at 4 major stops ISBT43, ISBT17 and M.Devi. Alighting happen recurringly in the last 5 stops on this route which are Saketri, Manimajra, Indracolony, MMtown and M/Complex. The passenger capacity on this route is close to the seating capacity showing moderate utilization of this fleet. In down direction the passenger flow is close to the bus capacity showing maximum utilization of the fleet. The average trip length is 7.7kms showing short trip lengths.

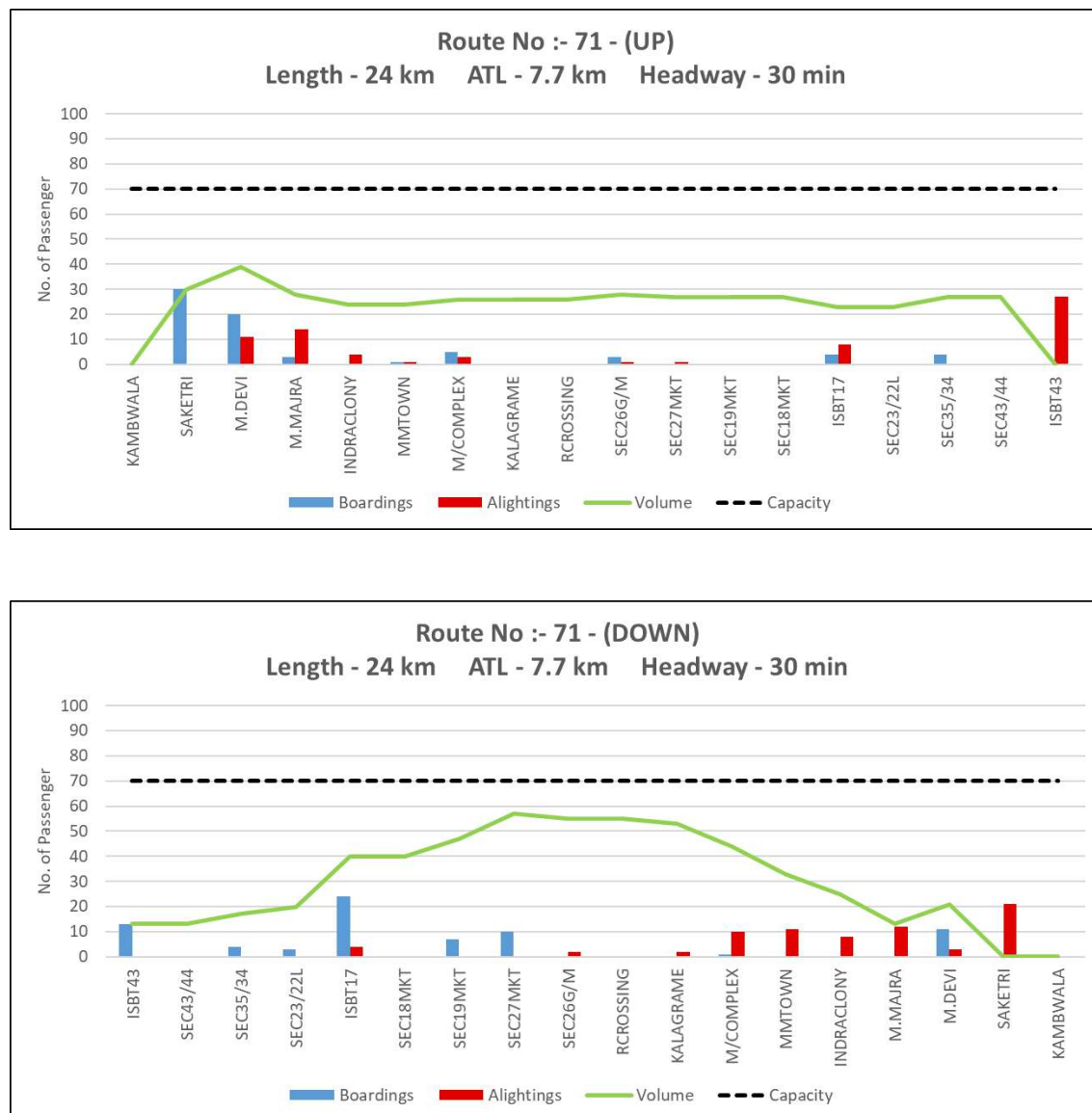


Figure 7: Peak hour Boarding & Alighting pattern along with line load of Route 71 (Up and Down)

Route no. 143: Dhanas to Airport via SP Barrier, Khuda Lahora, PGI , New OPD, GH –16 , ISBT –17, Aroma, Sector 21 Market, Sector 20- Market, Sector 30 Market, Sector 29 Market, Elante Mall, Poultry Farm, Industrial Area Ph-I and Hallo Majra

From figure below, it can be seen that in up direction around 90% of passengers board from Airport-Behilan, Airport-Chonk and ISBT17. Whereas the down direction all the intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load is quite low as compared to the seating capacity till, while for the rest trip, passenger load gradually decreases till Dhanas. In down direction, passenger volume in bus remains less than the seating capacity throughout the trip. The maximum load in both up and down direction is lower than the bus capacity indicating the fleet is underutilized during peak hours. The average trip length of passengers on this route is about 7.4 km indicating people travels shorter distances.

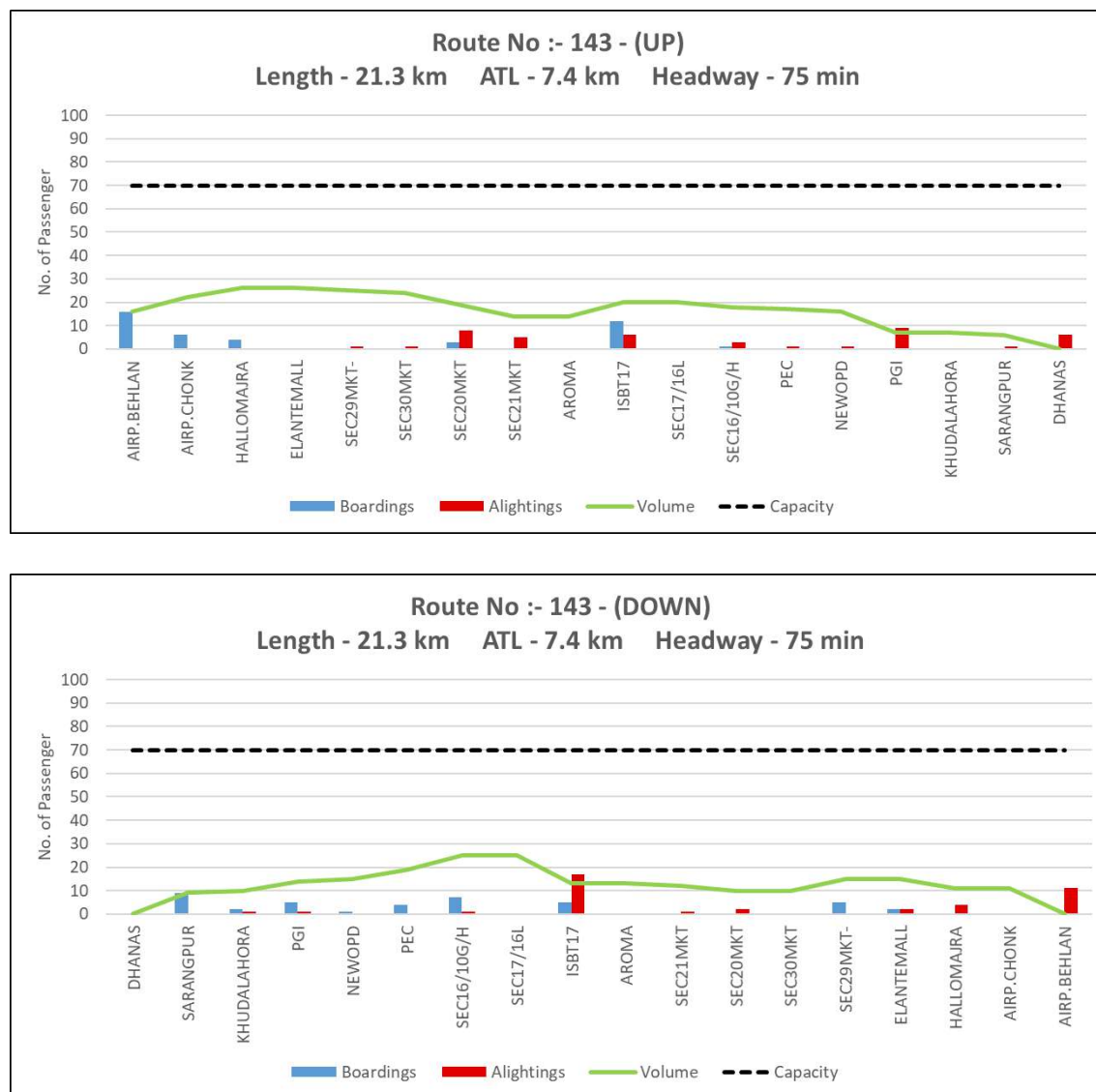


Figure 8: Peak hour Boarding & Alighting pattern along with line load of Route 143 (Up and Down)

Route no. 241: ISBT-43 to Mansa Devi via Sector 52/53, 52/61, YPS, 51 market, Sector 51/50, 44/45, 45 market, 45/46, 33/32, 20/30, 19/27, 27 market, 26 Grain Market, TPT Chowk, CTU workshop, Industrial.Area.Ph-1, Railway Station, Railway Crossing, Kalagram, Housing Board Chowk and Mani Majra

From figure below, it can be seen that in up direction around 85% of passenger are boarding from Manasa Devi, HB Chowk and Railway Station. While Alighting are recurring through the rest of the stops. Whereas in down direction intermediate stops show patterns of boarding and alighting which is close to the total boarding. In up direction, the passenger load on gradually decreasing to half of the bus seating capacity. In down direction, passenger volume trajectory is near to bus seating capacity which increases to the bus capacity gradually towards the end. The maximum load in both up and down direction is similar to bus seating capacity indicating moderate utilization of fleet during peak hours. The average trip length of passengers on this route is about 9.4km indicating people travels 42% of the total route length.

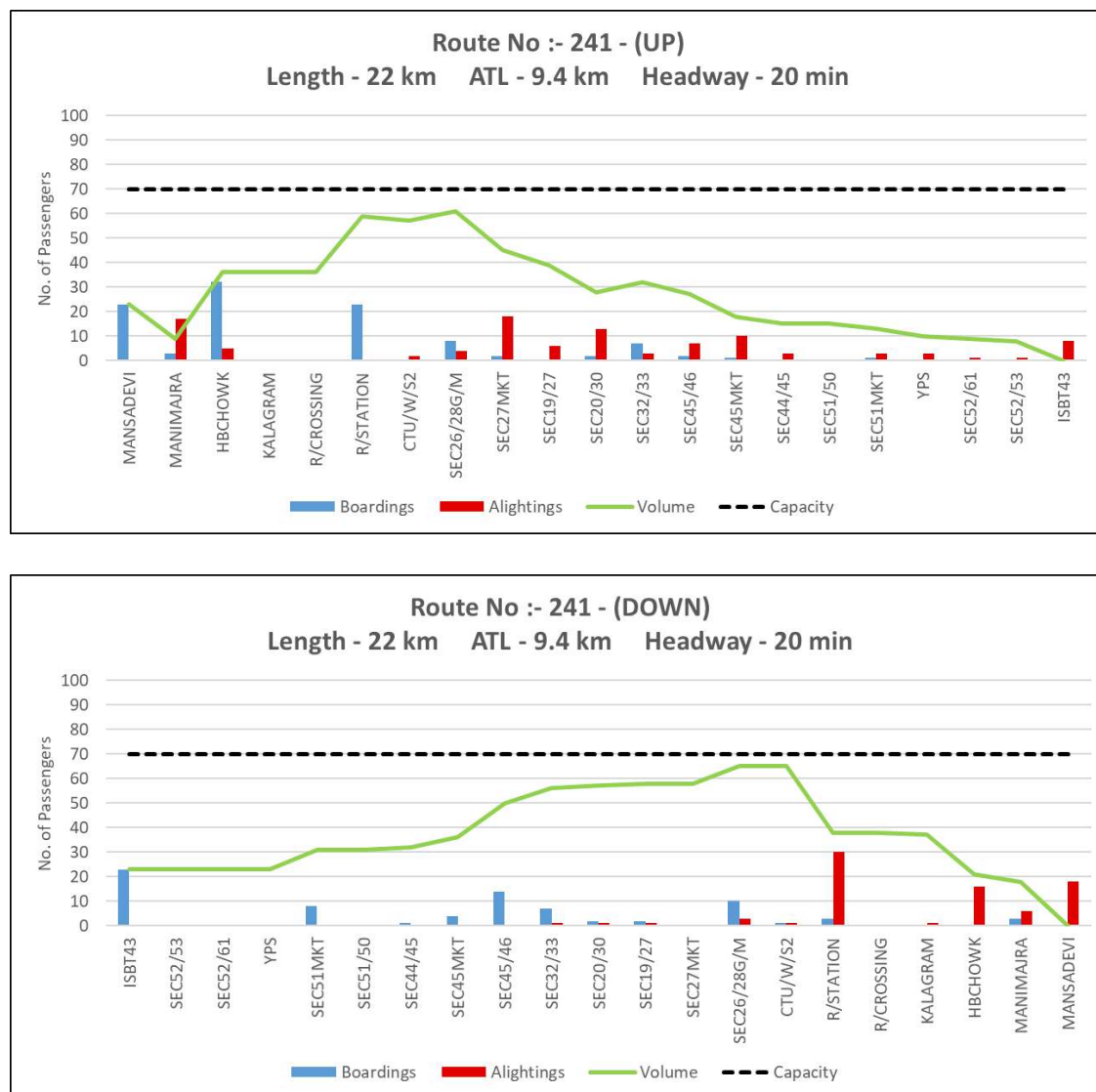


Figure 9: Peak hour Boarding & Alighting pattern along with line load of Route 241 (Up and Down)

Route no. 1C and 1A: PGI to PGI via 15/14, 15 market, 16 market, 17/16, ISBT- 17, 18 market, 19 market, 27 market, 28 market, CTU workshop, Railway Station, Railway Crossing, Housing Board Chowk, Mani Majra, Mani Majra Town, Municipal Complex, Railway Crossing, 28/26, SGGS College, 7 market, 8 market, 9 market, 10 market, 11market and OPD and vice-versa.

From figure below, it can be inferred that in circular route around passengers are boarding from PGI, Mani Majra and Chandigarh railway station, while the rest stops show zero BA activity. Also it is quite evident that 85% of the people alight at PGI, Sector 26 and ISBT17. Whereas in anti-clockwise direction trip, all the intermediate stops show similar pattern of boarding and alighting which is less than 15% of the total boarding. In clockwise direction, passenger volume is near to the seating capacity between Railway station to ISBT17 and PGI to Sector 26, while for the rest trip, passenger load remains steady occupying only 25% of the seating capacity. The maximum load in both up and down direction is near to the bus capacity indicating that optimum utilization of fleet. The average trip length of passengers on this route is 7.4 km indicating people travels shorter distances.

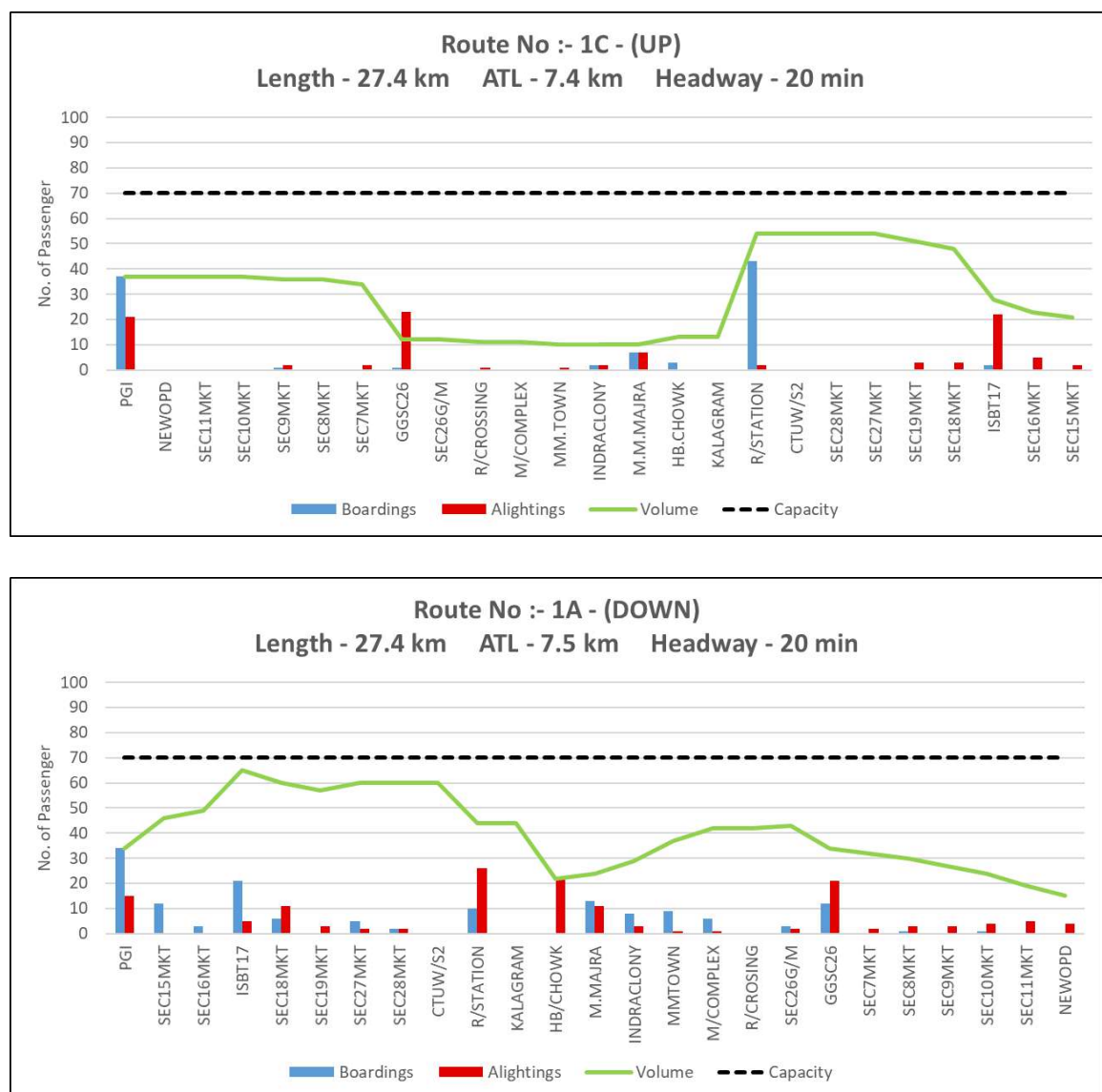


Figure 10: Peak hour Boarding & Alighting pattern along with line load of Route 1A and 1C

Route no 2C & 2A: Mani Majra – Mani Majra via Indira Colony, M.M. Town, M/Complex, R/Crossing, Sec-28/26 G. Mkt, 19/7, 18/8, 22/17, 16/10, 15/11, PGI, Khuda Lahora, S/P Barrier, Dhanas, DMC, CTU Workshop, Sec-25/38, 25/24, 24/15, 23/16, ISBT-17, 18/21, 19/20, 27/30, 28/29, Elante Mall, Colony No.- 4, CTU Workshop, R/Station, R/Crossing, H.B. Chowk and vice versa

From figure below, it can be inferred that in circular route around passengers show similar pattern of boarding and alighting which is less than 15% of the total boarding and 50% of the passengers alight at Mani Majra and ISBT17. Whereas in anti-clockwise direction trip, all the intermediate stops show similar pattern of boarding and alighting which is less than 15% of the total boarding. In anti-clockwise direction, passenger volume in bus near to bus seating capacity till Dhanas which is 50% of trip and the rest trip has passenger volume 75% of the bus seating. The maximum load in both up and down direction is near to the bus seating capacity indicating that optimum utilization of fleet. The average trip length of passengers on this route is 8.9 km indicating people travels shorter distances.

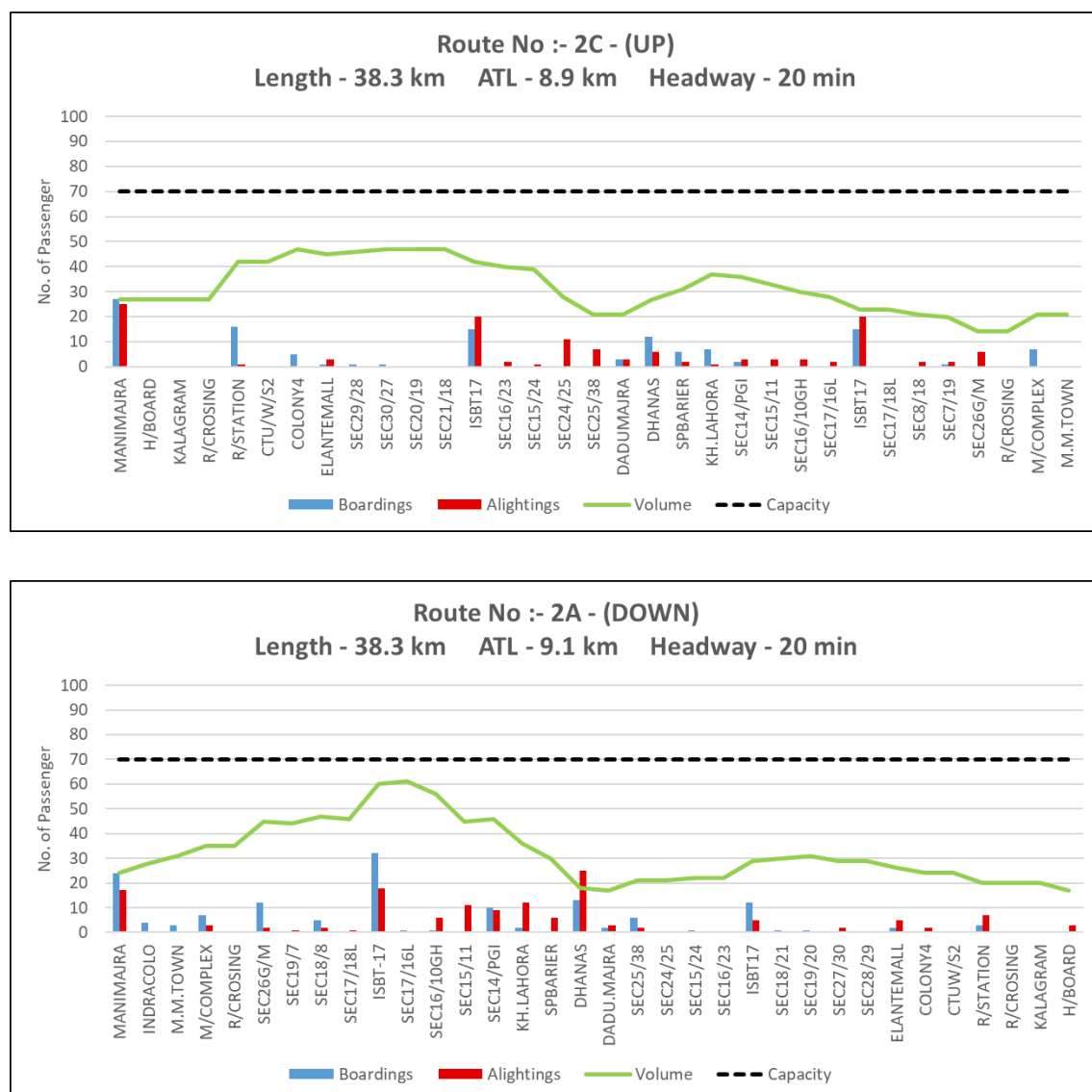


Figure 11: Peak hour Boarding & Alighting pattern along with line load of Route 2A and 2C

Route no 2D: PGI TO IT Park via New OPD, 10/11,10/16, ISBT-17, 8/18, 7/19, 26/27, Grain Market, CTU Workshop, Railway Station, Railway Crossing, Kalagram , Housing Board, Manimajra, Indira Colony and Kishangarh.

From figure below, it can be seen that in up direction all stops perform similar boarding which is less than 10% of the boarding and 80% of the passengers are alighting at housing board chowk, PEC and PGI. Whereas the down direction all intermediate stops perform similar boarding and alighting during the trip which is less than 10% of the boarding. In up direction, the passenger load exceeds bus capacity throughout the trip. While in down direction, passenger volume remains similar to total bus seating capacity overall the trip. The maximum load in both up direction is similar to bus capacity indicating optimum utilization of fleet during peak hours. While down direction trip indicates maximum passenger load similar to seating capacity. The average trip length of passengers on this route is about 11.2 km indicating people travel 40% of total route length.

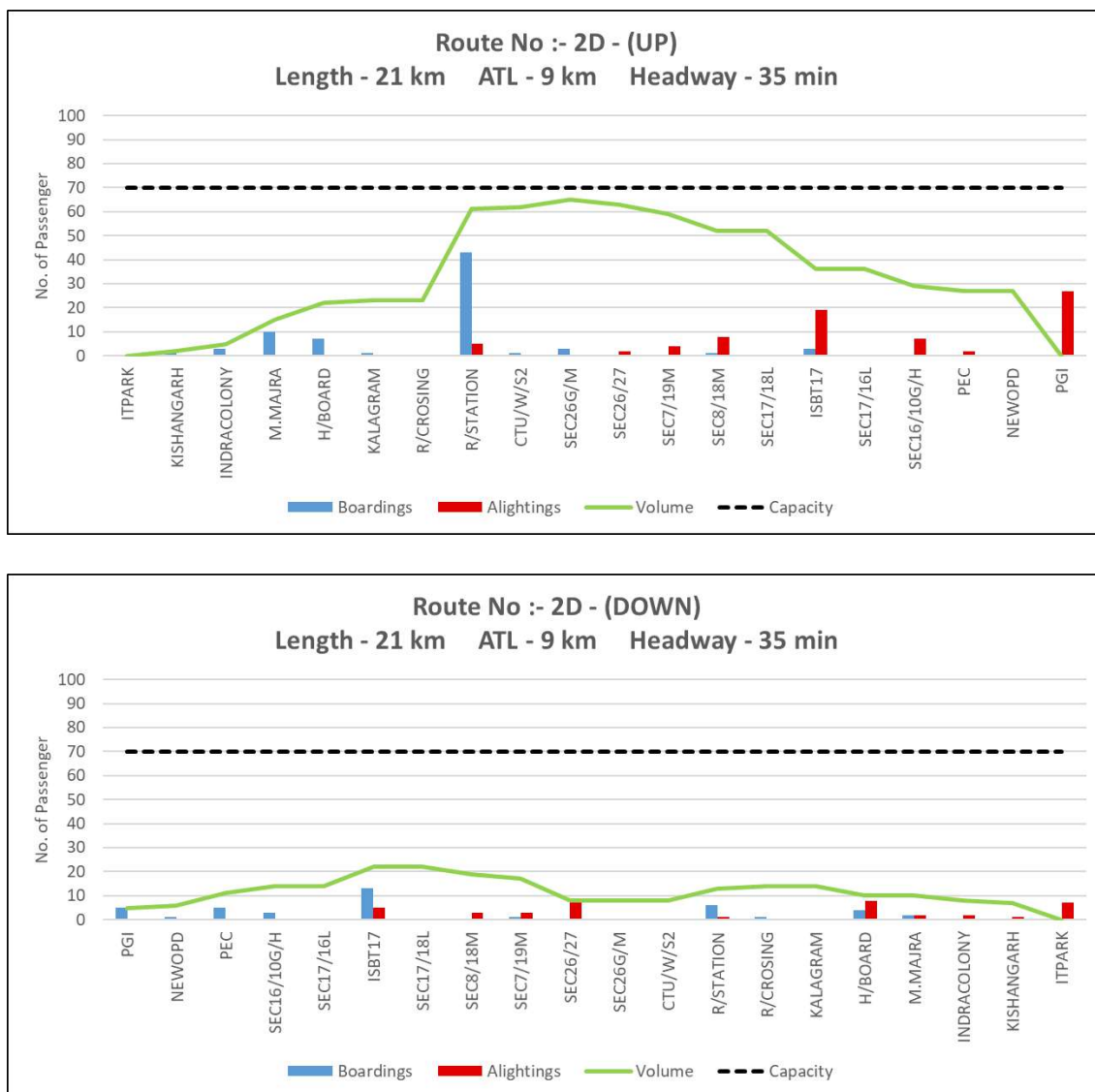


Figure 12: Peak hour Boarding & Alighting pattern along with line load of Route 2D (Up and Down)

Route no. 4C & 4A: ISBT-43 to ISBT-43 via Sector 43/44, 42 market, 41 market, 41/40, Badheri Chowk, Frenko Hotel, O/Barrier, Ph-6, Maloya, 38west, CTU Depot-III, DMC, Dhanas, S/P/Barrier, Khuda Lahora, PGI, OPD, 10/11, 10/16,17/16, ISBT-17, Aroma, 44/43 and vice versa.

From figure below, it can be inferred that in circular route around passengers show 90% boarding from 4 stops ISBT43, Sector 56, Dhanas and SP Barrier. 90% Alighting occur majorly from ISBT 43, Sector 56, Dhanas and SP Barrier. In up direction the passenger drop to less than 10 at KH.lhora stop and boarding go up from next stop which is PGI. The maximum load in both up and down direction is near to the bus capacity indicating that optimum utilization of fleet. The average trip length of passengers on this route is 8.1 km indicating people travels shorter distances.

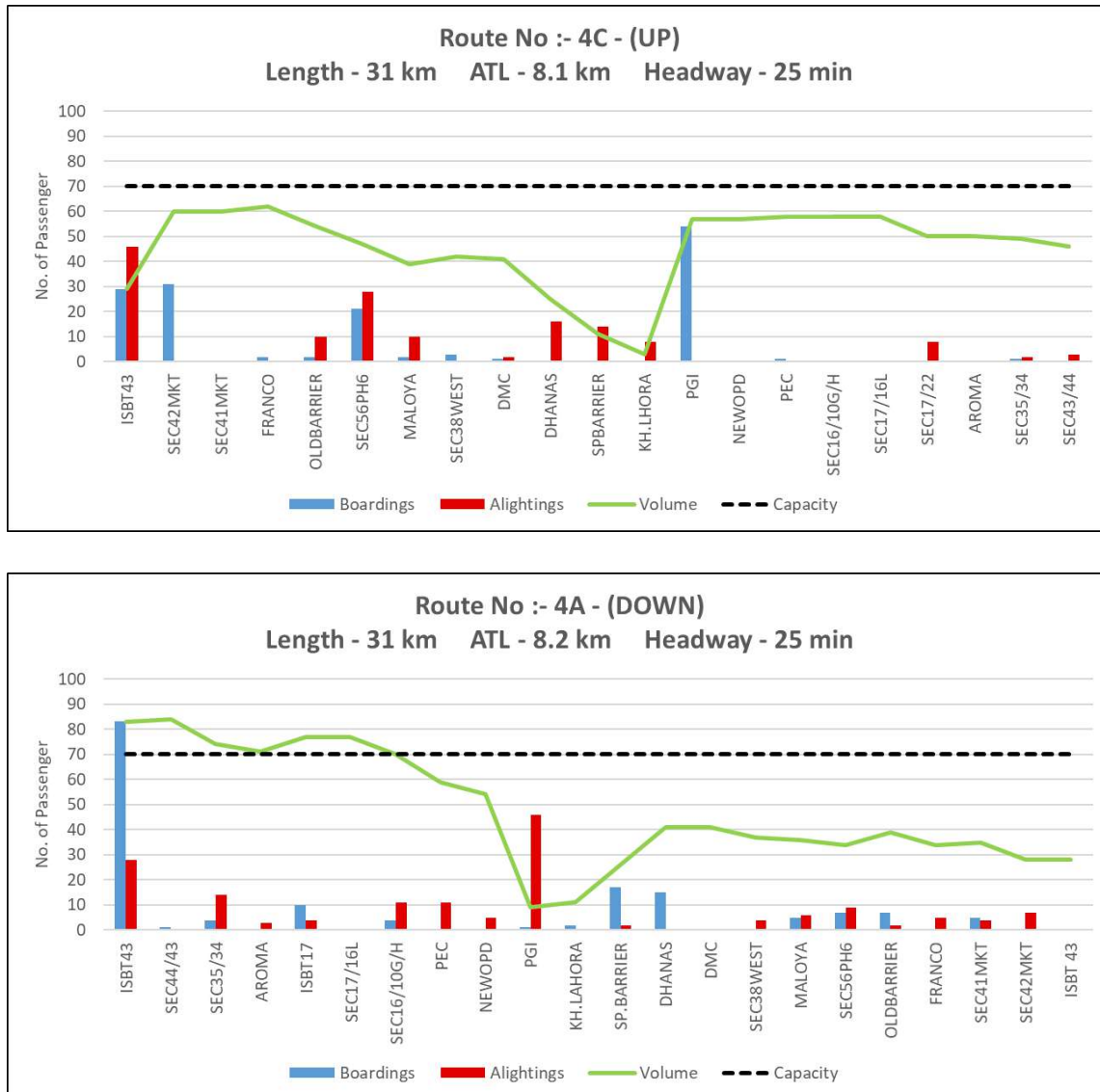


Figure 13: Peak hour Boarding & Alighting pattern along with line load of Route 4A and 4C

Route no. 5C & 5A: Ram Darbar to Ram Darbar via Sector 47 market, 46, 45, 44 markets, ISBT-43, 43/44, 42, 41, 40, 39 markets, Maloya, 38/39, 38/40, 38/37, 25/24, 14/15, University, PGI, 11/15, 10/16, 9/17, 8/18, 19/7, 26/27, 28/27, 29/30, 31/32, 47/46, 47 market

In this Circular route Boarding and alighting are seen to happen at all 27 stops in a recurring manner. 6 stops on this route show comparatively lesser B&A activities amongst others. In anti-circular route as well all stops are used for Boarding and alighting activities. Alighting are majorly from Sector 11/15, Ramdarbar, Maloya and IBST 43. The passenger volume in Up direction is more than the bus capacity which shows optimum utilization of the fleet and in down direction as well the capacity is more than the bus capacity which shows optimum utilization of this route as well. The average trip length is 7.3 km showing passengers travel for 25% of the trip length.

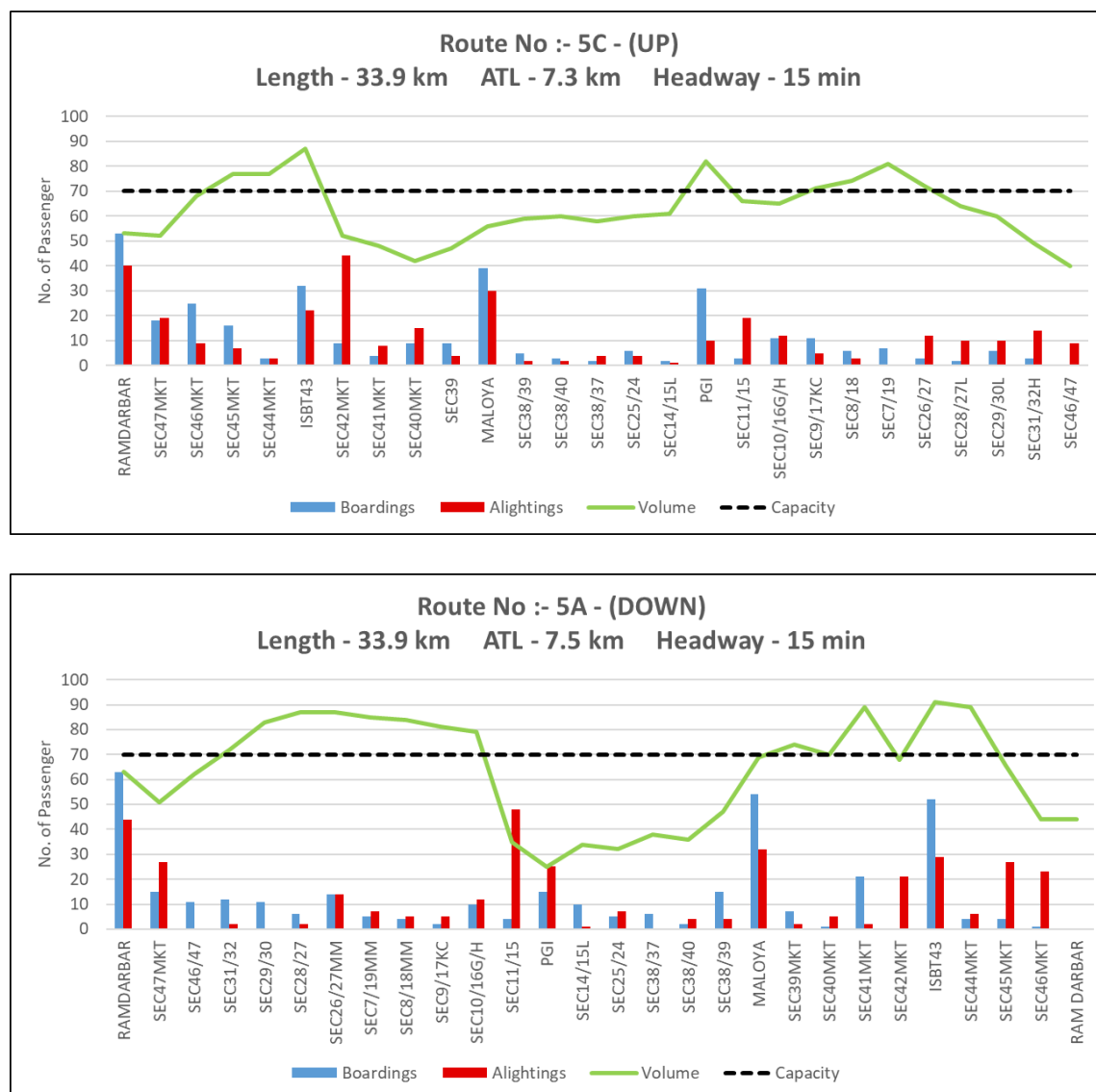


Figure 14: Peak hour Boarding & Alighting pattern along with line load of Route 5A and 5C

Route no. 7C & 7A: ISBT-17 to ISBT-17 via 22/23, 22 market, 21, 20, 30, 29 market, Tribune, Ram Darbar, 31, 32, 33, 34, 35 market, Attawa, ISBT-43, 43/44, 42 market, 41/42, 37/36, 37, 38 markets, 39/38 west. Maloya, 38 west, 25/38, 25/24, 24, 23 markets, 23/22.

From figure below, it can be inferred that in circular route around passengers show similar pattern of boarding and alighting which is less than 15% of the total boarding and 70% of the passengers alight at Sector 23 MKT. Whereas in anti-clockwise direction trip, all the intermediate stops show similar pattern of boarding and alighting which is less than 15% of the total boarding. In clockwise direction, passenger volume is near to the seating capacity throughout the trip. In anti-clockwise direction, passenger volume in bus near to bus seating capacity throughout the trip. The maximum load in both up and down direction is near to the bus seating capacity indicating that optimum utilization of fleet. The average trip length of passengers on this route is 6.8 km indicating people travels shorter distances.

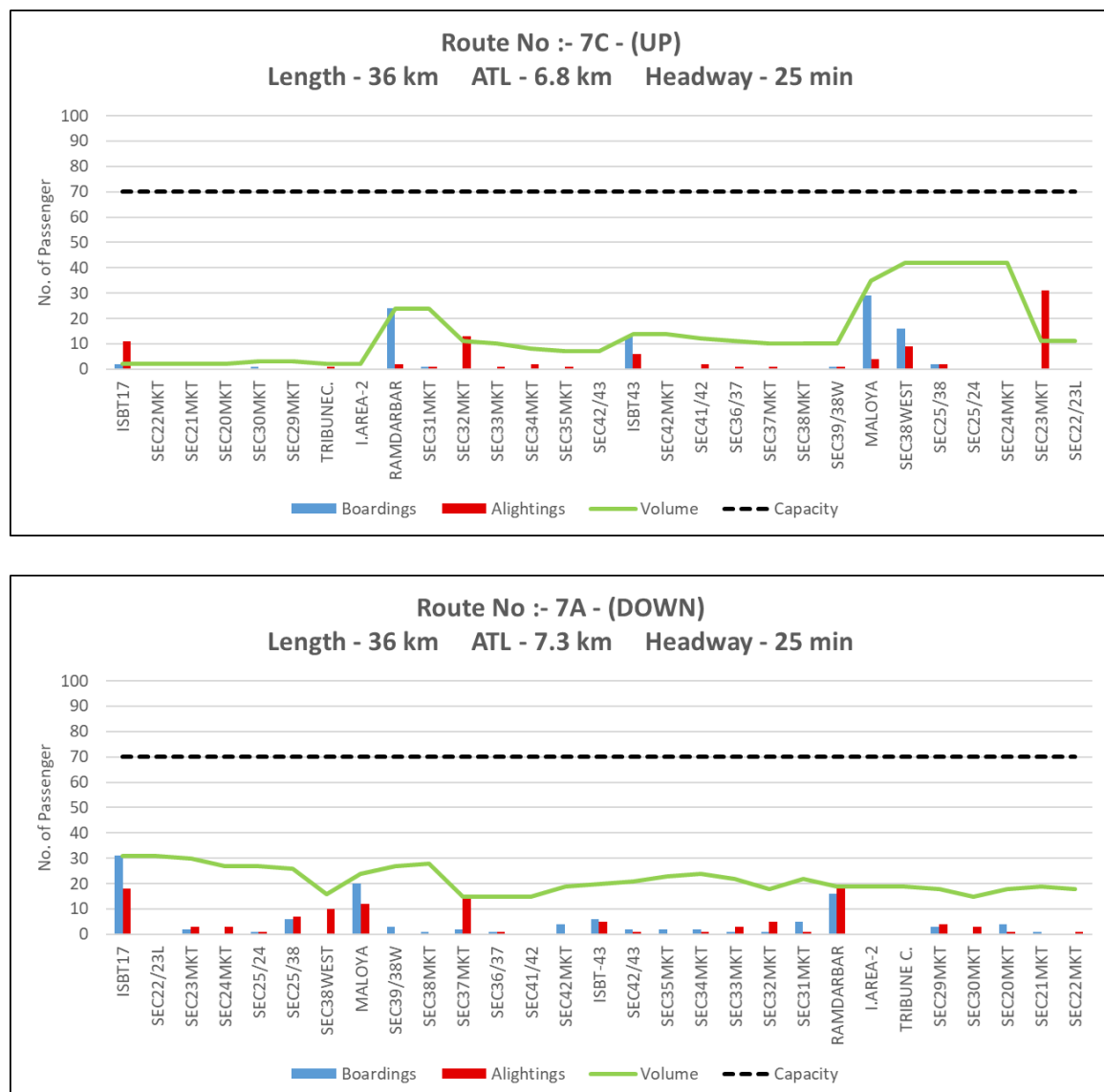


Figure 15: Peak hour Boarding & Alighting pattern along with line load of Route 7A and 7C

Route no 18B: ISBT-43/Session Court to High Court via Sector 43/44, 42 market, 41/42, 37/36, 37 market, 38/37, 24/37 Consumer/Court , 23/36, 23/22, Sector17 CAT, ISBT-17, 17/18, 9/8, 9 market, MLA Flats and High Court

From figure below, it can be seen that in up direction around all passengers are boarding from High court, Mini Sector and ISBT17. 95% passengers are alighting performed ISBT43. Whereas in down direction both the terminals perform boarding activity and 95% passengers alight at ISBT17 and High court. In both up and down direction passenger volume remains half of bus seating capacity throughout the trip. The maximum load in both up and down direction is half of bus seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 8.6 km indicating people trips similar to the route length.

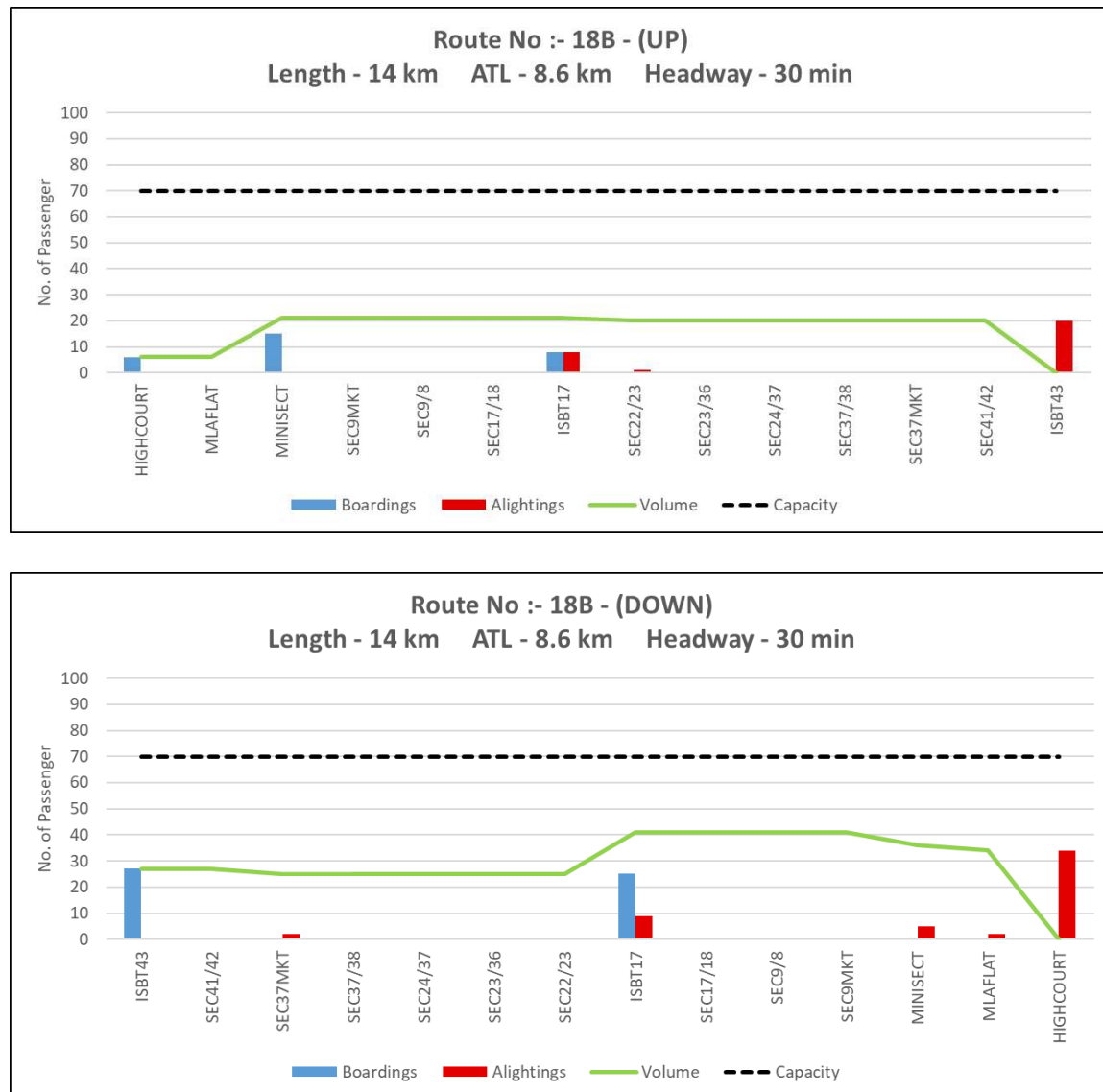


Figure 16: Peak hour Boarding & Alighting pattern along with line load of Route 18B (Up and Down)

Route no. 23A: ISBT-43 TO Khuda Ali Sher via. Sector 43/44, 35/34, Aroma, 22/17, Hospital 16/10, 15/11, PGI, New OPD, PEC and Naya Gaon

Route 23A shows Boarding from 4 stops out of total 13 stops, Passengers board from Kudalisher, Nayagaon, Engineering College and Sector 22/17 and Major Alighting at ISBT43, Sector 35/34 and PGI. Three of the total 13 stops have zero B&A activities. In down route 95% boarding are from only one stop which is ISBT 43 and alight at PGI, OPD and Sector 15 stops. The passenger density on these route is near to the seating capacity showing good usage of the fleet in both up and down direction. The Average Trip Length is 6.3kms showing that people travel 45% of the total route length.

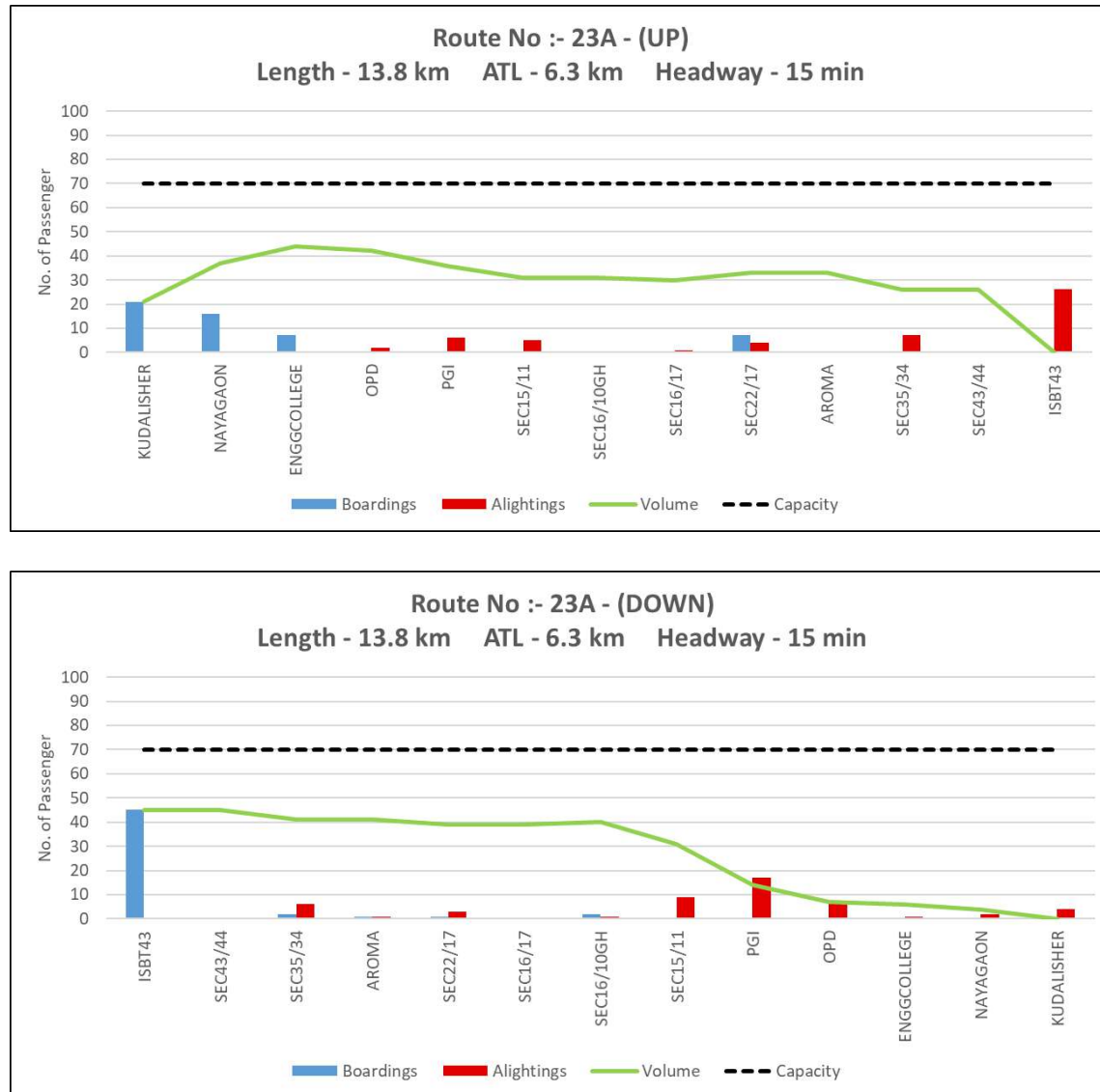


Figure 17: Peak hour Boarding & Alighting pattern along with line load of Route 23A (Up and Down)

Route no. 28C & 28A: ISBT-43 to ISBT-43 via Sector 35/34, Aroma, Sector 22/17, 16/10, OPD, PGI, Khuda Lahora, IRB Camp, S.P. Barrier, Dhanas, Dadu Majra, Sec-25/38, 24/37, Kissan Bhawan, Sector 35/36, Sector 43/42 and vice versa

In this Circular route Up direction shows Boarding from 3 stops majorly, which are ISBT 43, PGI and New OPD. Alighting are seen at 4 stops which are ISBT43, DMC, ISBT17 and Dhanas. Out of total 20 stops on this route, 14 stops show less to zero B&A activities happening. In Down route all stops show recurring Boarding and alighting activities. The passenger capacity in up direction is less than the seating capacity for the first half which drops to zero at Khuda Lahora stop and rises from PGI stop and stays put till the end with passenger flow close to the Bus capacity showing good utilization of the fleet. In Down direction the passenger flow is lesser than the seating capacity which drops even lower after PGI stop showing under utilization of the fleet. Average trip length is 7.7kms showing that passengers take short trips on this route.



Figure 18: Peak hour Boarding & Alighting pattern along with line load of Route 28A and 28C

Route no. 10AC: ISBT-43 to Mata Mansa Devi via. Sector 43/44, Sector 44/51, 45, 46, 47 markets, Ram Darbar, Tribune Chowk, Hallo Majra Chowk, Colony No.-4, CTU workshop, Railway Station, Kalagram , Housing Board Chowk and Mani Majra

In Up direction 90% Boarding happen at 3 stops which are Mansadevi, Housing Board and Railway Station stops. Maximum alighting is at ISBT43 along with some at R/Darbar and Tribune. Out of the total 16 stops on this route 7 stops show low to nil B&A activities. In down direction boarding are seen at 3 major stops ISBT43, Railway Station and Poltri Farm. Alighting happen in a recurring manner in the last 3 stops on this route which are Mansadevi, M.Majra and Housing Board. The passenger capacity in up direction on this route is close to the bus capacity showing maximum utilization of this fleet. In down direction the passenger flow is close to the seating capacity showing moderate utilization of the fleet. The average trip length is 9.4kms showing passengers travel for 48% of the trip lengths.

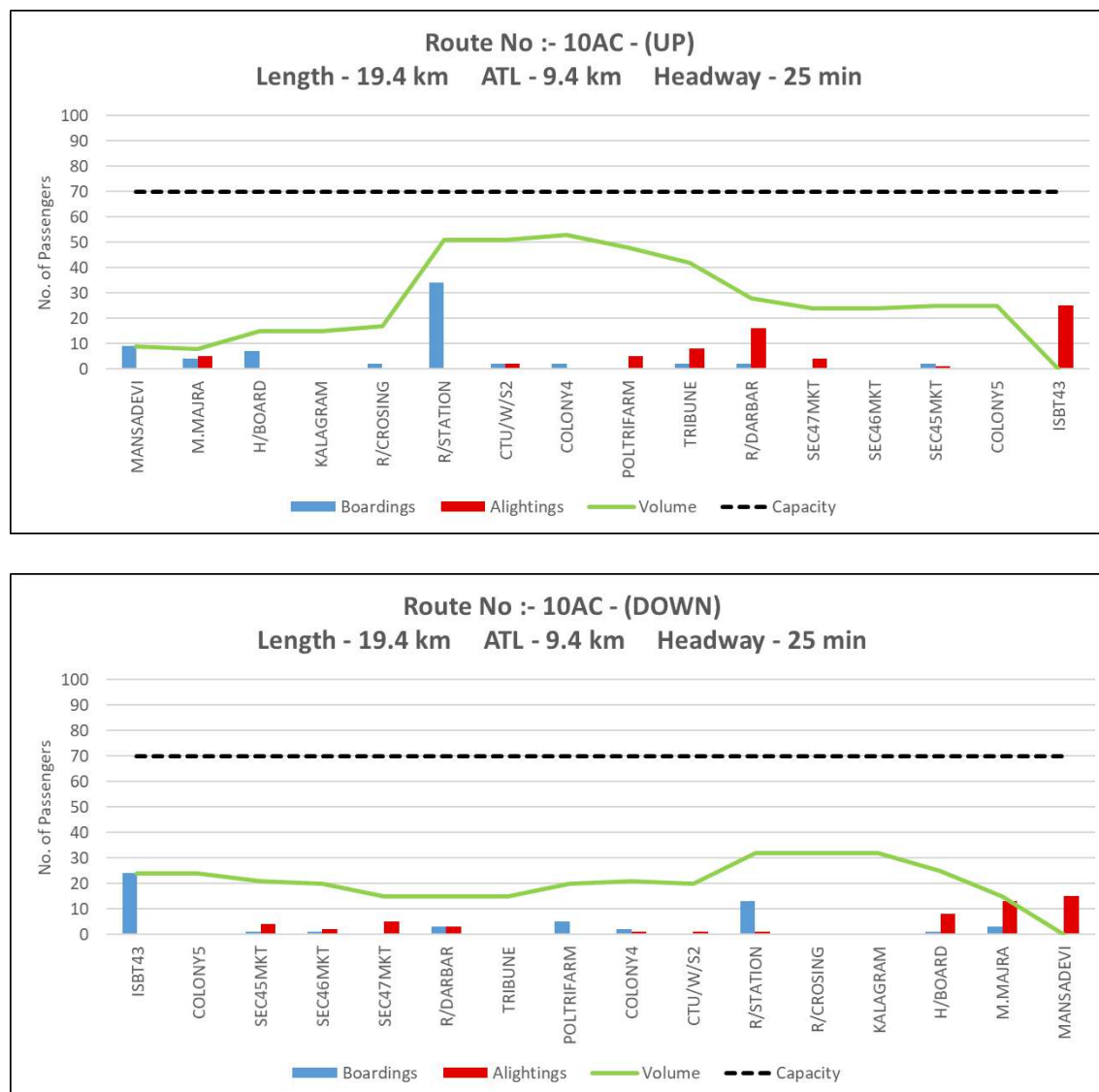


Figure 19: Peak hour Boarding & Alighting pattern along with line load of Route 10AC (Up and Down)

Route no. 37AC: ISBT 43 to Railway Station via Sector 43/44, 42/43, Sector 36/35, 23/22, ISBT-17, Sec 18 market, Sec19 market , Sec 27 market, 28 market and CTU workshop

From figure below, it can be seen that in up direction around all passengers are boarding from ISBT17 and railway station and 95% passengers are alighting performed ISBT43. Whereas in down direction both the terminals perform boarding activity and 95% passengers alight at ISBT17 and railway station. In both up and down direction passenger volume remains half of bus seating capacity throughout the trip. The maximum load in both up and down direction is half of bus seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 7.8 km indicating people trips similar to the route length.

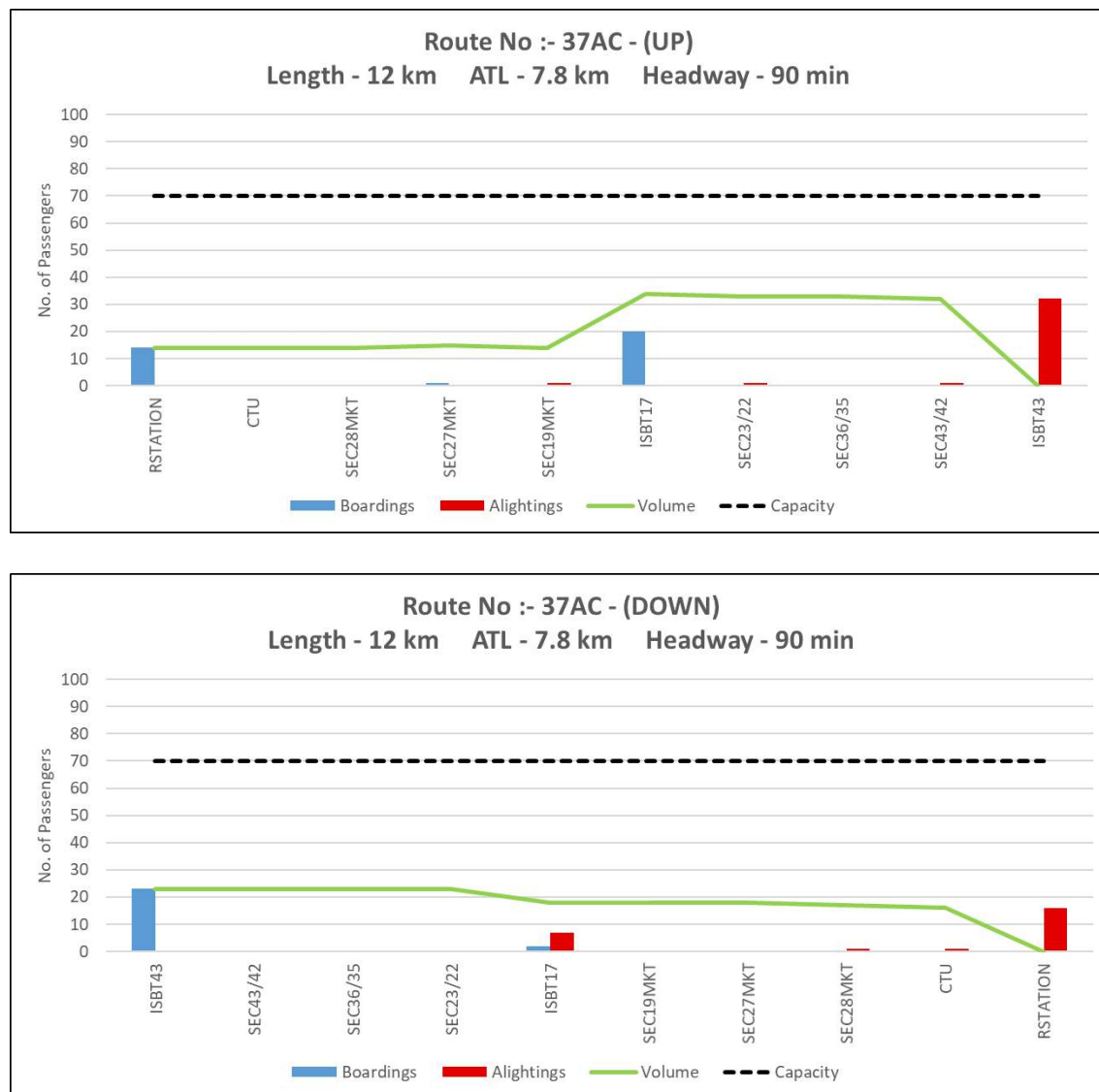


Figure 20: Peak hour Boarding & Alighting pattern along with line load of Route 37AC (Up and Down)

Route no. 202AC: ISBT-43 to PB. Civil. Sectt. (High Court) via Sector 43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, K.C,U.T.Sectt, MLA Flat, High Court and Pb civil. Sectt.

In the figure below, it can be seen that in up direction 4 stops out of 11 stops perform boarding from PB civil, high court, mini sector and ISBT17 and 75% are alighting at last stop i.e. ISBT43. Whereas in down direction 2 terminals perform boarding and passengers mainly alight at ISBT17, mini sector, high court and PB civil. In up direction passenger line load remains steady which is similar to bus seating capacity throughout the trip. While in down direction line load is similar to bus capacity throughout the trip. Maximum passenger load line in up direction almost reaches bus seating capacity and in down direction it exceeds bus capacity indicating optimum use of fleet during peak hours. The average trip lengths of passengers are 6.9 km indicating passenger travel around half of the route length.

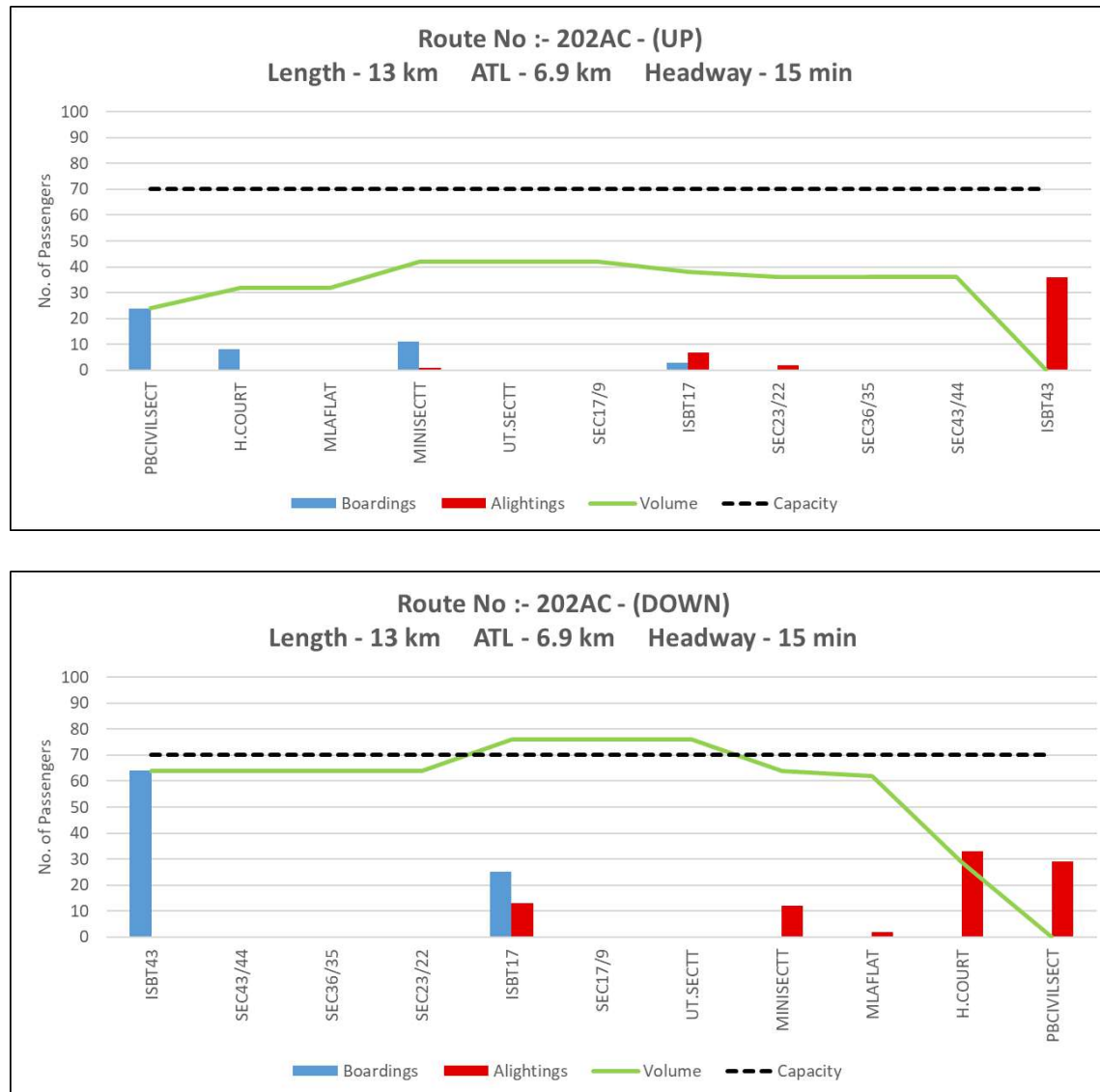


Figure 21: Peak hour Boarding & Alighting pattern along with line load of Route 202AC (Up and Down)

Route no. 203AC: ISBT-43 to Mata Mansa Devi via Sector 43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, 18, 19, 27, 28 markets, CTU workshop, Railway Station, Railway Crossing, Kalagram, Housing Board Chowk and Mani Majra

In the figure below, it can be seen that in up direction 90% passengers board from railway station and are alighting at last stop i.e. ISBT43. Whereas in down direction 2 terminals and sector 18 perform 90% of boarding and passengers mainly alight at ISBT17, railway station and Mansa Devi. In up direction passenger volume gradually increases and further remains steady which is above bus seating capacity throughout the trip. While in down direction line load is almost half of bus seating capacity till railway station. Maximum passenger load line in up direction exceeds bus seating capacity and in down direction it is half of the bus seating indicating moderate use of fleet during peak hours. The average trip lengths of passengers are 9.3 km indicating passenger travel around half of the route length.

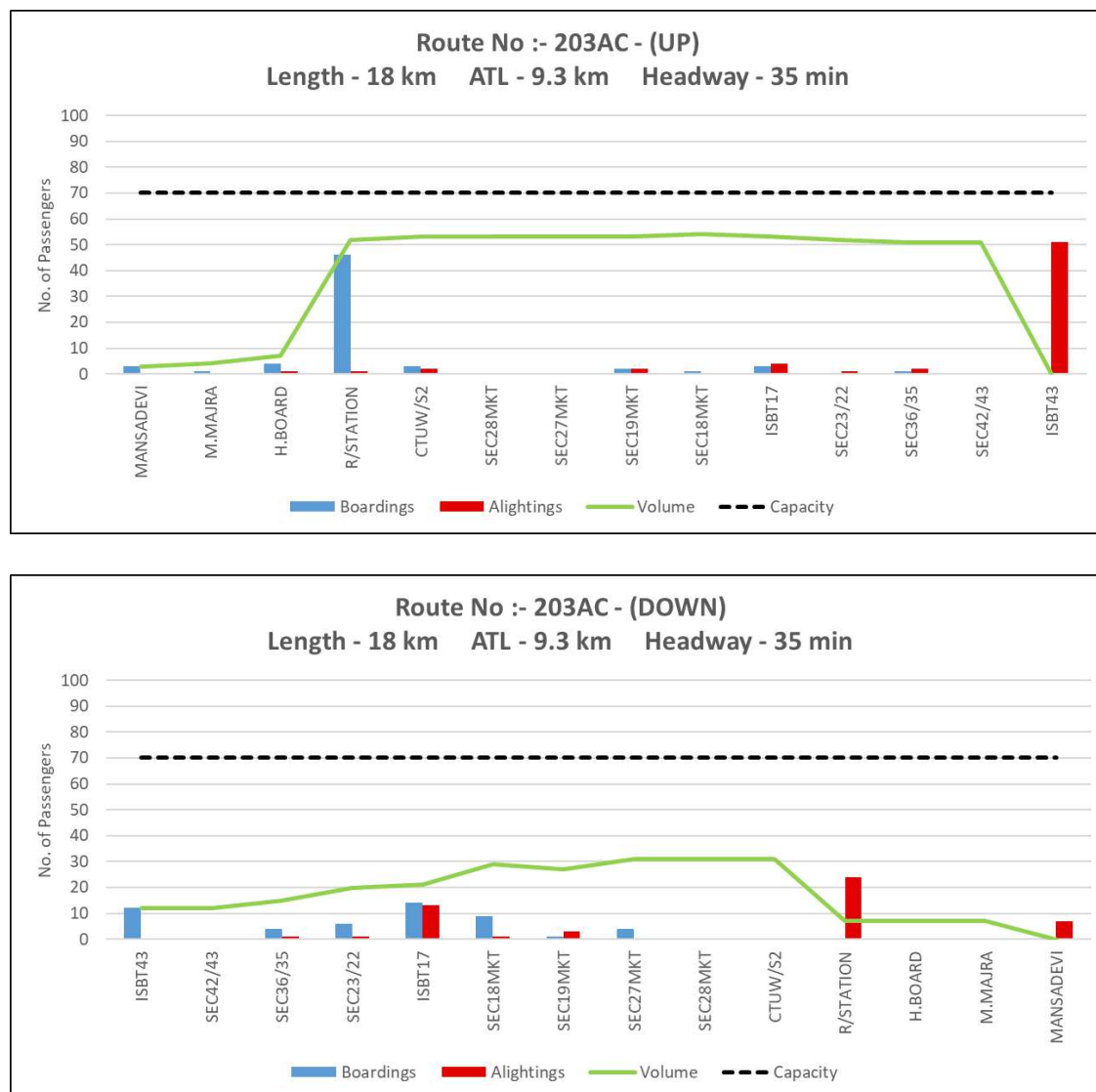


Figure 22: Peak hour Boarding & Alighting pattern along with line load of Route 203AC (Up and Down)

Route no. 205AC: ISBT-43 to PGI via Sector 43/44, 35/34, Aroma, 22/17,16/10, 11/10, 11/2 and OPD

In the figure below, it can be seen that in up direction 90% of the passengers are boarding from PGI and ISBT17 and more than 90% are alighting ISBT17 and ISBT43. Whereas in down direction, 2 stops out of 10 stops perform boarding activity and the rest indicating zero BA activity. Additionally, 97% of the passengers are alighting at last three stops namely PEC, OPD and PGI. Passenger volume in up direction remains steady throughout the trip which is similar to seating capacity. While in down direction passenger volume throughout the trip exceeds bus capacity. Maximum passenger load line in up direction is similar to seating capacity and in down direction it exceeds bus capacity indicating optimum utilization of fleet during peak hours. The average trip length of passengers travelling in this route is 7.6km which is more than half of the route length.

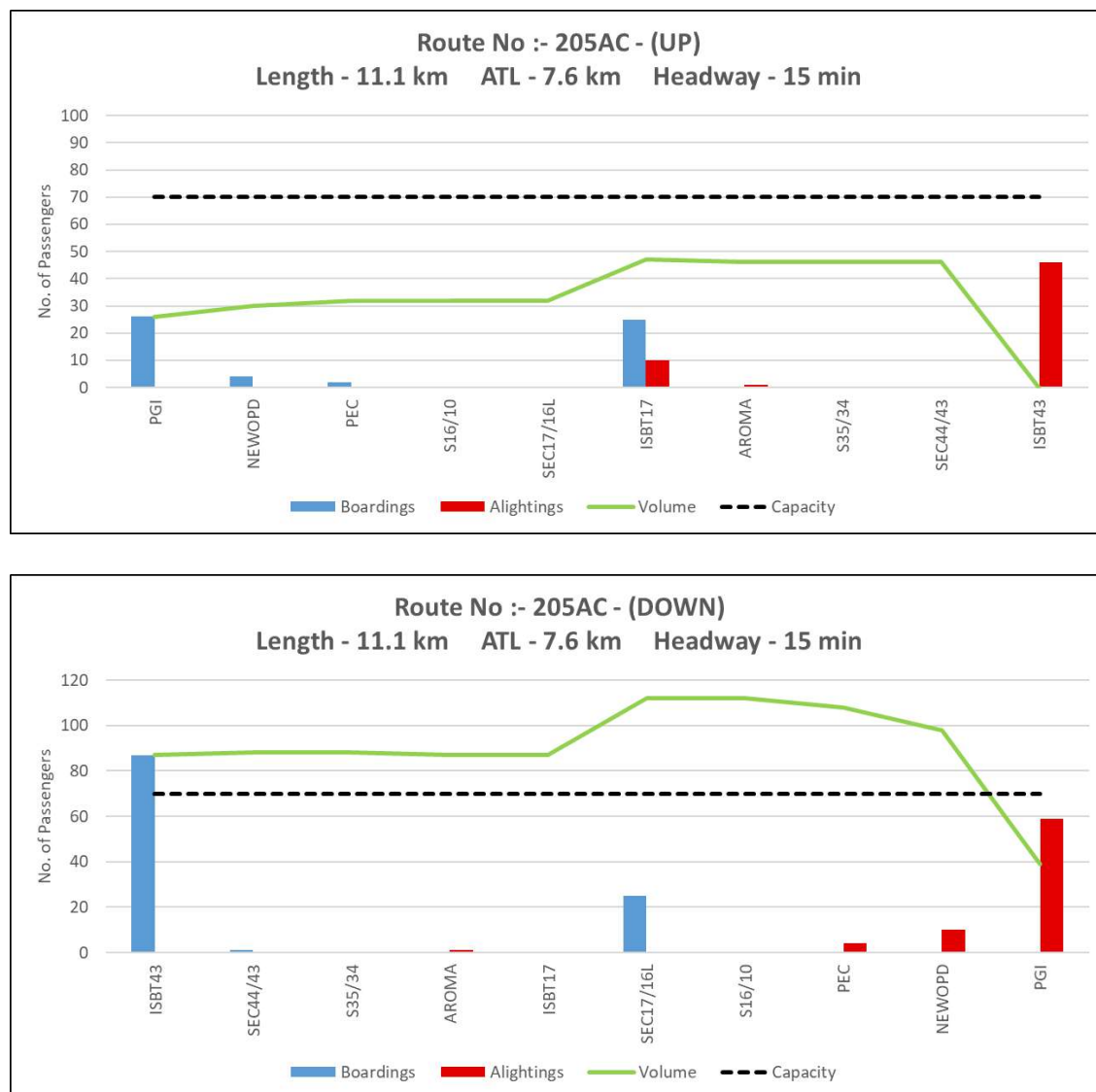


Figure 23: Peak hour Boarding & Alighting pattern along with line load of Route 205AC (Up and Down)

Route no. 206AC: ISBT-43 to I.T.Park via Sector 43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, 8/18, 7/19, 26/27, Grain Market, Railway Crossing, Kalagram, Housing Board Chowk, Mani Majra and Kishangarh

In the figure below, it can be seen that in up direction 90% passengers are boarding from IT Park, Mani Majra, housing board and ISBT17 and 80% of them are alighting at last stop i.e. ISBT43. Whereas in down direction 2 terminals perform 90% of boarding and passengers mainly alight at Sector26 and IT park. In up direction passenger volume gradually increases and further remains steady which is above bus seating capacity throughout the trip. While in down direction line load is similar to bus capacity from trip start and it gradually decreases to seating capacity till IT Park. Maximum passenger load line in both up and down direction is similar to bus capacity indicating optimum utilization of fleet during peak hours. The average trip lengths of passengers are 9.6 km indicating passenger travel around half of the route length.

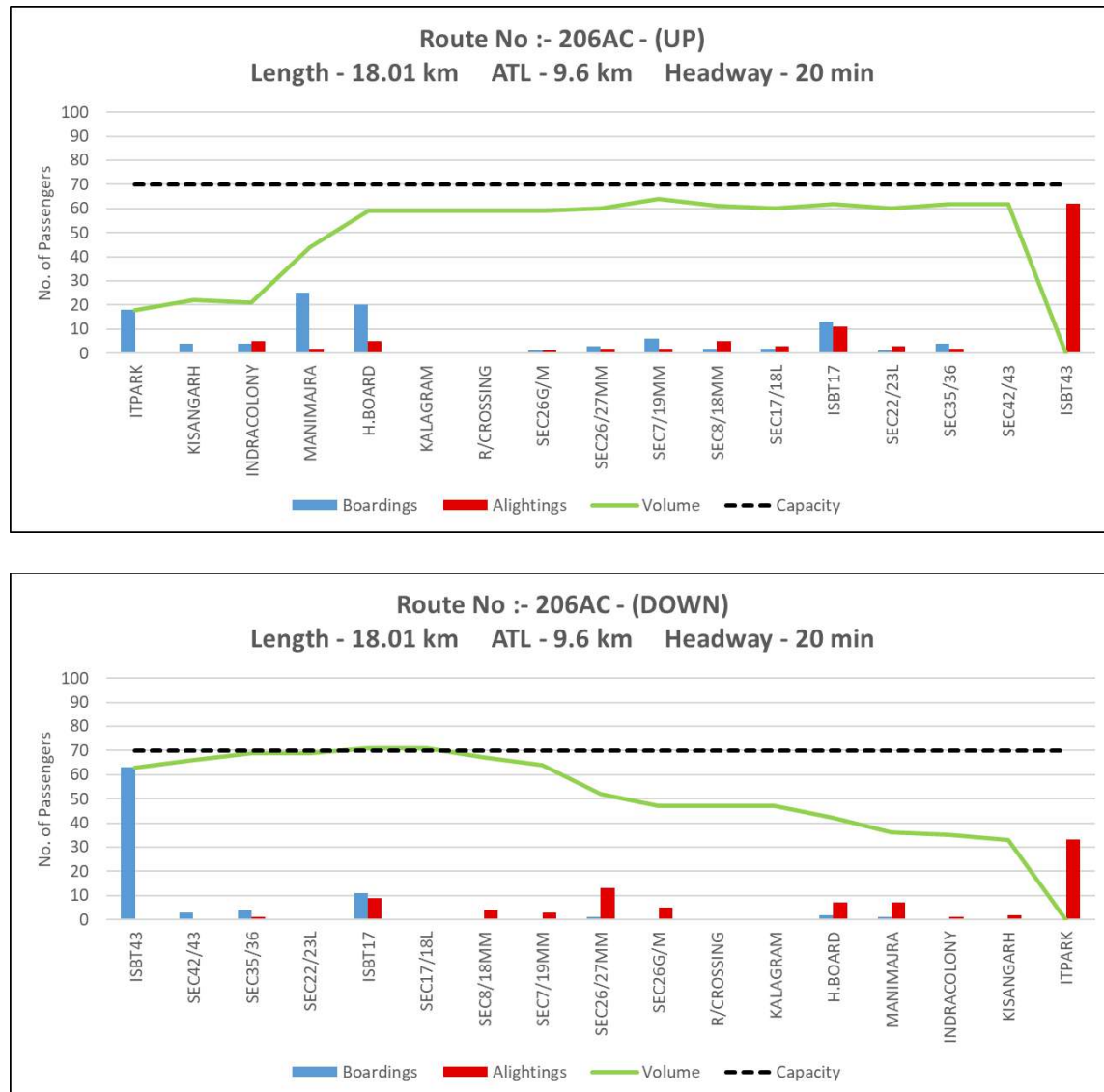


Figure 24: Peak hour Boarding & Alighting pattern along with line load of Route 206AC (Up and Down)

Route no. 240AC: ISBT-43 to Mata Mansa Devi via Sector 43/44, 35/34, Sector 22 market, 23/22, ISBT-17, Aroma, 21, 20, 30, 29 markets, Elante Mall, Colony No.-4, CTU workshop, Railway Station, Kalagram, Housing Board Chowk and Mani Majra

From figure below, it can be seen that in up direction almost all stops perform BA activity which is less than 15% of the boarding and 80% of passengers alight at end stop i.e. ISBT43. Whereas in down direction, 80% passengers are boarding from ISBT43 and railway station and 80% are alighting at last three stops namely housing board chowk, mani majra and mata devi. In both up and down direction passenger load line remains almost half of bus seating throughout the trip. The maximum load in both up and down direction is also 65% of the seating indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is 10.8 km indicating people travel almost half of route length.

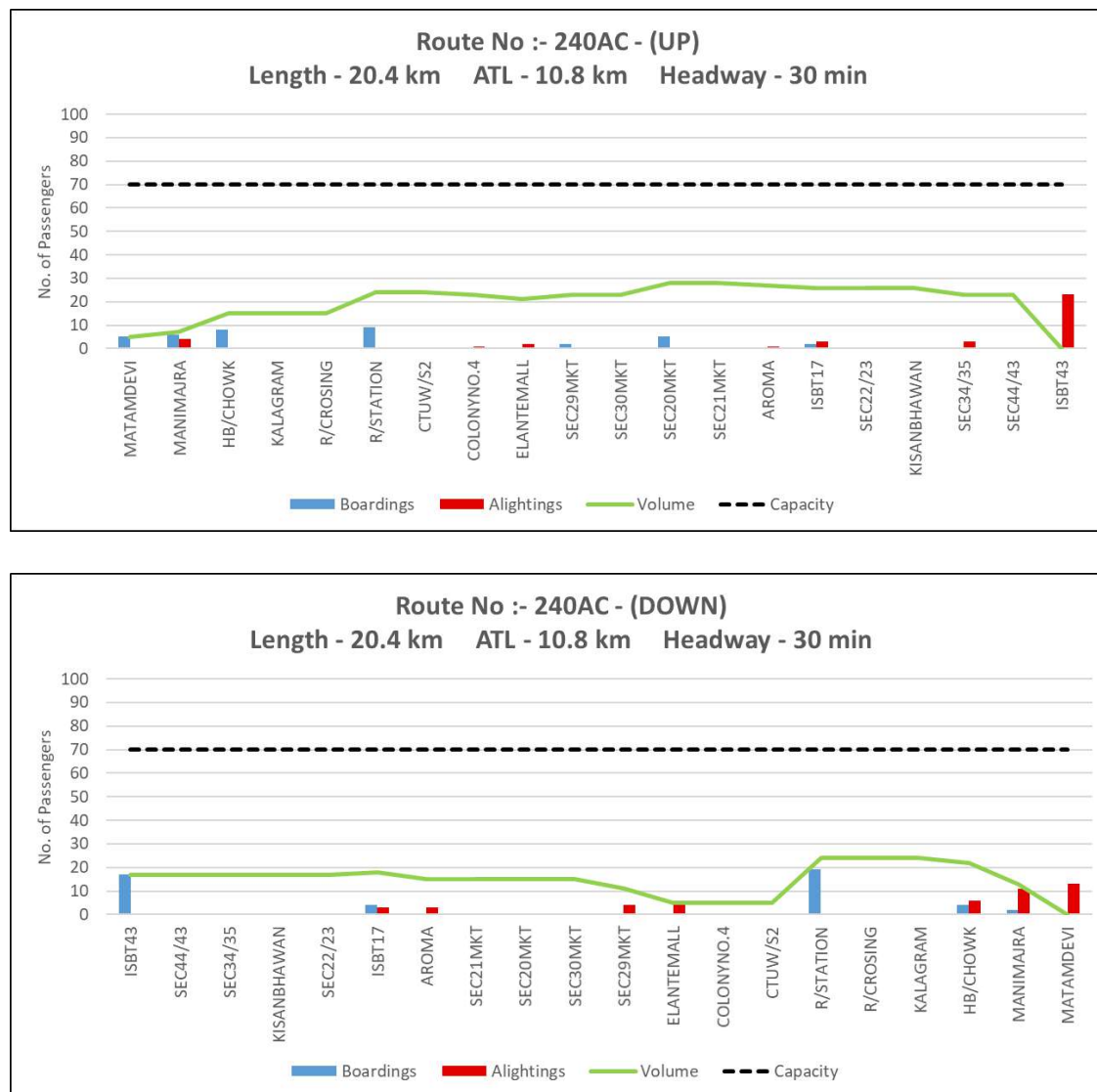


Figure 25: Peak hour Boarding & Alighting pattern along with line load of Route 240AC (Up and Down)

Route no. 24AAC and 24CAC: ISBT 43 to PGI via-.Sector 43/44, 35/34, Aroma,22/17, 16/10 (Hospital), 15/11 and vice versa

From the below figure it can be seen that all boarding is from ISBT 17 stop and all alighting at ISBT 43 in both Up and down direction. The passenger capacity is lesser than the seating capacity in both directions showing underutilization of the fleet. The average trip length is 5.6kms showing that passengers travel for 98% of the route.

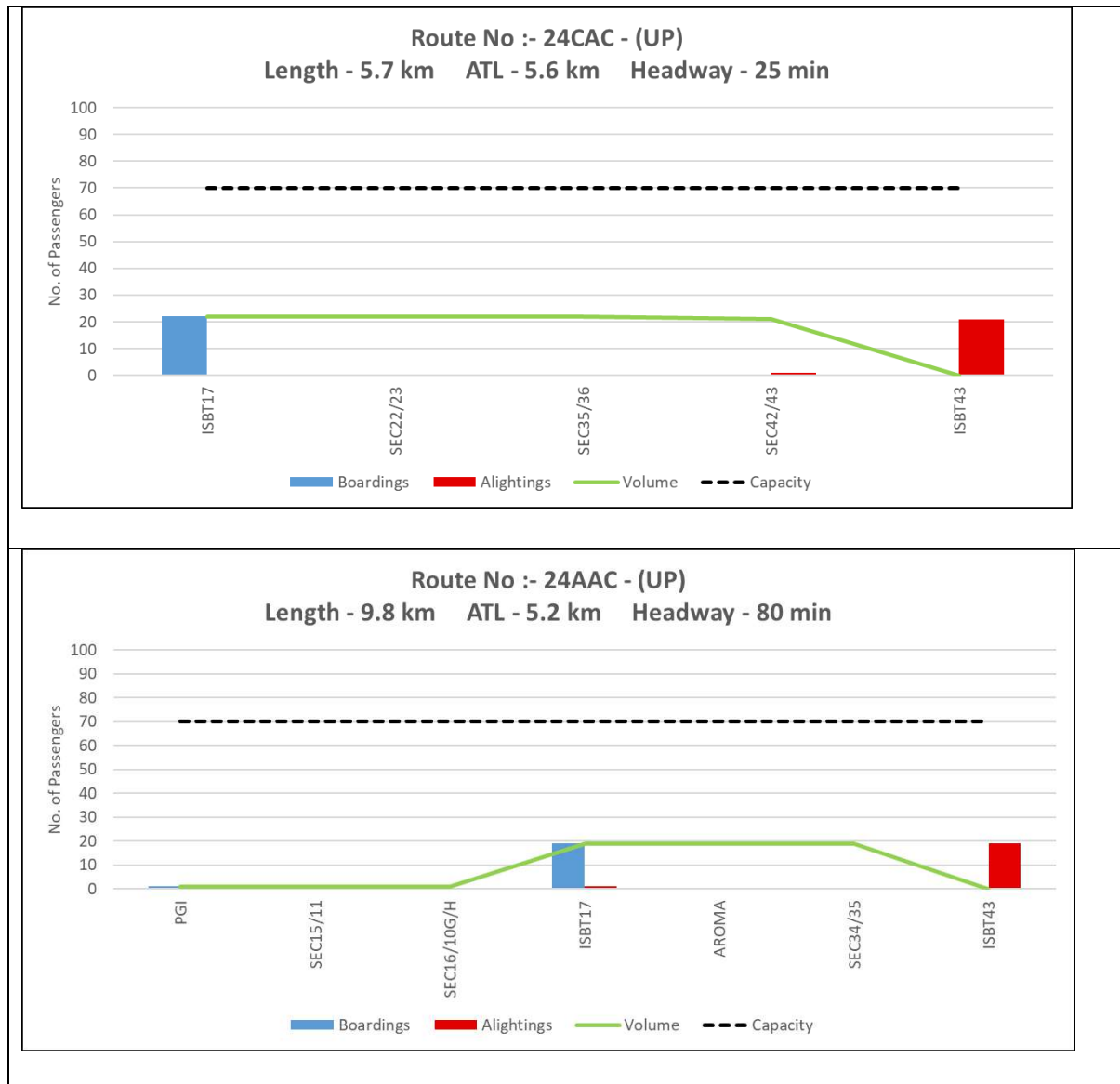


Figure 26: Peak hour Boarding & Alighting pattern along with line load of Route 24AAC and 24CAC

Route no. 254AC: ISBT 43 to IT Park via 43/44, 42/43, 36/35, 23/22, ISBT-17, 18 market., 19 market., 27 market., 27/28, 28 market., CTU workshop, Railway Station, Railway Crossing , Kalagram , Housing Board, Mani Majra and Kishangarh

From figure below, it can be seen that in up direction all intermediate stops perform similar boarding which is less than 15% of boarding and 80% of passengers alight at end stop i.e. ISBT43. Whereas in down direction, 3 stops out of 18 stops perform 96% boarding activity namely ISBT43, ISBT17 and railway station and similar alighting pattern is found in the last 6-8 stops which is less than 20% of the boarding. In both up and down direction passenger load line remains almost half of bus seating till throughout the trip. The maximum load in both up and down direction is 65% of the seating indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is 9.9 km indicating people travel almost half of route length.

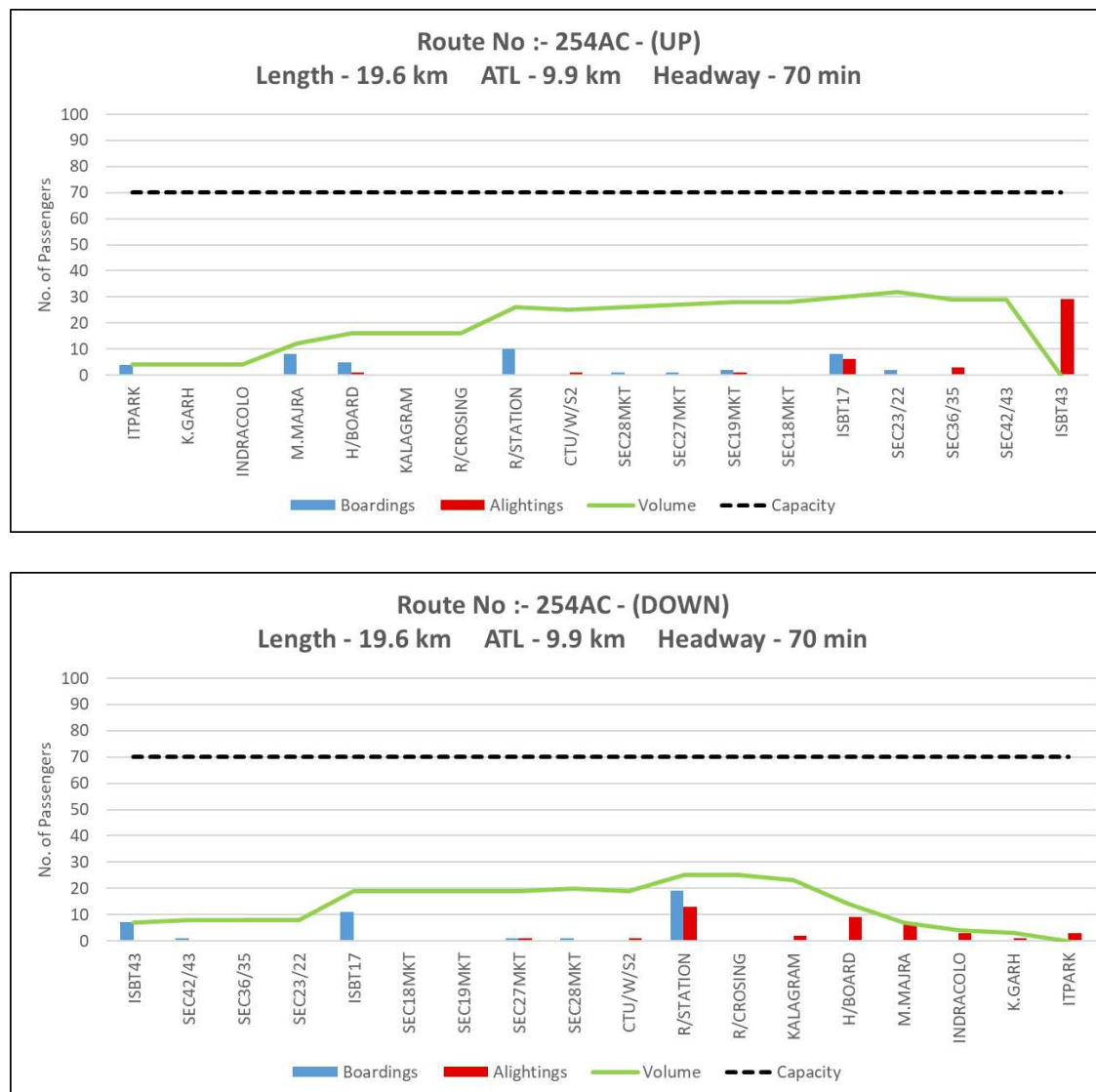


Figure 27: Peak hour Boarding & Alighting pattern along with line load of Route 254AC (Up and Down)

Route no. 239CAC & 239AAC: ISBT-43 to ISBT-43 via 43/44, 42, 41 markets, 40/41, 38/37, 25/24, 14/15, University, PGI, New OPD, PEC, Pb.Civil Sectt, High Court, MLA Flat, UT.Sectt, 17/16, ISBT-17, Aroma, 34/35, 44/43 and vice versa

From figure below, it can be seen that in circular direction all boarding are performed by both terminals, PGI and high court and 80% of the passengers alight at ISBT43, PGI and sector14. Whereas is down direction, 90% of the passenger's board from two terminals and majorly alight at Mini sector, high court, PB civil and terminals. In up direction passenger load remains steady which is above to bus seating throughout the trip. While in down direction passenger load from ISBT43 to high court passenger load remains similar to bus seating and rest trip has occupancy of 20% of seating capacity. The maximum load in both up and down direction is above bus seating capacity indicating moderate utilization of fleet during peak hours. The average trip length of passengers on this route is 9.8 km indicating people travel 40% of route length.

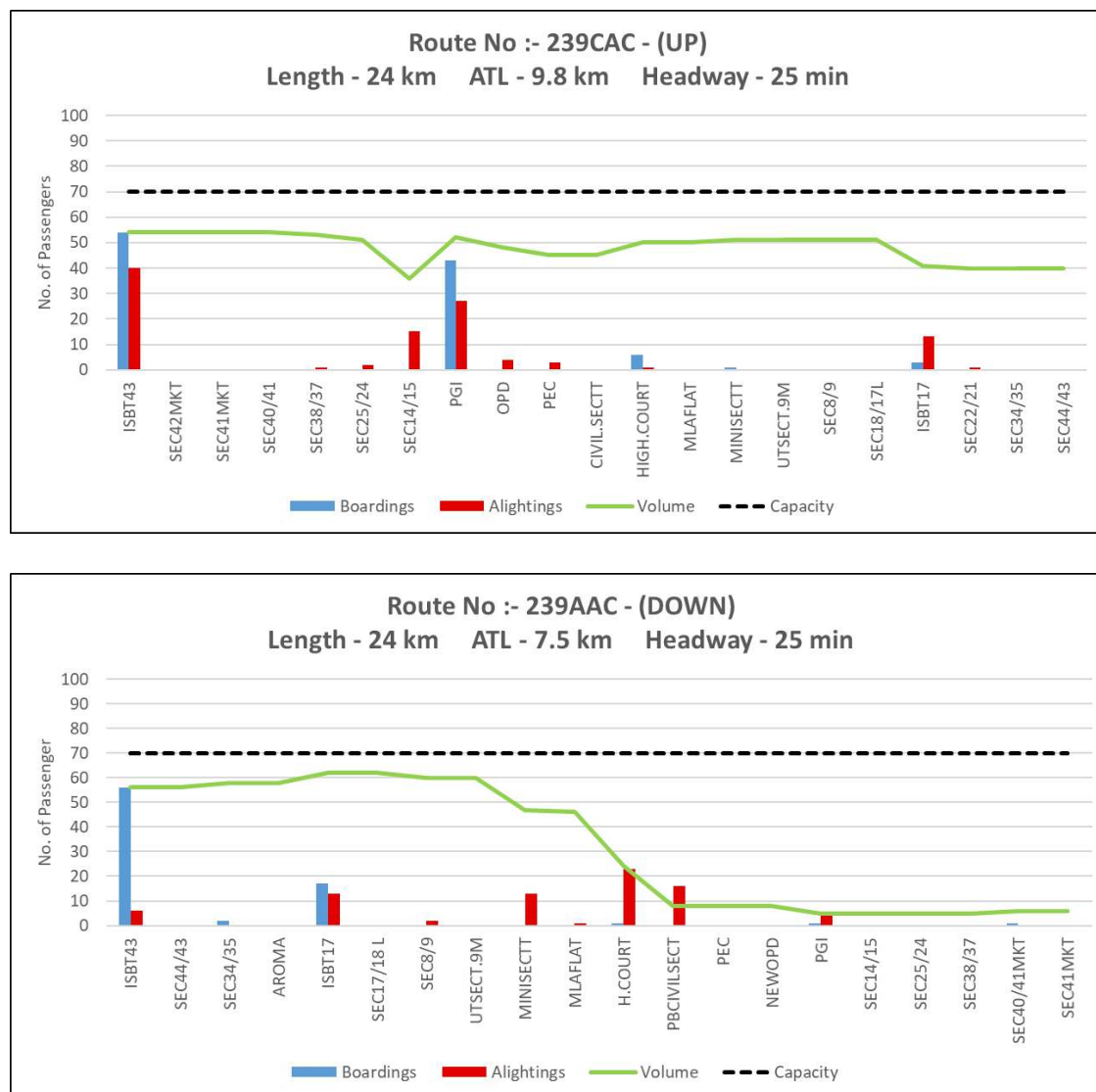


Figure 28: Peak hour Boarding & Alighting pattern along with line load of Route 239AAC and 239CAC

Route no. 8: Mohali Ph-XI to PGI via Sec-65 Market, 65/64, 49/48, Sector 48 Market, Sector 47 Market, Sector 46 Market, Sector 45 Market, 44 Market, ISBT-43,43/44, 35/34,. 22/17, 16/10 (Hospital) and New OPD

From figure below, it can be seen that in up direction around 95% of passengers board from Mohali phase-II and ISBT17, while the rest stops show zero BA activity. Whereas the down direction all the intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load is quite low as compared to the seating capacity till ISBT17, while for the rest trip, passenger load gradually decreases till Kiambwala. In down direction, passenger volume in bus remains steady at 50% of the seating capacity utilized throughout the trip. The maximum load in both up and down direction is lower than the bus capacity indicating the fleet is underutilized during peak hours. The average trip length of passengers on this route is about 6 km indicating people travels shorter distances.

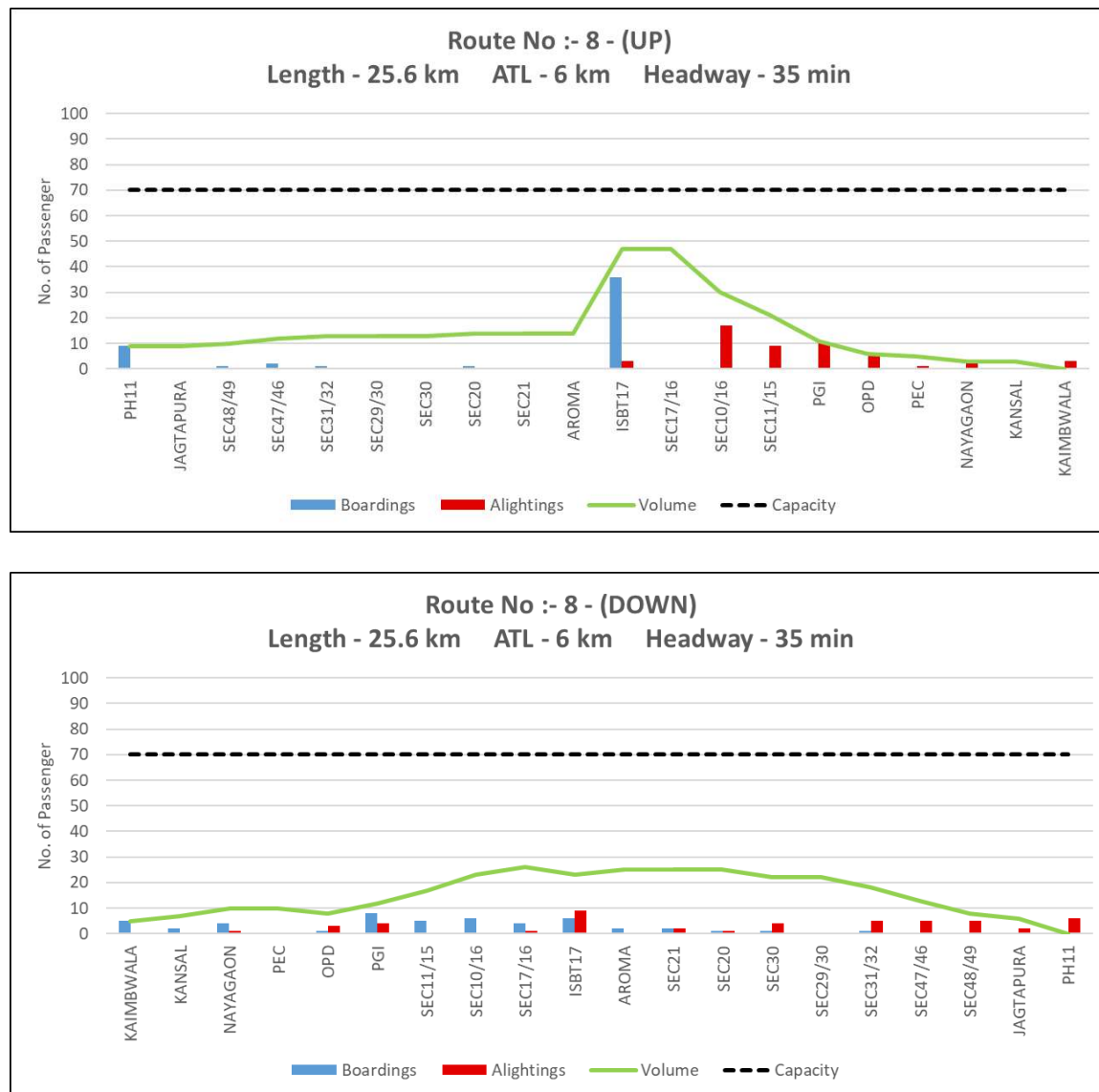


Figure 29: Peak hour Boarding & Alighting pattern along with line load of Route 8 (Up and Down)

Route no. 25: ISBT 17 to Khizarabad via Sector16/10, 15/11, PGI, Khuda Lahora, S/P Barrier, Mullanpur, Parol, Majra, Boothgarh, Block and Sialwa, Jhande Majra.

From figure below, it can be seen that in up direction around 85% of passenger are boarding from Khizarabad, Sialba and Black, while same pattern of alighting is seen which is 15% of the total boarding. Whereas the down direction all the intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load gradually increases and exceeds bus capacity from block till PGI which is more the 50% of the trip time. In down direction, passenger volume trajectory remains steady which is near to bus seating capacity. The maximum load in both up direction is exceeding the bus capacity indicating optimum use of fleet during peak hours. In down direction scenario is similar to the bus seating capacity. The average trip length of passengers on this route is about 14.2 km indicating people travels 50% of the total route length.

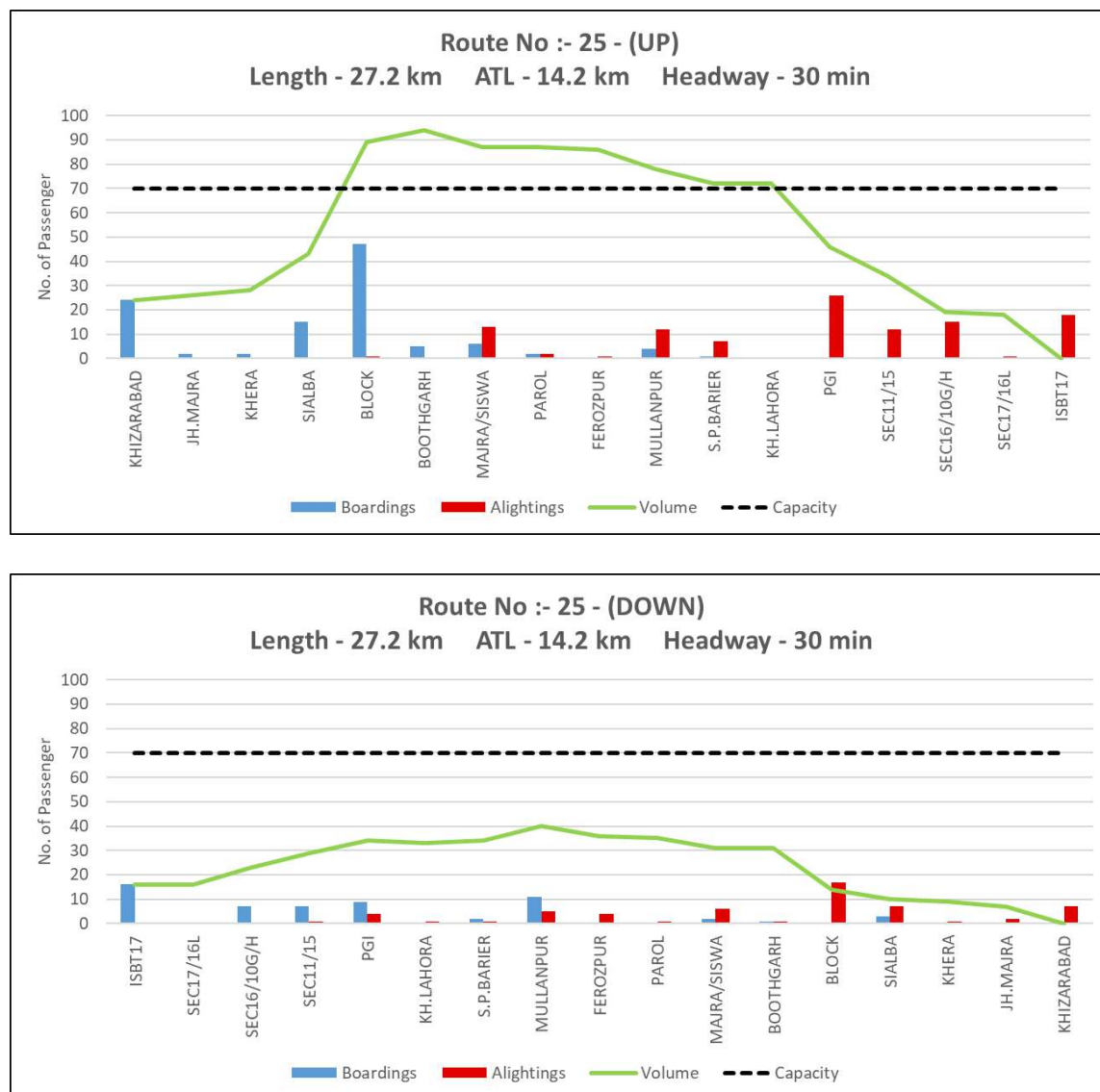


Figure 30: Peak hour Boarding & Alighting pattern along with line load of Route 25 (Up and Down)

Route no. 30: ISBT-43 TO Nada Sahib via Sector 43/44, 42/43, 36/35, 23/22, ISBT-17, Aroma, 21, 20 Markets, 19/27, 26/27, 26/28 Grain Market, Railway Crossing, Kalagram, Housing Board, Manimajra, Command Hospital and Majri Chowk

From figure below, it can be seen that in up direction around 85% of passenger are boarding from Nada saheb and 90% of the passengers are alighting at Mani Majra, ISBT17 and ISBT43. Whereas the down direction all the intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load on gradually decreasing to half of the bus seating capacity. In down direction, passenger volume trajectory is near to bus seating capacity throughout the trip. The maximum load in both up and down direction is similar to bus seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 8.7 km indicating people travels 40% of the total route length.

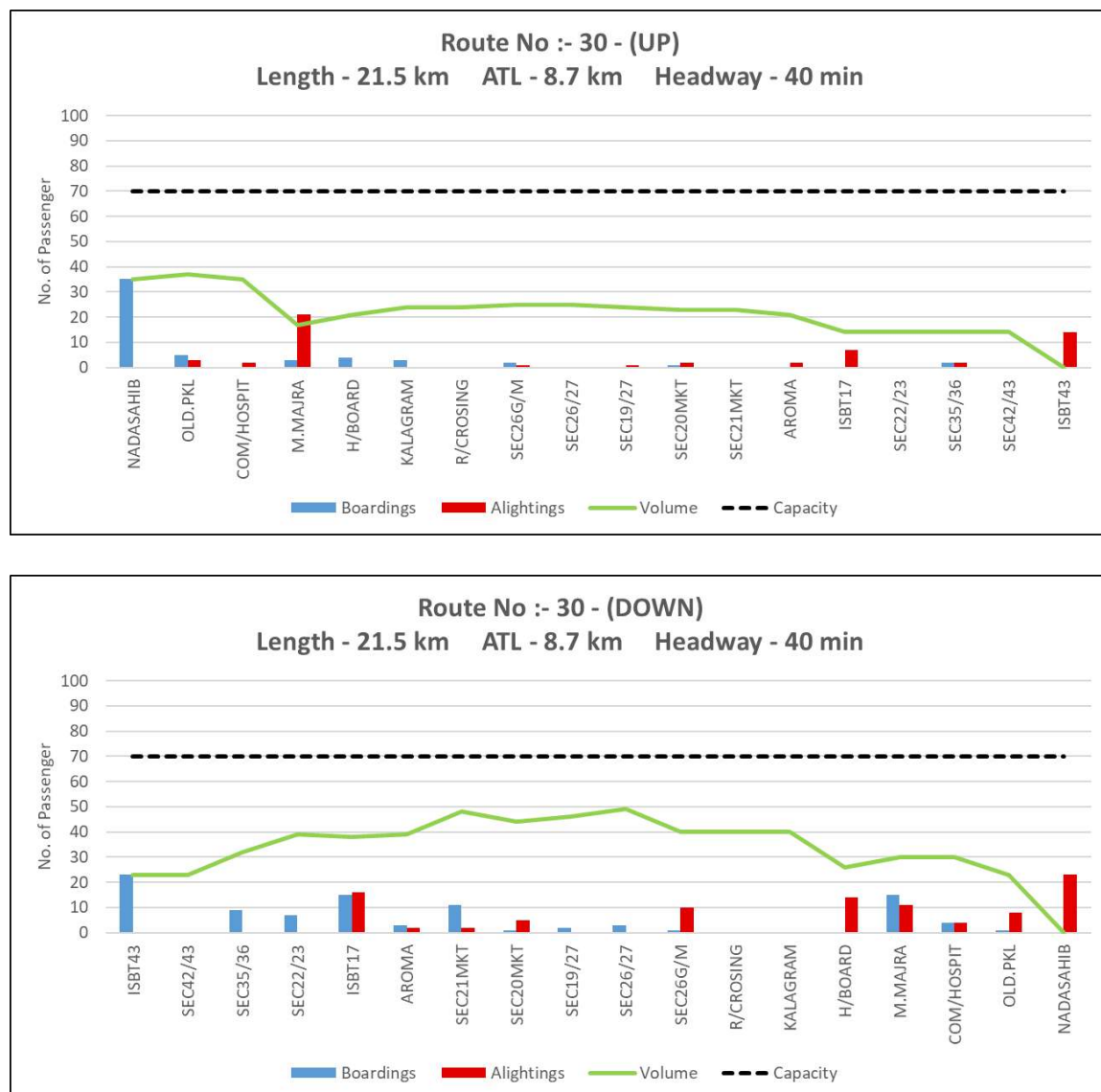


Figure 31: Peak hour Boarding & Alighting pattern along with line load of Route 30 (Up and Down)

Route no. 32: PGI to Derabassi via OPD, PEC, Sector 10/16, Sector 17-16, ISBT-17, Sector 8/18, 7/19, 26/27, Grain.Market, 26/28, Housing Board, Mani Majra, Old Panchkula, Nada Sahib, Ramgarh, Mubarakpur

From figure below, it can be seen that in up direction around 85% of passenger are boarding from Derabassi and 90% of the passengers are alighting at PGI, ISBT17, Manimajra and Old.PKL. Whereas the down direction all the intermediate stops show patterns of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load on gradually decreasing to half of the bus seating capacity. In down direction, passenger volume trajectory is near to bus seating capacity which increases to the bus capacity gradually towards the end. The maximum load in both up and down direction is similar to bus seating capacity indicating moderate utilization of fleet during peak hours. The average trip length of passengers on this route is about 12.5 km indicating people travels 35% of the total route length.

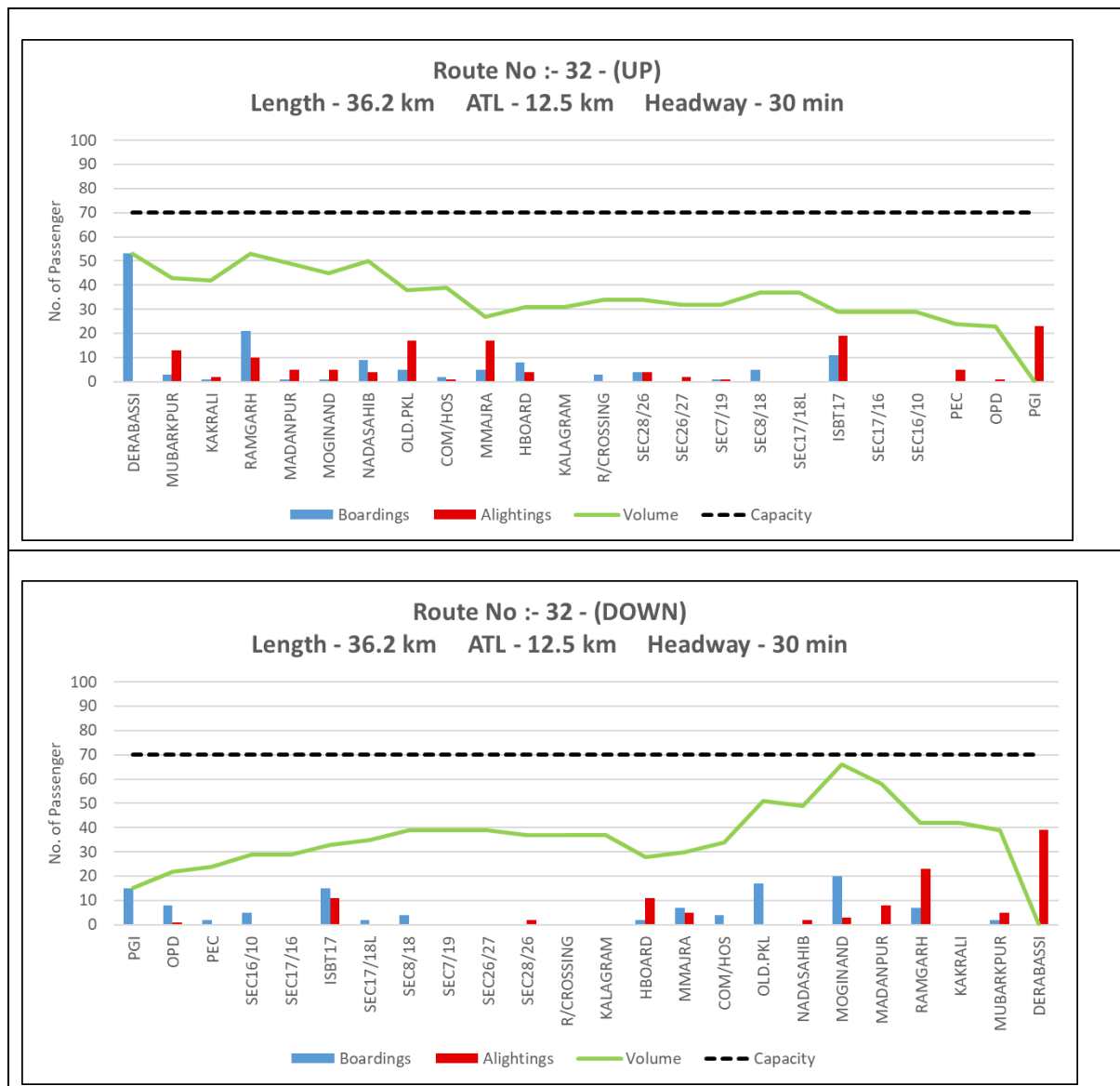


Figure 32: Peak hour Boarding & Alighting pattern along with line load of Route 32 (Up and Down)

Route no. 34: PGI to Derabassi. via OPD, 10/16, ISBT-17, Aroma, Piccadilly Chowk, 21/34, 20/33, 30/32, 29/31, Hallo Majra, Airport, Zirakpur, Bankharpur and Mubarakpur.

From below figure, it can be seen that in up direction boarding are from 5 stops out of total 18 stops and alighting are from the last 6 stops on this route. 70% of them alight at 3 terminals which are PGI, ISBT17 and sector 20. Whereas the down direction 90% of passengers board from PGI, New OPD and Zirakpur. While 90% alighting are from Derabassi, Airport and Zirakpur. All intermediate stops show almost same pattern which is less than 10% boarding. In up direction, the passenger load is equal to the bus capacity, while for the rest trip, passenger load is equal to the seating capacity even during peak hour. In down direction, passenger volume in bus is similar to the seating capacity throughout the trip. The maximum load in down direction trip is equal to the bus capacity indicating the fleet is well utilized during peak hours. The average trip length of passengers on this route is about 14.1 km indicating people travels for around 50% of the route length.

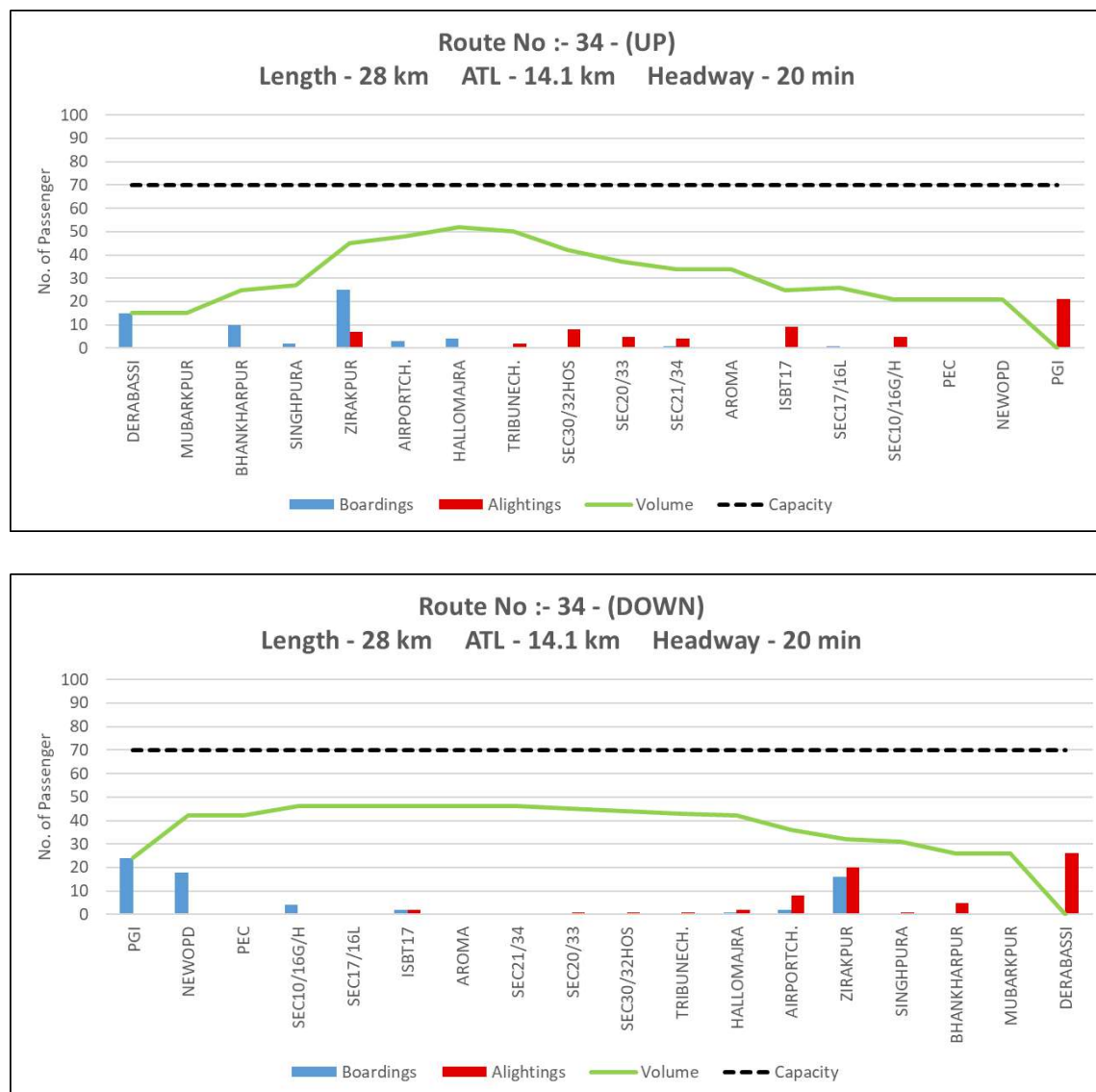


Figure 33: Peak hour Boarding & Alighting pattern along with line load of Route 34 (Up and Down)

Route no. 35: ISBT-17 to Kharar via Aroma, Sec-35/22, Kissan Bhawan, 35/36, 41/37, 41/40, Fountain Chowk, Ph-2, Ph-6, Balongi, Daun, Desu Majra and Mundi Kharar.

90% of boarding happen from 2 stops, Kharar and Balongi. And alighting happens from 4 stops, ISBT17, F.CH, Sector 55 and PH6 Mohali. Out of the total 14 stops 7 stops show B&A activities and 7 stops show near to null activities. The seating capacity in both up and down direction is close to the bus capacity which shows good utilization of the fleet. The maximum load in both up and down direction is near to the bus capacity indicating that optimum utilization of fleet. The average trip length is 11.5kms which shows passenger travel for 62% of the total route.

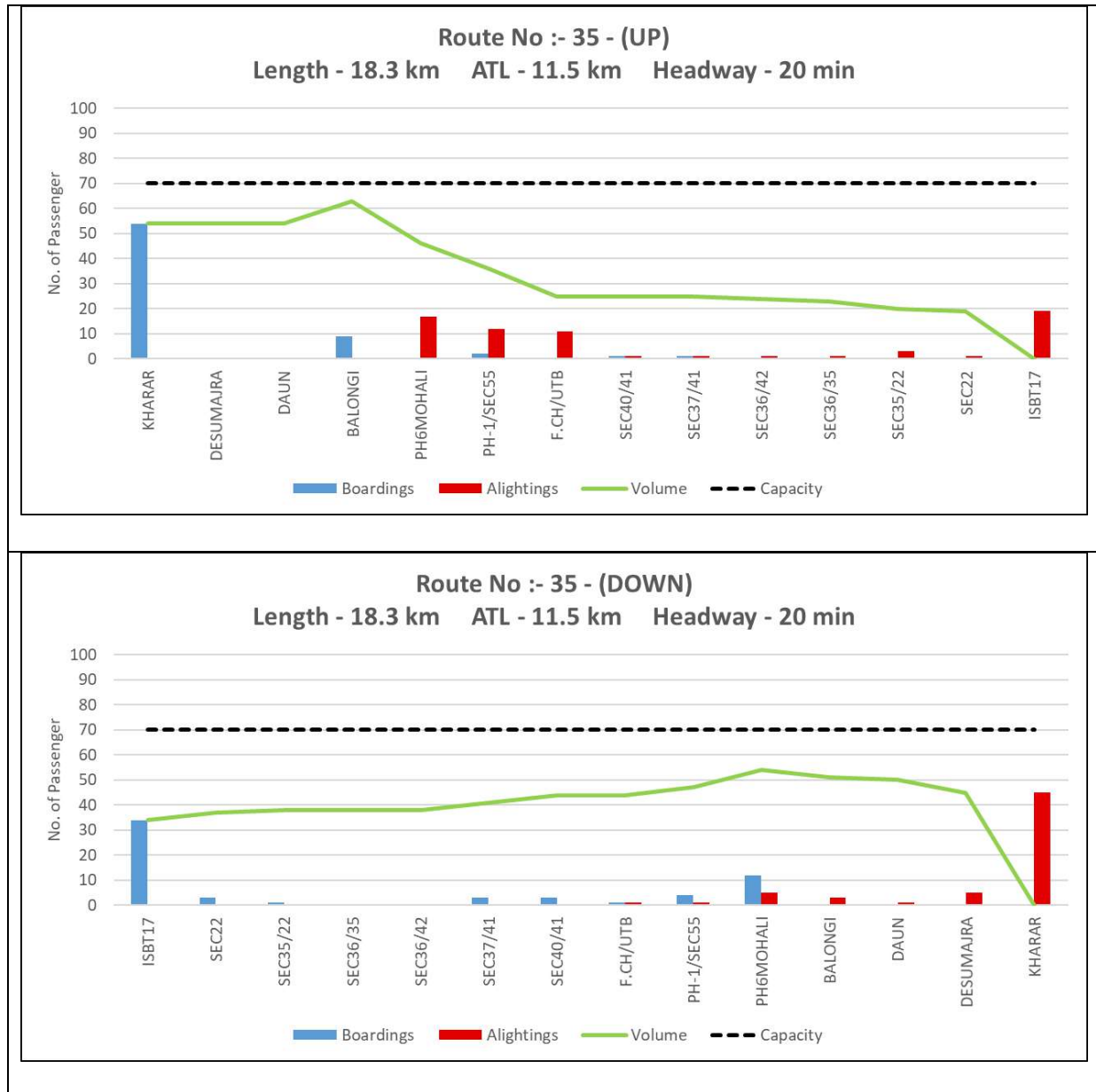


Figure 34: Peak hour Boarding & Alighting pattern along with line load of Route 35 (Up and Down)

Route no 39: Mohali Ph-11 to PGI via Industrial Area Ph-9, Mohali Sector 66/67, 67/68, 68 outer, 69/68, Kumbra, Education Board Punjab, Forties, B/S Mohali, Ph-7, Chawla Chowk, Ph-3/7, 52/61, YPS Chowk, ISBT-43, 43/44, 42/43, 36/35, 23/22, ISBT-17, 22/17, 16/10, 11/10, 11/2 and OPD

On this route in Up and Down both directions, 75% boarding are seen from ISBT43 and alighting at PGI. The up direction first half of route shows less passenger flow with consistent B&A activities on first 8 stops. The first half has a passenger capacity close to the seating capacity till ISBT43 stop after which the passenger flow doubles for the rest half increasing to go above the bus capacity showing optimum utilization of this fleet. In down direction first half has passenger flow more than bus capacity which decreases from ISBT43 stop and lowers to lesser than seating capacity showing moderate use of this fleet. The Average Trip Length is 9.5kms showing short trips taken by passengers.

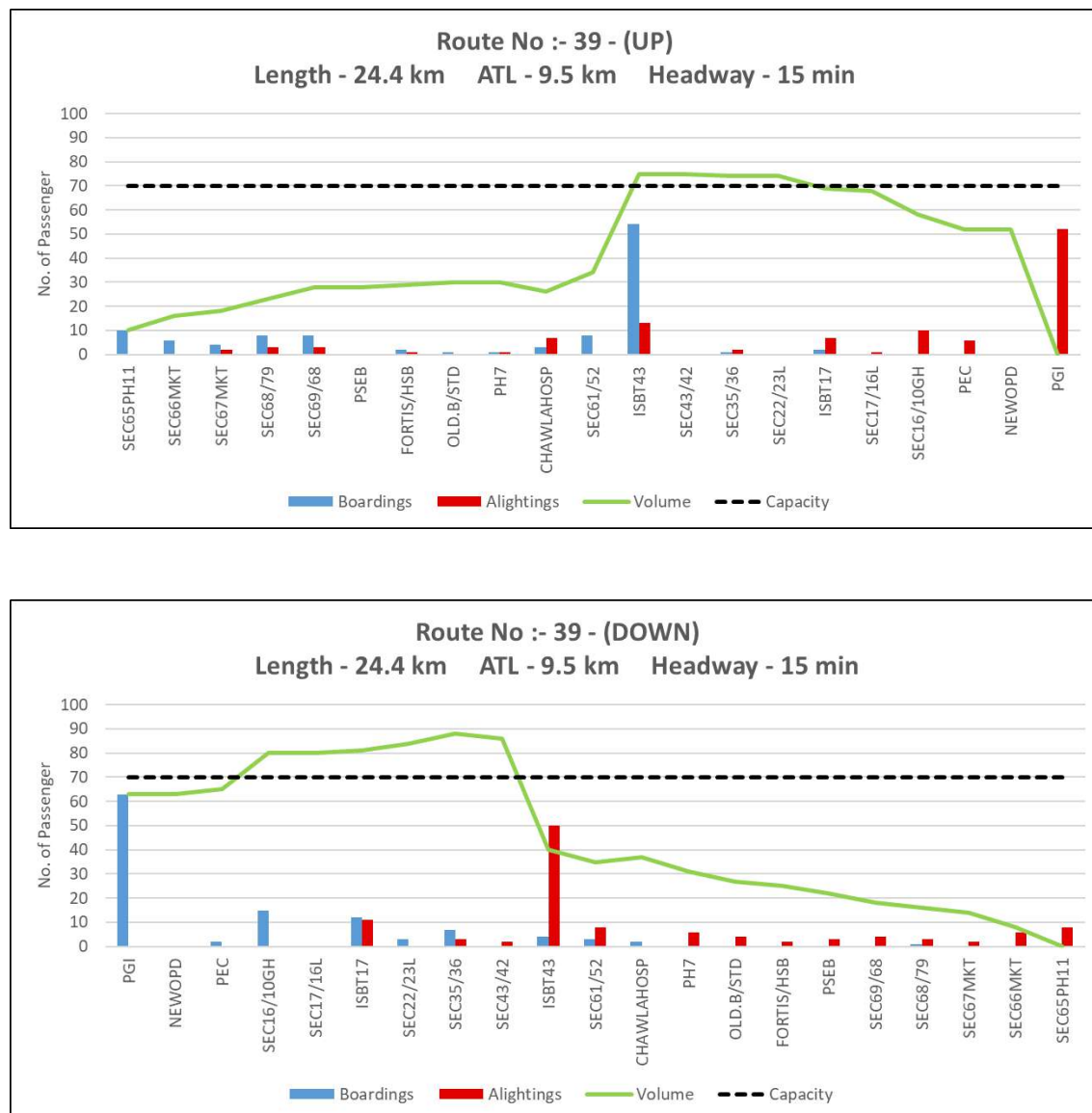


Figure 35: Peak hour Boarding & Alighting pattern along with line load of Route 39 (Up and Down)

Route no. 40: ISBT-17 TO KHIZRABAD via Sector 16/10, PGI, Khuda Lahora, S/P Barrier , Mullanpur, Parol, Majra, Balock, Manakpur, Sangatpura and Kubaheri

In route 40 major boarding are seen to happen at 4 stops from the total 17 stops, which are Block, Khderabad, Kubaheri and M.Pur/S.Pur. Alighting occur in the last 5 stops majorly PGI, Sector 11/15, Sector 16/10G, and ISBT17. Rest of the 8 stops on this route show minimal B&A activities. In down route 90% Boarding are seen from 3 stops which are ISBT17, PGI and Mullanpur. Alighting are from Block, Sialbamairi and Khderabad. The passenger volume in Up direction is more than the bus capacity which shows optimum utilization of the fleet and in down direction the capacity is close to the bus capacity which demarcates that the fleet is well utilized. The average trip length is 14.2 km showing passengers travel for 45% of the trip length.

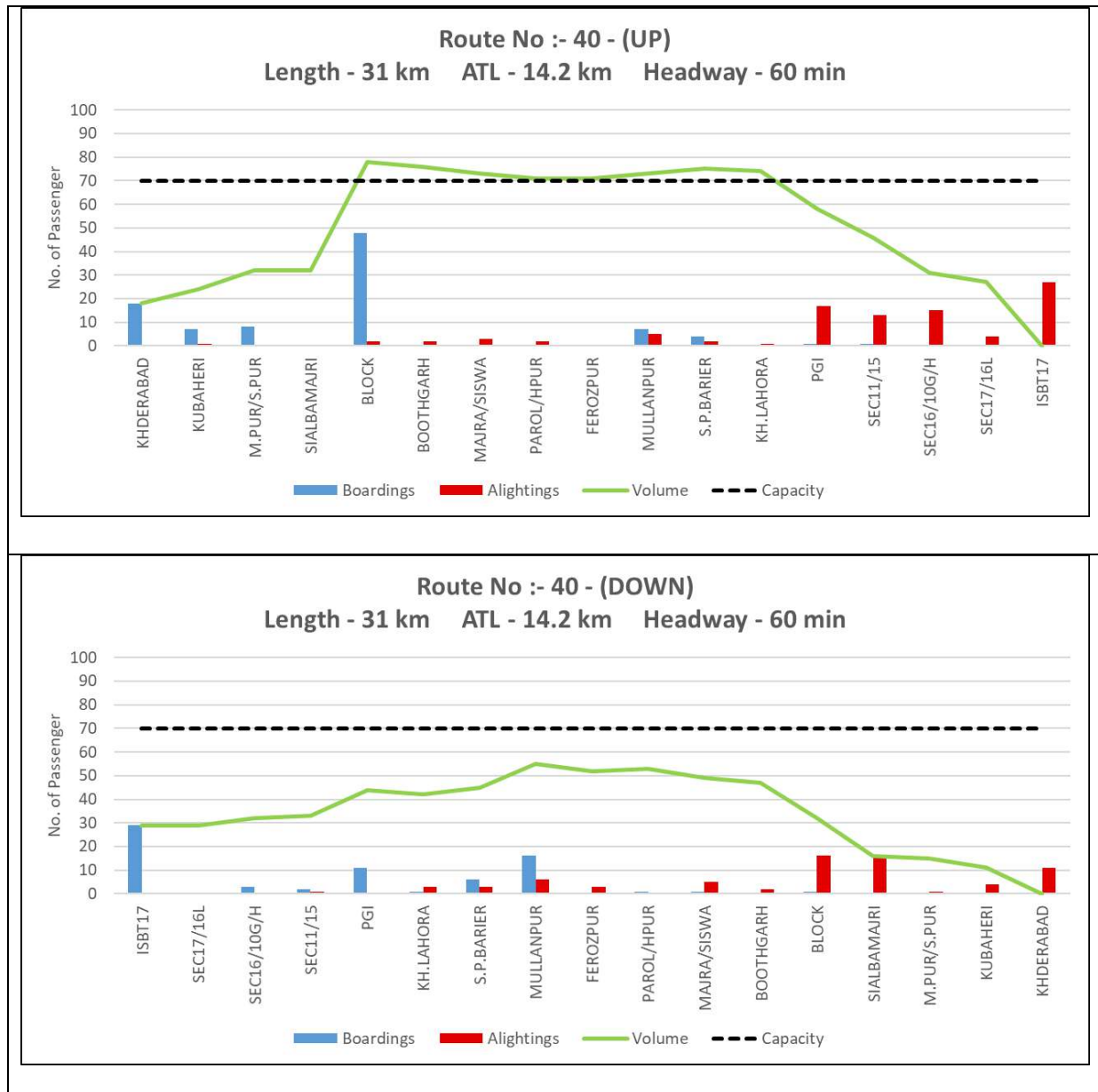


Figure 36: Peak hour Boarding & Alighting pattern along with line load of Route 40 (Up and Down)

Route no. 79: PGI TO Chhattbir Zoo via OPD, 10/16, ISBT-17, Aroma, Sector 21, 20 market, Sector 20/30, 32 Hospital, Tribune Chowk, Hallo Majra, Airport, Zirakpur and Dayalpura

In route 79, Up direction 95% boarding takes place at 4 stops, namely Zirakpur, Dayalpura, Chhatt and Chhattbir. 90% alighting are seen at the last 4 stops PGI, Sector 10, ISBT 17 and Sector 30. Out of the total 17 stops 7 stops show minimal B&A activities. In down direction major boarding are seen from PGI, New OPD and Sector 10. 90% Alighting are from Zirakpur, Dayalpura and Chhatt. The seating capacity in up direction is equal to the bus capacity showing optimum utilization of the fleet. In down direction The passenger flow is equal to the seating capacity showing good utilization of the fleet. The Average trip length is 12.5kms showing short trips.

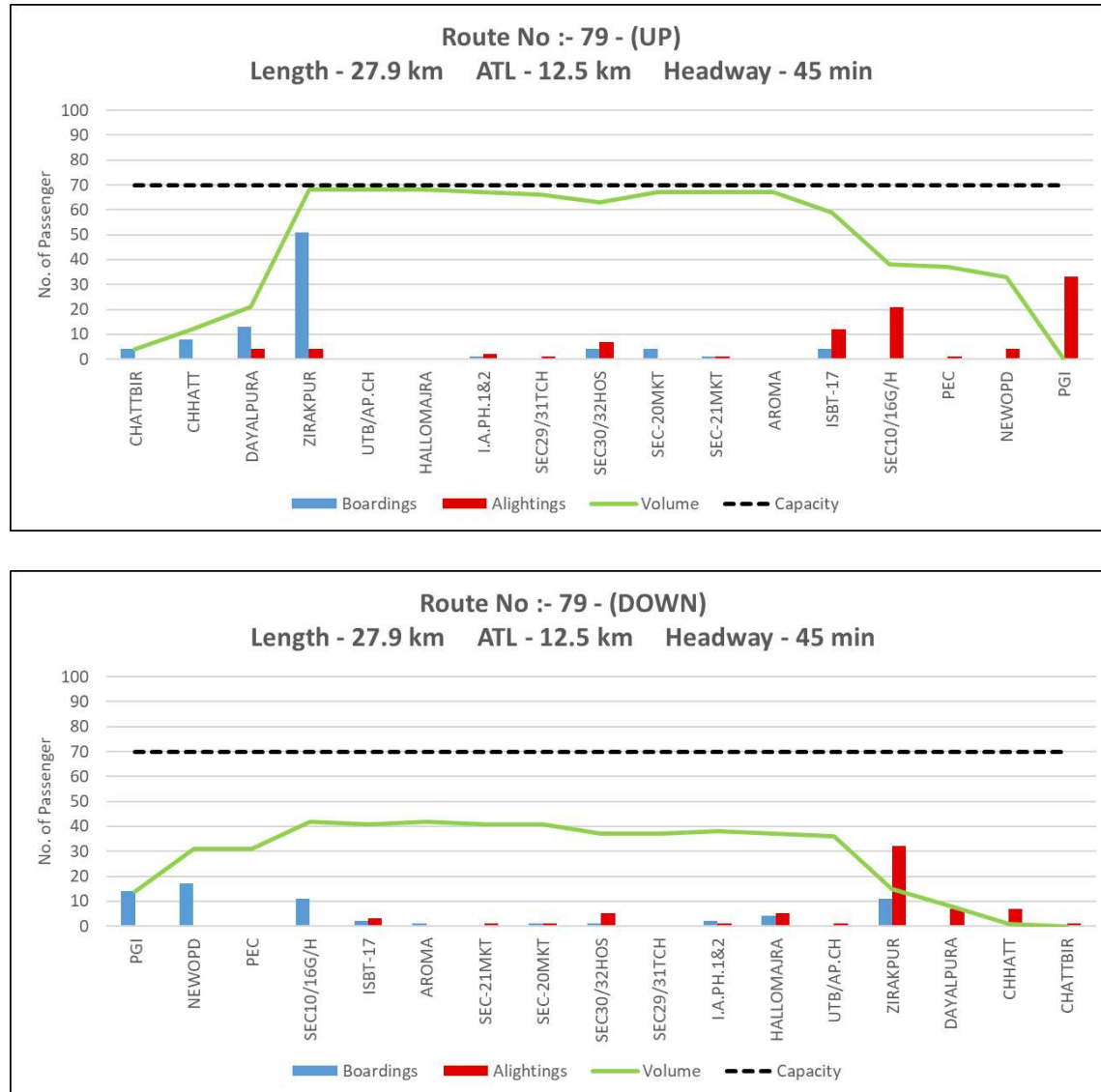


Figure 37: Peak hour Boarding & Alighting pattern along with line load of Route 79 (Up and Down)

Route no. 80: Zirakpur to PGI via Baltana, Raipur Kalan, Mouli Jagaran, Hallo Majra, Colony No.4, Elante Mall, Sector 29 market, 30 market, 20 market, Aroma, 22/17, 16/10, PEC and New OPD

In route 80, Up direction 95% boarding takes place at 4 stops, namely Zirakpur, Baltana, Vikasnagar and Sector 17. 90% alighting are seen at the last 4 stops PGI, PEC, Sector 16 and Sector 17. Out of the total 18 stops, 8 stops show minimal B&A activities. In down direction major boarding are seen from PGI, PEC and Sector 17. 90% Alighting are from Zirakpur, Baltana, Raipur and Halomajra. The seating capacity in up direction is equal to the bus capacity showing optimum utilization of the fleet. In down direction the passenger flow is equal to the seating capacity showing good utilization of the fleet. The Average trip length is 9.8kms showing short trips.

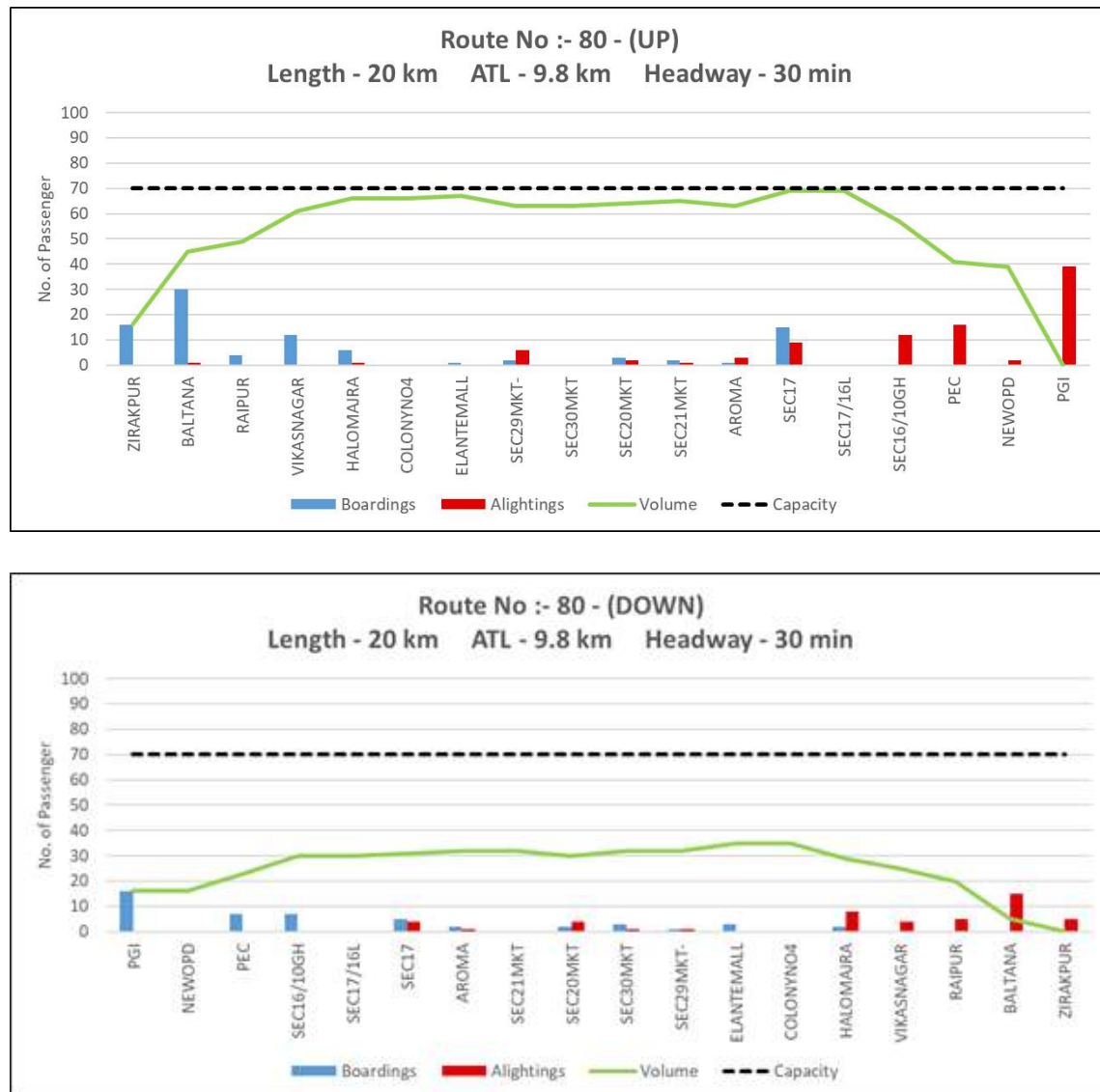


Figure 38: Peak hour Boarding & Alighting pattern along with line load of Route 80 (Up and Down)

Route no 212: PGI to Zirakpur (Singhpura Bhuda) via OPD, Sec 11/2, 10/16, ISBT-17, Aroma, Piccadilly Chowk, 21/34, 20/33, 30/32, 29/31, Hallo Majra, Airport and Zirakpur

Route 212 Up route graph shows 90% boarding activities from Derabassi, Singhpura, Zirakpur and Sector30/32Hos and major alighting are from last 3 stops which are PGI, PEC and IBST17. Out of the total 18 stops 8 stops have very less to zero B&A activity in up route. Down route shows 95% boarding from 4 stops which are PGI, PEC, IBST 17 and Zirakpur namely and Alighting are from Derabassi and Zirakpur mainly. From 18 stops on down route 5 stops have zero B&A activities. The passenger capacity in up route is way above than the bus capacity sighting a consistent high over usage of this fleet. In down route the passenger flow is slightly above the bus capacity showing optimum utilization of this fleet. The average trip length is 15kms showing that people travel 73% of the total route length.

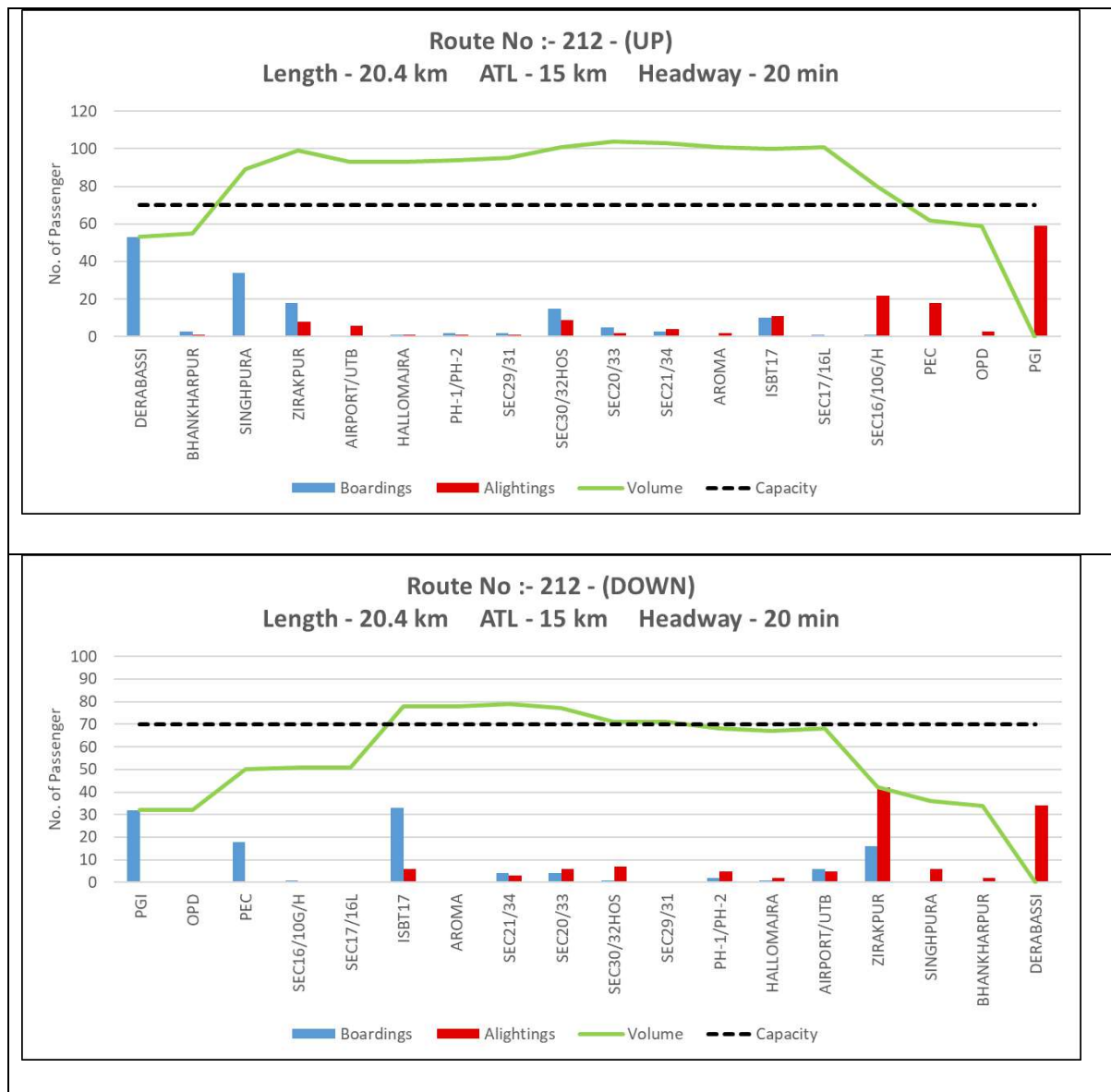


Figure 39: Peak hour Boarding & Alighting pattern along with line load of Route 212 (Up and Down)

Route no 213: Landran to Dera Bassi via Dc.office/ Lakhnaur, Sohana, Sector 69/68, Fortis/ PSEB, Ph-7/8, YPS Chowk , ISBT-43, Sector 44, Sector 45, Sector 46/47, Sector 32/31(Hospital), Tribune Chowk , Hallo Majra, Airport Chowk , Zirakpur, Singhpura, Bankharpur

From figure below, it can be seen that in up direction around 85% of passengers board from Derabassi and Zirakpur. Alighting are majorly at ISBT 43, Landran and Zirakpur. Whereas the down direction all boardings are from Landran, Sohana, ISBT 43, Sector 46 and Sector 31. The intermediate stops show almost same pattern of boarding and alighting which is less than 10% of the total boarding. In up direction, the passenger load is quite low as compared to the seating capacity, for the rest half of the trip, passenger load gradually decreases till Landran. In down direction, passenger volume in bus remains less than the seating capacity till Sector 47 where it rises till Zirakpur stop. The maximum load in both up and down direction is lower than the bus capacity indicating the fleet is underutilized during peak hours. The average trip length of passengers on this route is about 12.4 km indicating people travels shorter distances.

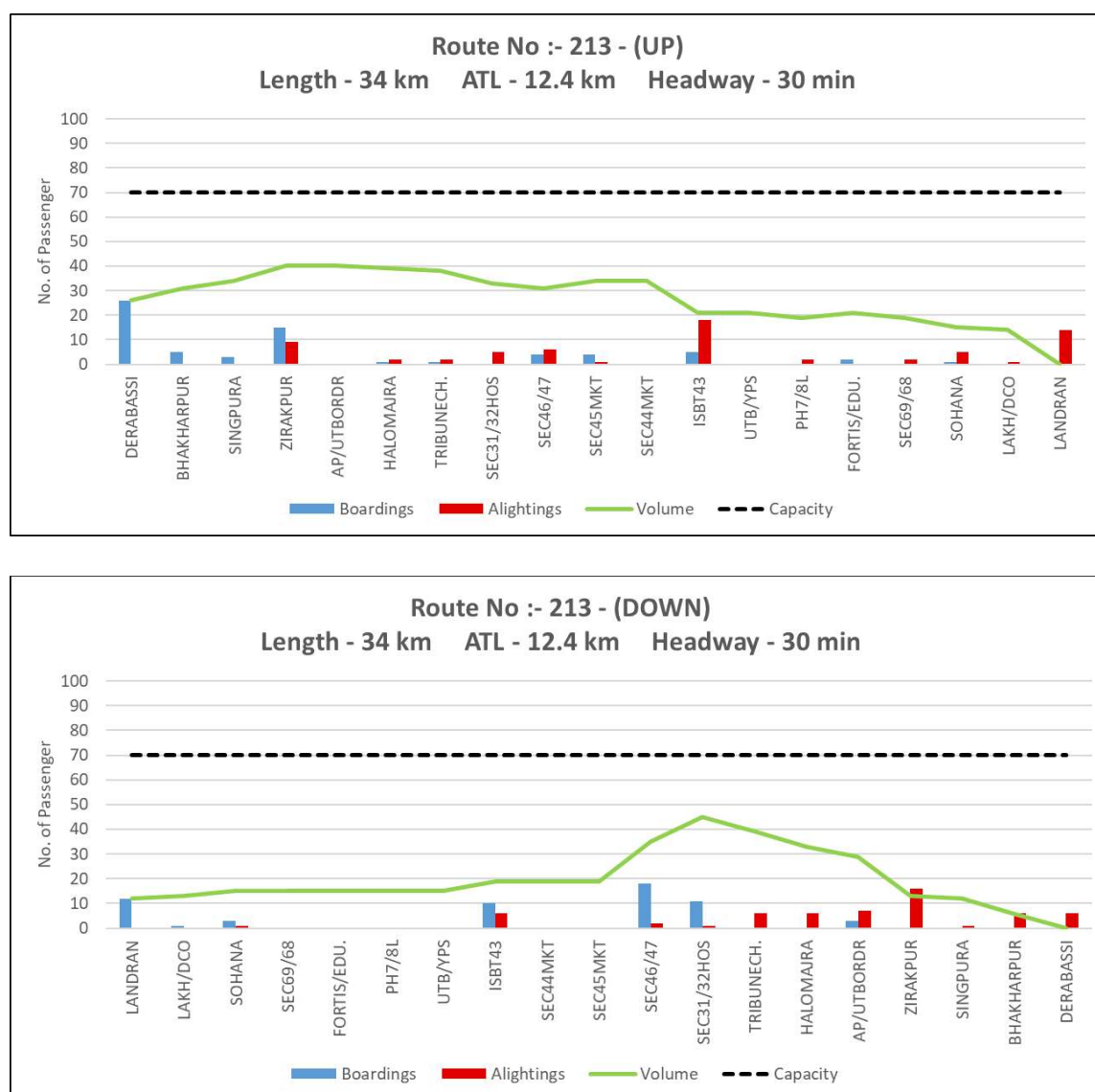


Figure 40: Peak hour Boarding & Alighting pattern along with line load of Route 213 (Up and Down)

Route no. 242: Landran to PGI via Lakhnaur, Sohana, Radha Swami/71, SCL, 71, IVY, Ph-5/3 B-II, Ph-4/3 B-I, Ph-2/3A, F/Chowk, 42/53, ISBT-43, 43/44, 35/34, 22/21, 17/22, 16 Hospital, 11/10, 11/2, PEC and N/OPD.

On route 242, 90% Boarding activity is seen from 4 major stops out of the total 20 stops, namely ISBT 43, Landran, Sohana and Saneta. Alighting are occurring at the last 5 stops which are PGI, PEC, ISBT17 and ISBT43. Passenger capacity is more than the bus capacity for the last half of the trip. In the first half of the trip the passenger capacity is slightly lesser than the bus capacity which still indicates an optimum utilization of the fleet. In down trip the passenger volume is close to the bus capacity for first half of the trip which falls from ISBT43 stop and stays lesser than the seating capacity for the later half. The Average trip length is 10.4kms which shows passengers taking trips for 34% of the route length.

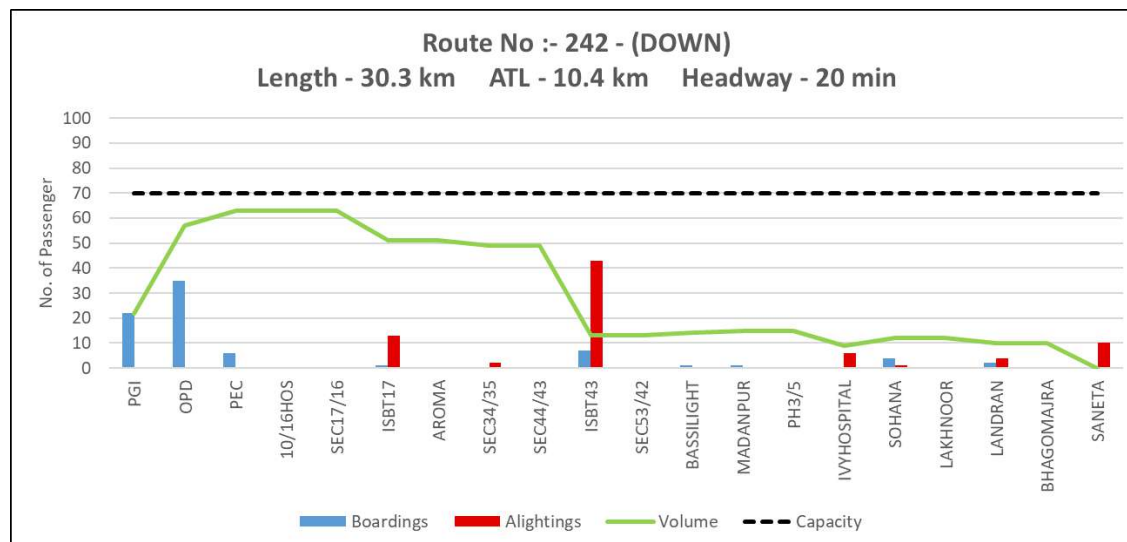
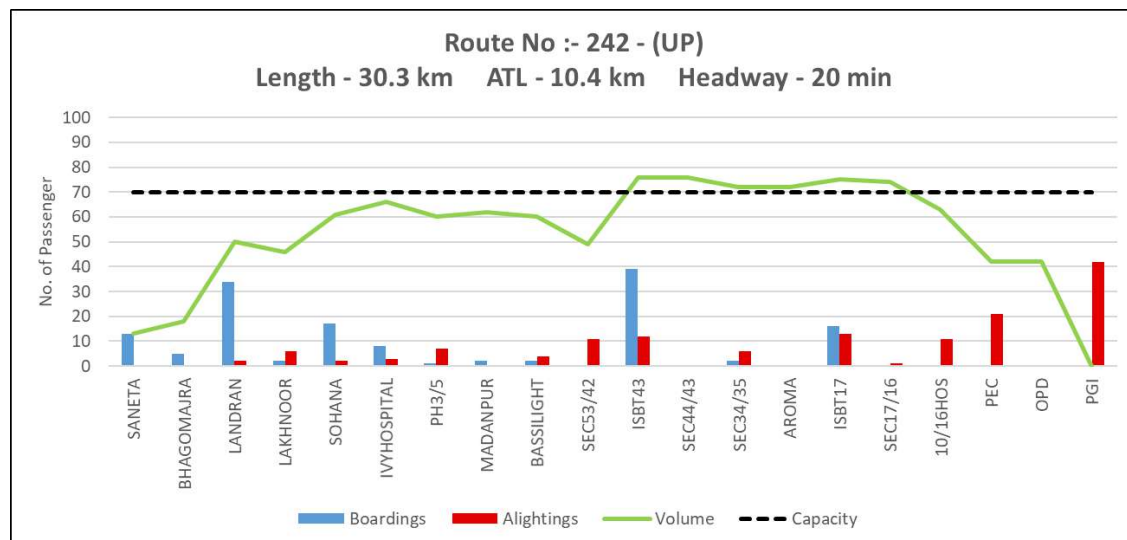


Figure 41: Peak hour Boarding & Alighting pattern along with line load of Route 242 (Up and Down)

Route no. 2B: PGI TO Panchkula Bus Stand via OPD, PEC, 10/11,10/16, 17/16, ISBT-17, 18/17, 8/18, 7/19, 26/27, 26/28, Grain Market, CTU workshop, Railway Station, Kalagram, Housing Board Chowk, 9/16, 10/15, 11/12 Really Chowk

From figure below, it can be seen that in up direction around 85% of passenger are boarding from railway station and ISBT17 and 80% of the passengers are alighting at ISBT17 and PGI. Whereas the down direction around 90% of passenger are boarding from 5 out of 19 stops namely, PGI, ISBT17, OPD, sector 16 and HBorder, while 90% of the passengers are alighting at railway station and PKL bus stand. In up direction, the passenger load is near to seating capacity from railway station to ISBT17 and rest trip has half the passenger as total seating. In down direction, passenger volume trajectory is gradually increasing up to total bus capacity till railway station. The maximum load in both up and down direction is similar to bus capacity indicating better utilization of fleet during peak hours. The average trip length of passengers on this route is about 10.2 km indicating people travel almost half of total route length.

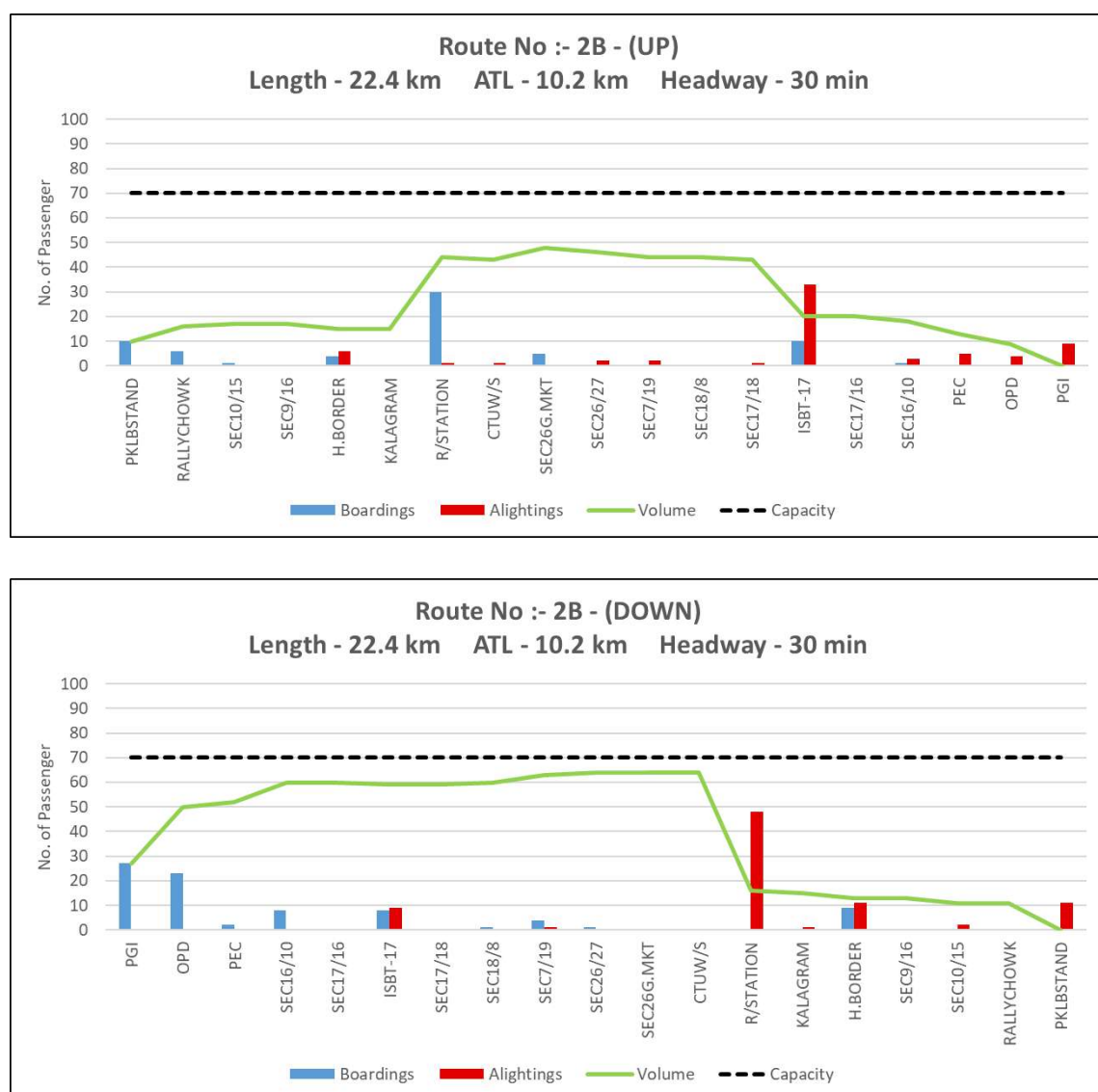


Figure 42: Peak hour Boarding & Alighting pattern along with line load of Route 2B (Up and Down)

Route no: 2F: PGI to Dhakoli Border via New OPD, PEC, 10/11, 10/16, 17/16, ISBT-17, 17/18, 8/18, 7/19, 26/27, 26/28, Grain Market., CTU workshop, Railway Station, Kalagram , Housing Board Chowk, Panchkula Sector -8/17, 9/16, 10/15, 10/11, 11/4, Sector 4 Panchkula., Sector 21 Panchkula, and Sector 20 Panchkula

From figure below, it can be seen that in up direction all stops perform similar boarding which is less than 10% of the boarding and 80% of the passengers are alighting at housing board chowk, PEC and PGI. Whereas the down direction all intermediate stops perform similar boarding and alighting during the trip which is less than 10% of the boarding. In up direction, the passenger load exceeds bus capacity throughout the trip. While in down direction, passenger volume remains similar to total bus seating capacity overall the trip. The maximum load in both up direction is similar to bus capacity indicating optimum utilization of fleet during peak hours. While down direction trip indicates maximum passenger load similar to seating capacity. The average trip length of passengers on this route is about 11.2 km indicating people travel 40% of total route length.

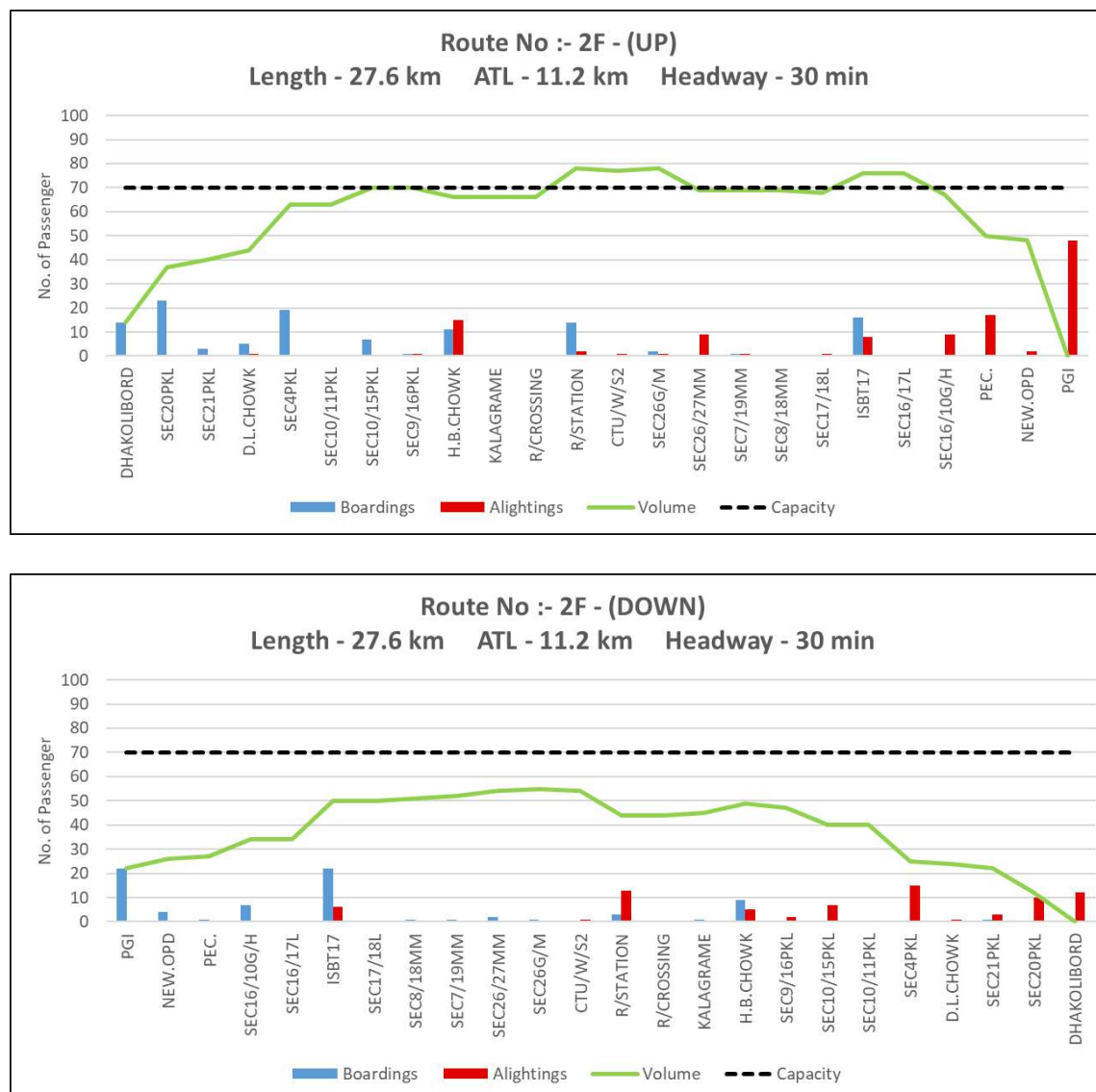


Figure 43: Peak hour Boarding & Alighting pattern along with line load of Route 2F (Up and Down)

Route no. 9C & 9A: Mata Mansa Devi to Mata Mansa Devi via –Mani majra, Housing Board Chowk, Kalagram, Railway Crossing, Railway Station, CTU Workshop, 28 Petrol Pump, Industrial. Area ph-1, Tribune Chowk, Industrial. Area Ph-2, Ramdarbar, Outer 47, Jagatpura, Ph-11, Police Chowki Sector 66, Mandikaran Houses, Sector 67, 67/68, 63/64, 49/50, 45/46, 33/32, 20/30, 27/19, 26/27, 26/28, Transport Chowk, Railway Crossing, Housing Board Colony and Manimajra

In this Circular route Boarding and alighting are seen to happen at all stops in a recurring way from the total 25 stops. Rest of the 6 stops on this route show minimal B&A activities. In anti-circular route 25% stops are used for Boarding activities from 4 stops which are M.Majra, Housing Board and R.darbar. Alighting are from Mansadevi Sector30, Sector 32 and Sector 46. The passenger volume in Up direction is more than the bus capacity which shows optimum utilization of the fleet and in down direction the capacity is close to the bus capacity which drops at Sector 46 and goes up after PH11. The average trip length is 9.6 km showing passengers travel for 25% of the trip length.

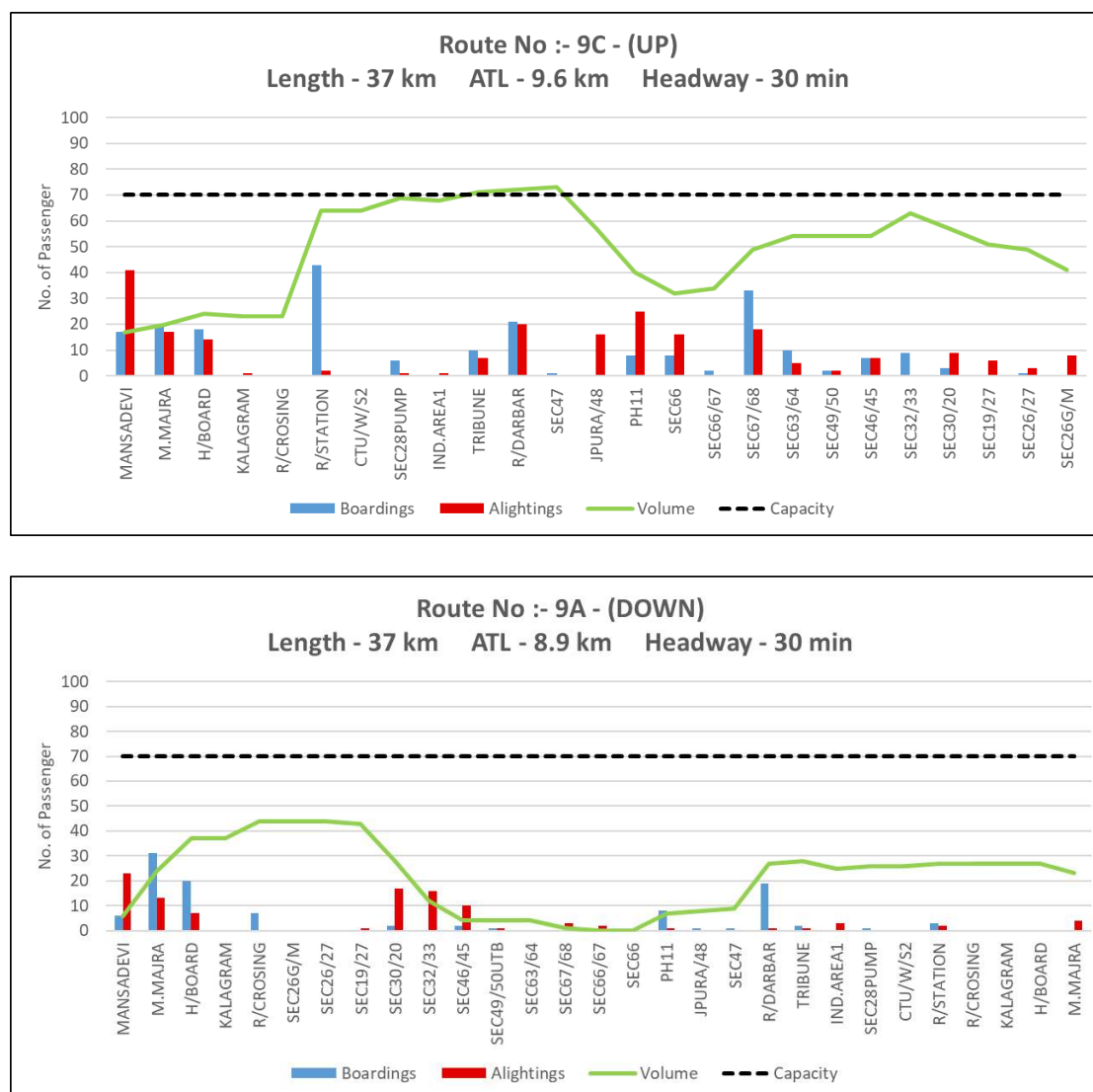


Figure 44: Peak hour Boarding & Alighting pattern along with line load of Route 9A and 9C

Route no. 20A: ISBT 17 to Kharar via Aroma, Sector 21 Market, 20/21,33/34, 44/45, ISBT-43, YPS, Ph-7, PSEB, Sector 68/69 Sohana, Lakhnaur, Landran, Chappar Chiri, Sante Majra

On Route 20 A the graph below for up route shows 85% boarding is done from Kharar stop and rest from 5 other stops out of the total 16 stops. Alighting is seen at 4 stops namely Sohana, LKHNOR.DCO, Landran and ISBT17. Rest of the 11 stops show less to nil B&A activities in the up route. In down route Boarding are from ISBT17, ISBT43, Sohana and LKHNOR.DCO stops. 90% of alighting is at Kharar stop. The passenger flow in up route is higher than the bus capacity for first half till Sohanan stop and then is lesser than the seating capacity for rest of the trip showing moderate utilization of this fleet. In Down trip the first major length till Sohana has lesser than seating capacity passengers which later on increases till Kharar stop showing moderate utilization of this fleet. The average trip length is 10.3kms showing short trips by passengers on this route.

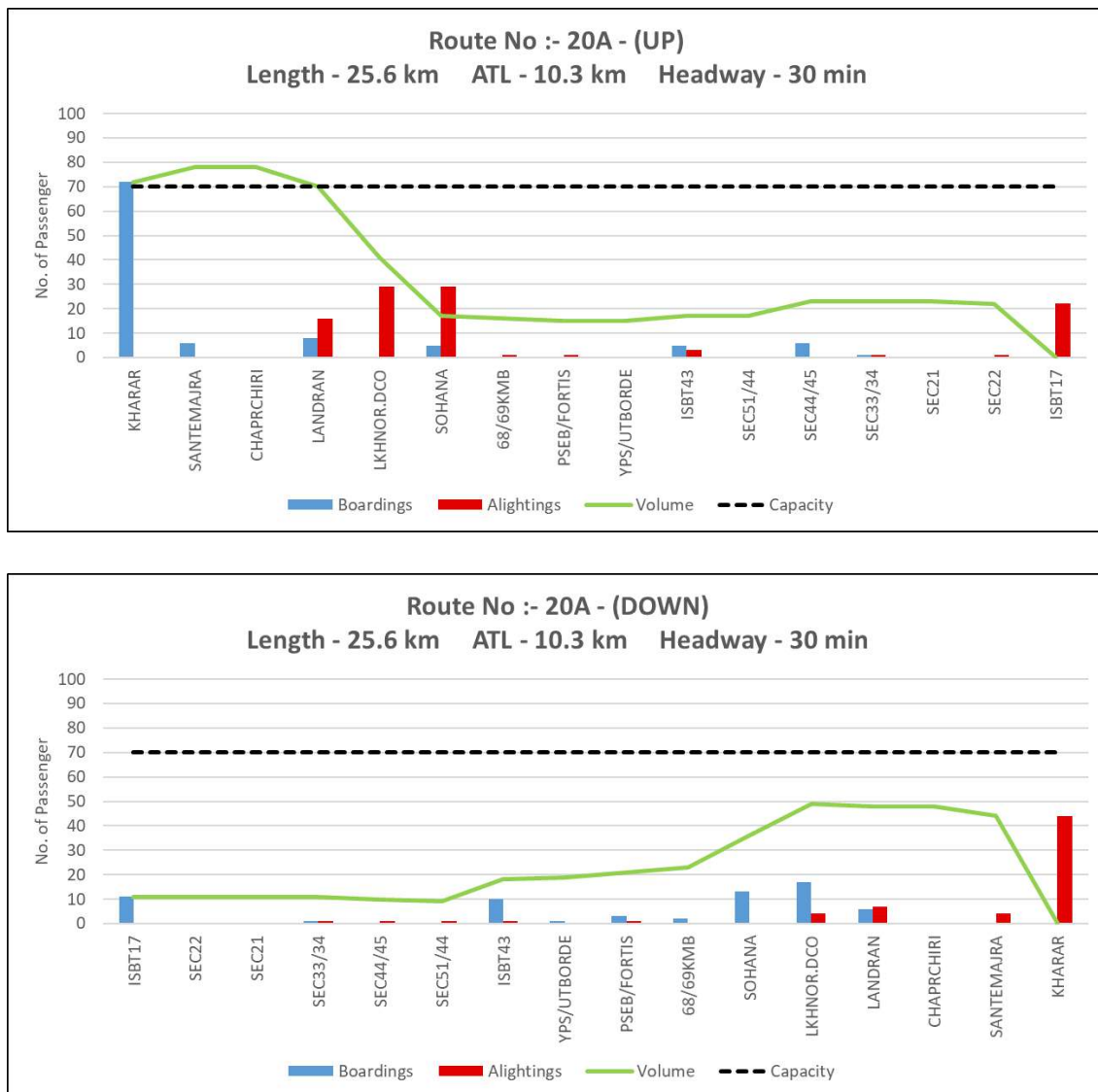


Figure 45: Peak hour Boarding & Alighting pattern along with line load of Route 20A (Up and Down)

Route no. 25A: ISBT 17 to Khizerabad via Sector16/10, 15/11, PGI, Khuda Lahora, S/P Barrier, Mullanpur, Parol, Majra, Balock, Sialwa.

This 27 km long Khizerabad to ISBT17 route shows 90% boarding in up direction from Khizerabad, Block and Sialba stops Alighting at ISBT 17, PGI, Sialba and Sector 11/15 stops. Out of the total 16 stops on this route 7 stops show low to nil B&A activities happening. Whereas in down direction 90% of the passengers are boarding from stops ISBT17, sector15, PGI and Mullanpur and majorly alighting at Block and Khizerabad. The passenger capacity in up direction exceed bus capacity from Sialba till PGI and remains close to half of the seating capacity till trip end at ISBT17. While in down direction passenger's capacity gradually increase from ISBT17 to Block to nearly bus capacity. As in both up and down direction, the maximum passenger load is reaching bus capacity on this route, there is optimum utilization of fleet. The Average trip length is 13.8kms showing passengers travel for 51% of this route length.

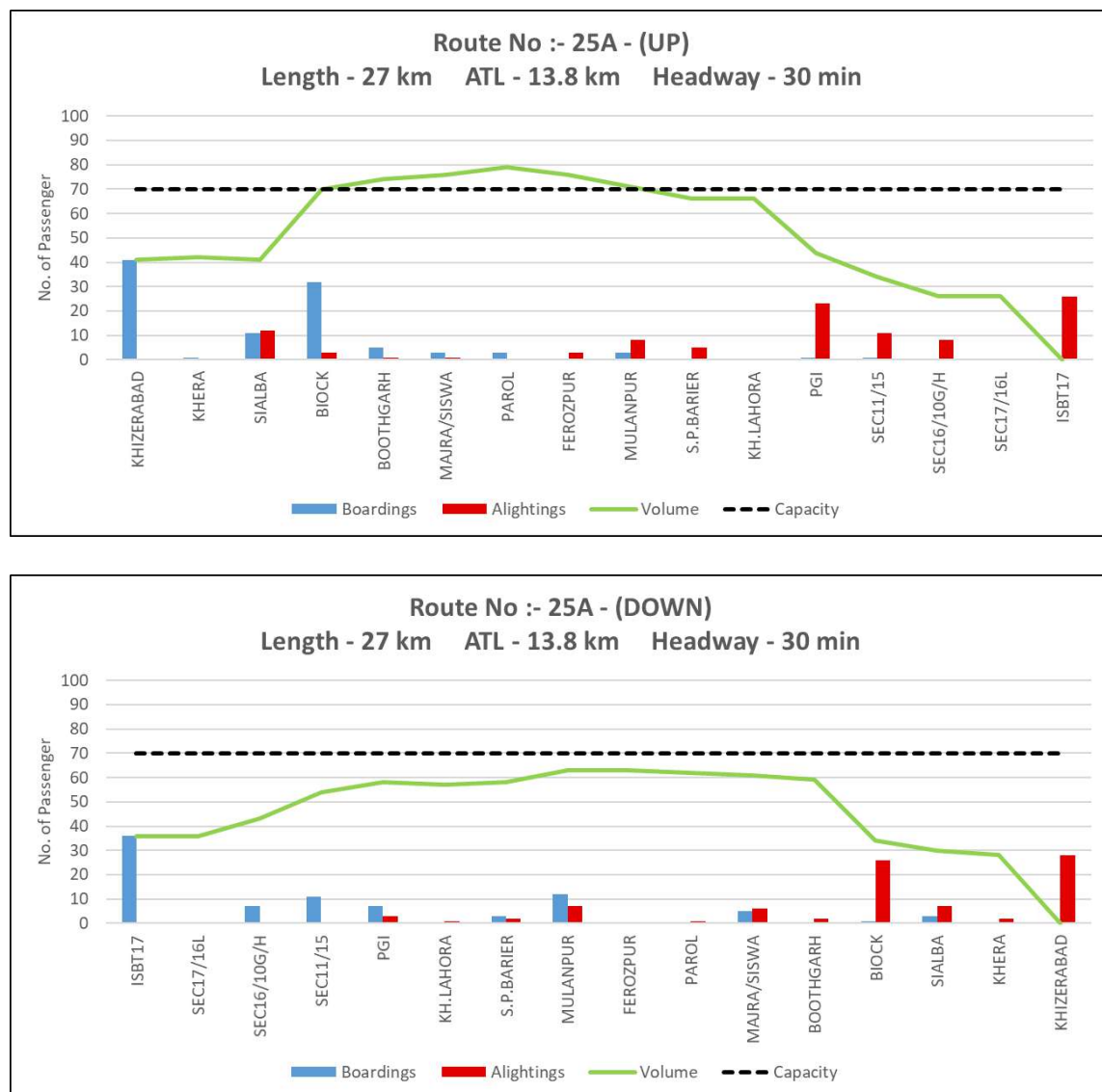


Figure 46: Peak hour Boarding & Alighting pattern along with line load of Route 25A

Route no. 30A: ISBT-43 to Cantonment via. Sector 43/44, 42/43, 36/35, Kissan Bhawan, 22/35, 21/34, 20/33, 30/32, 29/31 Tribune, Hallo Majra, Colony No.4, CTU workshop, Railway Station, Railway Crossing, Kalagram, Housing Board Chowk, Mani Majra, Command Hospital. Tank Chowk.

Route 30A shows 90% boarding activity from Cantonment stop, Military house and Housing board stops. Most of the Alighting are seen at Manimajra, ISBT43, Tank.Chowk, Sector 21/34 and Halomajra in up direction. Whereas in down direction 90% of the passengers are boarding from ISBT43, railway station and mani majra and more than 90% are alighting at sector32, housing board, military house and cantonment. This route has 19 stops out of which 9 stops show less B&A activities. Talking about Passenger capacity on this route the passenger flow remains close to the bus capacity showing good utilization of this fleet. The average trip length is 9kms long showing passenger travel for 42% of this route length.

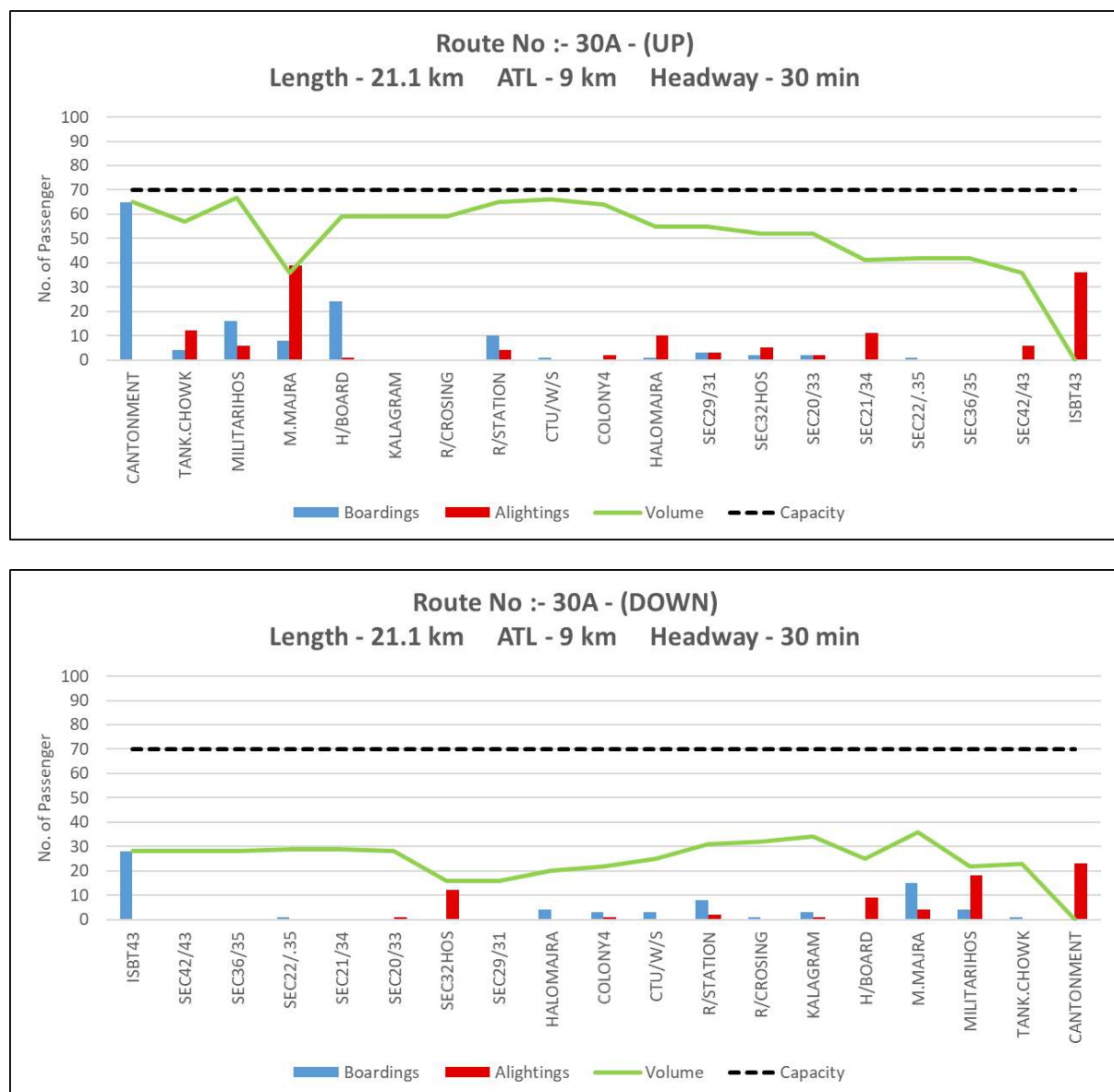


Figure 47: Peak hour Boarding & Alighting pattern along with line load of Route 30A

Route no. 30B: ISBT-43 to Panchkula Sector 4 via Sector 43/44,42/43, 36/35, 23/22, ISBT-17, Aroma, 21 Market., 20 Market., 19/27, 27 Market., 26/28 Grain Market., Railway Crossing, Kalagram Housing Board Chowk, 9/16 Panchkula, 10/15,11/12, Railway Chowk and Sector-4 Panchkula

From figure below, it can be seen that in up direction the first five stops perform 90% of boarding and all intermediate stops show almost similar alighting which is less than 15% of total boarding. Whereas in down direction almost all stops perform BA activity which is less than 15% of the boarding. In up direction passenger load lower than bus capacity. While in down direction, line load gradually increase till more than half of the seating capacity remains steady till trip end. The maximum load in up direction is above the bus seating capacity indicating and in down direction maximum line load equals to half of the seating capacity indicating moderate utilization of fleet during peak hours. The average trip length of passengers on this route is 9.3 km indicating people travel almost 45% of total route length.

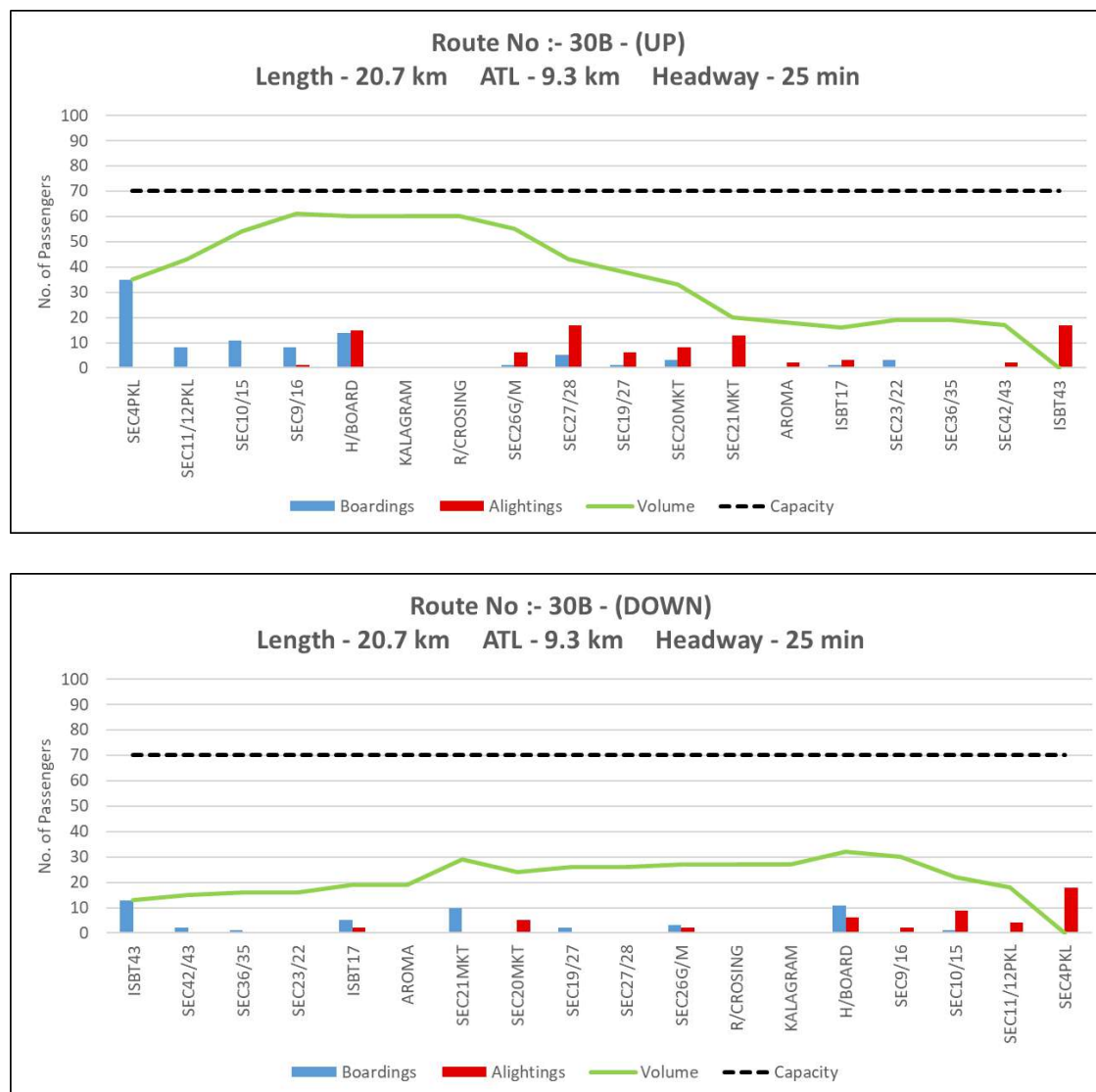


Figure 48: Peak hour Boarding & Alighting pattern along with line load of Route 30B (Up and Down)

Route no. 32A: New ISBT-Mohali to Mata Mansa Devi via Ph-6, Frenko, Sector 40/41, 41 Market, 42 Market, ISBT- 43, 43/44, 42/43, 36/35, 23/22, ISBT-17, 8/18, 7/19,26/27, 26 Grain Market, Railway Crossing, Kalagram , Housing Board and Mani Majra.

From figure below, it can be seen that in up direction around 5 stops out of 22 stops perform boarding and alighting while rest perform zero BA activity. Whereas in down direction most of the boarding and alighting are performed by ISBT17, Mansa Devi and housing board. In up down direction passenger load is 60% of the bus seating capacity till ISBT17 and decreases to 5 passengers on board afterwards. Whereas in down direction passenger volume remains null till ISBT17 indicating loss of revenue km till the half trip and the rest half performs 60% of the seating capacity. The maximum load in both up and down direction is half of bus seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 7.6 km indicating people travel shorter trips.

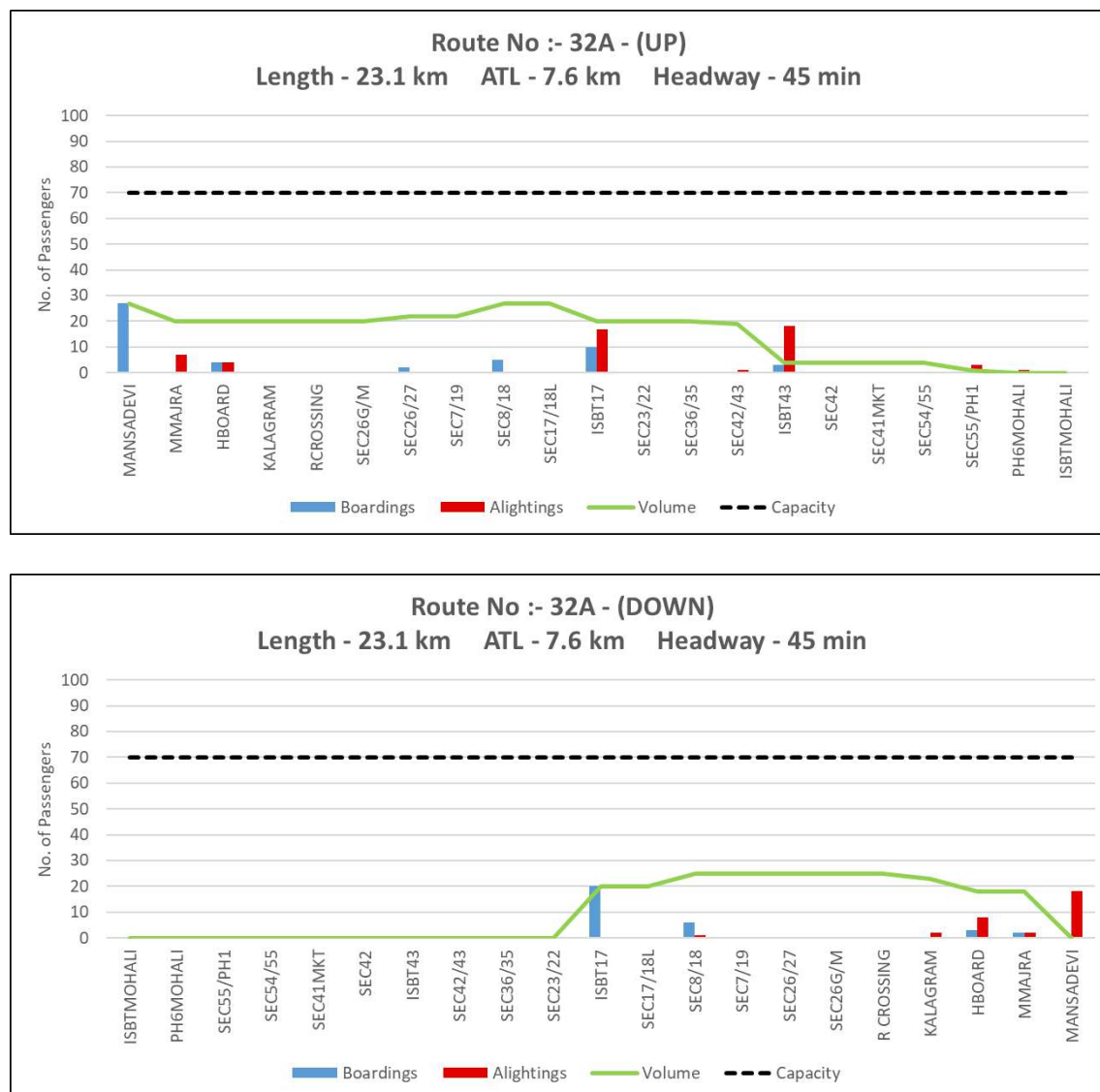


Figure 49: Peak hour Boarding & Alighting pattern along with line load of Route 32A (Up and Down)

Route no. 35B: ISBT-17 to Kharar via Aroma, Sector 35/22, Kissan Bhawan, 35/36, Sector 43/42, ISBT-43, 53/42, F/Chowk, Ph-2, Ph-6, Balongi, Daun, Desu Majra and Mundi Kharar

From figure below, it can be seen that in up direction around 75% of passenger are boarding from Kharar and mostly alighting at Phase 6 Mohali, Phase1 Mohali and the two terminals. Whereas in down direction all the intermediate stops same type of boarding which is less than 15% of the boarding and 90% are alighting at Phase-6 Mohali and Kharar. In up direction passenger volume trajectory is similar to the bus seating capacity from that trip start and gradually decreases from Phase-6 Mohali. While in down direction passenger load gradually increases to 60 passengers till Phase-6 Mohali and remains constant till the trip end. The maximum load in up direction exceeds bus capacity indicating optimum utilization of fleet during peak hours, while load remains near to bus capacity in the down direction. The average trip length of passengers on this route is about 11.2 km indicating people travel almost 60% of total route length.

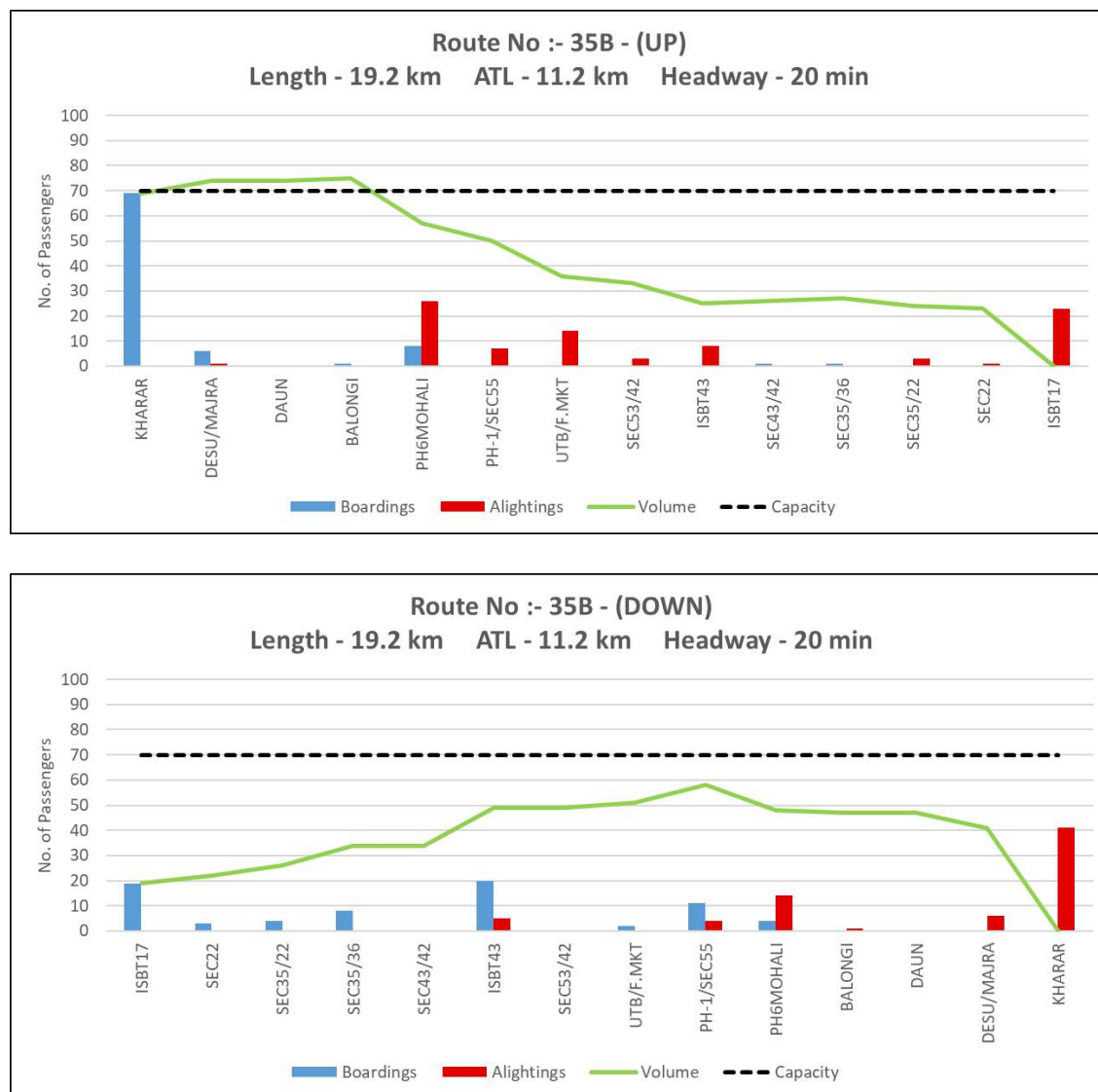


Figure 50: Peak hour Boarding & Alighting pattern along with line load of Route 35B (Up and Down)

Route no. 123A: Mohali Railway Station to PGI via Sector 65 market, 64 market, 63 market, 51/63, 51 market, ISBT-43, Sector 43/44, 35/34, Aroma, 22/17, 16 Hospital, Sec-11 market, OPD and PGI

From figure below, it can be seen that in up direction around 85% of passenger are boarding from Sector 65 market, ISBT43 and Sector 17/22 and all intermediate stops show similar alighting which is less than 15% of total boarding. Whereas in down direction around 90% of passenger are boarding PGI, PEC and ISBT43, while 90% of the passengers are alighting at ISBT43 and Sector17/22. In both up and down direction passenger load is similar to the bus seating capacity. The maximum load in both up and down direction is near to bus seating capacity indicating better utilization of fleet during peak hours. The average trip length of passengers on this route is about 7.7 km indicating people travel almost 35% of total route length.

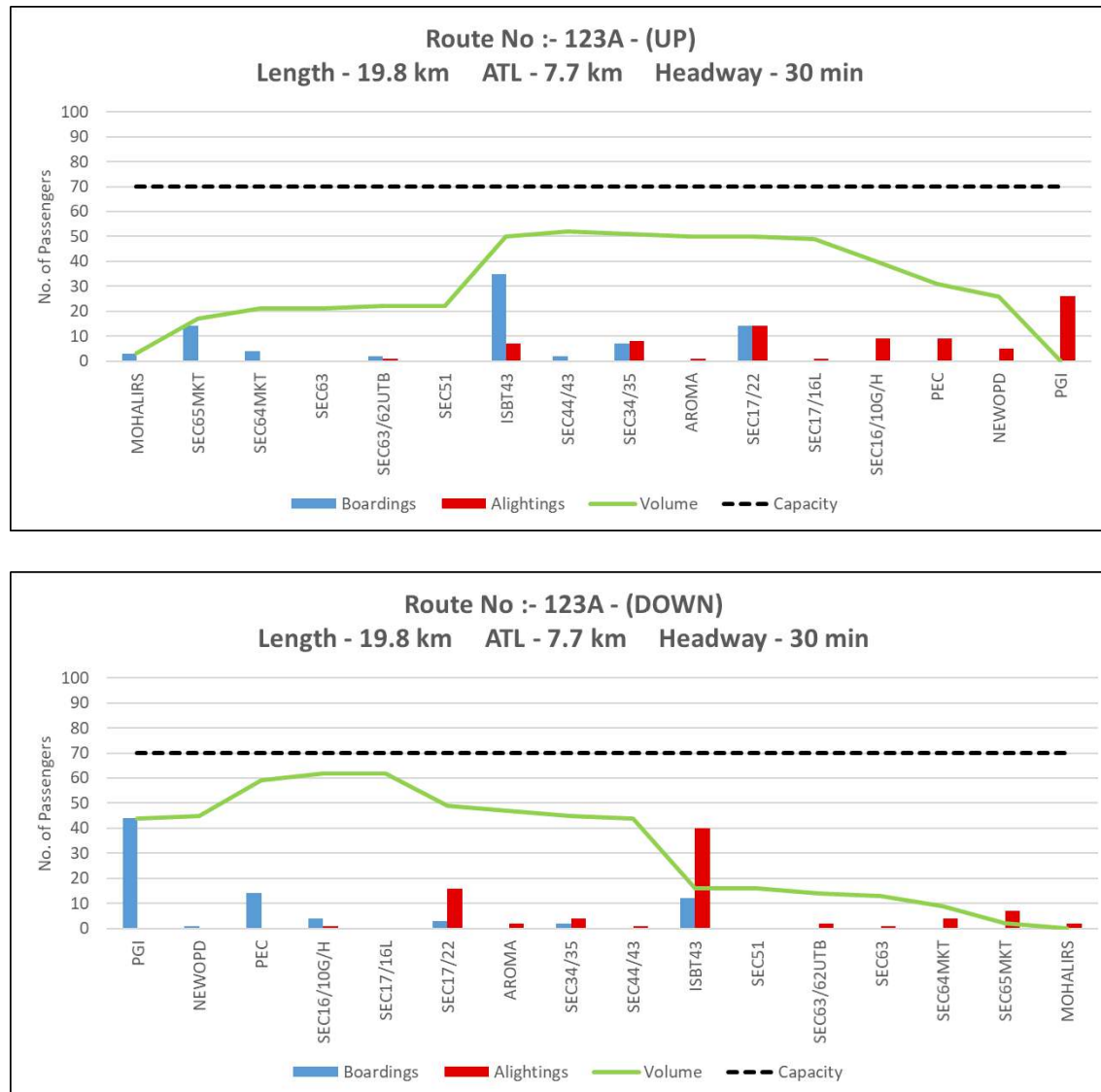


Figure 51: Peak hour Boarding & Alighting pattern along with line load of Route 123A (Up and Down)

Route no. 20AC: ISBT 17 to Kharar via Aroma, Sec-21 Market, 20/21, 33/34, 44/45, ISBT-43, YPS, 69/70, Ph-7, Sohana, Lakhnaur, Landran, Chappar Chiri and Sante Majra

On this Route Up direction shows 95% Boarding from Kharar, Santemajra and Landran. 90% of Alighting on this route are from ISBT17, Sohana and Lkhnor.Dco stops. In down direction as well major Boarding are from ISBT17, ISBT43 and Sohana stops. 90% alighting are at Kharar stop. Talking about the passenger flow in up direction for first half of the trip the passenger capacity is more than the bus capacity which lowers down after Sohana stop and stays close to the seating capacity for the later half of the trip showing good utilization of this fleet. In down direction for first half of the trip, passenger flow is close to bus capacity and in the later half is above the bus capacity showing over utilization of the fleet. The average trip length is 12.4kms showing that people travel for 48% of the trip length.

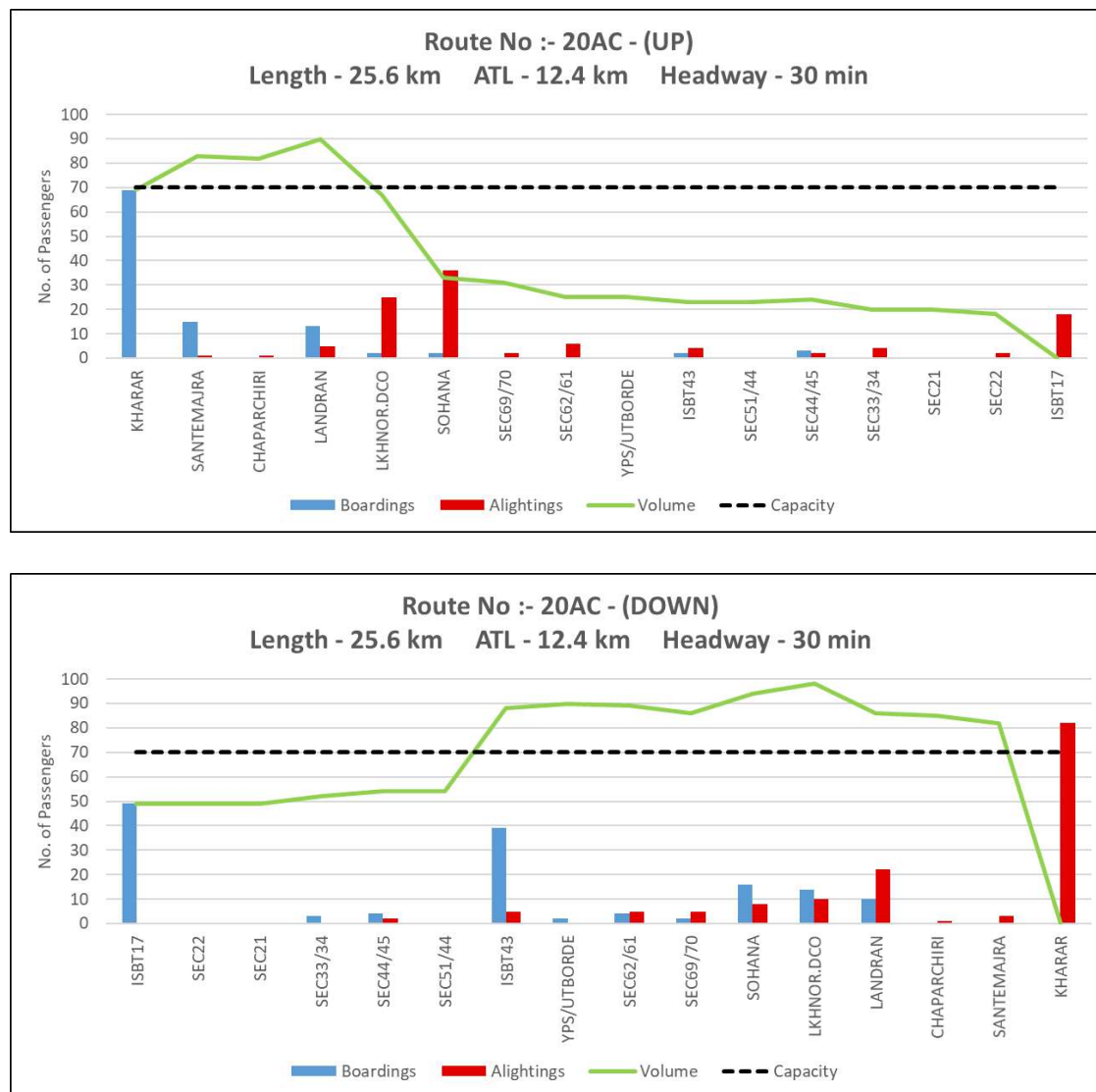


Figure 52: Peak hour Boarding & Alighting pattern along with line load of Route 20AC (Up and Down)

Route no. 26AC: Dhanas to Dhakoli via S.P. Barrier, Khuda Lahora, PGI, OPD, 10/11, 10/16, ISBT-17, 19/20, 27/30, 28/29, 28 P.P., CTU workshop, Railway Station, Colony No. 4, Hallo Majra, Airport, Zirakpur and Dhakoli

The figure below shows that in up route 95% of boarding are from Dhakoli and Zirakpur stops. And Alighting from Dhanas, PGI and ISBT17 stops. In down route major boarding are from PGI, New OPD and ISBT17 and 90% alighting are at Dhakoli and Railway station. Out of the total 22 stops in up direction only 4 stops show B&A activities and rest of the 18 stops have very less to nil activity. In down direction intermediate stops show recurring B&A activities. The passenger capacity in both up and down direction are close to the bus capacity showing optimum utilization of this fleet. The Average trip length is 17km showing passenger travel for 60% of the route length.

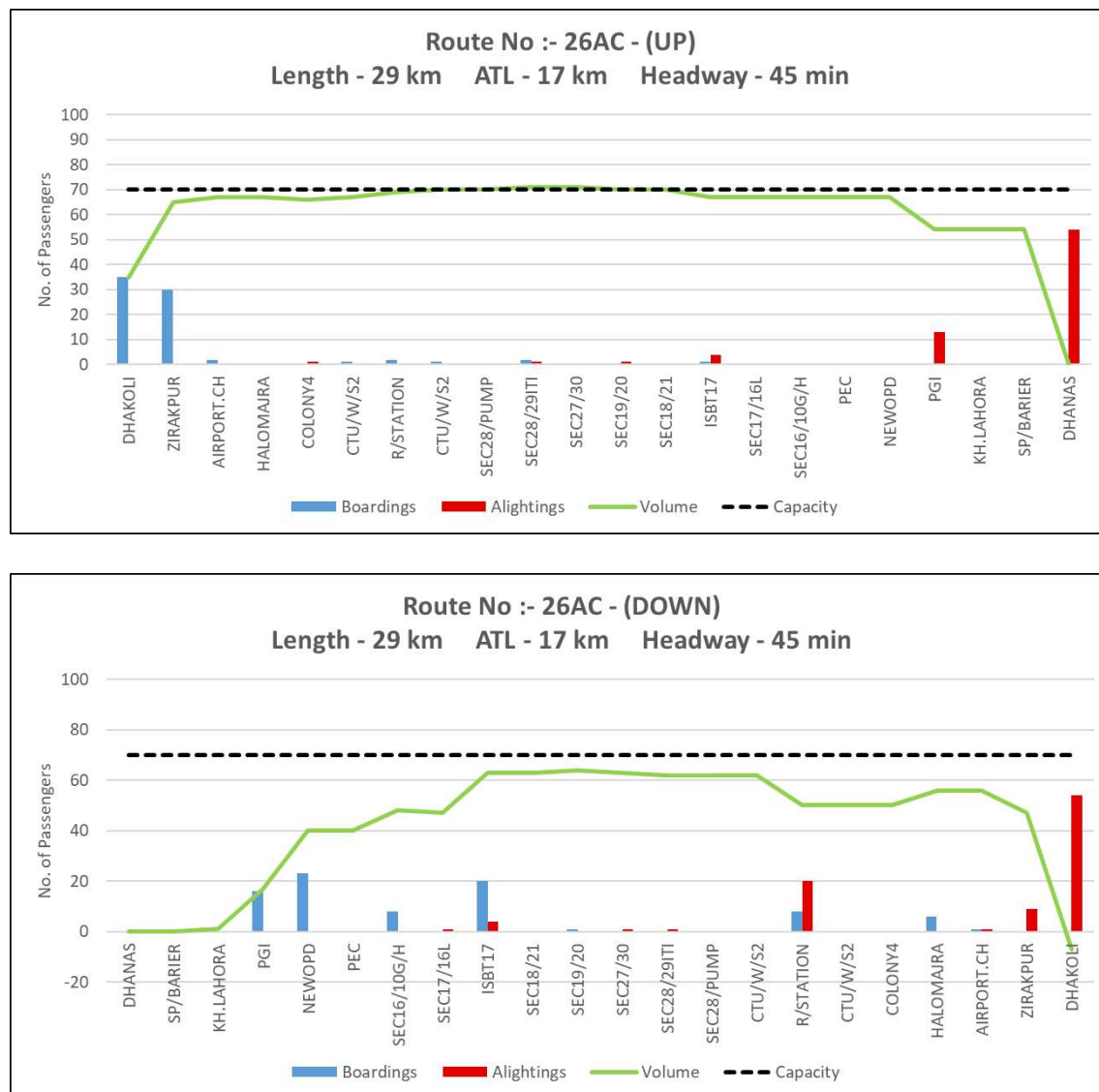


Figure 53: Peak hour Boarding & Alighting pattern along with line load of Route 26AC (Up and Down)

Route no. 36AC: PGI to New Airport via New OPD , Sector 11/2, 10/16 (Hospital), ISBT-17, Aroma, 21 market, 20 market, Sector 30/20 divided, 30/32 Hospital, 29/31, Tribune Chowk, Hallo Majra, old Airport Chowk, Zirakpur, Nabha Sahib and Dialpura outer

From figure below, it can be seen that in up direction around 5 stops out of 20 stops perform boarding namely Zirakpur, Nabhasahib, Jhungain, Hallo Majra and ISBT17 and 90% of the passengers alight at Sector16, PGI and ISBT17. Whereas in down direction all the intermediate stops perform similar boarding which is less than 15% of the boarding and 90% alight at Nanasahib, Zirakpur and new airport. In up direction passenger load exceeds bus capacity from Zirakpur till the trip end. Whereas in down direction passenger volume gradually increases and remains similar to seating till Zirakpur while the rest trip has load 40% of the seating capacity. The maximum load in both up direction exceeds bus capacity indicating optimum utilization of fleet during peak hours and maximum load in down trip is near to seating capacity. The average trip length of passengers on this route is about 13.9 km indicating people travel 45% of total route length.

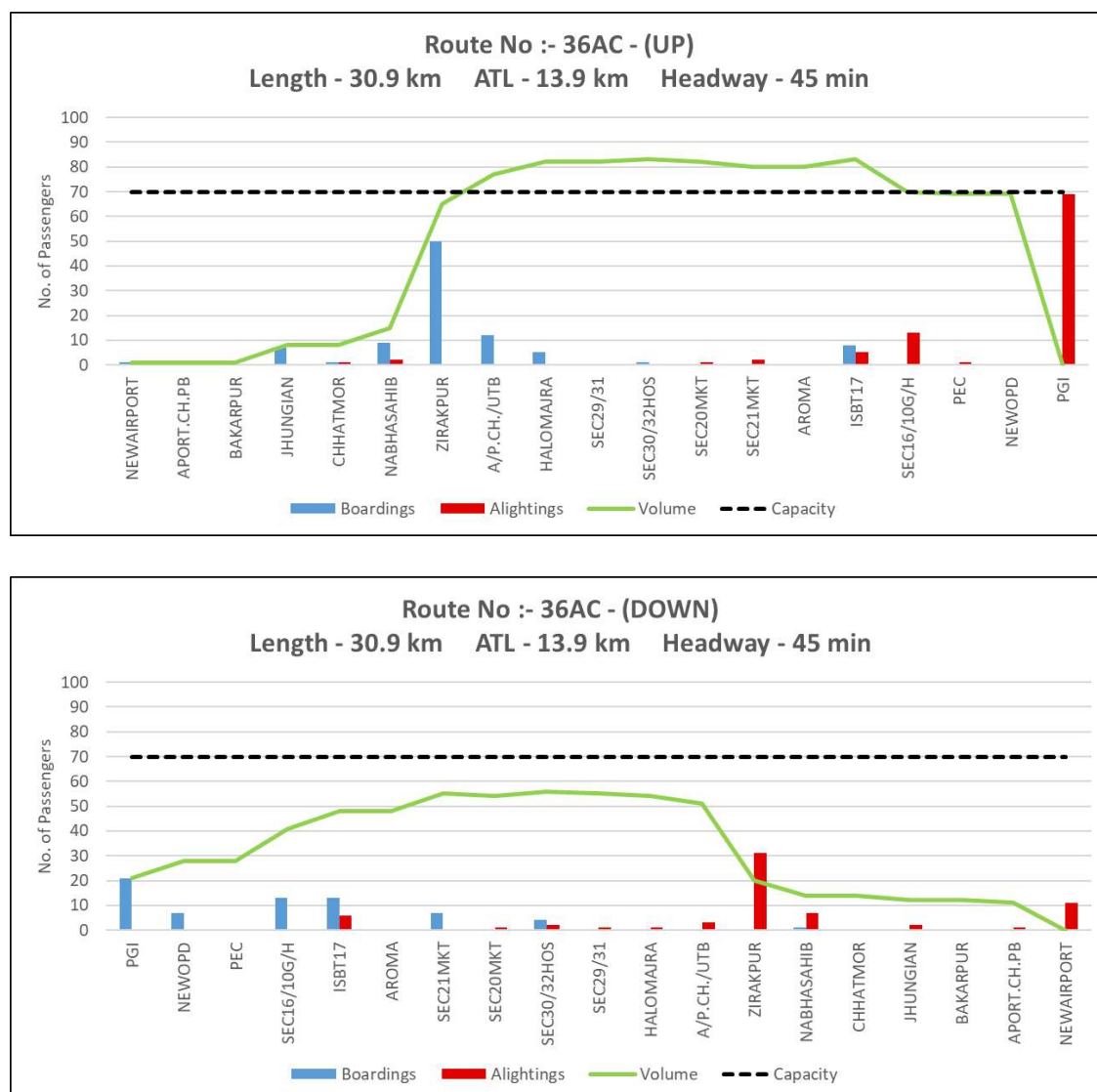


Figure 54: Peak hour Boarding & Alighting pattern along with line load of Route 36AC (Up and Down)

Route no.38AC: ISBT-17 to New Airport via Sector 22/23, 35/36, 43/42, ISBT-43, 51, 50, 49, 48 mkt, Jagatpura, 65 market outer, White House, Industrial. Area Ph-9, Sector 82 market and Airport

From figure below, it can be seen that in up direction 4 stops out of 14 stops perform boarding and rest shows zero BA activity and 70% of passengers alight at end stop i.e. ISBT17. Whereas in down direction, only 2 stops out of 14 stops perform all boarding activity namely ISBT43 and ISBT17 and 90% of the passengers are alighting at ISBT43 and new airport. In both up and down direction passenger load line remains almost half of bus seating throughout the trip. The maximum load in both up and down direction is also 65% of the seating indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is 10.8 km indicating people travel almost half of route length.

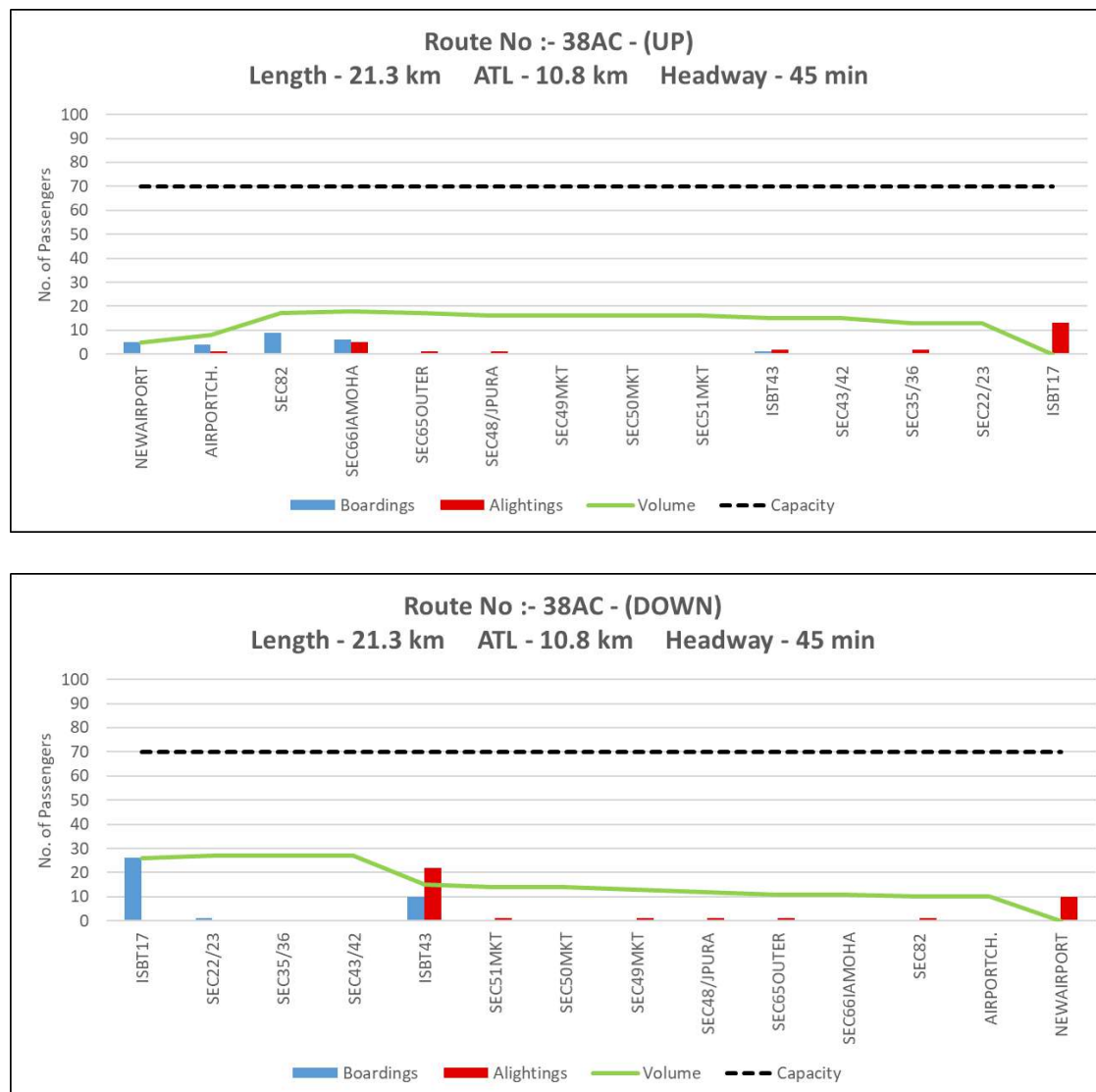


Figure 55: Peak hour Boarding & Alighting pattern along with line load of Route 38AC (Up and Down)

Route no. 211AC: PGI to Cantonment via OPD, Sector 11/2 , 16/10 Hospital, 17 Bus Stand, Aroma, 21, 20 Markets, 19/27, 27 market, 28 market, CTU workshop, Railway Station, Railway Crossing , Kalagram, Housing Board, Mani Majra and Command Hospital

In the figure below, it can be seen that in up direction 90% of the passengers are boarding from cantonment, housing board and ISBT17 and alighting at sector16, PEC and PGI. Whereas in down direction 80% of the passengers are boarding from sector16, sector21 market and mini majra and alighting at railway station, housing board and cantonment. In both up and down direction passenger volume gradually increases, stays more than and similar to seating capacity throughout the trip respectively. Maximum passenger load line in up direction almost reaches bus capacity and in down direction it is similar to seating capacity indicating moderate use of fleet during peak hours. The average trip lengths of passengers are 10.8 km indicating passenger travel 45% of the route length.

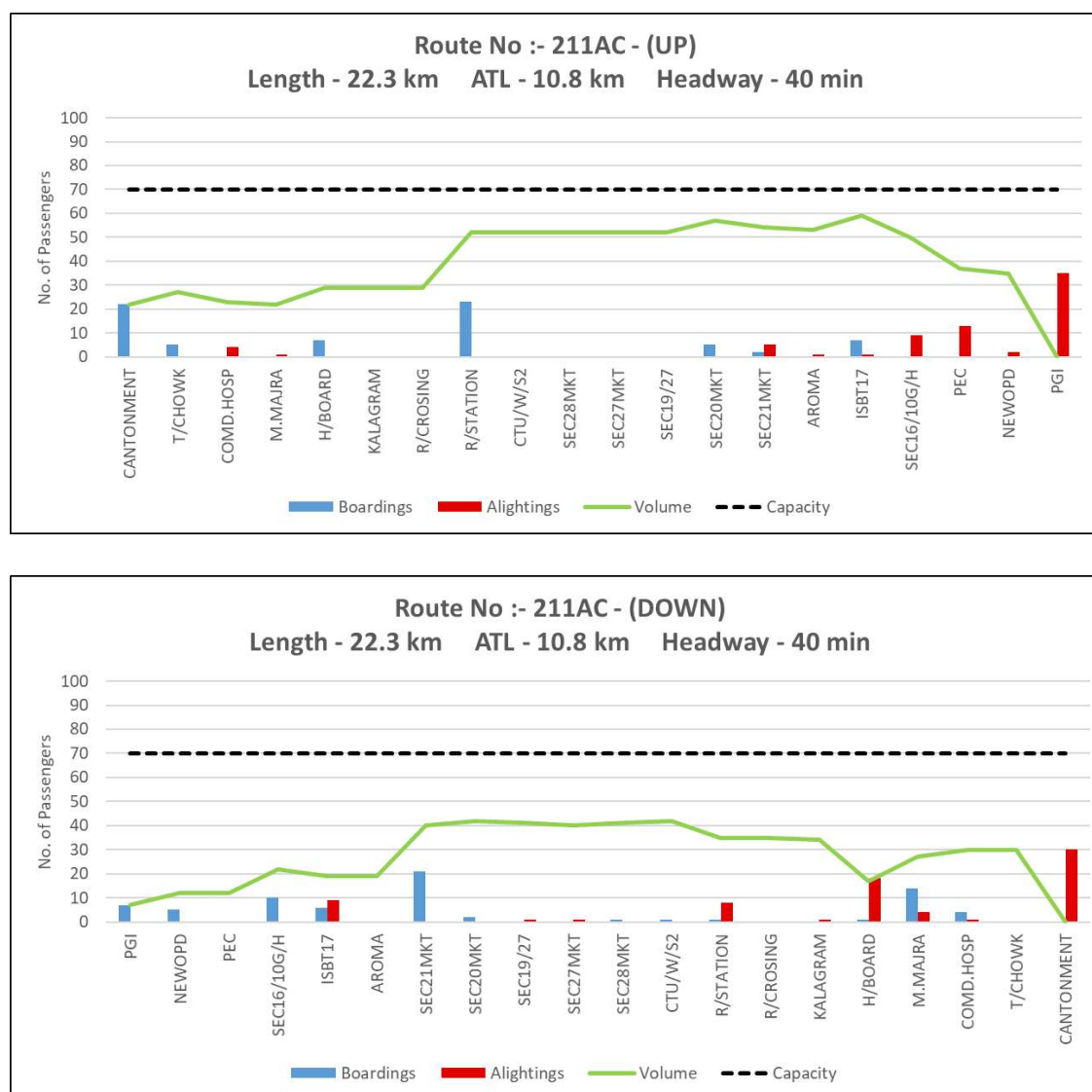


Figure 56: Peak hour Boarding & Alighting pattern along with line load of Route 211AC (Up and Down)

Route no. 214AC: SP Barrier to Nada Sahib via Khuda Lahora, PGI, OPD, PEC, Sector 10/16, 17/16, ISBT-17, 18 Market., 19 Market., 27 Market., 28 Market., Industrial Area Ph.1, Railway. Station, Railway. Crossing, Kalagram, Housing Board Chowk, Mani Majra, M. Hospital, old Panchkula and Majri chowk

In the figure below, it can be seen that in up direction 90% of the passengers are boarding from Nada sahib, housing board, railway station and ISBT17 and 65% are alighting at ISBT17, PGI and Sarangpur bridge. Whereas in down direction 90% of the passengers are boarding from PGI and ISBT17 and 70% are alighting at railway station and Nada Sahib. In up direction passenger volume gradually increases, stays 75% of seating capacity from PGI to railway station and further remains half of the seating till trip end. Maximum passenger load line in up direction almost reaches bus seating capacity and in down direction it is half of seating capacity indicating moderate use of fleet during peak hours. The average trip lengths of passengers are 9.4 km indicating passenger travel around 45% of the route length.

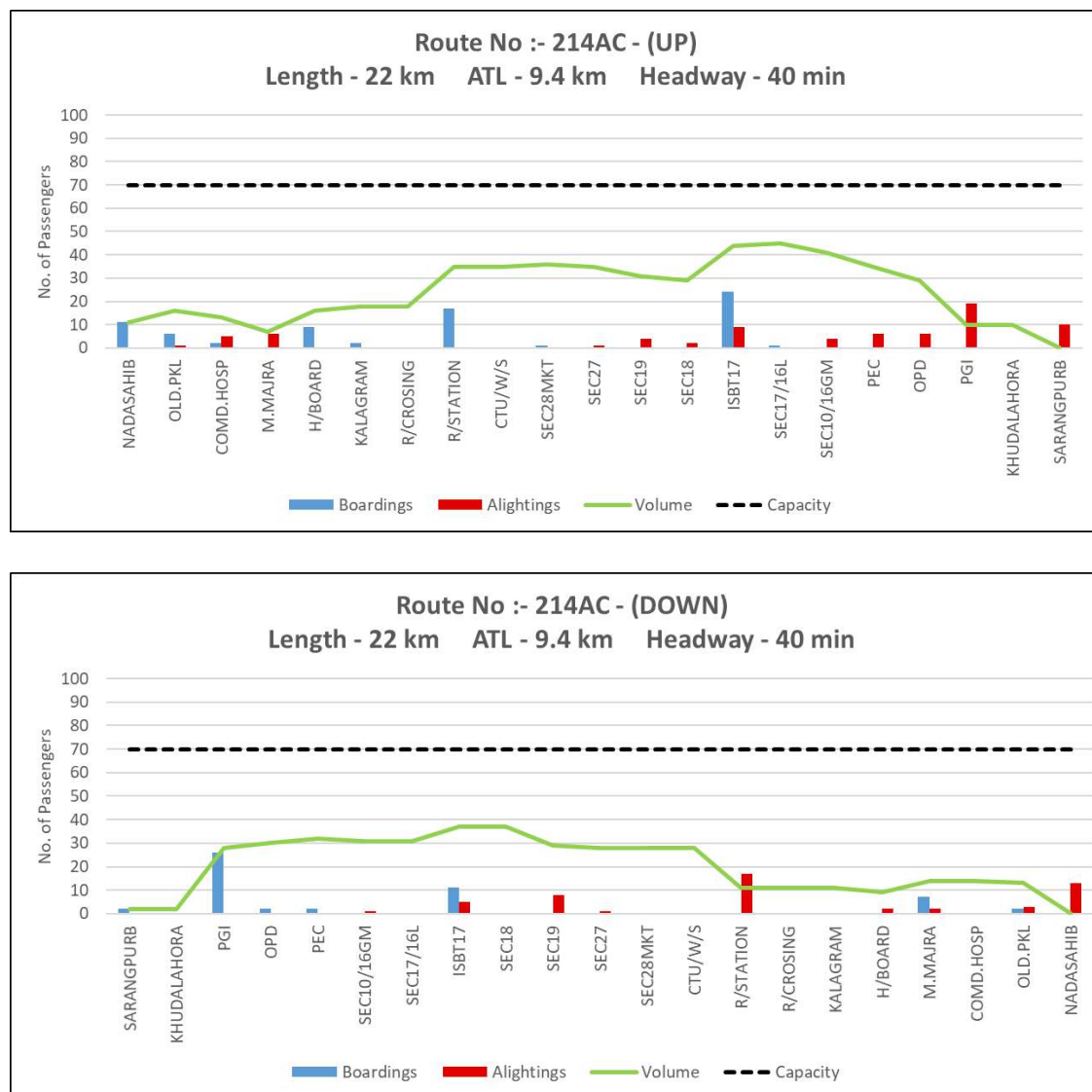


Figure 57: Peak hour Boarding & Alighting pattern along with line load of Route 214AC (Up and Down)

Route no. 216AC: ISBT-43 to Derabassi via Sector 43/44, 42/43, 36/35, Kissan Bhawan, 23/22, ISBT-17, Aroma, Sector 21/34, Sector20/33, Sector30/32, Tribune Chowk, Hallo Majra, Airport Chowk, Zirakpur and Bankharpur

In the figure below, it can be seen that in up direction 85% of the passengers are boarding from Derabassi and Singhpur and alighting at Singhpur, ISBT17 and ISBT43. Whereas in down direction direction 85% of the passengers are boarding from termianls and Zirakpur and alighting at Derabassi and Zirakpur. In both up and down direction passenger volume exceeds bus capacity throughout the trip. Maximum passenger load line of both up and down direction route exceeds the bus capacity showing optimum use of fleet during peak hours. The average trip lengths of passengers are 14.5 km that indicated passenger travel more than half of the route length.

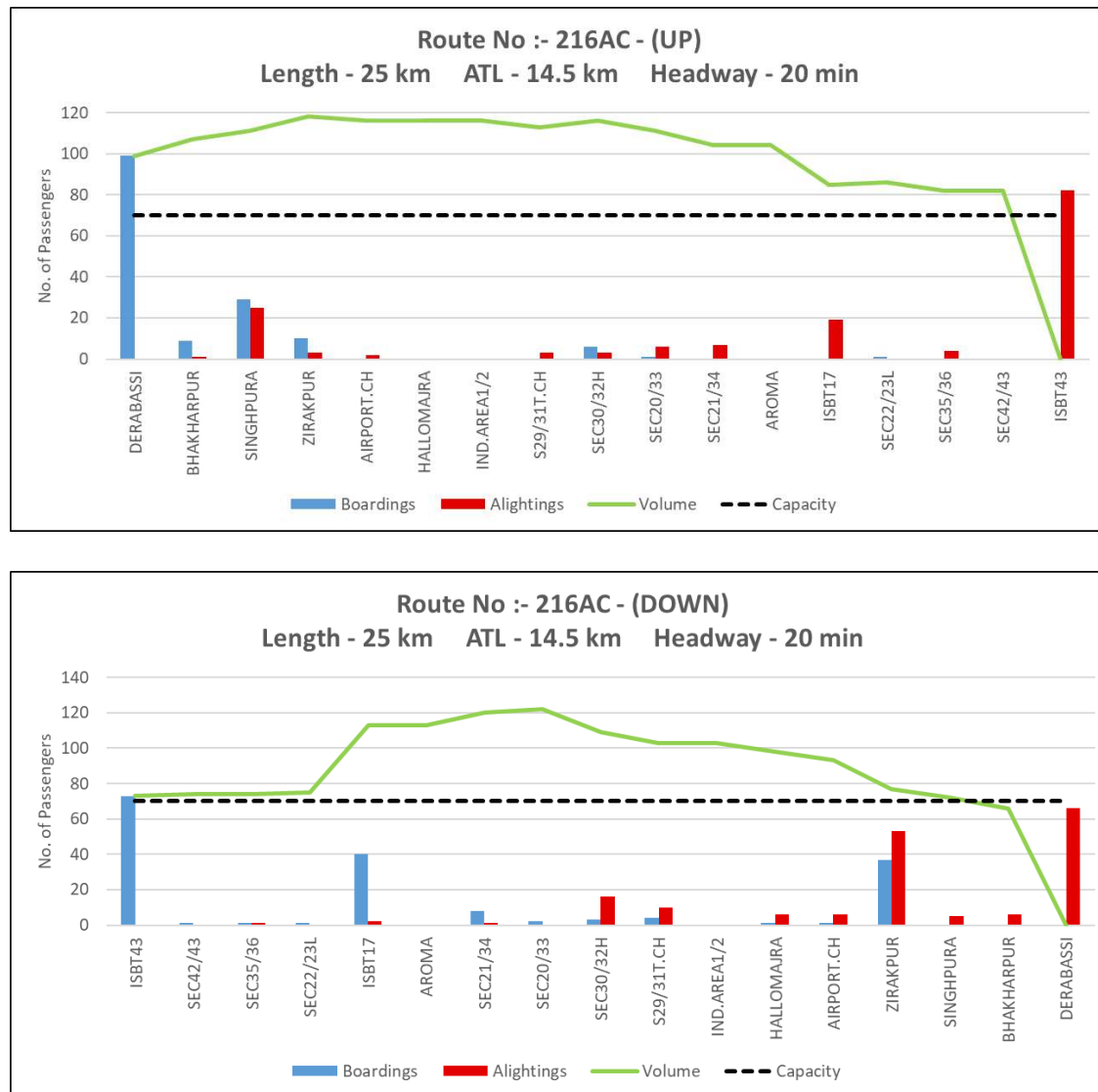


Figure 58: Peak hour Boarding & Alighting pattern along with line load of Route 216AC (Up and Down)

Route no. 38AAC: ISBT-17 to New Airport via Sector 22/23, 35/36, 43/42, ISBT-43, YPS Chowk, Ph-7, Sector 69/70, Sohana Gurdwara Sahib, 69/79, 68/80, 67/81, 82 market and Airport Chowk.

From figure below, it can be seen that in up direction all intermediate stop performs similar boarding and alighting activity which is less than 15% of the trip ridership. Boarding and alighting pattern is also similar in down direction which is less than 10% of the boarding. In both up and down direction passenger volume remains almost half of bus seating capacity throughout the trip. The maximum load in both up and down direction is half of bus seating capacity indicating underutilization of fleet during peak hours. The average trip length of passengers on this route is about 10 km indicating people travel 50% of route length.

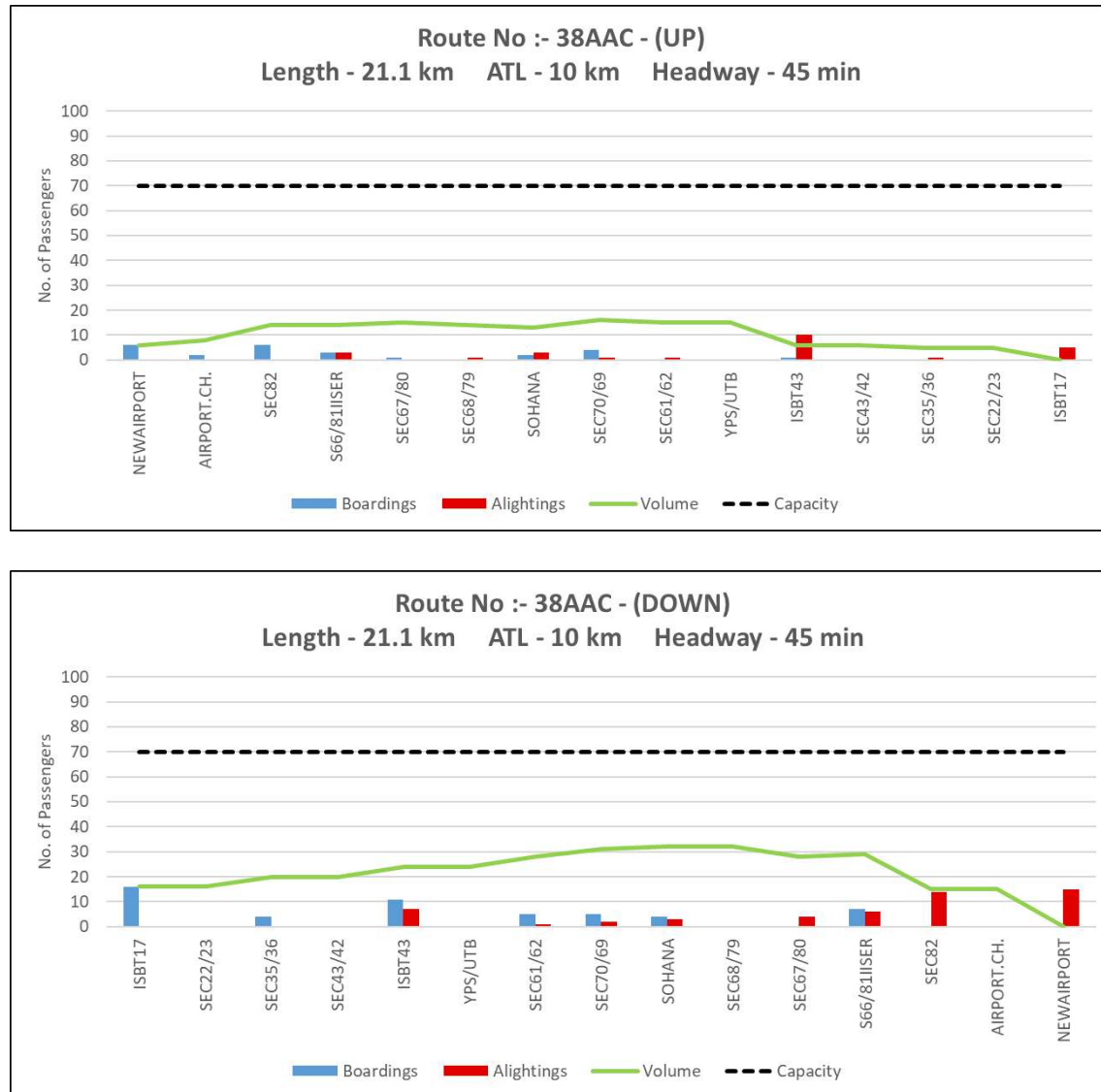


Figure 59: Peak hour Boarding & Alighting pattern along with line load of Route 38AAC (Up and Down)

Table 2: Details of route wise average trip length, maximum load and load factor

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
3	MOULIJAGRAN - PGI	17	60	8.5	UP	Morning	86	76	1.1
3	PGI - MOULIJAGRAN	17	60	8.2	DOWN	Morning	75	54	0.8
8	PH 11 - KAIMBWALA	25.6	35	6.0	UP	Afternoon	50	47	0.7
8	KAIMBWALA - PH 11	25.6	35	6.1	DOWN	Morning	48	26	0.4
11	PBCIVIL SEC - ISBT17	7.1	60	4.8	UP	Morning	30	30	0.4
11	ISBT17 - PBCIVIL SEC	7.1	60	5.2	DOWN	Morning	48	48	0.7
17	AORTBEHLAN - ISBT43	15.3	20	7.6	UP	Afternoon	82	60	0.9
17	ISBT43 - AORTBEHLAN	15.3	20	6.9	DOWN	Afternoon	59	42	0.6
18	KAIMBWALA - ISBT43	15.5	30	7.6	UP	Morning	54	50	0.7
18	ISBT43 - KAIMBWALA	15.5	30	6.8	DOWN	Morning	24	23	0.3
22	IT PARK - ISBT 43	19.6	15	8.5	UP	Afternoon	128	98	1.4
22	ISBT 43 - IT PARK	19.6	15	8.1	DOWN	Afternoon	115	77	1.1
25	KHIZARABAD - ISBT17	27.2	30	14.2	UP	Morning	108	94	1.3
25	ISBT17 - KHIZARABAD	27.2	30	13.1	DOWN	Evening	58	40	0.6
28	MANSADEVI - JUJHARNAGAR	25.9	45	8.9	UP	Afternoon	47	25	0.4
28	JUJHARNAGAR - MANSADEVI	25.9	45	10.2	DOWN	Morning	64	50	0.7
30	NADA SAHIB - ISBT43	21.5	40	8.7	UP	Afternoon	55	37	0.5
30	ISBT43 - NADA SAHIB	21.5	40	9.2	DOWN	Morning	95	49	0.7
32	DERABASSI - PGI	36.2	30	12.5	UP	Afternoon	133	53	0.8
32	PGI - DERABASSI	36.2	30	12.9	DOWN	Afternoon	110	66	0.9
34	DERABASSI - PGI	28	20	14.1	UP	Afternoon	61	52	0.7
34	PGI - DERABASSI	28	20	12.7	DOWN	Afternoon	67	46	0.7
35	KHARAR - ISBT17	18.3	20	11.5	UP	Afternoon	67	63	0.9
35	ISBT17 - KHARAR	18.3	20	11.8	DOWN	Afternoon	61	54	0.8
39	SEC65PH11 - PGI	24.4	15	9.5	UP	Afternoon	108	75	1.1
39	PGI - SEC65PH11	24.4	15	9.3	DOWN	Afternoon	112	88	1.3
40	KHDERABAD - ISBT17	31	60	14.2	UP	Afternoon	94	78	1.1

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
40	ISBT17 - KHDERABAD	31	60	13.4	DOWN	Afternoon	71	55	0.8
71	KAMBWALA - ISBT43	24	30	7.7	UP	Afternoon	70	39	0.6
71	ISBT43 - KAMBWALA	24	30	8.2	DOWN	Afternoon	73	57	0.8
79	CHATTBIR - PGI	27.9	45	12.5	UP	Morning	90	68	1.0
79	PGI - CHATTBIR	27.9	45	10.3	DOWN	Morning	64	42	0.6
80	ZIRAKPUR - PGI	20	30	9.8	UP	Afternoon	92	69	1.0
80	PGI - ZIRAKPUR	20	30	9.4	DOWN	Morning	48	35	0.5
143	AIRP.BEHLAN - DHANAS	21.3	75	7.4	UP	Morning	42	26	0.4
143	DHANAS - AIRP.BEHLAN	21.3	75	6.8	DOWN	Morning	40	25	0.4
212	DERA BASSI - PGI	20.4	20	15.0	UP	Afternoon	148	104	1.5
212	PGI - DERA BASSI	20.4	20	12.1	DOWN	Afternoon	118	79	1.1
213	DERABASSI - LANDRAN	34	30	12.4	UP	Morning	67	40	0.6
213	LANDRAN - DERABASSI	34	30	10.9	DOWN	Afternoon	58	45	0.6
241	MANSA DEVI - ISBT43	22	20	9.4	UP	Afternoon	104	61	0.9
241	ISBT43 - MANSA DEVI	22	20	9.6	DOWN	Afternoon	78	65	0.9
242	SANETA - PGI	30.3	20	10.4	UP	Afternoon	141	76	1.1
242	PGI - SANETA	30.3	20	9.8	DOWN	Afternoon	79	63	0.9
10AC	MANSA DEVI - ISBT43	19.4	25	9.4	UP	Morning	66	53	0.8
10AC	ISBT43 - MANSA DEVI	19.4	25	9.3	DOWN	Afternoon	53	32	0.5
123A	MOHALI RS - PGI	19.8	30	7.7	UP	Afternoon	81	52	0.7
123A	PGI - MOHALI RS	19.8	30	8.1	DOWN	Afternoon	80	62	0.9
123AAC	MOHALI RS - PGI	19.8	95	9.3	UP	Afternoon	82	52	0.7
123AAC	PGI - MOHALI RS	19.8	95	8.9	DOWN	Afternoon	46	36	0.5
143AC	APORTBEHLAN - DHANAS	21.3	150	11.3	UP	Morning	92	79	1.1
143AC	DHANAS - APORTBEHLAN	21.3	150	7.9	DOWN	Morning	44	32	0.5
17AC	APORTBEHLAN - ISBT43	15.3	50	8.8	UP	Afternoon	55	47	0.7
17AC	ISBT43 - APORTBEHLAN	15.3	50	8.2	DOWN	Afternoon	40	29	0.4
18AC	KAIMBWALA - ISBT43	15.5	40	7.5	UP	Afternoon	34	30	0.4

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
18AC	ISBT43 - KAIMBWALA	15.5	40	8.1	DOWN	Afternoon	30	28	0.4
18B	HIGH COURT - ISBT43	14	30	8.6	UP	Morning	29	21	0.3
18B	ISBT43 - HIGH COURT	14	30	7.7	DOWN	Morning	52	41	0.6
1A	PGI - PGI	27.4	20	7.5	DOWN	Afternoon	146	65	0.9
1C	PGI - PGI	27.4	20	7.4	UP	Afternoon	96	54	0.8
202AC	PBCIVILSECT - ISBT 43	13	15	6.9	UP	Afternoon	46	42	0.6
202AC	ISBT 43 - PBCIVILSECT	13	15	8.2	DOWN	Afternoon	89	76	1.1
203AC	MANSA DEVI - ISBT43	18	35	9.3	UP	Morning	64	54	0.8
203AC	ISBT43 - MANSA DEVI	18	35	8.5	DOWN	Morning	50	31	0.4
205AC	PGI - ISBT 43	11.1	15	7.6	UP	Morning	57	47	0.7
205AC	ISBT 43 - PGI	11.1	15	8.2	DOWN	Afternoon	113	112	1.6
206AC	IT PARK - ISBT 43	18.01	20	9.6	UP	Afternoon	103	64	0.9
206AC	ISBT 43 - IT PARK	18.01	20	10.0	DOWN	Afternoon	85	71	1.0
20A	KHARAR - ISBT17	25.6	30	10.3	UP	Afternoon	103	78	1.1
20A	ISBT17 - KHARAR	25.6	30	11.4	DOWN	Afternoon	64	49	0.7
20AC	KHARAR - ISBT17	25.6	30	12.4	UP	Afternoon	106	90	1.3
20AC	ISBT17 - KHARAR	25.6	30	13.3	DOWN	Afternoon	143	98	1.4
211AC	CANTONMENT - PGI	22.3	40	10.8	UP	Afternoon	71	59	0.8
211AC	PGI - CANTONMENT	22.3	40	9.6	DOWN	Morning	73	42	0.6
212AC	DERA BASSI - PGI	20.4	75	13.5	UP	Afternoon	90	74	1.1
212AC	PGI - DERA BASSI	20.4	75	13.4	DOWN	Morning	88	68	1.0
214AC	NADA SAHIB - SARANGPUR B	22	40	9.4	UP	Afternoon	73	45	0.6
214AC	SARANGPUR B - NADA SAHIB	22	40	9.2	DOWN	Afternoon	52	37	0.5
216AC	DERABASSI - ISBT 43	25	20	14.5	UP	Afternoon	155	118	1.7
216AC	ISBT 43 - DERABASSI	25	20	14.0	DOWN	Afternoon	172	122	1.7
239AAC	ISBT 43 - ISBT 43	24	25	7.5	DOWN	Morning	78	62	0.9
239CAC	ISBT 43 - ISBT 43	24	25	9.8	UP	Morning	107	54	0.8
23A	KUDALISHER - ISBT43	13.8	15	6.3	UP	Afternoon	51	44	0.6

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
23A	ISBT43 - KUDALISHER	13.8	15	7.3	DOWN	Afternoon	51	45	0.6
240AC	MATA M DEVI - ISBT43	20.4	30	10.8	UP	Afternoon	37	28	0.4
240AC	ISBT43 - MATA M DEVI	20.4	30	9.7	DOWN	Afternoon	46	24	0.3
242AC	SANETA - PGI	30.3	50	10.5	UP	Afternoon	133	85	1.2
242AC	PGI - SANETA	30.3	50	11.1	DOWN	Afternoon	144	116	1.7
24AAC	PGI - ISBT43	9.8	80	5.2	UP	Evening	20	19	0.3
24AAC	ISBT43 - PGI	9.8	80	6.7	DOWN	Evening	36	36	0.5
24CAC	ISBT17 - ISBT43	5.7	25	5.6	UP	Evening	22	22	0.3
24CAC	ISBT43 - ISBT17	5.7	25	5.5	DOWN	Evening	43	43	0.6
254AC	IT PARK - ISBT43	19.6	70	9.9	UP	Afternoon	41	32	0.5
254AC	ISBT43 - IT PARK	19.6	70	8.4	DOWN	Afternoon	40	25	0.4
25A	KHIZERABAD - ISBT17	27	30	13.8	UP	Afternoon	101	79	1.1
25A	ISBT17 - KHIZERABAD	27	30	13.7	DOWN	Afternoon	85	63	1.0
26AC	DHAKOLI - DHANAS	29	45	17.0	UP	Afternoon	74	71	1.0
26AC	DHANAS - DHAKOLI	29	45	13.2	DOWN	Afternoon	84	64	0.9
28A	ISBT43 - ISBT43	25.7	30	7.4	DOWN	Afternoon	68	30	0.4
28C	ISBT43 - ISBT43	25.7	30	7.7	UP	Afternoon	84	63	0.9
2A	MANI MAJRA - MANIMAJRA	38.3	20	9.1	DOWN	Afternoon	142	61	0.9
2B	PKLBSTAND - PGI	22.4	30	10.2	UP	Morning	67	48	0.7
2B	PGI - PKLBSTAND	22.4	30	9.2	DOWN	Morning	83	64	0.9
2BAC	PKLBSTAND - PGI	22.4	110	11.0	UP	Morning	58	48	0.7
2BAC	PGI - PKLBSTAND	22.4	110	8.8	DOWN	Morning	52	38	0.5
2C	MANIMAJRA - MANI MAJRA	38.3	20	8.9	UP	Afternoon	119	47	0.7
2CAC	MANIMAJRA - MANI MAJRA	38.3	60	10.9	UP	Afternoon	67	29	0.4
2D	IT PARK - PGI	21	35	9.0	UP	Morning	74	65	0.9
2D	PGI - IT PARK	21	35	8.4	DOWN	Morning	41	22	0.3
2F	DHAKOLIBORD - PGI	27.6	30	11.2	UP	Morning	116	78	1.1
2F	PGI - DHAKOLIBORD	27.6	30	10.2	DOWN	Afternoon	75	55	0.8

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
2FAC	DHAKOLIBORD - PGI	27.6	120	12.8	UP	Afternoon	83	74	1.1
2FAC	PGI - DHAKOLIBORD	27.6	120	12.2	DOWN	Afternoon	55	40	0.6
30A	CANTONMENT - ISBT43	21.1	30	9.0	UP	Afternoon	137	67	1.0
30A	ISBT43 - CANTONMENT	21.1	30	8.4	DOWN	Afternoon	71	36	0.5
30AAC	CANTONMENT - ISBT43	21.1	85	10.7	UP	Afternoon	70	43	0.6
30AAC	ISBT43 - CANTONMENT	21.1	85	9.2	DOWN	Afternoon	57	43	0.6
30AC	NADA SAHIB - ISBT43	21.5	95	9.9	UP	Morning	72	41	0.6
30AC	ISBT43 - NADA SAHIB	21.5	95	10.2	DOWN	Afternoon	70	44	0.6
30B	SEC4PKL - ISBT43	20.7	25	9.3	UP	Afternoon	90	61	0.9
30B	ISBT43 - SEC4PKL	20.7	25	10.6	DOWN	Afternoon	48	32	0.5
30BAC	SEC4PKL - ISBT43	20.7	45	11.6	UP	Afternoon	51	32	0.5
30BAC	ISBT43 - SEC4PKL	20.7	45	10.8	DOWN	Afternoon	59	42	0.6
32A	MANSA DEVI - ISBT MOHALI	23.1	45	7.6	UP	Afternoon	51	27	0.4
32A	ISBT MOHALI - MANSA DEVI	23.1	45	7.9	DOWN	Afternoon	31	25	0.4
34AC	DERABASSI - PGI	28	105	16.7	UP	Afternoon	96	85	1.2
34AC	PGI - DERABASSI	28	105	14.0	DOWN	Afternoon	61	47	0.7
35AC	KHARAR - ISBT17	18.3	65	12.0	UP	Afternoon	88	76	1.1
35AC	ISBT17 - KHARAR	18.3	65	12.3	DOWN	Evening	116	114	1.6
35B	KHARAR - ISBT17	19.2	20	11.2	UP	Afternoon	86	75	1.1
35B	ISBT17 - KHARAR	19.2	20	10.9	DOWN	Afternoon	71	58	0.8
35BAC	KHARAR - ISBT17	19.2	20	11.5	UP	Afternoon	131	117	1.7
35BAC	ISBT17 - KHARAR	19.2	20	11.8	DOWN	Afternoon	125	106	1.5
36AC	NEWAIRPORT - PGI	30.9	45	13.9	UP	Afternoon	94	83	1.2
36AC	PGI - NEWAIRPORT	30.9	45	14.1	DOWN	Afternoon	66	56	0.8
37AC	R STATION - ISBT43	12	90	7.8	UP	Evening	35	34	0.5
37AC	ISBT43 - R STATION	12	90	8.1	DOWN	Evening	25	23	0.3
38AAC	NEWAIRPORT - ISBT17	21.1	45	10.0	UP	Afternoon	25	16	0.2
38AAC	ISBT17 - NEWAIRPORT	21.1	45	8.9	DOWN	Afternoon	52	32	0.5

Route No	Route Description	Length (km)	Headway (min)	Avg._Tl (km)	Trip Direction	Peak Hour	Total B/A	Max. Load Link	Load Factor (at max pass volume)
38AC	NEWAIRPORT - ISBT17	21.3	45	10.8	UP	Afternoon	25	18	0.3
38AC	ISBT17 - NEWAIRPORT	21.3	45	8.5	DOWN	Afternoon	37	27	0.4
4A	ISBT 43 - ISBT 43	31	25	8.2	DOWN	Afternoon	161	84	1.2
4C	ISBT43 - ISBT43	31	25	8.1	UP	Afternoon	147	62	0.9
5A	RAM DARBAR - RAM DARBAR	33.9	15	7.5	DOWN	Afternoon	354	91	1.3
5C	RAM DARBAR - RAM DARBAR	33.9	15	7.3	UP	Afternoon	318	87	1.2
71AC	KAMBWALA - ISBT43	24	30	8.7	UP	Morning	35	26	0.4
71AC	ISBT43 - KAMBWALA	24	30	9.0	DOWN	Morning	52	25	0.4
79AC	CHHATBIR - PGI	27.9	45	12.7	UP	Morning	109	78	1.1
79AC	PGI - CHHATBIR	27.9	45	12.7	DOWN	Morning	61	45	0.6
7A	ISBT17 - ISBT17	36	25	7.3	DOWN	Afternoon	114	31	0.4
7C	ISBT17 - ISBT17	36	25	6.8	UP	Morning	89	42	0.6
80AC	ZIRAKPUR - PGI	20	70	11.7	UP	Morning	84	70	1.0
80AC	PGI - ZIRAKPUR	20	70	9.9	DOWN	Afternoon	78	58	0.8
8AC	PH 11 - KAIMBALA	25.6	140	7.4	UP	Morning	19	13	0.2
8AC	KAIMBALA - PH 11	25.6	140	9.1	DOWN	Morning	14	10	0.1
9A	MANSA DEVI - MANSA DEVI	37	30	8.9	DOWN	Afternoon	105	44	0.6
9C	MANSA DEVI - MANSA DEVI	37	30	9.6	UP	Afternoon	219	73	1.0

Source: ETM data from CTU/CCBSS, August 2019

Annexure 3: Route-wise Operations Performance Parameters

Table 3: Operations Performance parameters of existing routes (2019)

Route No	Route Length (km)	Morning Peak Headway	Afternoon Headway	Daily Ridership	Average Pax Trip Length (km)	On road Fleet	Pax/Bus/Day	Actual Veh. Km	Pass. Km	Load Factor	IPK	Total Revenue (Rs)	Total Expenditure (Rs)	Operating Ratio
Within Chandigarh														
3	17	45	45	764	7.1	2	382	187	5462	0.42	4.09	9799	11317	0.87
10	19.4	20	21	1083	7.2	5	217	776	7816	0.14	1.40	21913	46964	0.47
11	7.1	30	34	240	4.5	1	240	92	1073	0.17	2.60	2679	5586	0.48
17	15.3	10	10	2695	6.9	6	449	1239	18591	0.21	2.17	40664	75002	0.54
18	15.5	16	16	1290	5.8	4	323	698	7512	0.15	1.85	24943	42213	0.59
22	19.6	10	10	4597	7.2	7	657	1548	33035	0.30	2.97	66174	106948	0.62
28	25.9	26	27	1094	8.3	4	274	751	9040	0.17	1.46	14990	40822	0.37
37	12	0	0	179	6.1	1	179	108	1095	0.14	1.66	3780	6536	0.58
71	24	15	15	2755	6.9	6	459	1200	19143	0.23	2.30	36250	72624	0.50
143	21.3	30	54	501	7.6	2	251	234	3792	0.23	2.14	7735	14180	0.55
203	18	26	30	653	7.1	4	163	288	4660	0.23	2.27	13909	17430	0.80
206	18.1	10	10	3629	8.7	7	518	1412	31746	0.32	2.57	69948	97513	0.72
240	20.4	15	25	1455	8.7	5	291	775	12670	0.23	1.88	29678	46915	0.63
241	22	11	11	3876	8.0	9	431	1474	31058	0.30	2.63	55597	101809	0.55
254	19.6	0	270	356	7.4	2	178	196	2637	0.19	1.82	7137	11862	0.60
239A	23.2	36	30	722	6.0	3	241	348	4323	0.18	2.07	13395	24036	0.56
239C	23.2	36	30	959	7.3	3	320	348	7022	0.29	2.76	19852	24036	0.83
205	11.1	9	9	2245	6.0	6	374	666	13539	0.29	3.37	45882	46001	1.00
202	13	8	8	2144	6.5	6	357	806	13894	0.25	2.66	42135	55670	0.76
1C	27.4	23	30	1407	6.6	5	281	712	9253	0.19	1.98	18480	38719	0.48
2C	38.3	23	14	2627	9.1	6	438	1379	24011	0.25	1.91	42317	83445	0.51
2D	21	23	23	992	7.4	4	248	420	7366	0.25	2.36	13981	25418	0.55
4C	31	26	34	1454	6.7	4	364	496	9716	0.28	2.93	22337	34259	0.65
7C	36	45	45	819	6.1	4	205	540	4979	0.13	1.52	10544	29349	0.36
18B	14	45	45	326	6.4	4	82	140	2091	0.21	2.33	4843	8473	0.57
23A	16	11	10	2565	5.7	7	366	1008	14576	0.21	2.54	34521	61004	0.57
28C	25.7	30	30	1217	6.2	3	406	720	7524	0.15	1.69	17246	39110	0.44
24A	9.8	0	0	176	4.8	1	176	108	841	0.11	1.63	2882	6524	0.44
24C	5.7	0	0	447	4.6	2	224	160	2058	0.18	2.80	8519	9659	0.88
1A	27.4	23	30	1125	6.9	5	225	767	7713	0.14	1.47	13829	41697	0.33
4A	31	30	39	1303	7.2	4	326	496	9430	0.27	2.63	17569	34259	0.51
7A	36	30	27	1163	6.5	4	291	792	7582	0.14	1.47	14194	43045	0.33
28A	25.7	30	30	1054	6.0	3	351	617	6332	0.15	1.71	15305	33523	0.46
5C	33.9	20	21	5642	6.5	8	705	1424	36902	0.37	3.96	74044	98342	0.75
2A	38.3	23	21	2591	8.4	6	432	1341	21662	0.23	1.93	36456	81127	0.45
5A	33.9	15	14	5622	6.8	8	703	1898	38103	0.29	2.96	75358	131122	0.57
Chandigarh to Mohali														
8	25.6	18	25	613	5.5	4	153	717	3365	0.07	0.86	8067	43381	0.19
32	35	12	19	2577	10.7	6	430	1505	27466	0.26	1.71	39317	81797	0.48
36	30.9	33	38	622	12.4	4	156	494	7740	0.22	1.26	13153	26871	0.49
38	21.3	15	25	757	8.4	3	252	724	6343	0.13	1.05	15068	39360	0.38
39	24.4	5	7	6426	8.0	12	536	2830	51638	0.26	2.27	102375	195496	0.52

Route No	Route Length (km)	Morning Peak Headway	Afternoon Headway	Daily Ridership	Average Pax Trip Length (km)	On road Fleet	Pax/Bus/Day	Actual Veh. Km	Pass. Km	Load Factor	IPK	Total Revenue (Rs)	Total Expenditure (Rs)	Operating Ratio
123A	19.8	10	17	1855	6.9	5	371	1049	12817	0.17	1.77	31000	63510	0.49
9C	37	21	30	1587	8.5	4	397	814	13412	0.24	1.95	23484	49263	0.48
32A	23.1	17	25	742	6.4	4	186	670	4758	0.10	1.11	10516	36409	0.29
38A	21.1	15	25	884	8.0	3	295	760	7038	0.13	1.16	19191	41284	0.46
9A	37	21	30	1320	7.7	4	330	740	10210	0.20	1.78	18198	44785	0.41
Chandigarh to Panchkula														
30	21.5	9	14	2266	7.5	4	567	1312	17050	0.19	1.73	36931	79372	0.47
214	25.9	21	75	721	7.6	6	120	311	5474	0.25	2.32	13760	18810	0.73
211	22.3	21	25	909	8.9	4	227	535	8049	0.21	1.70	17192	32390	0.53
2B	22.4	14	38	1429	8.4	5	286	605	11939	0.28	2.36	22976	36602	0.63
30A	21.1	10	15	2395	7.5	5	479	1224	17937	0.21	1.96	40483	74064	0.55
30B	20.7	9	14	2379	9.3	6	397	1283	22223	0.25	1.85	44424	77671	0.57
2F	27.6	12	15	2350	9.9	6	392	1380	23334	0.24	1.70	37805	83518	0.45
Chandigarh to Zirakpur														
26	29	14	19	1445	13.2	4	361	1073	19027	0.25	1.35	30032	64938	0.46
80	20	11	25	1644	9.0	5	329	780	14874	0.27	2.11	27410	47206	0.58
212	20.4	8	17	3071	11.9	6	512	1163	36438	0.45	2.64	60141	70373	0.85
213	34	28	38	1181	10.1	7	169	544	11911	0.31	2.17	21097	32923	0.64
216	25.5	7	11	5158	12.6	8	645	2168	65189	0.43	2.38	116047	149709	0.78
79	27.9	24	38	1128	10.4	4	282	530	11738	0.32	2.13	20187	32082	0.63
Chandigarh to Outside														
20	25.6	10	15	3093	11.0	6	516	1229	33891	0.39	2.52	62947	66785	0.94
25	27.2	33	38	1037	11.0	3	346	381	11454	0.43	2.72	20055	20696	0.97
34	28	10	17	2135	12.5	9	237	1316	26709	0.29	1.62	43486	79644	0.55
35	18.3	8	11	3624	10.2	6	604	1244	36786	0.42	2.91	71951	67633	1.06
40	31	21	30	1271	11.4	3	424	806	14544	0.26	1.58	23978	43806	0.55
242	24.6	7	15	4526	8.8	8	566	1648	40045	0.35	2.75	81356	113841	0.71
20A	25.6	10	15	2145	9.6	6	358	1152	20672	0.26	1.86	35123	62611	0.56
25A	27	21	30	1246	11.3	3	415	702	14135	0.29	1.77	24273	38154	0.64
35B	19.2	8	13	3327	9.8	6	555	1286	32706	0.36	2.59	62844	69916	0.90
Total/ Average				127630		330		59139	1079156	0.26	2.16	2143749	3631441	0.59

Source: Route wise compilation from ETM data from 5th August 2019

Annexure 4: Alignment of Bus Routes – Business As Usual

Route 3: MAULI JAGRAN TO PGI via Sec- 18/7 Pk1,H.B.Chowk, Kalagram, R/Crossing, R/S, CTU w/s, 28/26 Grain Mkt, 19/7, 18/8, 22/17, 16/10 Hospital, 11/10, PEC, OPD

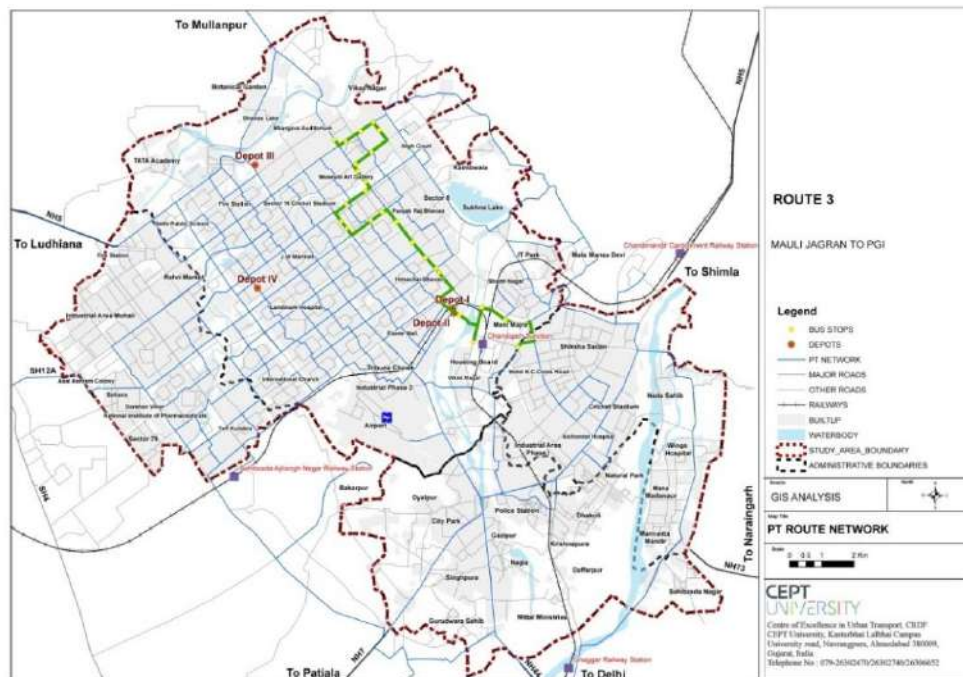


Figure 60: Existing Route3 (MAULI JAGRAN TO PGI via)

Route 11: ISBT-17 TO PB.CIVIL SECTT. Via Sec-17/18, 9/8,9mkt, MLA Flat, H/Court, Pb.Civil.Sectt, Spl.Trip School Sec-18, Ramdarbar

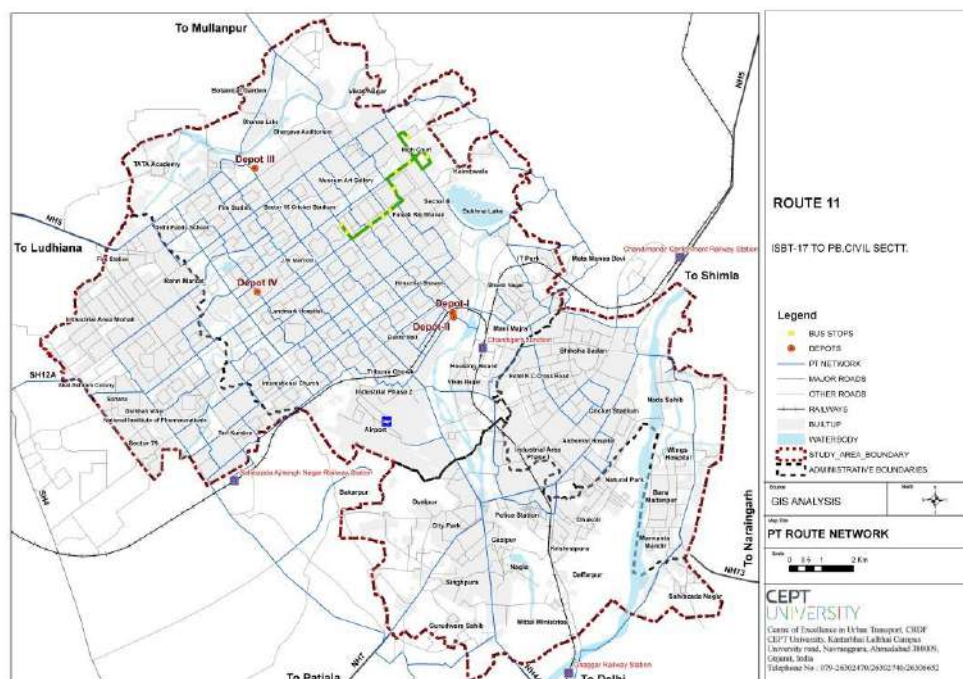


Figure 61: Existing Route 11 (ISBT-17 TO PB.CIVIL SECTT)

Route 17/ 17AC: ISBT-43 TO AIR PORT Via Sec 43/44, 42/43, Sec- 36/35, 23/22, ISBT-17, Aroma, 21, 20, 30, 29 mkts, Elante Mall, H/Majra, Behlana

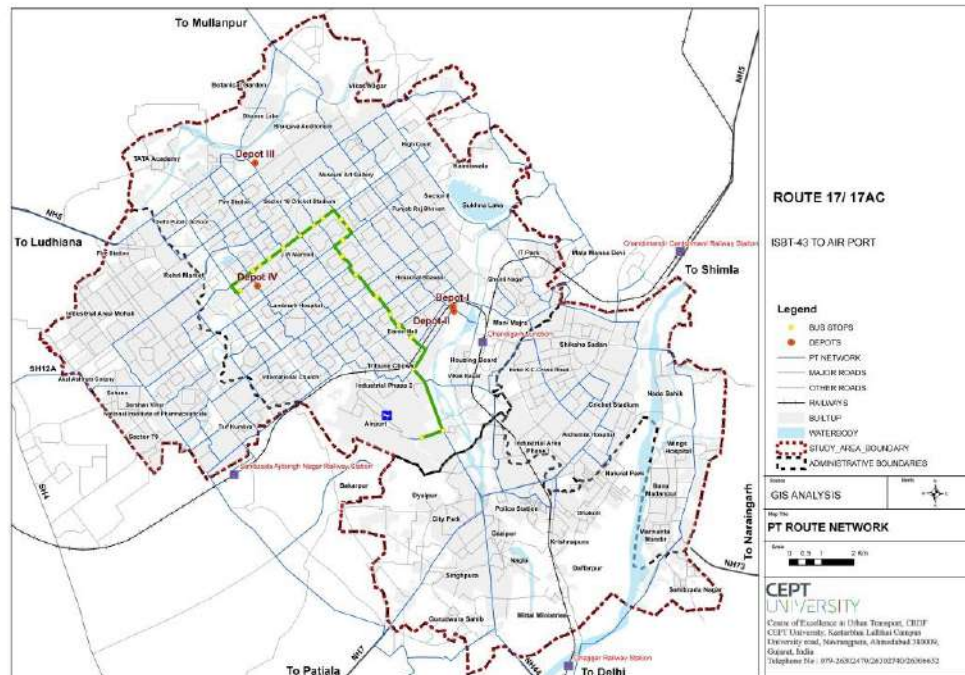


Figure 62: Existing Route 17/ 17AC (ISBT-43 TO AIR PORT)

Route 18/ 18AC: ISBT-43 TO KAIMBWALA via Sec 43/44, 42/43, Sec-36/35, 23/22, ISBT-17, 17 Mkt., U.T., Sectt., Mini Sectt., MLA Flat, High Court, Lake

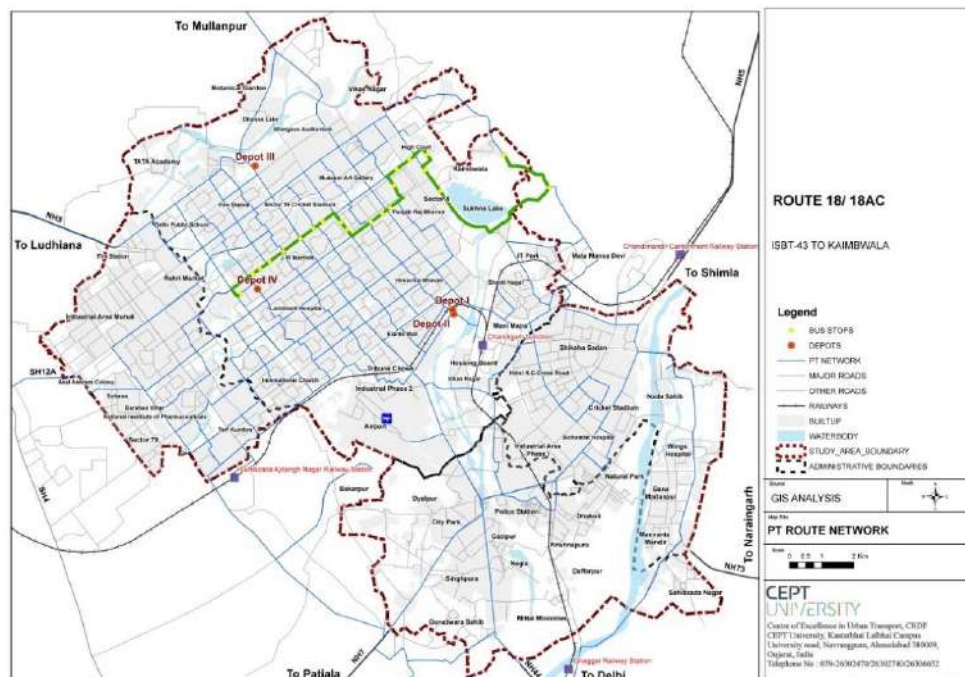


Figure 63: Existing Route 18/ 18AC (ISBT-43 TO KAIMBWALA)

Route 22/ 22AC: ISBT-43 TO MATA MANSA DEVI via Sec 43/44, 42/43, 36/35, 23/22, ISBT-17, 18, 19, 27, 28 mkts,CTU w/s, R/Station, R/Crossing, Kalagram, H/B Chowk, M/Majra, Kishangarh, I.T.Park

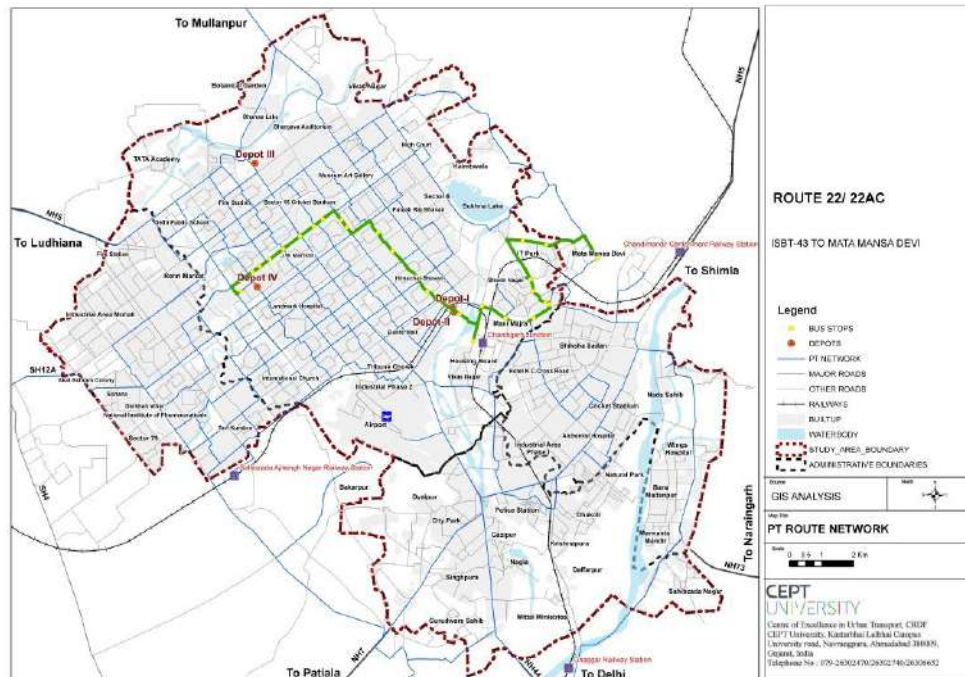


Figure 64: Existing Route 22/ 22AC (ISBT-43 TO MATA MANSA DEVI)

Route 28: MALOYA TO MATA MANSA DEVI via DMC, 38w, 38 Mkt, 37 Mkt, 36 Mkt, 36/35, 23/22, ISBT-17, 8/18,7/19, 26/27, 26/28 G/Mkt, CTU/w/s, R/Station, Kalagram ,H.B, M/Majra

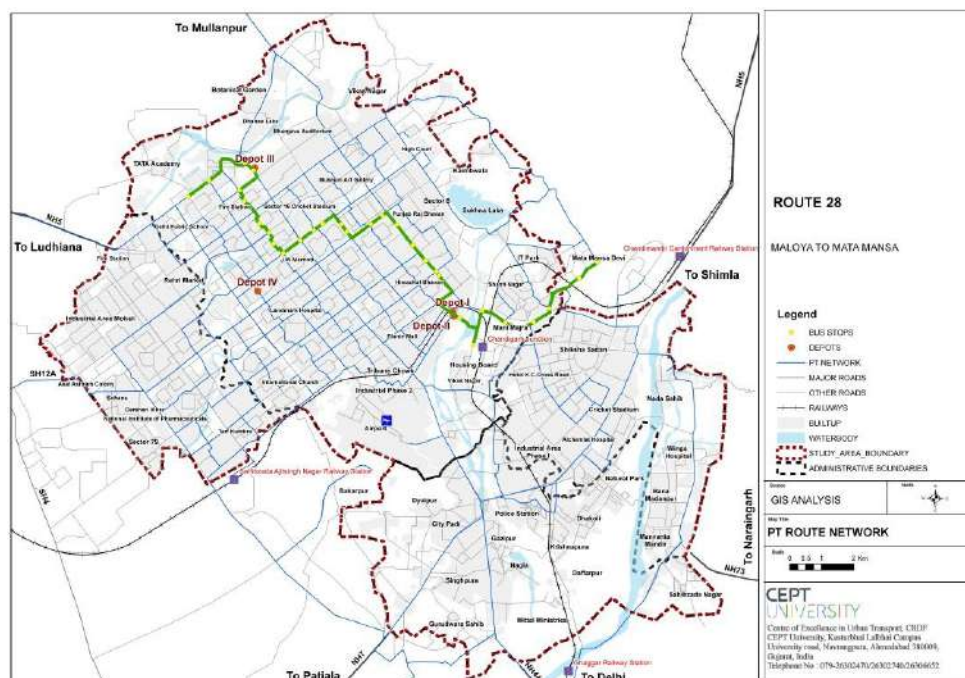


Figure 65: Existing Route 28 (MALOYA TO MATA MANSA DEVI)

Route 71/ 71AC: SBT-43 TO SAKETRI (KAIMBWALA) via Sec- 43/44, 35/34, K. Bhawan, 23/22, ISBT-17, 18, 19, 27 Mkts, G/Mkt, R/Crossing, M/M Complex, M/M Town, M/Majra, Mansa Devi, Saketri

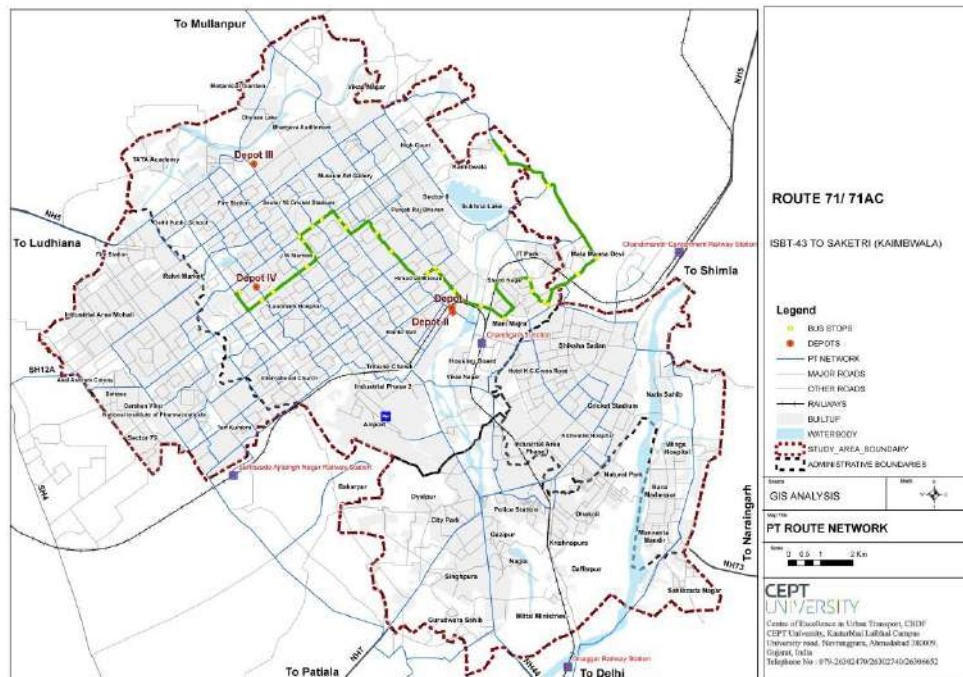


Figure 66: Existing Route 71/ 71AC(SBT-43 TO SAKETRI (KAIMBWALA))

Route 143/ 143AC: DHANAS TO AIRPORT via SP Barrier, KhudaLahora, PGI , New OPD, GH –16 , ISBT – 17,Aroma, Sec-21 Mkt. Sec-20-Mkt,Sec – 30 Mkt. , SEC-29 Mkt., Elante Mall, Poultry Farm, Ind. Area Ph-I, Hallo Majra

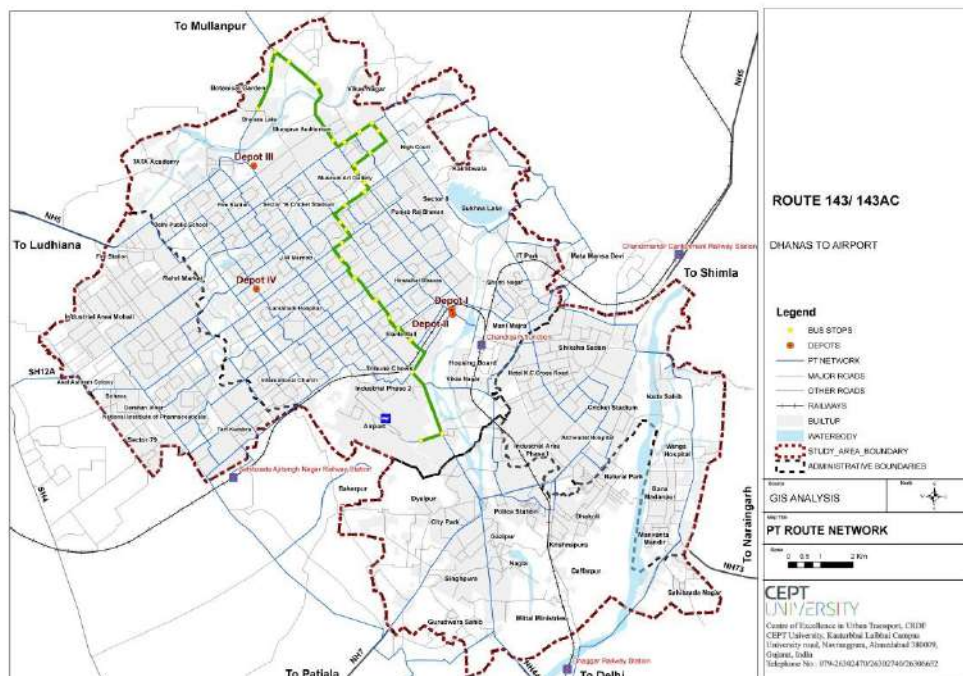


Figure 67: Existing Route 143/ 143AC(DHANAS TO AIRPORT)

Route 241/ 241AC: ISBT-43 TO Mansa Devi via Sec-52/53, 52/61, YPS, 51 mkt, Sec-51/50, 44/45, 45 mkt, 45/46, 33/32, 20/30, 19/27,27 mkt, 26 G/Mkt, TPT Chowk, CTU w/s.Indl.Area.Ph-1, R/Station, R/Crossing, Kalagram, H/B Chowk, M/Majra

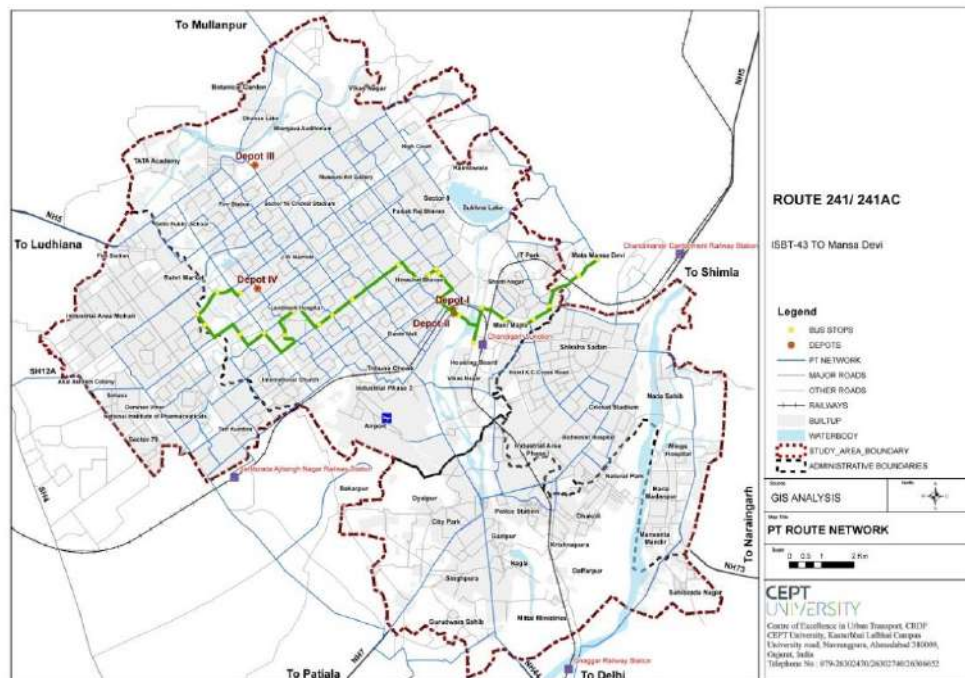


Figure 68: Existing Route 241/ 241AC(ISBT-43 TO Mansa Devi)

Route 1A/1C: PGI TO PGI via 15/14, 15mkt, 16 mkt, 17/16, ISBT- 17, 18 mkt., 19 mkt, 27 mkt,28 mkt, CTU w/s, R/Station, R/Crossing, H/B Chowk, M/Majra, M/Majra Town, M/Complex, R/Crossing, 28/26, SGGS College, 7 mkt, 8mkt, 9 mkt, 10 mkt, 11mkt, OPD

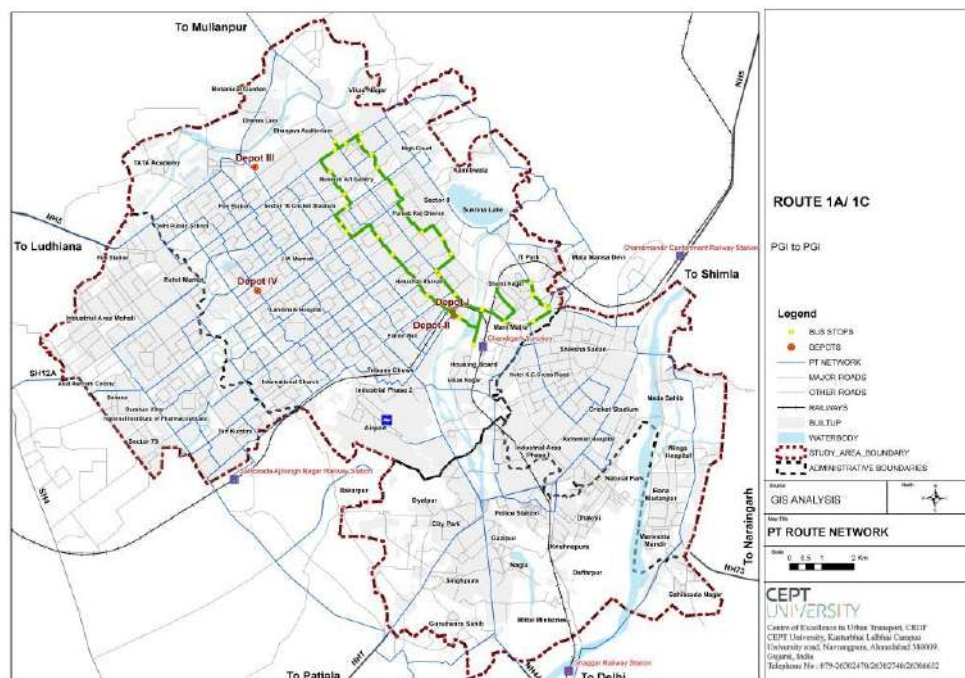


Figure 69: Existing Route 1A/ 1C (PGI to PGI)

Route 2A/2AAC: MANI MAJRA TO MANI MAJRA Via Indira Colony, M.M. Town, M/Complex, R/Crossing, Sec- 28/26 G. Mkt, 19/7, 18/8, 22/17, 16/10, 15/11, PGI, Khuda Lahora, S/P Barrier, Dhanas, DMC, CTU Workshop, Sec-25/38, 25/24, 24/15, 23/16, ISBT-17,18/21, 19/20, 27/30, 28/29, Elante Mall, Colony No.- 4, CTU Workshop, R/Station, R/Crossing, H.B. Chowk

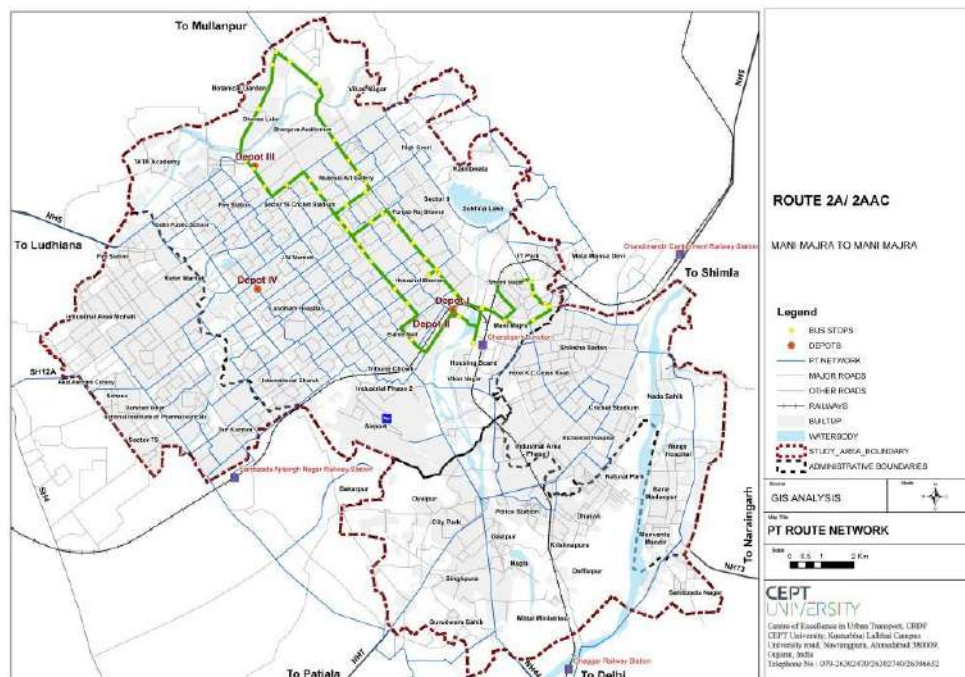


Figure 70: Existing Route 2A/ 2AAC (MANI MAJRA TO MANI MAJRA)

Route 2C/ 2CAC: MANI MAJRA TO MANI MAJRA Via H.B. Chowk, Kalagram, R/Station, CTU w/shop, Colony.No-4, Elante Mall, Sec-29/28, 30/27, 20/19, 21/18, 22/17, 16/23, 15/24, 24/25, 38/25, CTU w/shop, DMC, Dhanas, SP Barrier, K.Lahora, PGI, 11/15, 10/16, ISBT-17, 8/18, 7/19, 26/28 G.Mkt. R/Crossing, M/Complex, M.M. Town, Indra Colony

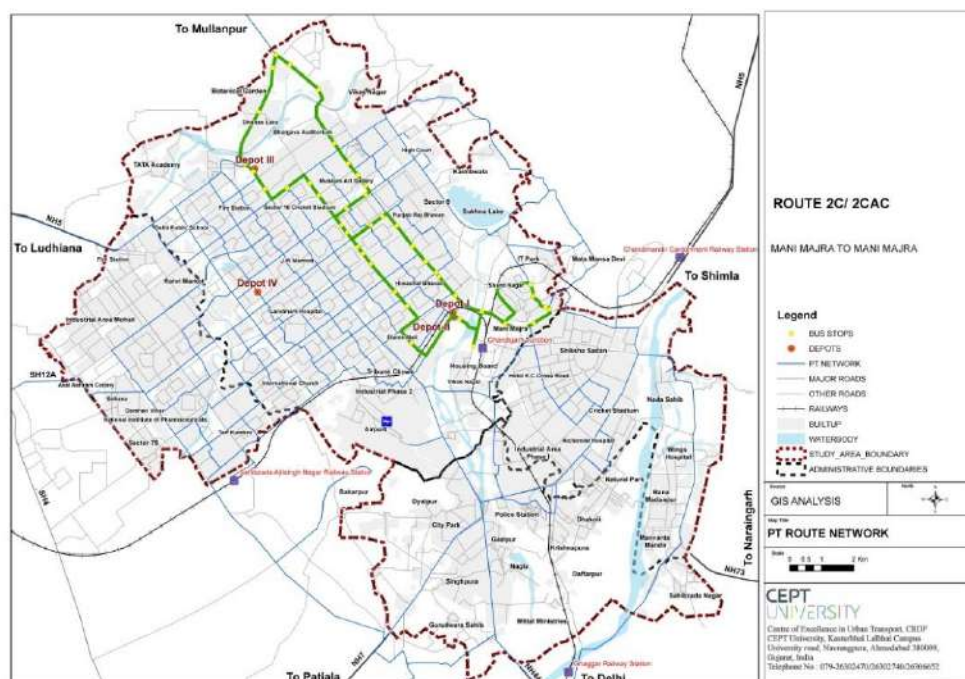


Figure 71: Existing Route 2C/ 2CAC (MANI MAJRA TO MANI MAJRA)

Route 2D: PGI TO IT Park Via New OPD, 10/11,10/16,ISBT-17, 8/18, 7/19, 26/27, Grain Market, CTU Workshop, R. Station, R. Crossing, Kalagram , H. Board, Manimajra, Indira Colony, Kishangarh

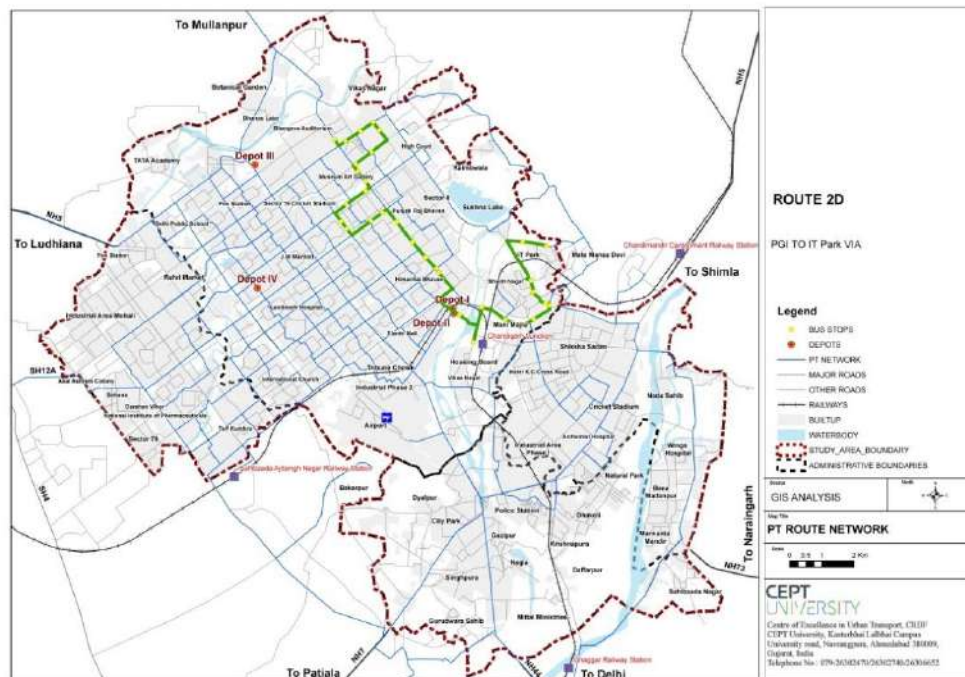


Figure 72: Existing Route 2D (PGI TO IT Park)

Route 4A/4AAC/4C/4CAC: SBT-43 to ISBT-43 via 43/44, 35/34, Aroma, 22/17,16/17, R/Garden, 16/10 Hospital, OPD, PGI, K/Lahora, S/P/Barrier, Dhanas, DMC, CTU Depot-III.38west, Maloya, Ph-6, Old Barrier, Frenko Hotel, BadheriChowk, 40/41, 41 mkt, 42 mkt

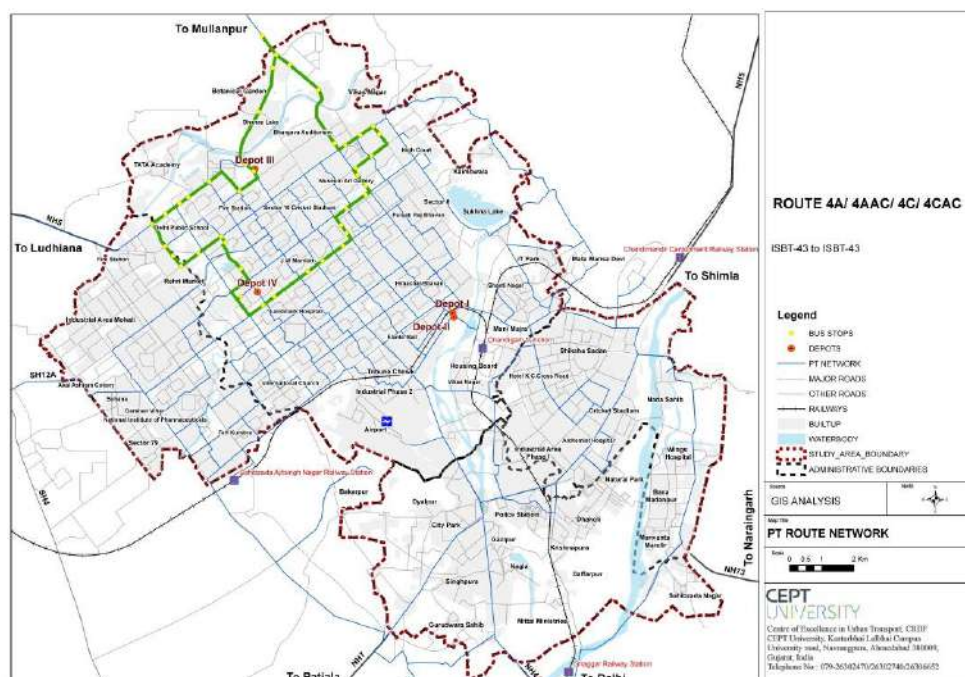


Figure 73: Existing Route4A/4AAC/4C/4CAC (SBT-43 to ISBT-43)

Route 5A/ 5C/ 5AAC: RAM DARBAR TO RAM DARBAR via Sec-47 mkt,46/47, 31/32, 30/29, 27/28, 27/26, 19/7, 18/8, 17/9,16/10, 15/11, PGI, Uni., 15/14, 24/25, 37/38, 40/38,39/38west, Maloya, 39, 40, 41, 42 mkts, ISBT-43, 44,45, 46, 47 mkts, Ram Darbar

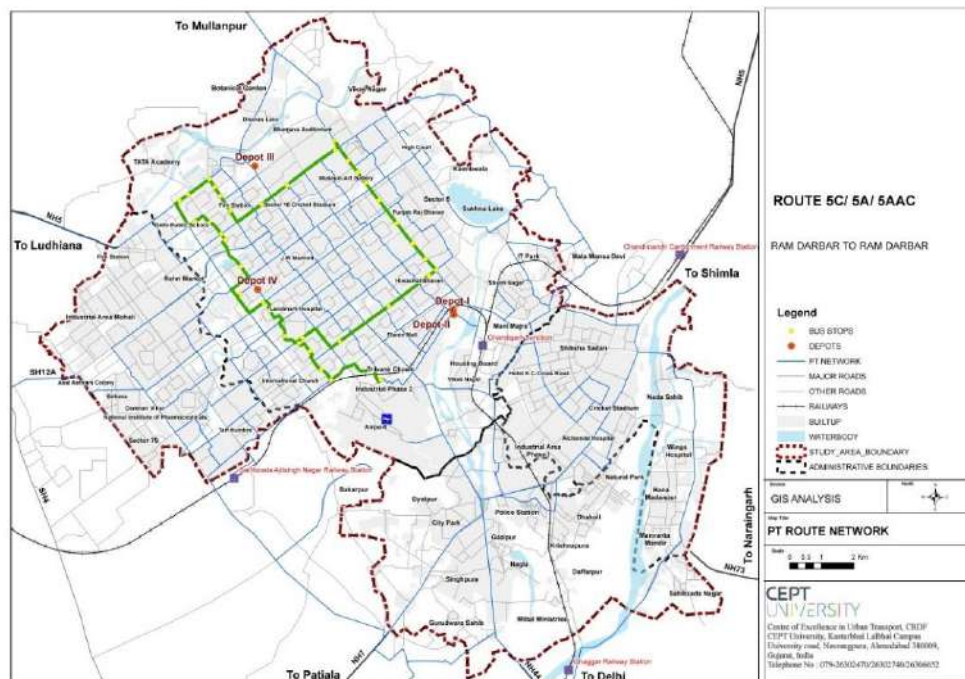


Figure 74: Existing Route5A/ 5C/ 5AAC (RAM DARBAR TO RAM DARBAR)

Route 7A/ 7C/ 7AAC: ISBT-17 to ISBT-17 via Sec-22/17, 22/23, 23, 24mkts, 24/25, 38/25, 38west, Maloya, 38/39, 38 mkt, 37 mkt, 36/37, 42/41, ISBT-43, Attawa, 35mkt, 34 mkt, 34, 33, 32, 31mkts, Ram Darbar, Indl.AreaPh-II, TribuneChowk, 29, 30, 20, 21 mkts, 22 mkt, 23/22

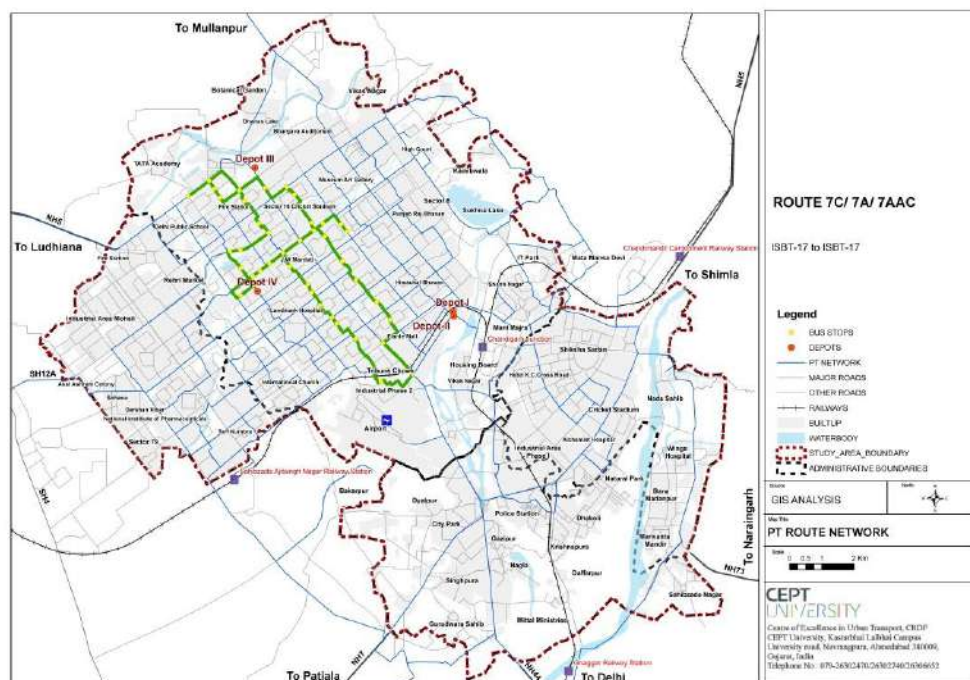


Figure 75: Existing Route7A/ 7C/ 7AAC (ISBT-17 to ISBT-17)

Route 18B: ISBT-43/SESSION COURT TO H/COURT via Sec43/44, 42 Mkt, 41/42, 37/36, 37 Mkt, 38/37, 24/37 Consumer/Court , 23/36, 23/22, Sec-17 CAT, ISBT-17, 17/18, 9/8, 9 Mkt, MLA/F

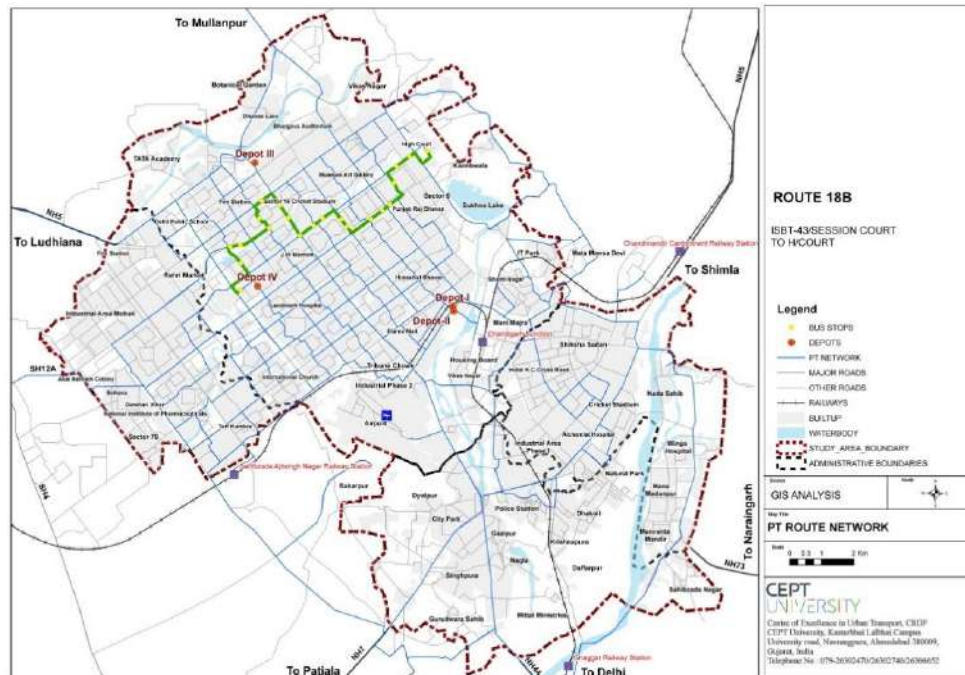


Figure 76: Existing Route 18B (ISBT-43/SESSION COURT TO H/COURT)

Route 23A/ 23AAC: ISBT-43 TO KHUDA ALI SHER via Sec-43/44,35/34, Aroma, 22/17, Hospital16/10, 15/11, PGI, New OPD, PEC, NayaGaon

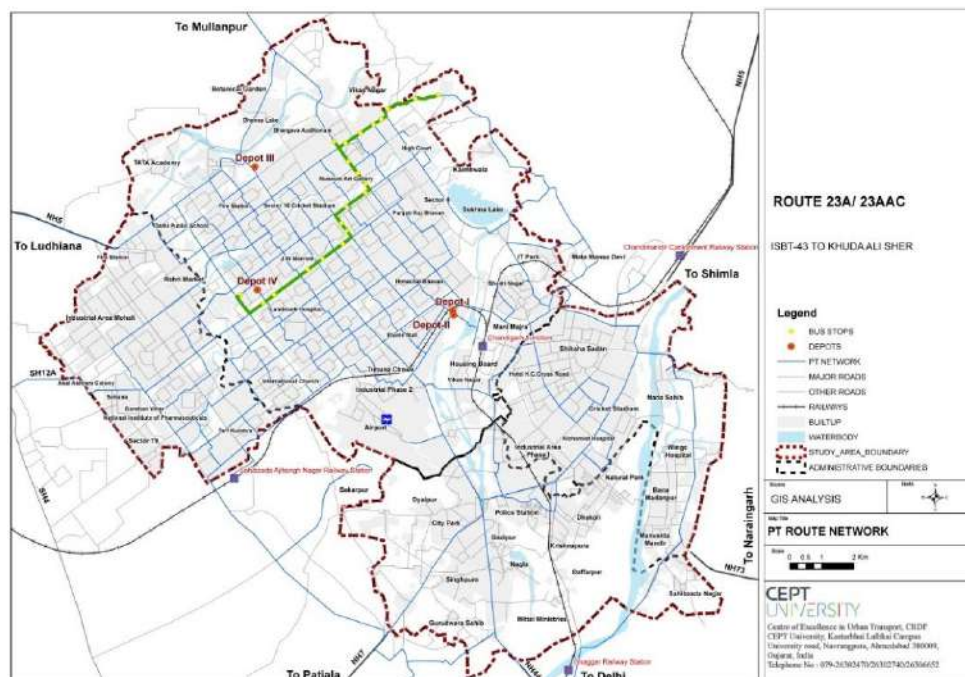


Figure 77: Existing Route 23A/ 23AAC (ISBT-43 TO KHUDA ALI SHER)

Route 28A/ 28C: ISBT-43 to ISBT-43 Via Sec-43/44, 42/43, Sec-36/35, 37/24, 38/25, CTU w/s, DaduMajra, Dhanas, S/P Barrier, IRB, KhudaLahora, PGI, OPD, Sec- 10/16, ISBT-17, Aroma, Sec-34/35, Sec-44/43

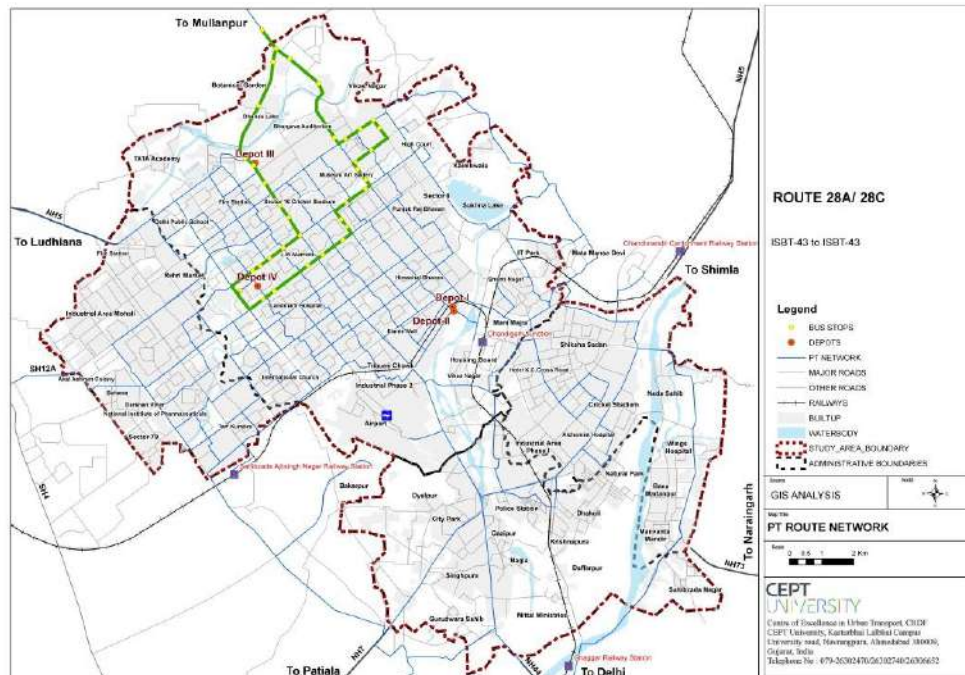


Figure 78: Existing Route 28A/ 28C (ISBT-43 to ISBT-43)

Route 10/ 10AC: ISBT-43 TO MATA MANSA DEVI via Sec 43/44, Sec-44/51, 45, 46, 47mkt, Ram Darbar, Tribune Chowk, H/MajraChowk, Colony No.-4, CTU w/s, R/Station, Kalagram, H/B Chowk, M/Majra

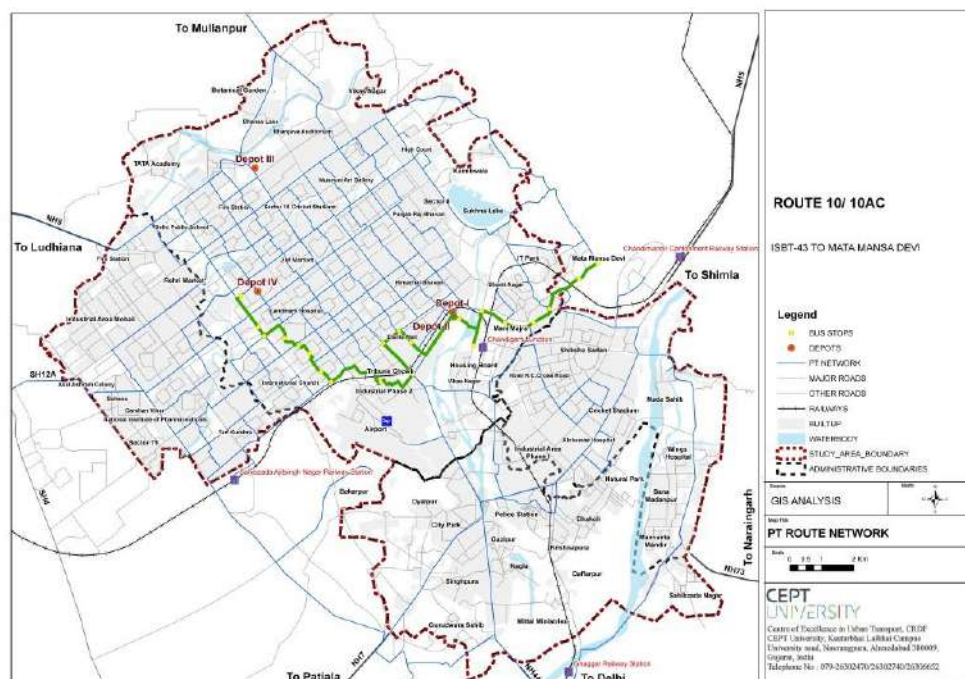


Figure 79: Existing Route 10/ 10AC (ISBT-43 TO MATA MANSA DEVI)

Route 37/ 37AC: ISBT 43 TO RAILWAY STATION via Sec 43/44, 42/43, Sec-36/35 23/22, ISBT-17, Sec 18 mkt, Sec19 mkt, Sec 27 mkt, 28 mkt, CTU

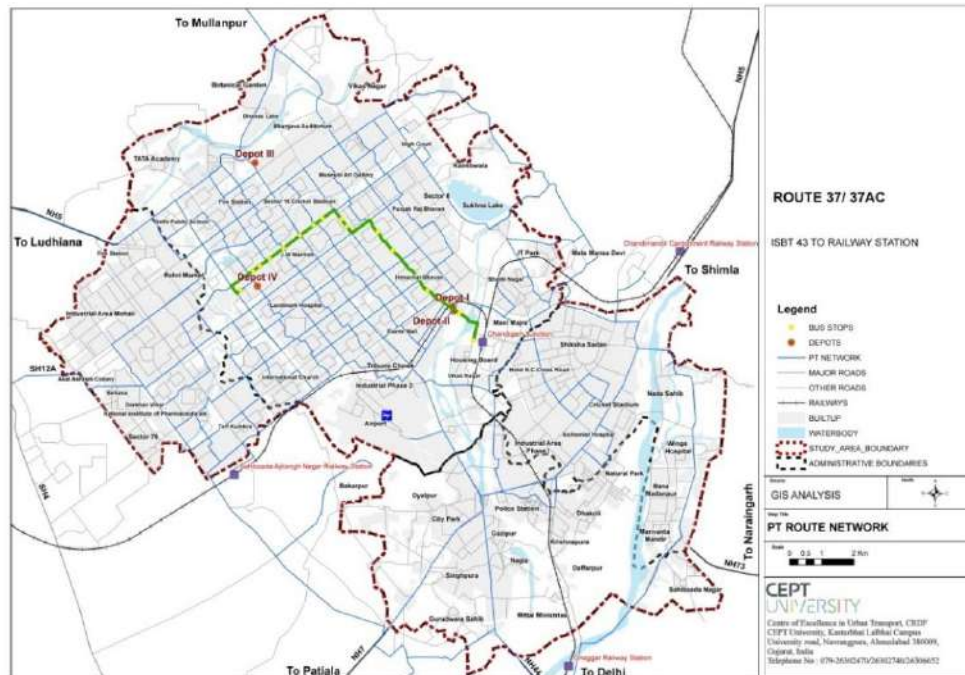


Figure 80: Existing Route 37/ 37AC(ISBT 43 TO RAILWAY STATION)

Route 202/ 202AC: ISBT-43 TO PB. CIVIL. SECTT. (HIGH COURT) via Sec-43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, K.C,U.T.Sectt, MLA Flat, H/Court, Pb civil. Sectt.

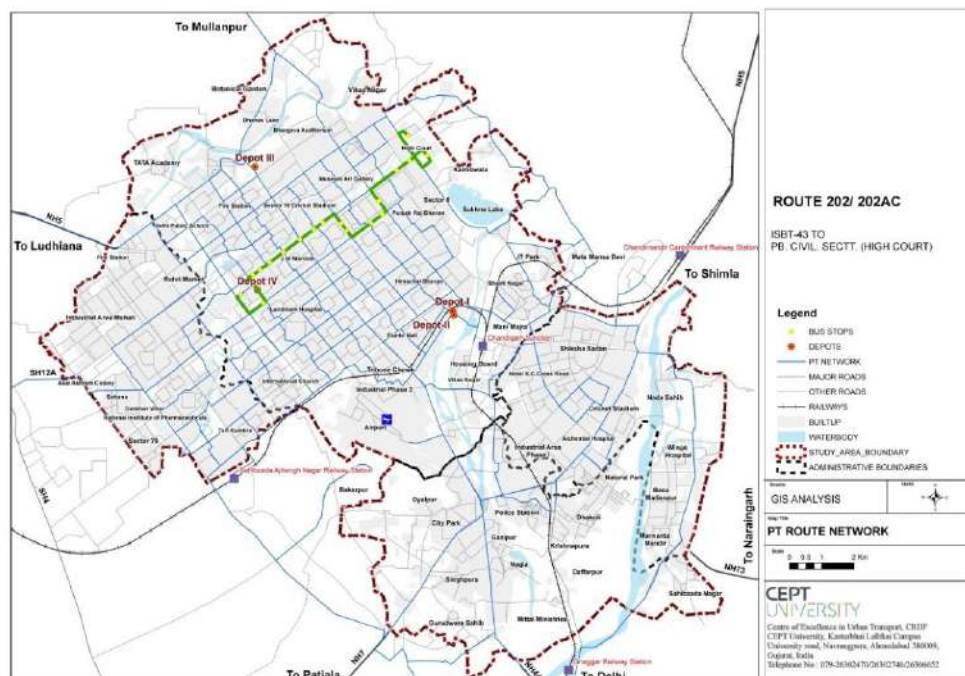


Figure 81: Existing Route 202/ 202AC(ISBT-43 TO PB. CIVIL. SECTT. (HIGH COURT))

Route 203/ 203AC: ISBT-43 TO MATA MANSA DEVI via Sec-43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, 18, 19, 27, 28 mkts, CTU w/s, R/Station, R/Crossing, Kalagram, H/B Chowk, M/Majra

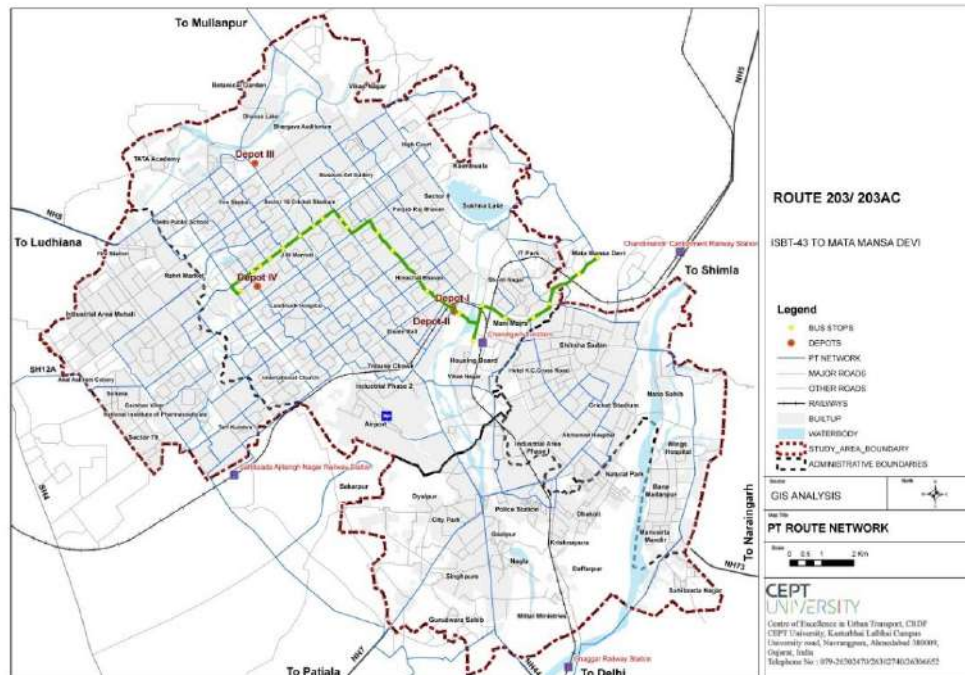


Figure 82: Existing Route 203/ 203AC(ISBT-43 TO MATA MANSA DEVI)

Route 205/ 205AC: ISBT-43 to PGI via Sec-43/44, 35/34, Aroma, 22/17,16/10, 11/10, 11/2, OPD

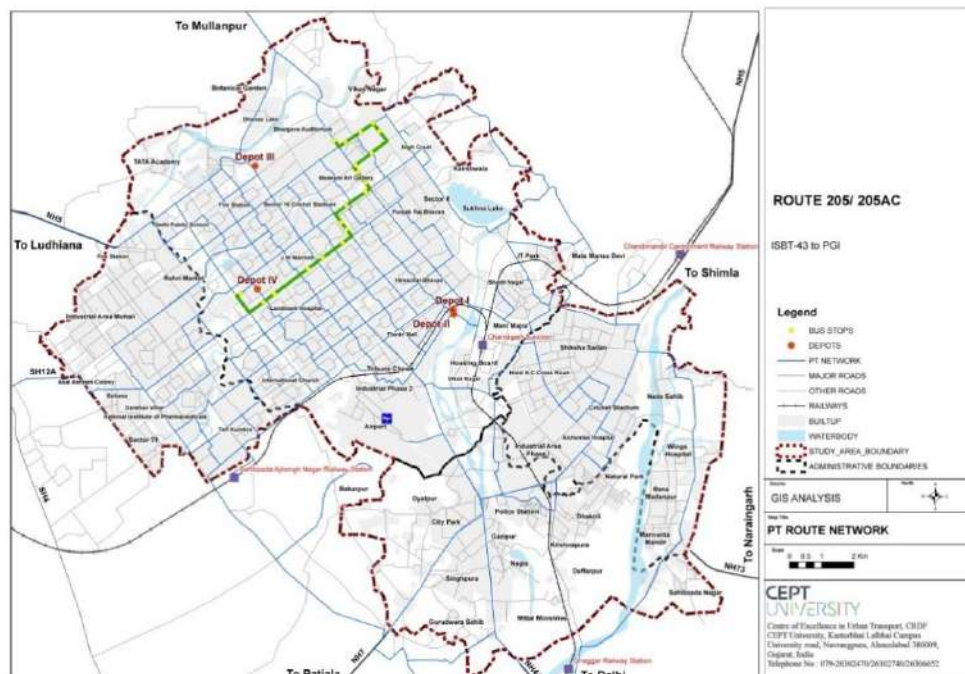


Figure 83: Existing Route 205/ 205AC(ISBT-43 to PGI)

Route 206/ 206AC: ISBT-43 TO I.T.Park via Sec-43/44, 42/43, 36/35, 23/22, ISBT-17, 17/18, 8/18, 7/19,26/27, G/Mkt, R/Crossing, Kalagram , H/B Chowk, M/Majra, Kishangarh

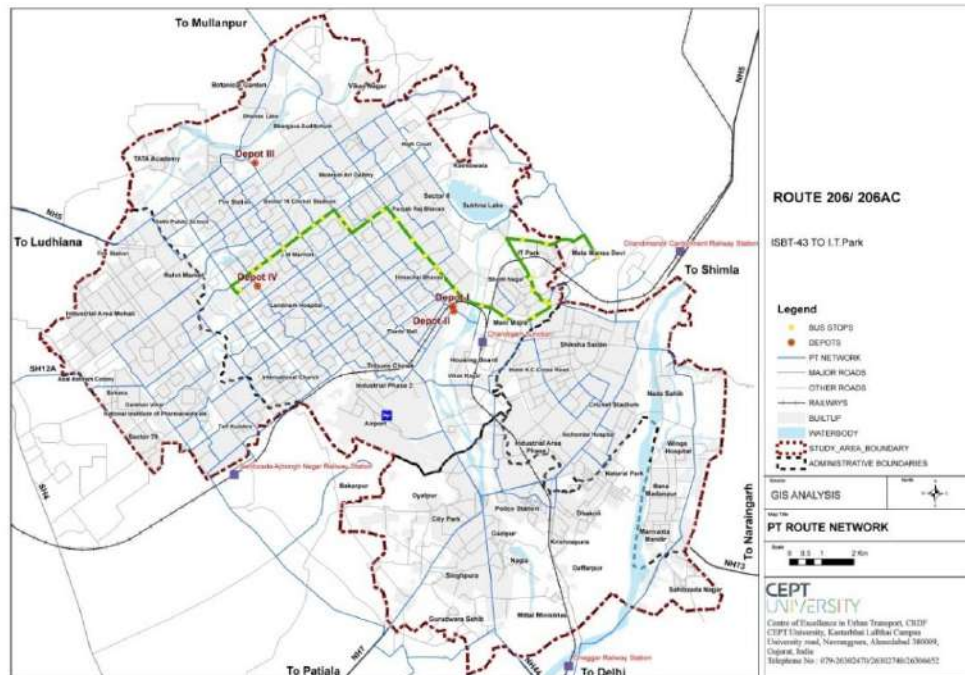


Figure 84: Existing Route 206/ 206AC(ISBT-43 TO I.T.Park)

Route 240/ 240AC: ISBT-43 TO MATA MANSA DEVI via Sec-43/44, 35/34, Sec 22 mkt, 23/22, ISBT-17, Aroma, 21, 20, 30, 29 mkts, Elante Mall, Colony No.-4, CTU w/s, R/StationKalagram, H/B Chowk, M/Majra

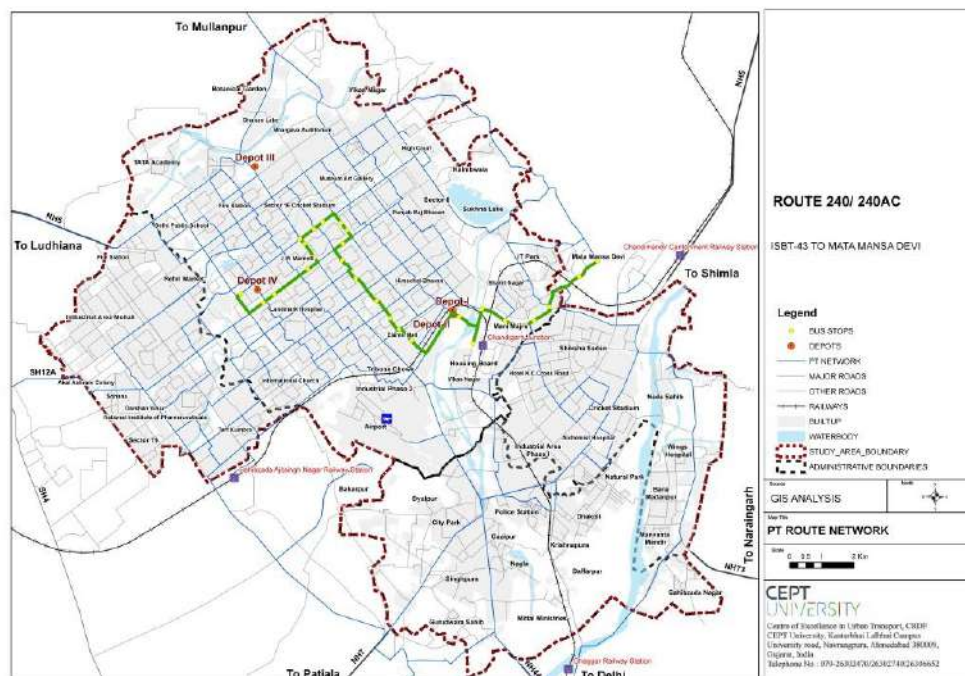


Figure 85: Existing Route 240/ 240AC(ISBT-43 TO MATA MANSA DEVI)

Route 24A/ 24AAC: ISBT 43 TO PGI Via Sec- 43/44, 35/34, Aroma,22/17, 16/10 (Hospital), 15/11

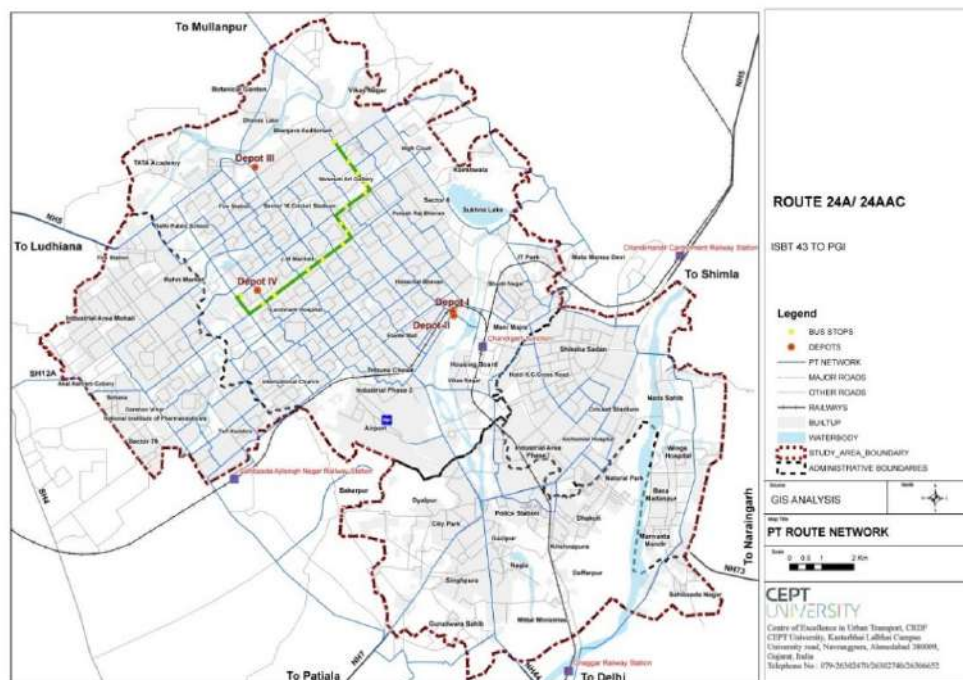


Figure 86: Existing Route 24A/ 24AAC (ISBT 43 TO PGI)

Route 24C/ 24CAC: ISBT 43 to ISBT-17 via Sec .43/44, 42/43, Sec-36/35 23/22, ISBT-17

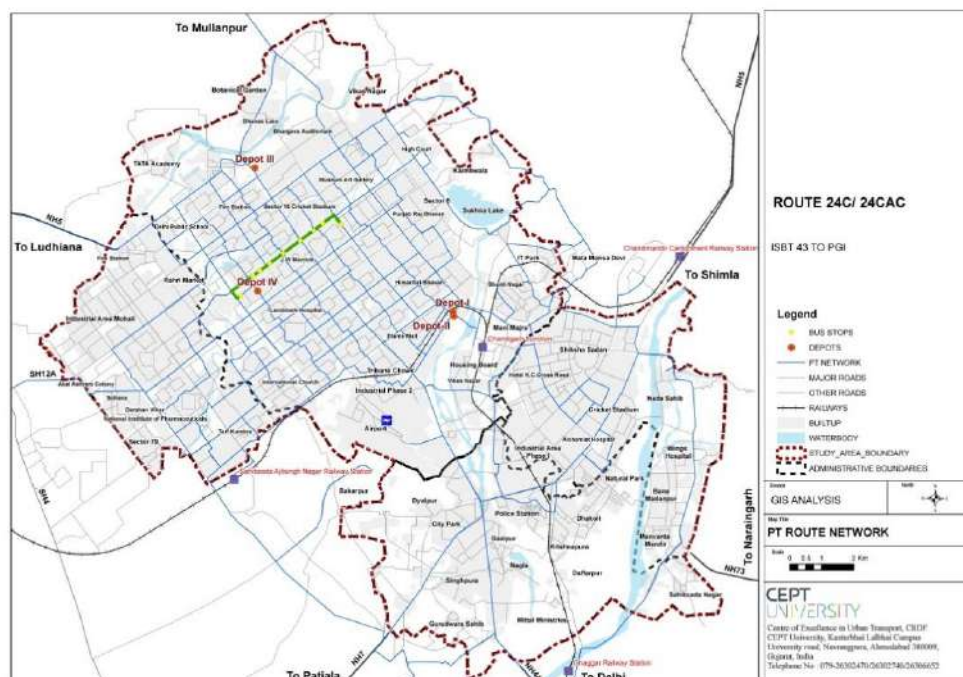


Figure 87: Existing Route24C/ 24CAC(ISBT 43 to ISBT-17)

Route 254/ 254AC: ISBT 43 TO IT PARK via 43/44, 42/43, 36/35, 23/22, ISBT-17, 18 mkt., 19 mkt., 27 mkt., 27/28, 28 Mkt., CTU workshop, Railway Station, Railway Crossing , Kalagram , H/B, M/Majra, Kishangarh

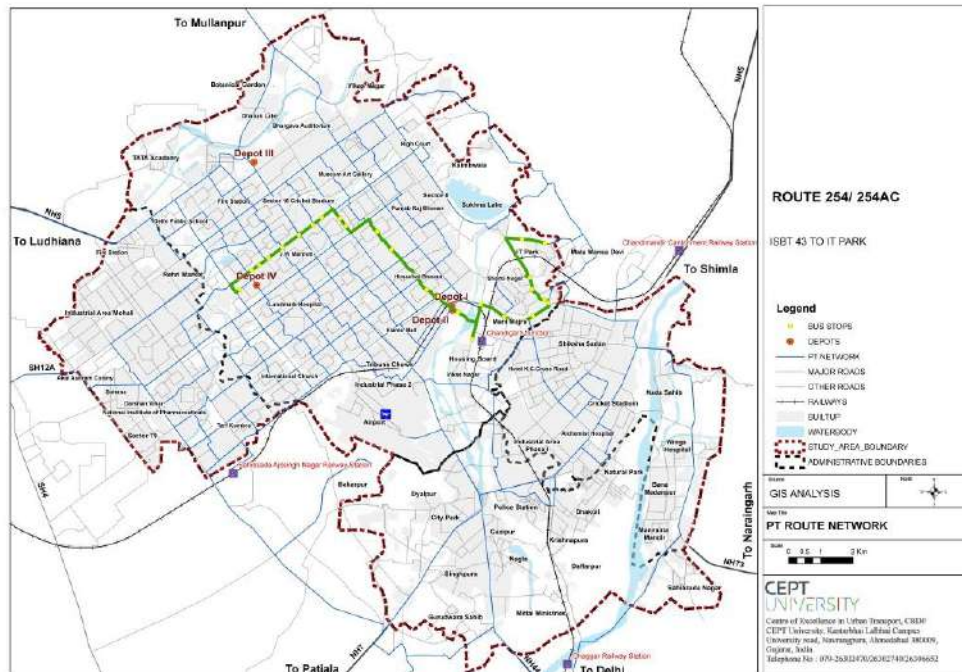


Figure 88: Existing Route 254/ 254AC(ISBT 43 TO IT PARK)

Route 239A/ 239AAC/ 239C/ 239CAC: ISBT-43 to ISBT-43 via ISBT-43, 43/44, 35/34,22/21, ISBT-17, 17/18, 9/8, 9 mkt, UT.Sectt, MLA Flat, H/Cout, Pb.CivilSectt, 3/2, PEC, N/OPD, PGI, 15/14, 24/25, 37/38, 41/40, 41mkt, 42mkt, ISBT-43 H/B Chowk, M/Majra

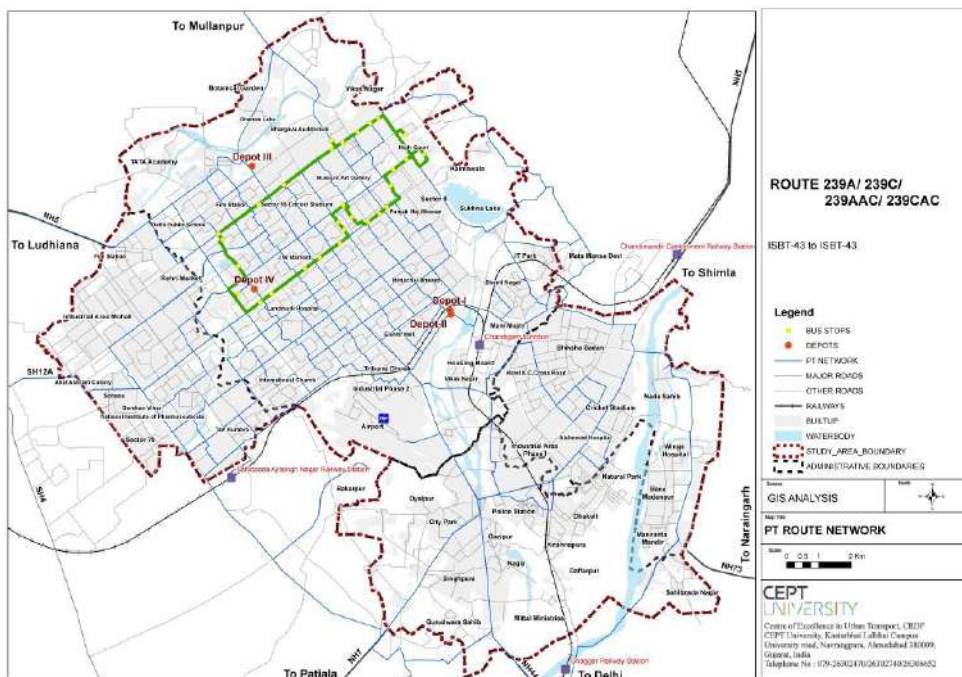


Figure 89: Existing Route 239A/ 239AAC/ 239C/ 239CAC(ISBT-43 to ISBT-43)

Route 8/ 8AC: MOHALI Ph-XI to PGI via Sec-65 Mkt, 65/64, 49/48, 48 Mkt., 47 Mkt., 46Mkt., 45 Mkt., 44 Mkt., ISBT-43,43/44, 35/34,. 22/17, 16/10 (Hospital), New OPD

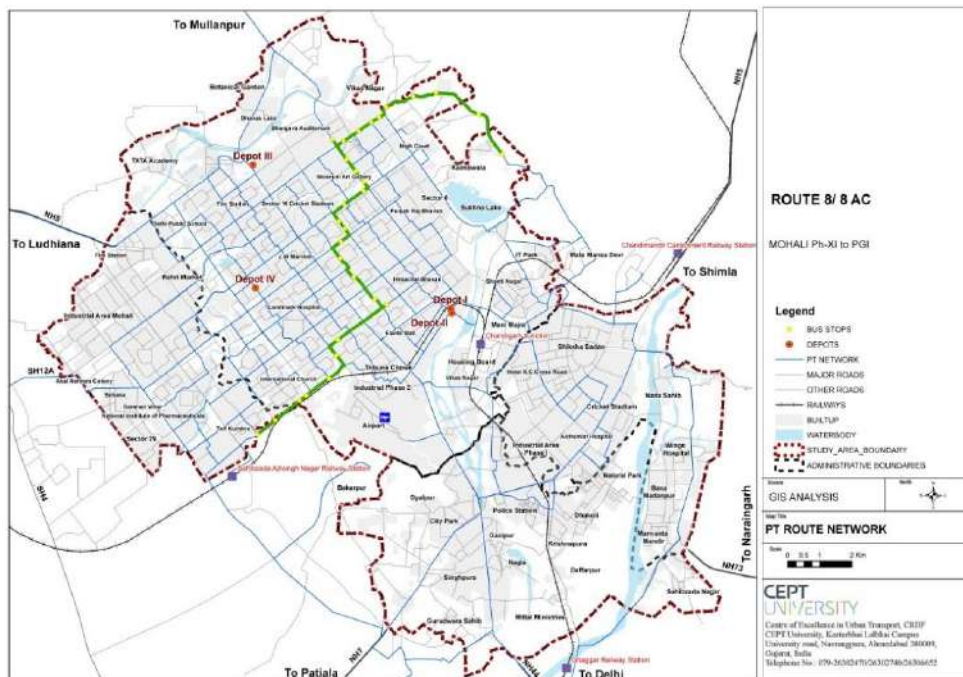


Figure 90: Existing Route8/ 8AC (MOHALI Ph-XI to PGI)

Route 25/ 25AC/ 25A: CHD TO KHIZRABAD via Sec-16/10, 15/11, PGI,K/Lahora, S/P Barrier, Mullanpur, Parol, Majra, Boothgarh, Block, Sialwa, JhandeMajra

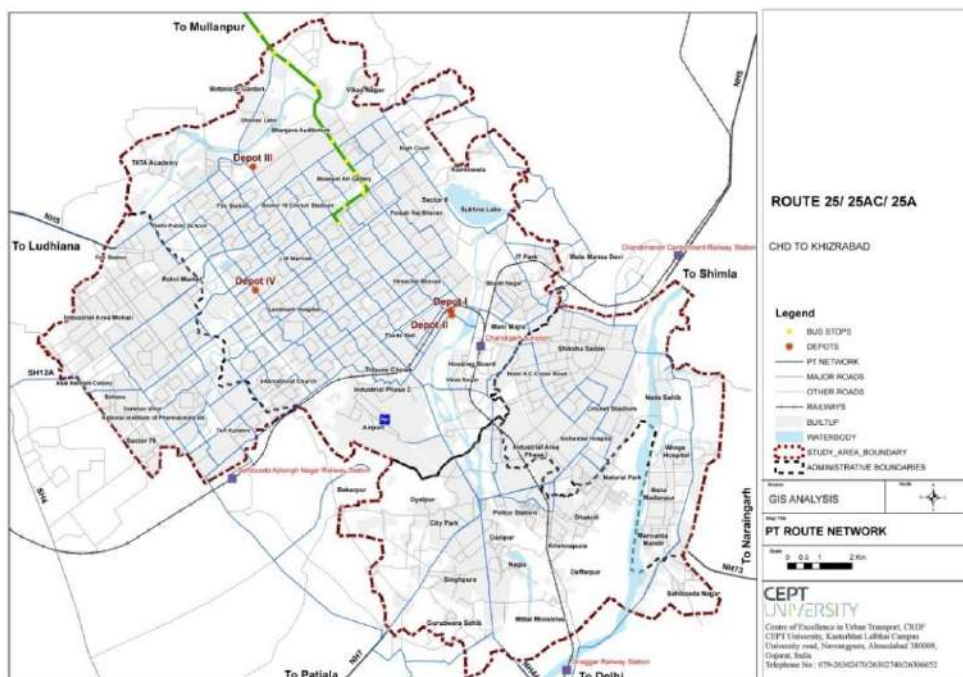


Figure 91: Existing Route 25/ 25AC/ 25A (CHD TO KHIZRABAD)

Route 30/ 30AC: ISBT-43 TO NADA SAHIB via Sec 43/44, 42/43, 36/35, 23/22, ISBT-17, Aroma, 21, 20 Mkts., 19/27,26/27, 26/28 G.Mkt, R/C, Kalagram, H. Board, Manimajra, Command Hospital, MajriChowk

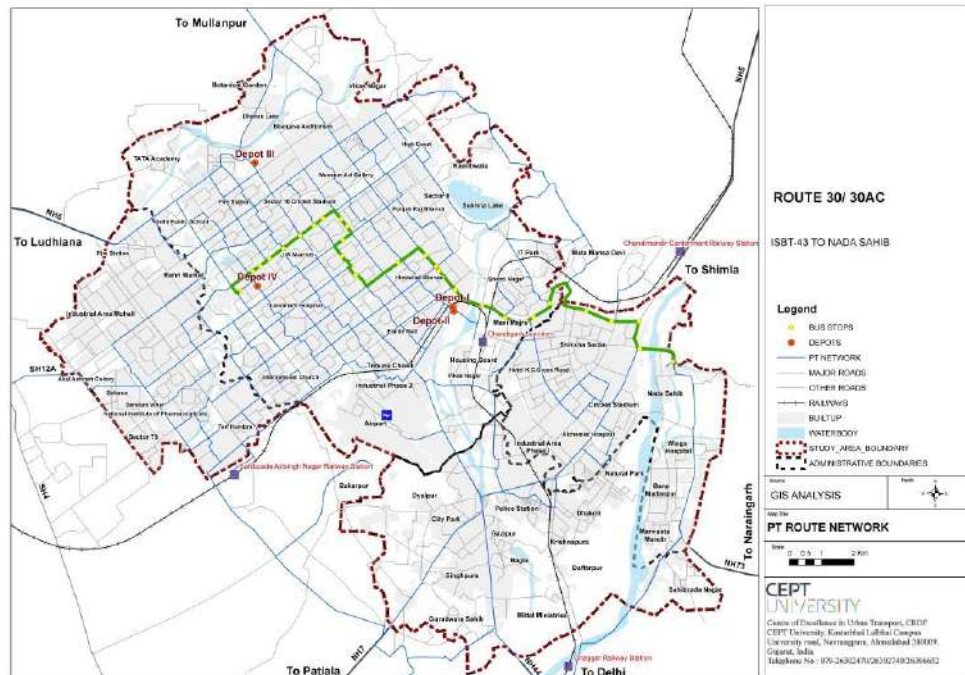


Figure 92: Existing Route 30/ 30AC(ISBT-43 TO NADA SAHIB)

Route 32: PGI TO DERABASSI Via OPD, PEC, Sec-10/16, Sec-17-16, ISBT-17,Sec-8/18, 7/19,26/27, G.Mkt,- 26/28, H/B, M/Majra, Old Pkl, Nada Sahib, Ramgarh, Mubarakpur

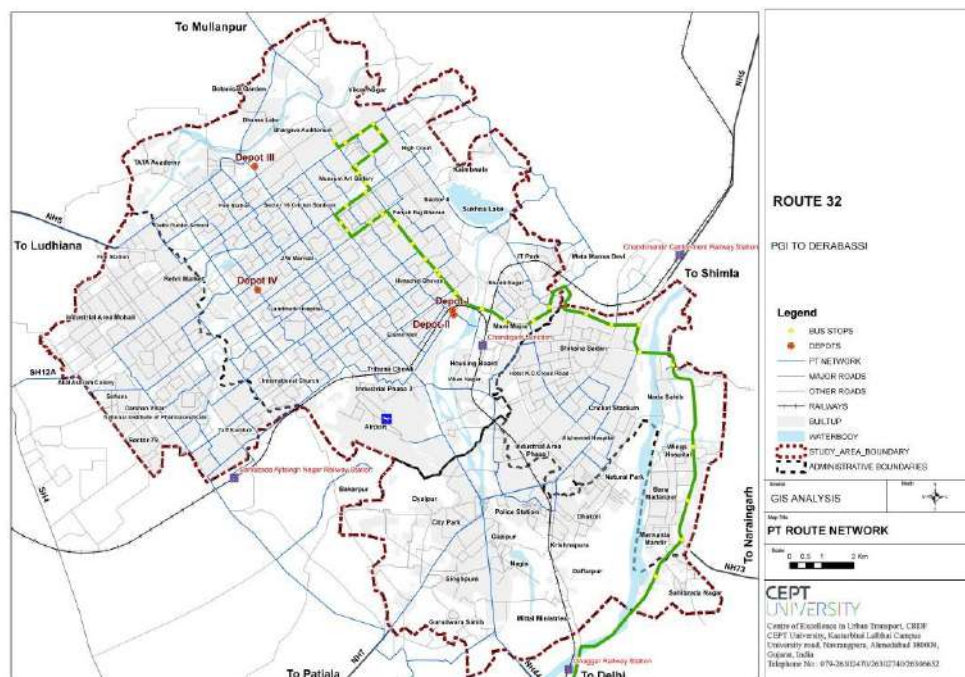


Figure 93: Existing Route 32 (PGI TO DERABASSI)

Route 34/ 34AC: PGI TO DERABASSI via OPD, 10/16, ISBT-17,Aroma, Piccadilly Chowk, 21/34, 20/33, 30/32, 29/31,HalloMajra, Airport, Zirakpur, Bankharpur,Mubarakpur

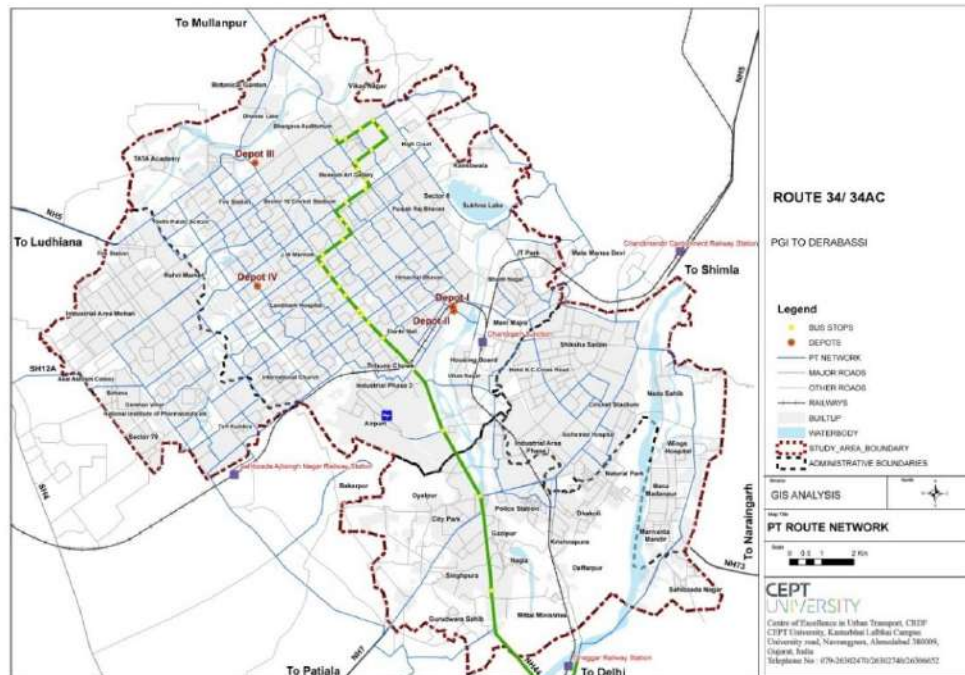


Figure 94: Existing Route 34/ 34AC(PGI TO DERABASSI)

Route 35/ 35AC/ 35B/ 35BAC: ISBT-17 TO KHARAR via Aroma, Sec-35/22,KissanBhawan, 35/36, 41/37, 41/40, F/Chowk, Ph-2,Ph-6, Balongi, Daun, DesuMajra, Mundi Kharar

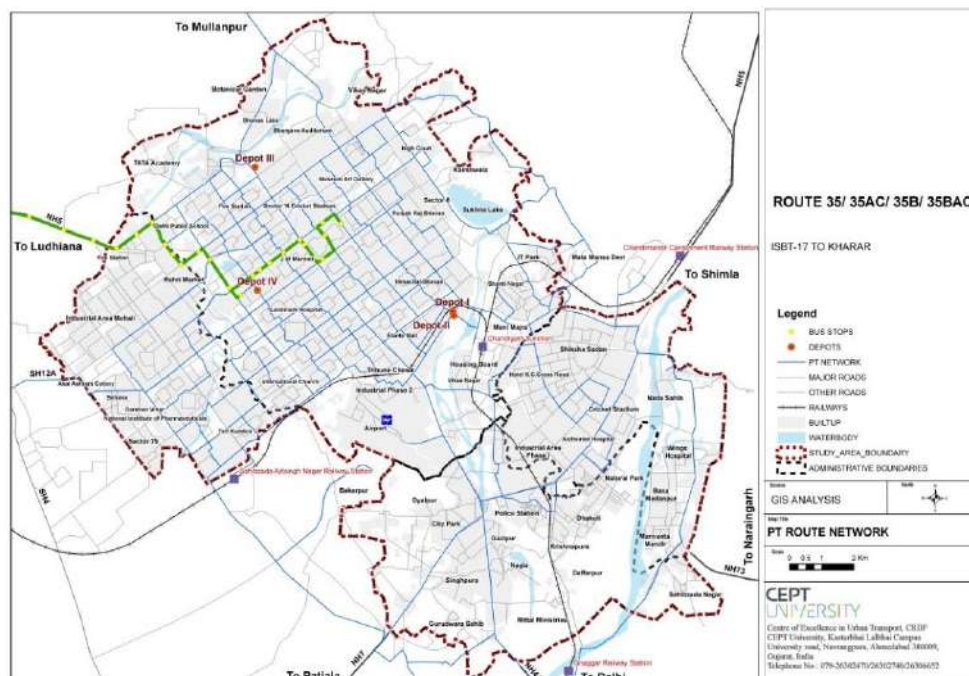


Figure 95: Existing Route 35/ 35AC/ 35B/ 35BAC(ISBT-17 TO KHARAR)

Route 39/ 39AC: MOHALI PH-11 TO PGI Via Indl.Area Ph-9, Mohali Sec-66/67, 67/68, 68 outer, 69/68, Kumbra,Edu.BoardPb., Forties, B/S Mohali, Ph-7, ChawlaChowk, Ph-3/7, 52/61, YPS Chowk, ISBT-43, 43/44,42/43, 36/35, 23/22, ISBT-17, 22/17, 16/10, 11/10,11/2, OPD

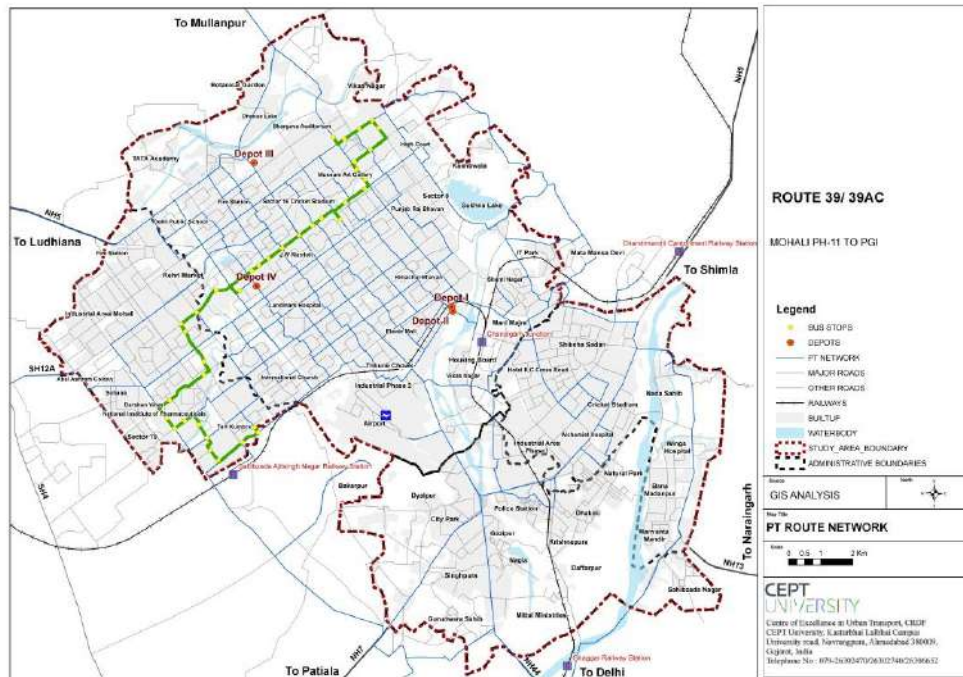


Figure 96: Existing Route 39/ 39AC(MOHALI PH-11 TO PGI)

Route 40: CHD TO KHIZRABAD via Sec-16/10, PGI, K/Lahora, S/P Barrier ,Mullanpur, Parol, Majra, Balock, Manakpur, Sangatpura, Kubaheri

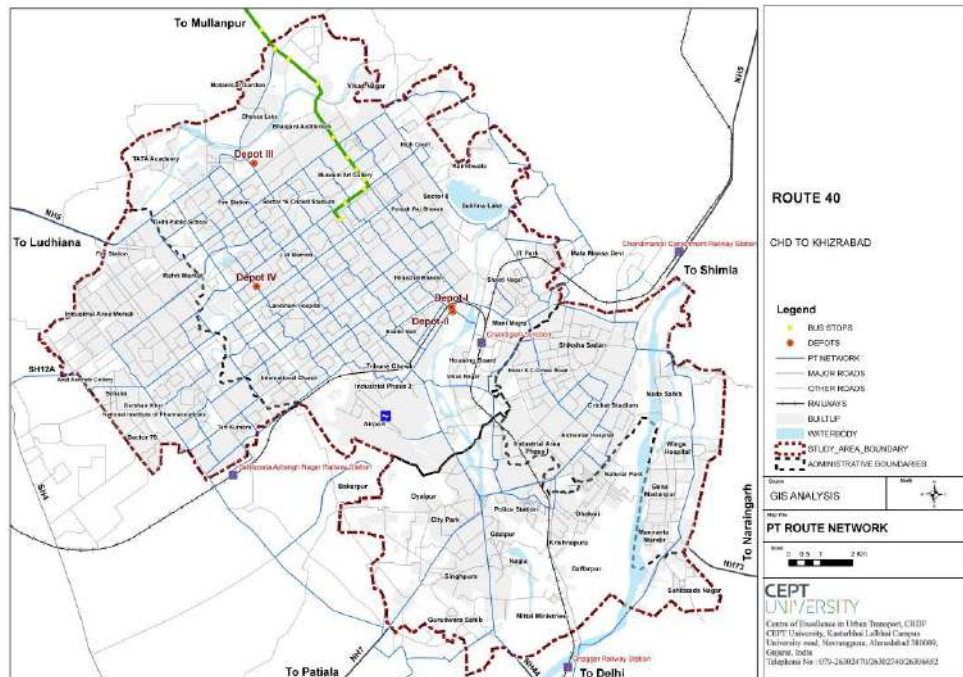


Figure 97: Existing Route 40 (CHD TO KHIZRABAD)

Route 79/ 79AC: PGI TO CHHATTBIR ZOO via OPD, 10/16, ISBT-17, Aroma, Sec-21, 20 Mkt., Sec-20/30, 32 Hospital, Tribune Chowk, Hallo Majra, Airport, Zirakpur, Dayalpura

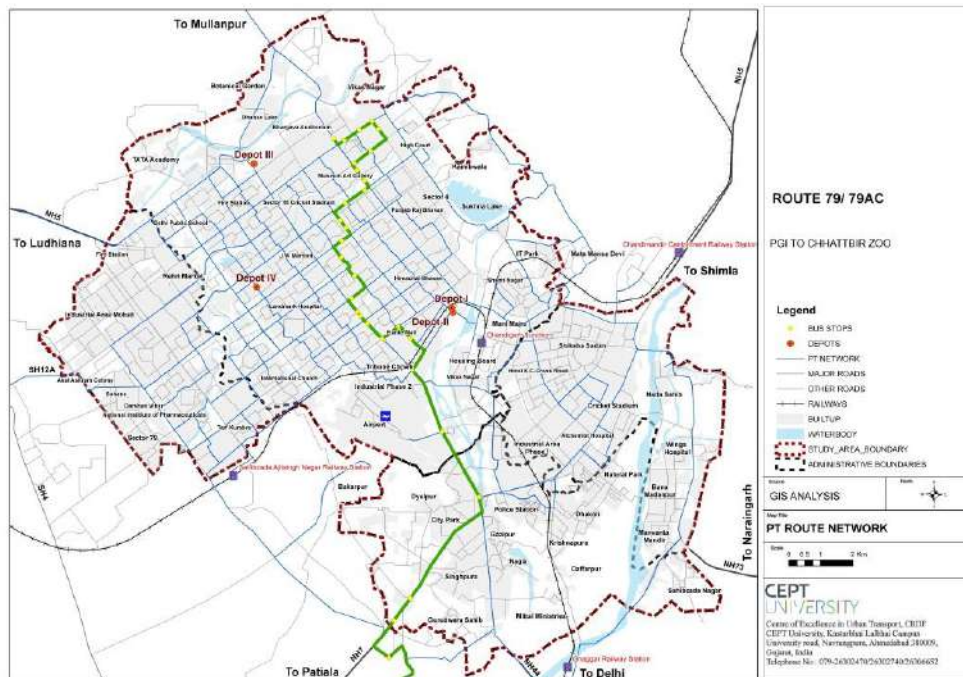


Figure 98: Existing Route 79/ 79AC(PGI TO CHHATTBIR ZOO)

Route 80/ 80AC: ZIRAKPUR TO PGI via Baltana, Raipur Kalan, MouliJagaran, Hallo Majra, Colony No.4, Elante Mall, Sec- 29 mkt, 30 mkt, 20 mkt, Aroma, 22/17, 16/10, PEC, New OPD

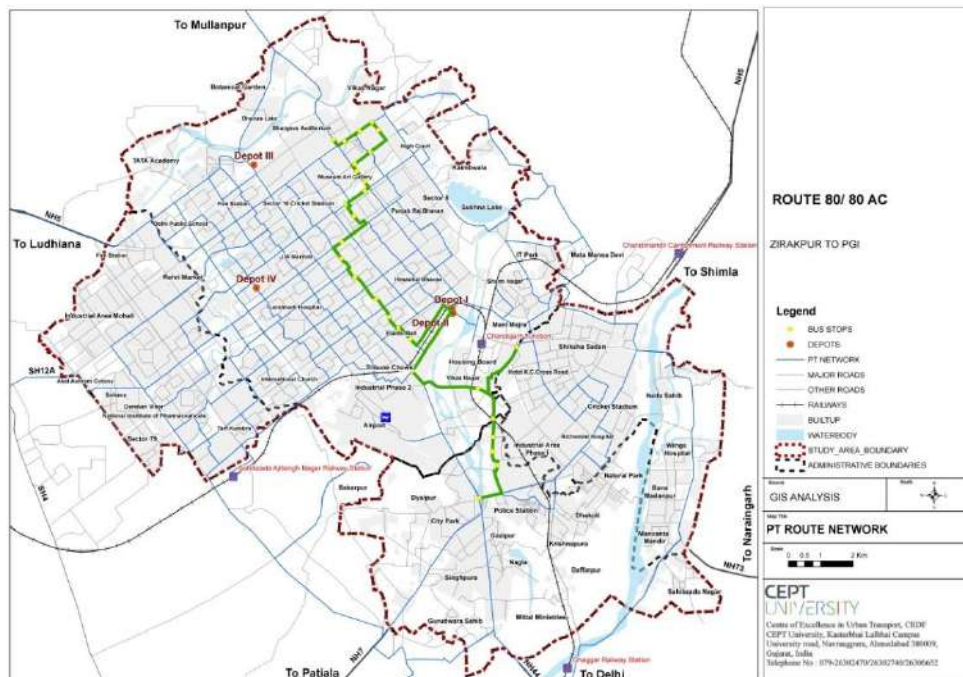


Figure 99: Existing Route 80/ 80AC(ZIRAKPUR TO PGI)

Route 212/ 212AC: PGI TO ZIRAKPUR (SINGHPURA BHUDA) via OPD, Sec 11/2, 10/16, ISBT-17, Aroma, Piccadilly Chowk, 21/34, 20/33, 30/32, 29/31, Hallo Majra, Airport, Zirakpur

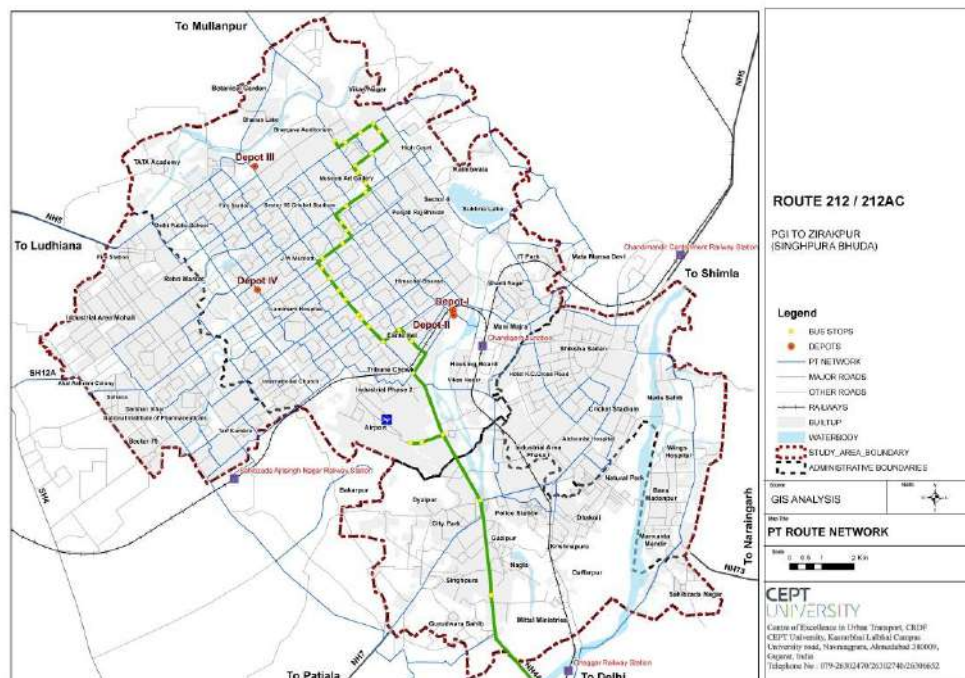


Figure 100: Existing Route 212/ 212AC(PGI TO ZIRAKPUR (SINGHPURA BHUDA))

Route 213: LANDRAN to DERA BASSI Via Dc.office/ Lakhnaur, Sohana, Sec-69/68, Fortis/ PSEB, Ph-7/8, YpsChowk , ISBT-43, Sec-44, Sec-45, Sec-46/47, Sec-32/31(Hospital), Tribune Chowk , Hallo Majra, Airport Chowk, Zirakpur, Singhpura, Bankharpur

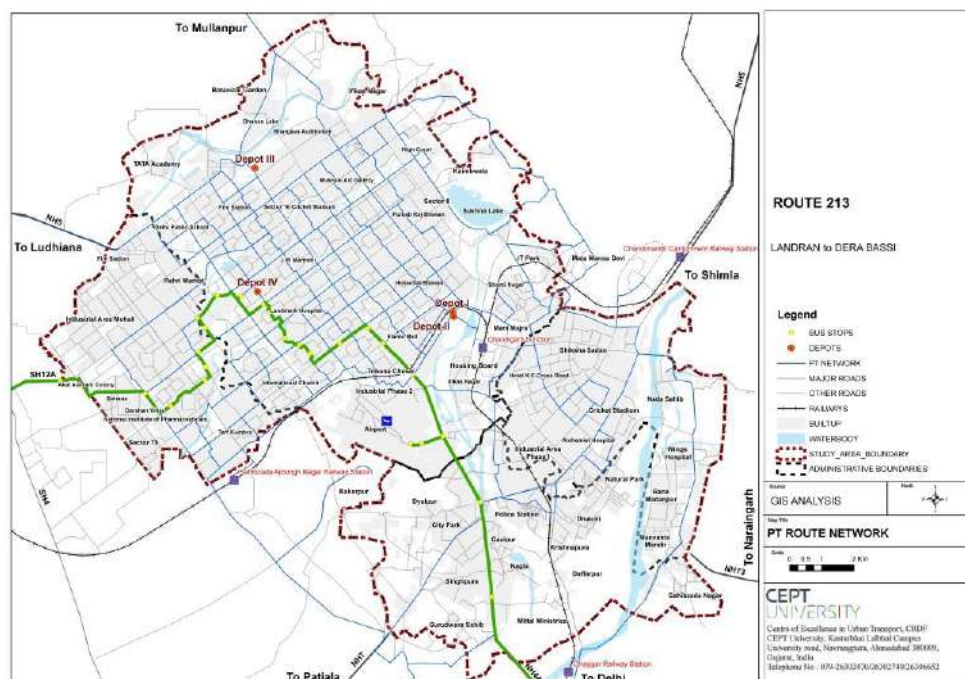


Figure 101: Existing Route 213 (LANDRAN to DERA BASSI)

Route 242/ 242AC: LANDRAN TO PGI via Lakhnaur, Sohana,R/Swami/71, SCL, 71, IVY, Ph-5/3 B-II, Ph-4/3 B-I,Ph-2/3A, F/Chowk, 42/53, ISBT-43, 43/44, 35/34,22/21, 17/22, 16 Hospital, 11/10, 11/2, PEC, N/OPD

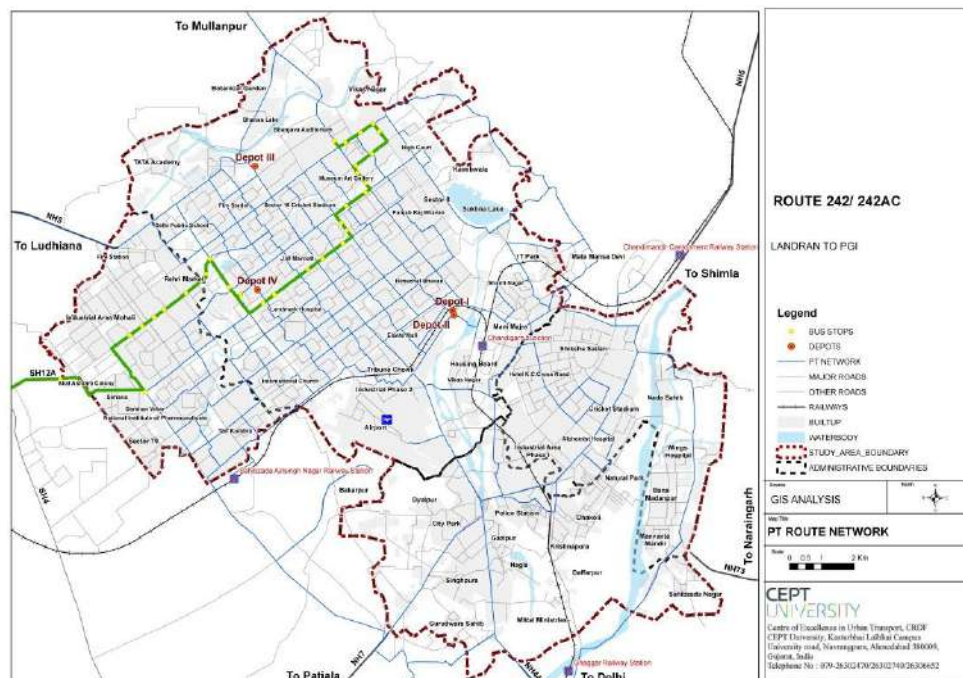


Figure 102: Existing Route 242/ 242AC(LANDRAN TO PGI)

Route 2B/ 2BAC: PGI TO PANCHKULA BUS STAND via OPD, PEC, 10/11,10/16, 17/16, ISBT-17, 18/17, 8/18, 7/19, 26/27, 26/28G/Mkt, CTU w/s, R/Station, Kalagram ,H.B. Chowk, 9/16, 10/15, 11/12

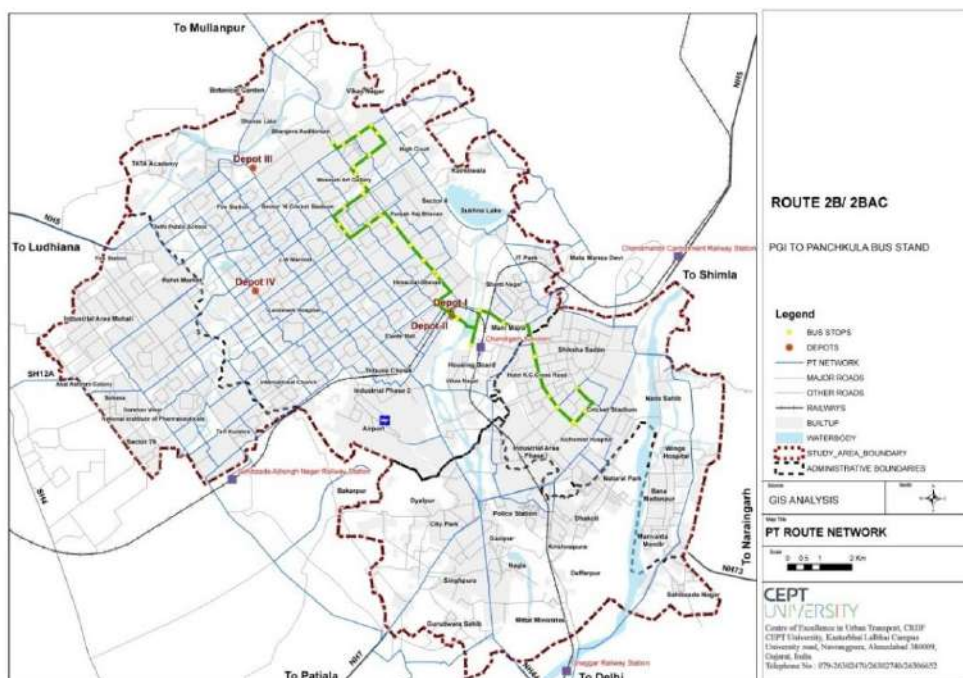


Figure 103: Existing Route 2B/ 2BAC (PGI TO PANCHKULA BUS STAND)

Route 2F/ 2FAC: PGI TO DHAKOLI BORDER via New OPD, PEC, 10/11, 10/16, 17/16, ISBT-17, 17/18, 8/18, 7/19, 26/27, 26/28, G. Mkt., CTU w/s, R/S, Kalagram ,H.B. Chowk, Panchkula Sector -8/17, 9/16, 10/15, 10/11, 11/4, Sec- 4 Pkl., Sec-21 Pkl., Sec-20 Pkl

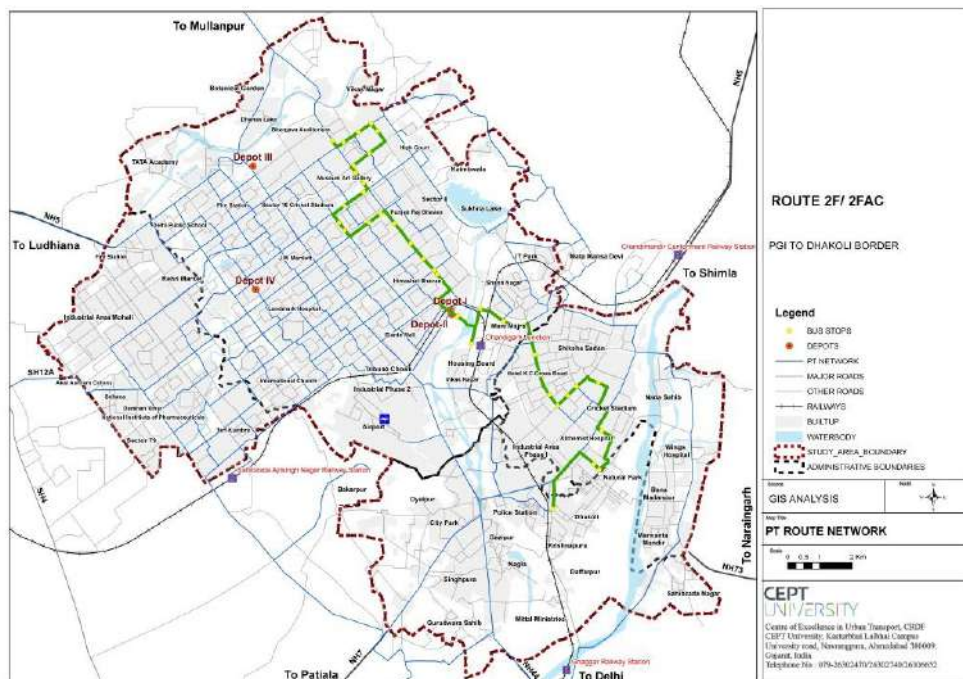


Figure 104: Existing Route2F/ 2FAC (PGI TO DHAKOLI BORDER)

Route 9A/ 9C/ 9CAC: MATA MANSA DEVI TO MATA MANSA DEVI Via M/Majra, H.Board, Kalagram, R/Cross., 28/26G/Mkt. 27/26, 27/19, 30/20, 32/33, 46/45, 50/49, 64/63,68/67, Sec-67, Mandikaran Houses, Sec-66 P/Chowki, Ph-11, Jagatpura, 47 outer, R/Darbar, Ind.Area Ph-2, Trib.Chowk, 29 Nursery, 28 P/Pump, CTU w/s, R/S, R/C, H.Board, M/Majra

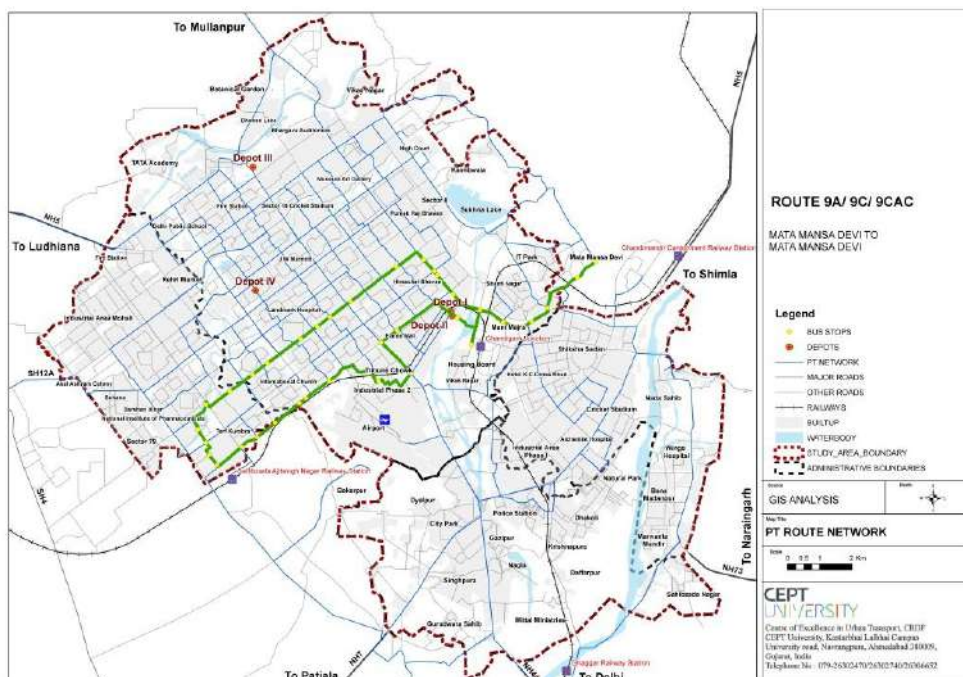


Figure 105: Existing Route9A/ 9C/ 9CAC (MATA MANSA DEVI TO MATA MANSA DEVI)

Route 20A/ 20AAC: CHD TO KHARAR Via Aroma, Sec-21 Mkt, 20/21,33/34, 44/45, ISBT-43, YPS, Ph-7, PSEB, Sec-68/69 Sohana, Lakhnaur, Landran, ChapparChiri, SanteMajra

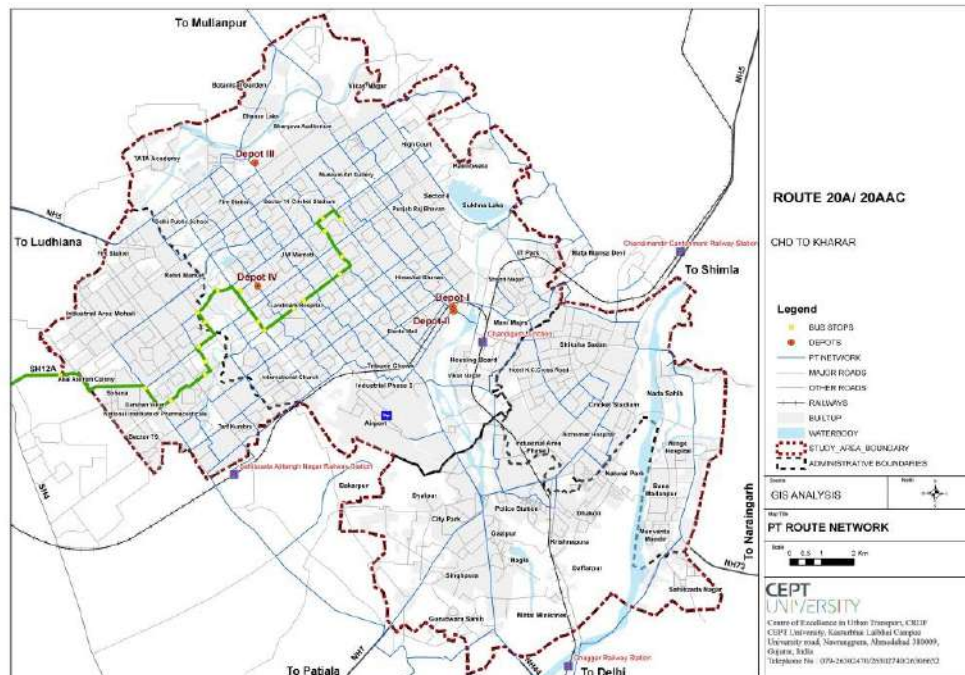


Figure 106: Existing Route 20A/ 20AAC (CHD TO KHARAR)

Route 30A/ 30AAC: ISBT-43 TO CANTONMENT via Sec 43/44, 42/43, 36/35, K/Bhawan, 22/35, 21/34, 20/33, 30/32, 29/31Tribune, H/Majra, Colony No.4, CTU w/s,R/Station, Railway Crossing, Kalagram,H.B Chowk, M/Majra, Command Hospital. Tank Chowk

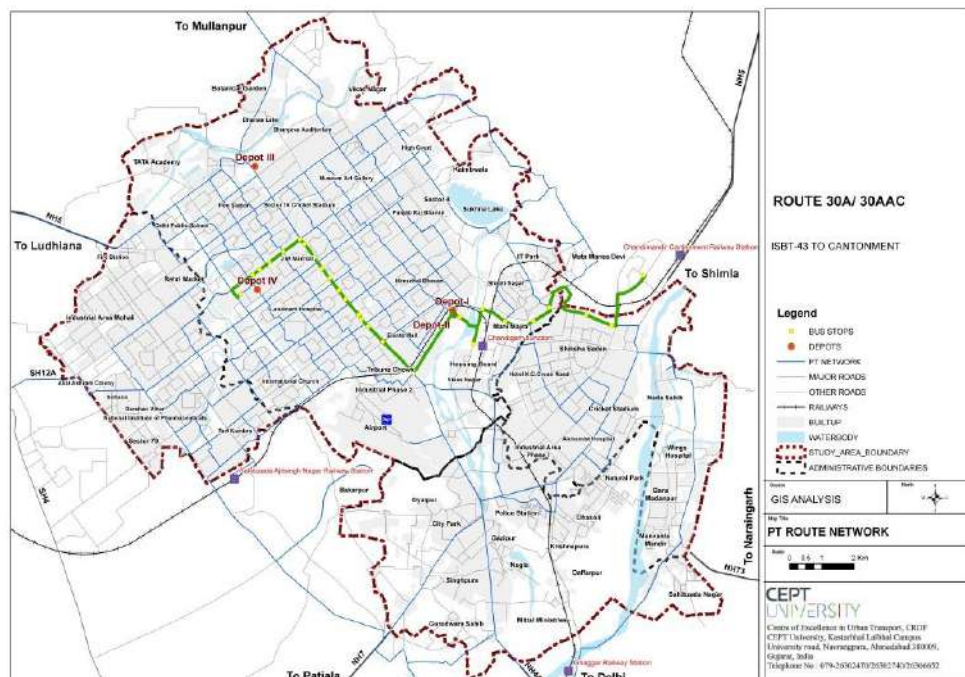


Figure 107: Existing Route 30A/ 30AAC (ISBT-43 TO CANTONMENT)

Route 30B/ 30BAC: ISBT-43 TO PANCHKULA SEC-4 via Sec 43/44,42/43, 36/35, 23/22, ISBT-17, Aroma, 21 Mkt., 20Mkt., 19/27, 27 Mkt., 26/28 Grain Mkt., Railway Crossing, Kalagram H. B. Chowk, 9/16 Pkl, 10/15,11/12, Raily Chowk, Sector-4 Panchkula

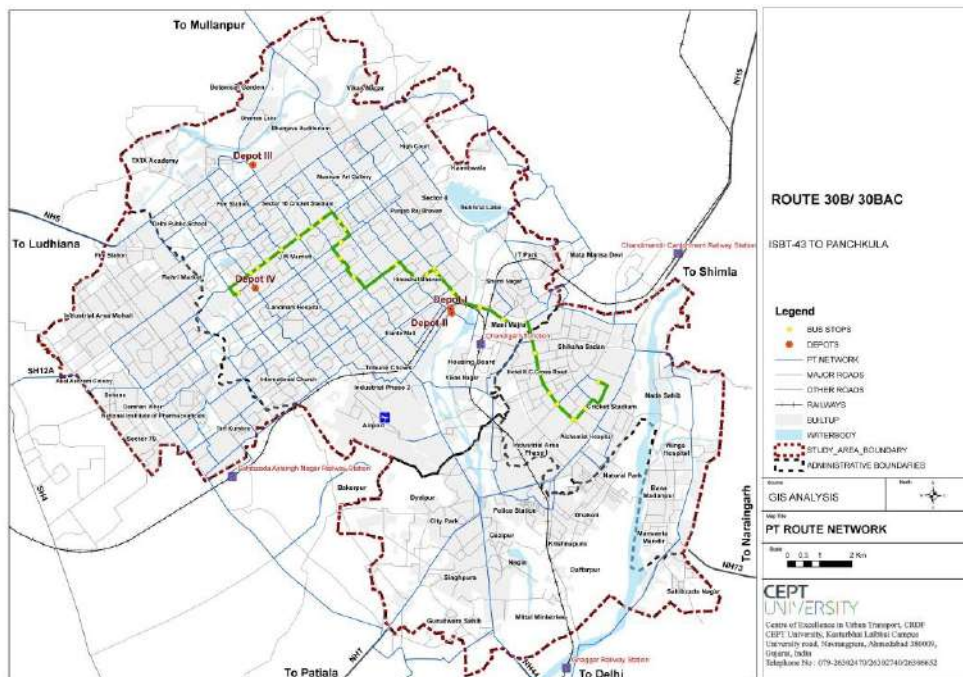


Figure 108: Existing Route 30B/ 30BAC(ISBT-43 TO PANCHKULA SEC-4)

Route 32A/ 32AAC: NEW ISBT-MOHALI TO MATA MANSA DEVI via Ph-6, Frenko, Sec-40/41, 41 Mkt, 42 Mkt, ISBT- 43,43/44, 42/43, 36/35, 23/22, ISBT-17, 8/18, 7/19,26/27, 26 G/Mkt, R/Crossing, Kalagram , H/B, M/Majra

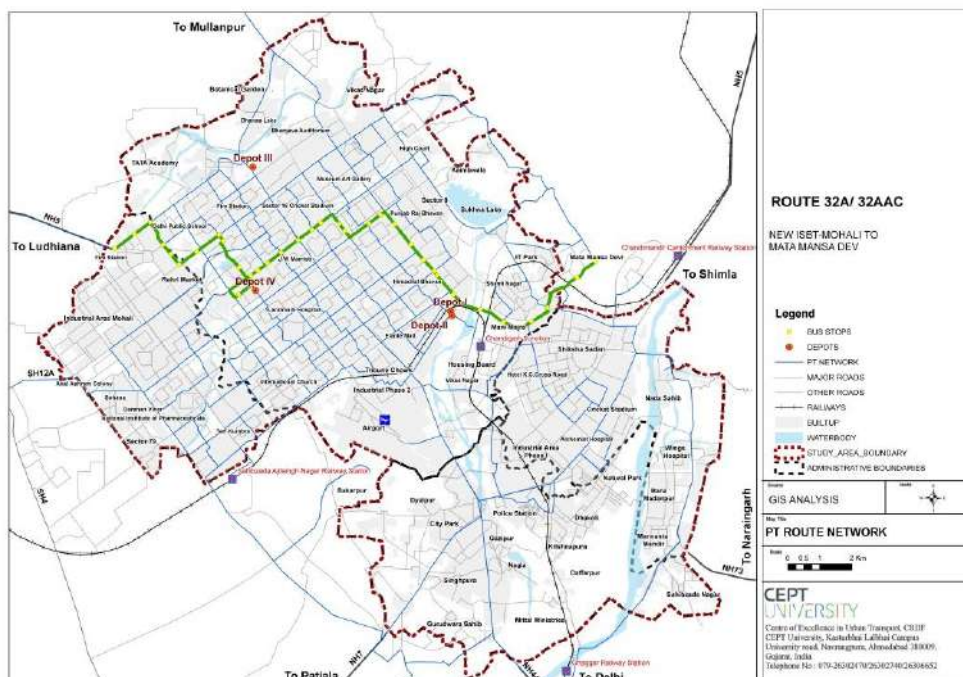


Figure 109: Existing Route 32A/ 32AAC(NEW ISBT-MOHALI TO MATA MANSA DEVI)

Route 123A/ 123AAC: MOHALI R/STATION TO PGI via Sec-65 mkt, 64 mkt, 63 mkt, 51/63, 51 mkt, ISBT-43, Sec-43/44,35/34, Aroma, 22/17, 16 Hospital, Sec-11 mkt, OPD,PGI

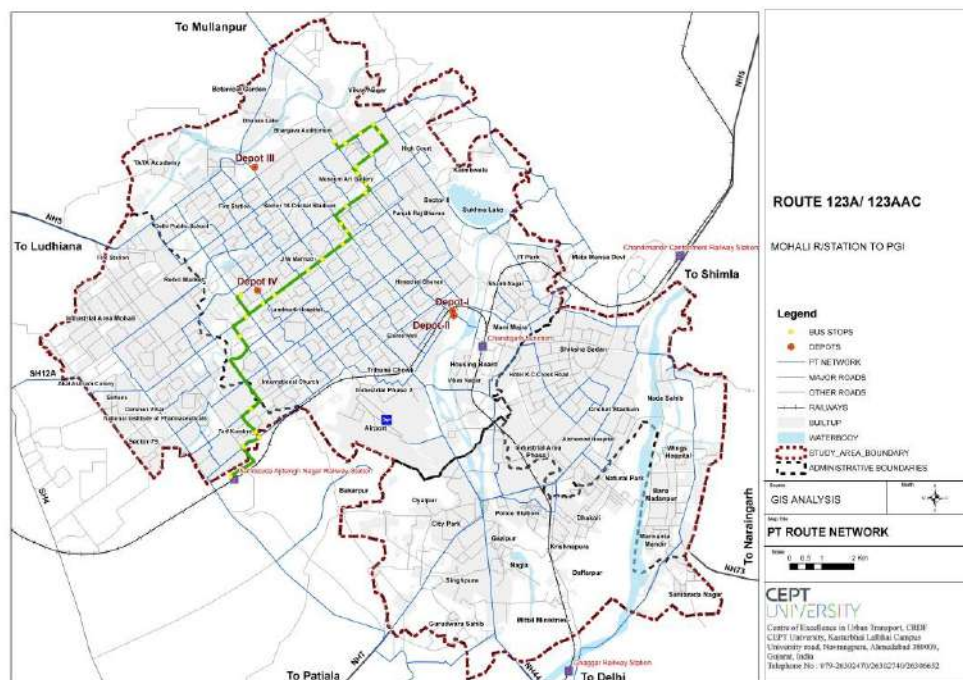


Figure 110: Existing Route 123A/ 123AAC(MOHALI R/STATION TO PGI)

Route 20/ 20AC: CHD TO KHARAR Via Aroma, Sec-21 Mkt, 20/21, 33/34, 44/45, ISBT-43, YPS, 69/70, Ph-7, Sohana,Lakhnaur, Landran, ChapparChiri, SanteMajra

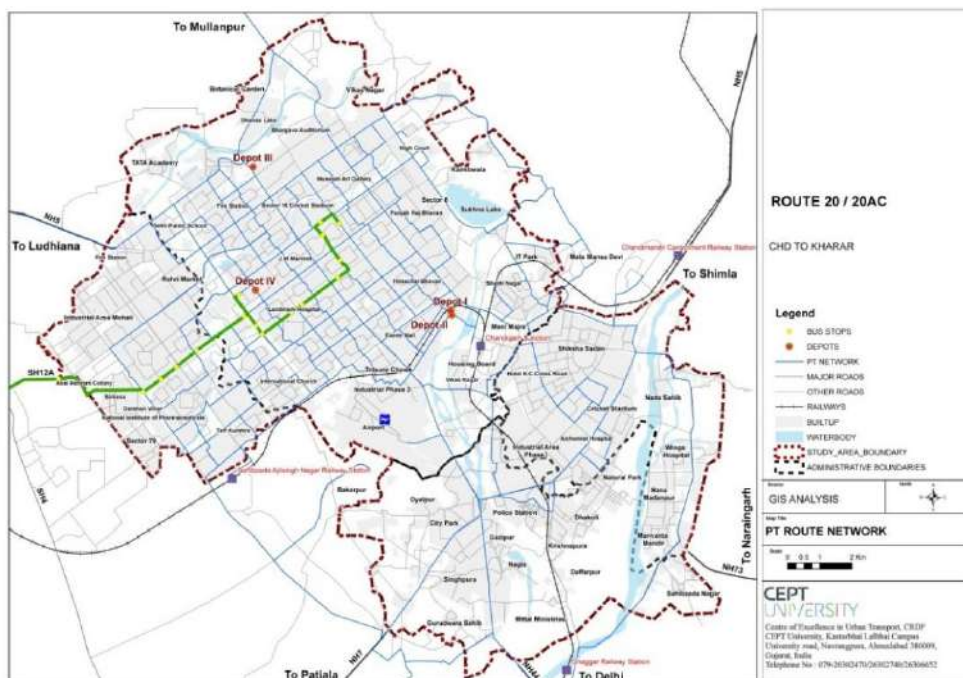


Figure 111: Existing Route 20/ 20AC (CHD TO KHARAR)

Route 26/ 26AC: DHANAS TO DHAKOLI via S.P. Barrier, K.Lahora, PGI, OPD, 10/11, 10/16, ISBT-17, 19/20, 27/30,28/29, 28 P.P., CTU w/s, R/S, Colony No. 4, Hallo Majra, Airport, Zirakpur, Dhakoli

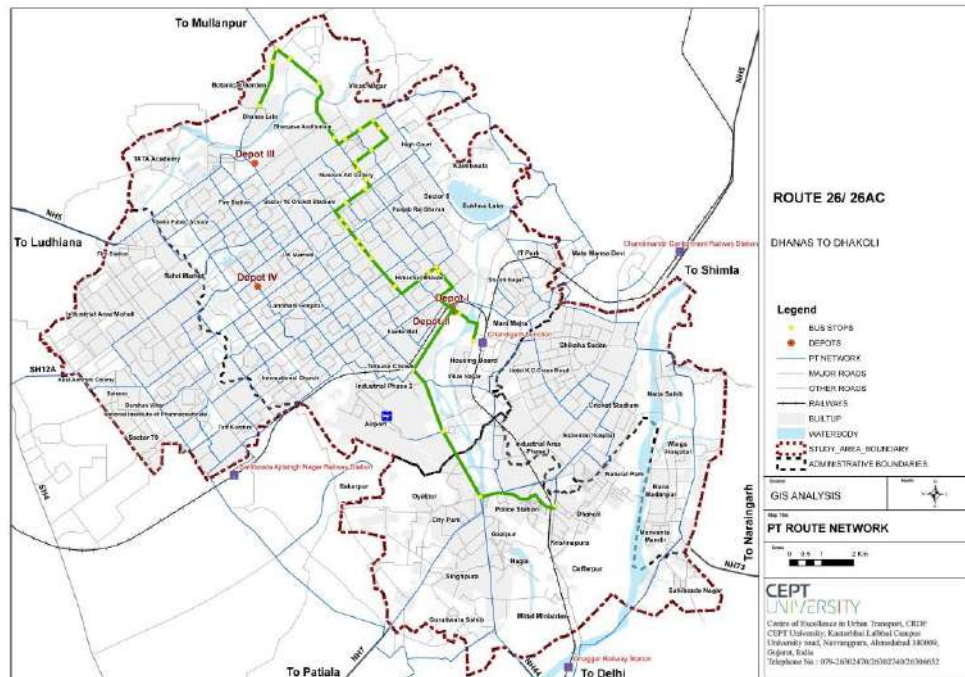


Figure 112: Existing Route 26/ 26AC (DHANAS TO DHAKOLI)

Route 36/ 36AC: PGI TO NEW AIRPORT Via New OPD ,Sec 11/2,10/16 (Hospital), ISBT-17, Aroma, 21 mkt, 20 mkt, Sec-30/20 divided, 30/32 Hospital, 29/31, Tribune Chowk, Hallo Majra, old Airport Chowk, Zirakpur, Nabha Sahib, Dialpura outer

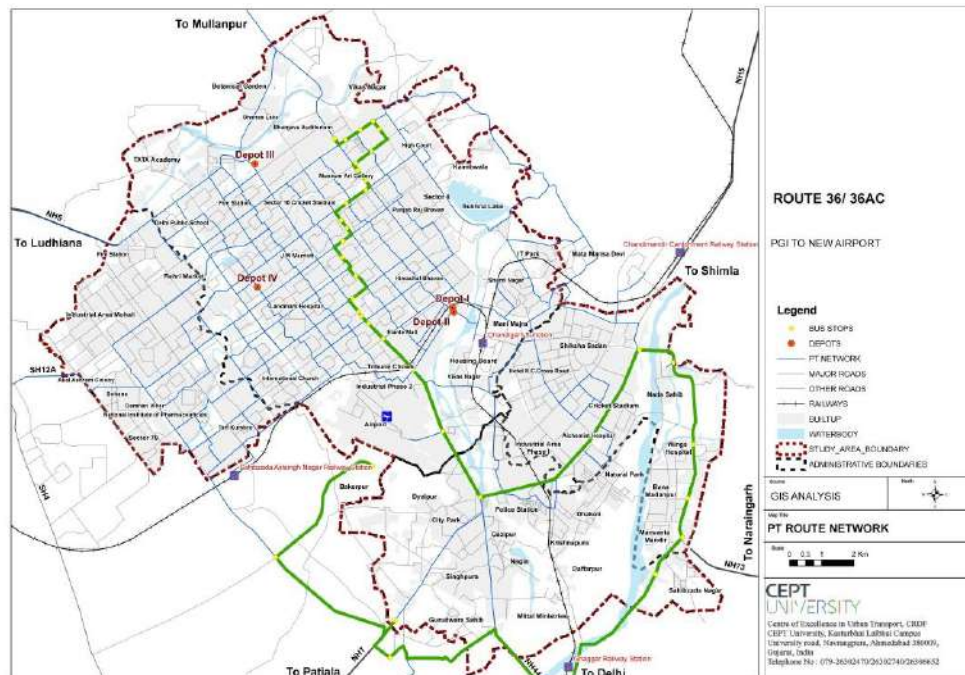


Figure 113: Existing Route 36/ 36AC (PGI TO NEW AIRPORT)

Route 38/ 38AC: ISBT-17 TO NEW AIRPORT Via Sec-22/23, 35/36, 43/42, ISBT-43, 51, 50, 49, 48 mkt, Jagatpura, 65mkt outer, White House, Indl. Area Ph-9, Sec-82 mkt, Airport

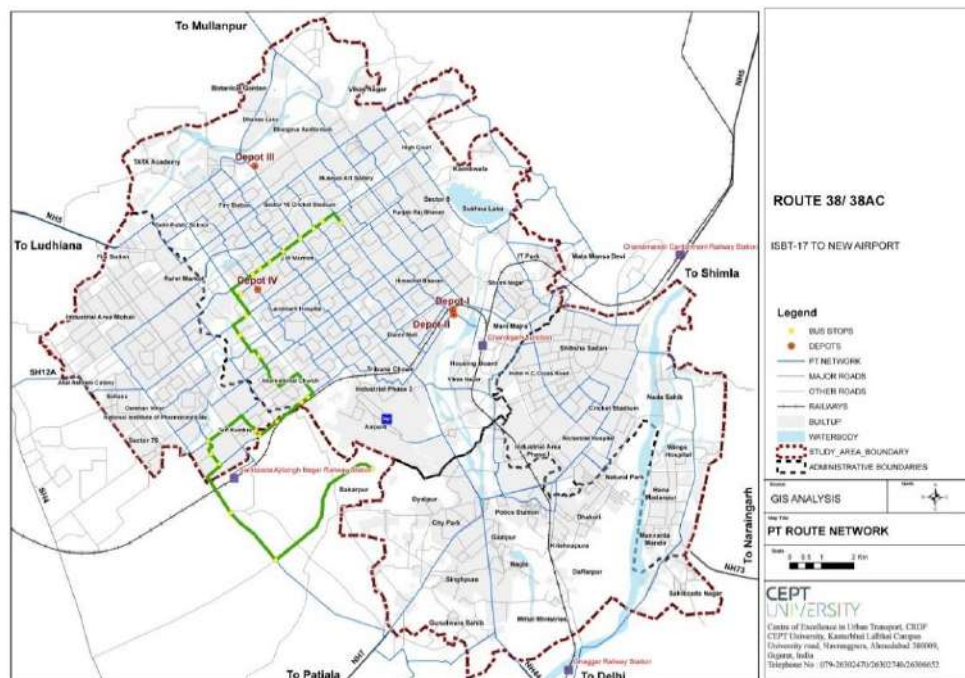


Figure 114: Existing Route 38/ 38AC(ISBT-17 TO NEW AIRPORT)

Route 211/ 211AC: PGI TO CANTONMENT via OPD, Sec 11/2 ,16/10 Hospital, 17 B/Stand, Aroma, 21, 20 Mkts, 19/27, 27 mkt, 28 mkt, CTU w/s, R/S, Railway Crossing , Kalagram, H/B, M/Majra, Command Hosp

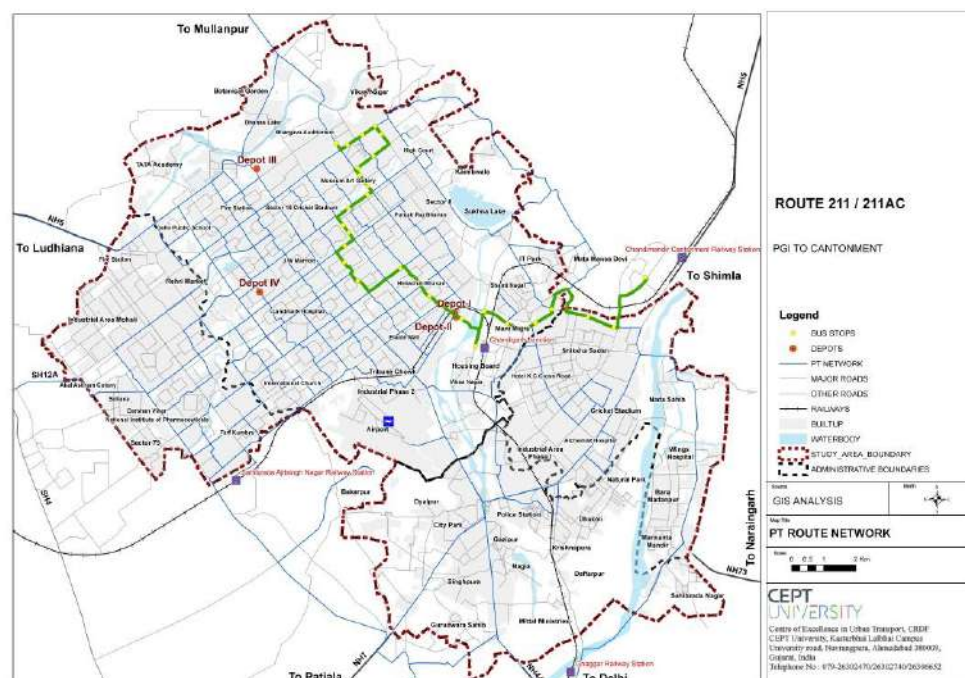


Figure 115: Existing Route 211/ 211AC (PGI TO CANTONMENT)

Route 214/ 214AC: SP Barrier TO NADA SAHIB via KhudaLahora, PGI, OPD, PEC, Sec-10/16, 17/16, ISBT-17, 18 Mkt., 19 Mkt., 27 Mkt., 28 Mkt., Indl.area/Ph.1, Rly. Station, Rly. Crossing, KalagramH/B. Chowk, M. Majra, M. Hospital, old PKL, Majrichowk

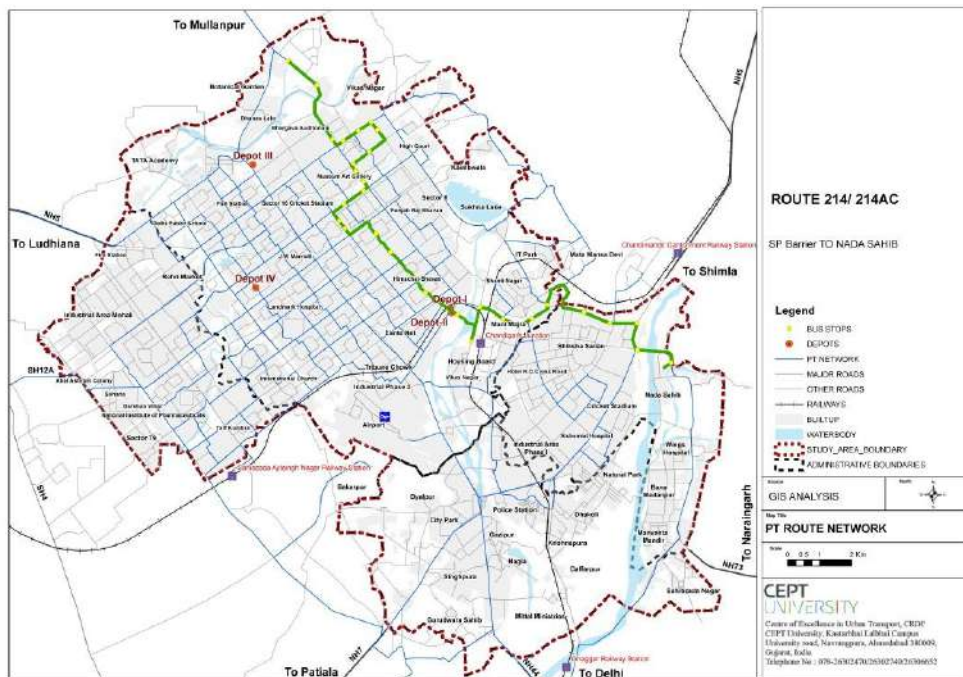


Figure 116: Existing Route 214/ 214AC(SP Barrier TO NADA SAHIB)

Route 216/ 126AC: ISBT-43 TO DERABASSI via Sec 43/44, 42/43,36/35, KissanBhawan, 23/22, ISBT-17, Aroma, Sec- 21/34, Sec-20/33, Sec-30/32, Tribune Chowk, Hallo Majra, Airport Chowk, Zirakpur, Bankharpur

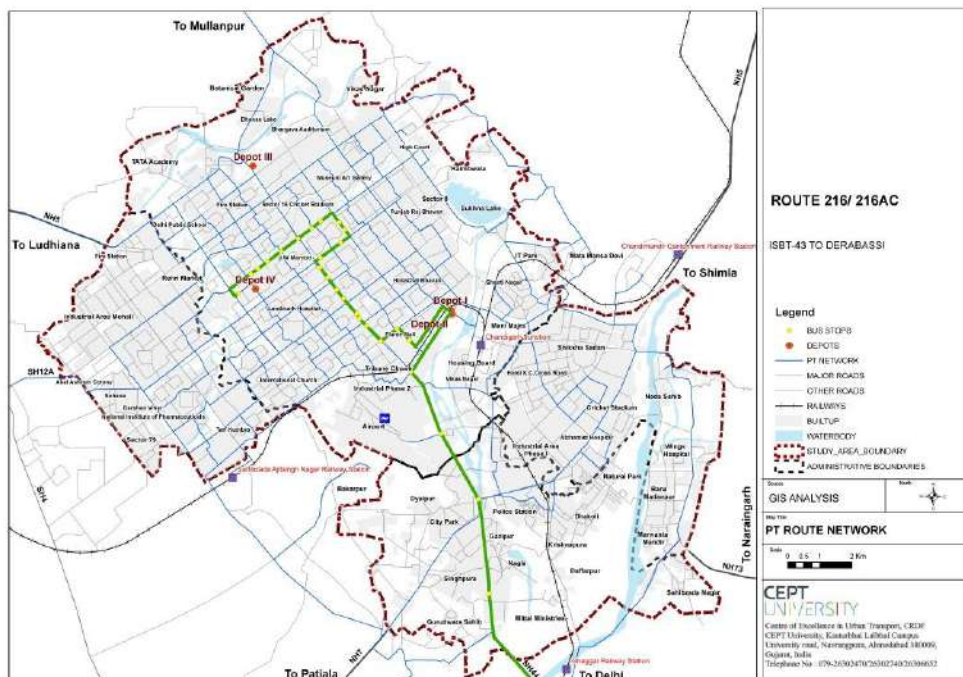


Figure 117: Existing Route 216/ 126AC(ISBT-43 TO DERABASSI)

Route 38A/ 38AAC: ISBT-17 TO NEW AIRPORT Via Sec-22/23, 35/36, 43/42, ISBT-43, YPS Chowk, Ph-7, Sec-69/70, Sohana Gurdwara Sahib, 69/79, 68/80, 67/81, 82 mkt, Airport Chowk

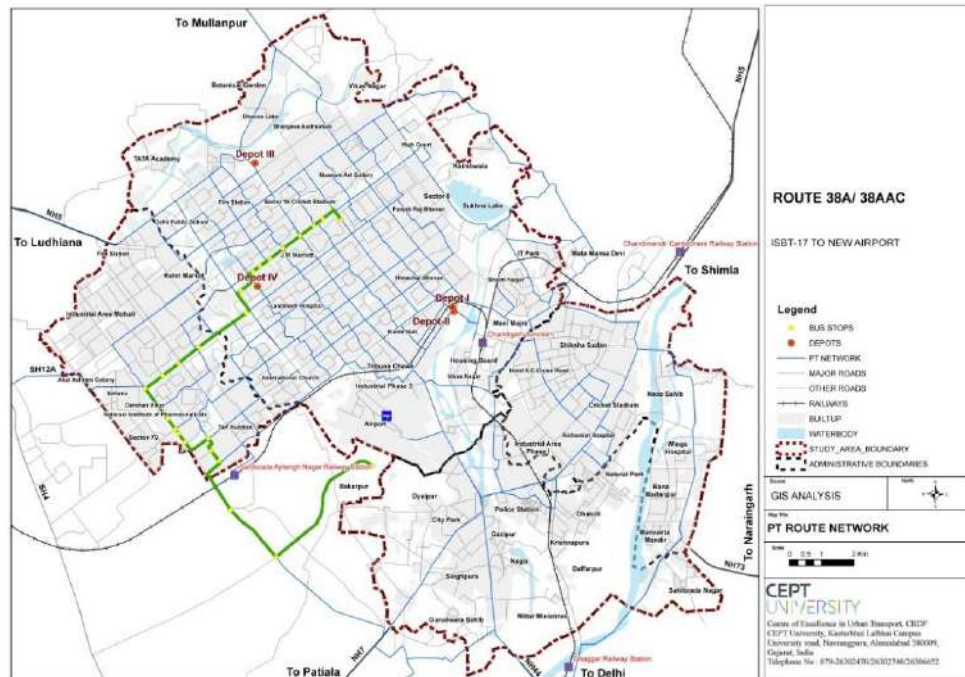


Figure 118: Existing Route 38A/ 38AAC (ISBT-17 TO NEW AIRPORT)

Annexure 5: Alignment of Proposed Bus Routes– Alternative Scenario 2A and 2B

Suburban Routes

Route 1: ISBT Mohali - Sukteri - ISBT 17 via Phase 1, Sector 40/39, Sector 40 Market, Sector 40/41, Sector 37 Market, Sector 36/37, Sector 36 Market, Sector 35/36, Kisan Bhawan, Sector 23/22, ISBT 17, Sector 22/17, Sector 18 Press, Sector 26 Guru Gobind School, Sector 26 Market, Kishangarh, It Park

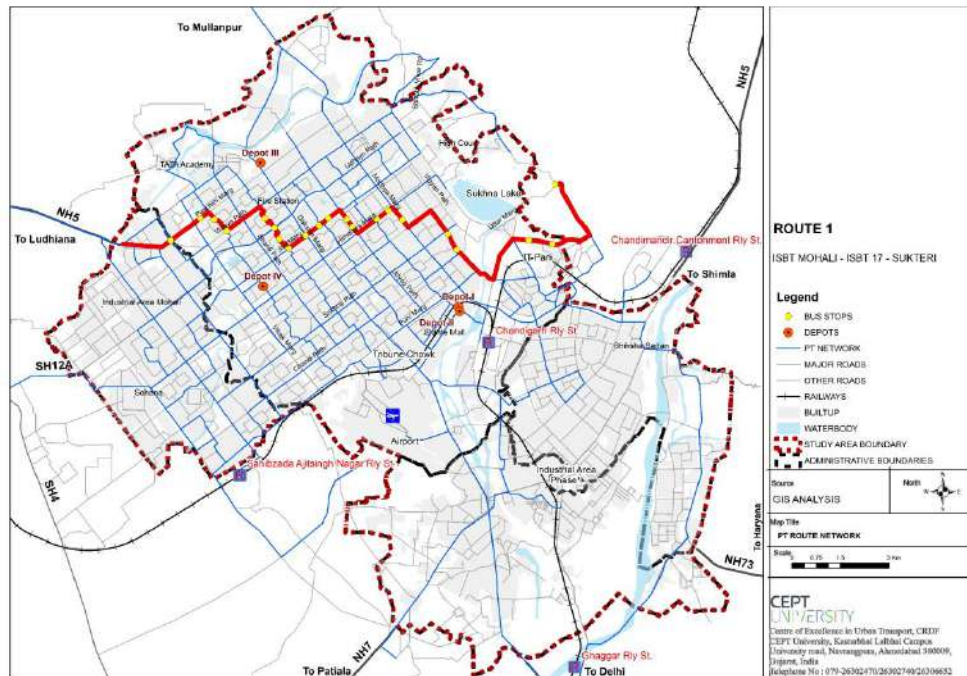


Figure 119: Proposed Route 1: (ISBT Mohali - Sukteri - ISBT 17)

Route 2 : Sohna Gurudwara - ISBT 43 - ISBT 17 - PGI - Khizrabad Via Sector 69/70, Sector 61/70, Phase 7, Sector 52/51, ISBT 43, Sector 43 Market, Sector 42/43, Attawa Chowk, Sector 35/36, Kisan Bhawan, Sector 22/23, ISBT 17, Sector 22/17, Sector 18/17, Sector 18 Press, Sector 9/17, Rose Garden, Sector 11/15, PGI, Khudda Lahora, Sarangpur Barrier, Mullanpur Barrier, Sarangpur Tax Barrier, Ferozpur, Paroul, Siswan Junction, Majra, Bhootgarh, Majri, Sialba, Khera, Jhande Majra, Kubaheri

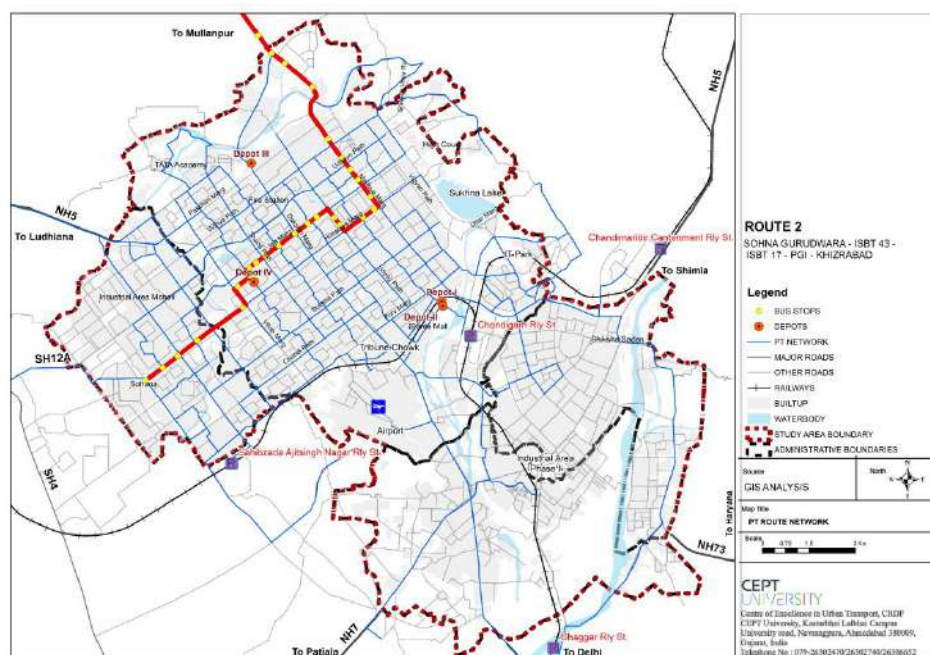


Figure 120: Proposed Route 2 (Sohna Gurudwara - ISBT 43 - ISBT 17 - PGI - Khizrabad)

Route 3: ISBT 17 - ISBT 43 - Airport via Sector 22/23, Kisan Bhawan, Sector 35/36, Attawa Chowk, Sector 42/43, Sector 43 Market, Sector 43/44, Sector 52/51, Phase 7, Sector 61/70, Sector 69/70, Gurudwara Singh Saheeda Sahib, Sector 68/79, Sector 76, Sector 80/67, Mauli, Sector 66, Phase 1 Ind area Mohali, Sector 82, New Airport Chowk

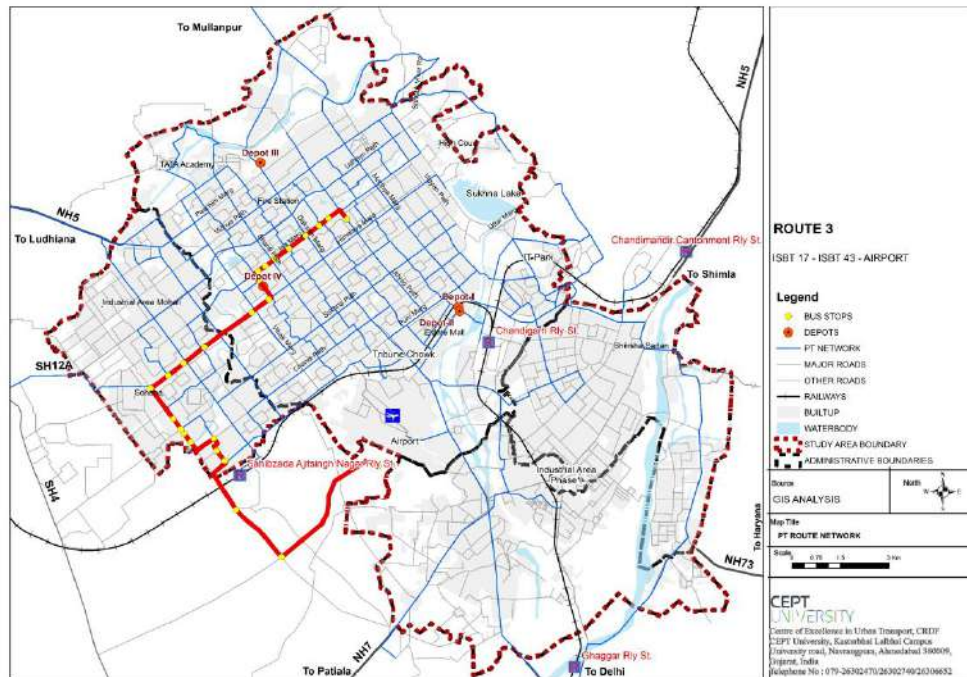


Figure 121: Proposed Route 3 (ISBT 17 - ISBT 43 - Airport)

Route 4: ISBT 17 - Railway Station – Cantonment Via Sector 22/17, Sector 18/21, Sector 21/18, Sector 19 Market, Sector 27 Market, Sector 28 Market, CTU Workshop, Railway Station, Railway Crossing, Kalagram, Housing Board Mani Majra, Command Hospital, Tank Chowk

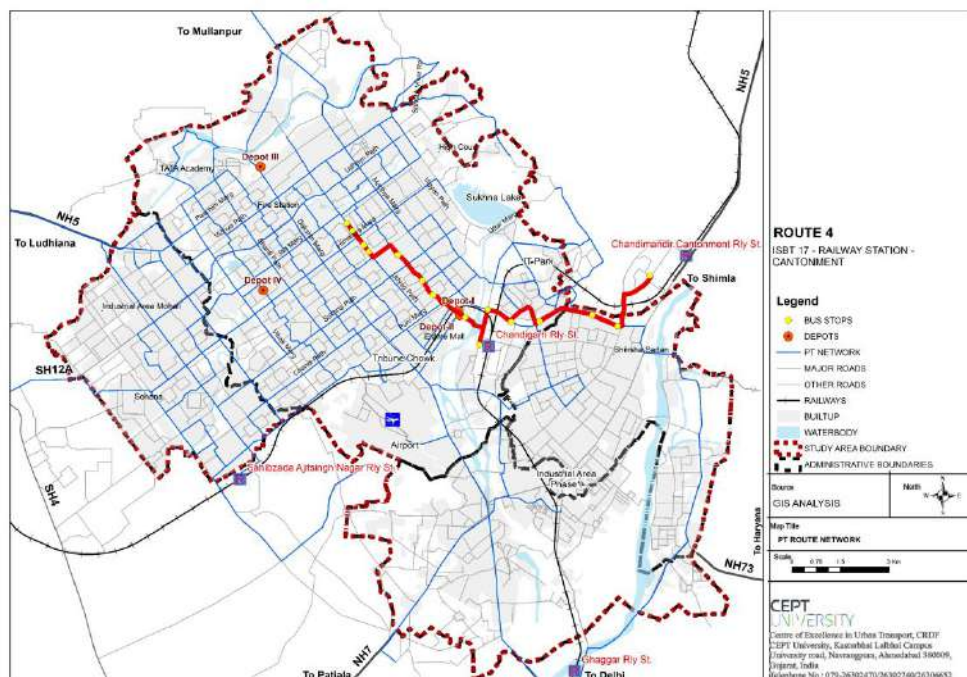


Figure 122: Proposed Route 4 (ISBT 17 - Railway Station – Cantonment)

Route 5: ISBT 43 - Tribune Chowk - Derabassi Via Sector 43/44, Sector 44 Market, Sector 45 Market, Sector 46, Sector 46 Market, Sector 32/31, Sector 29/31, Tribune Chowk, Hallo Majra, Behlana, Airport, Old Airport, Airport Chowk, Zirakpur bus stand, Singhpura, Bhankharpur

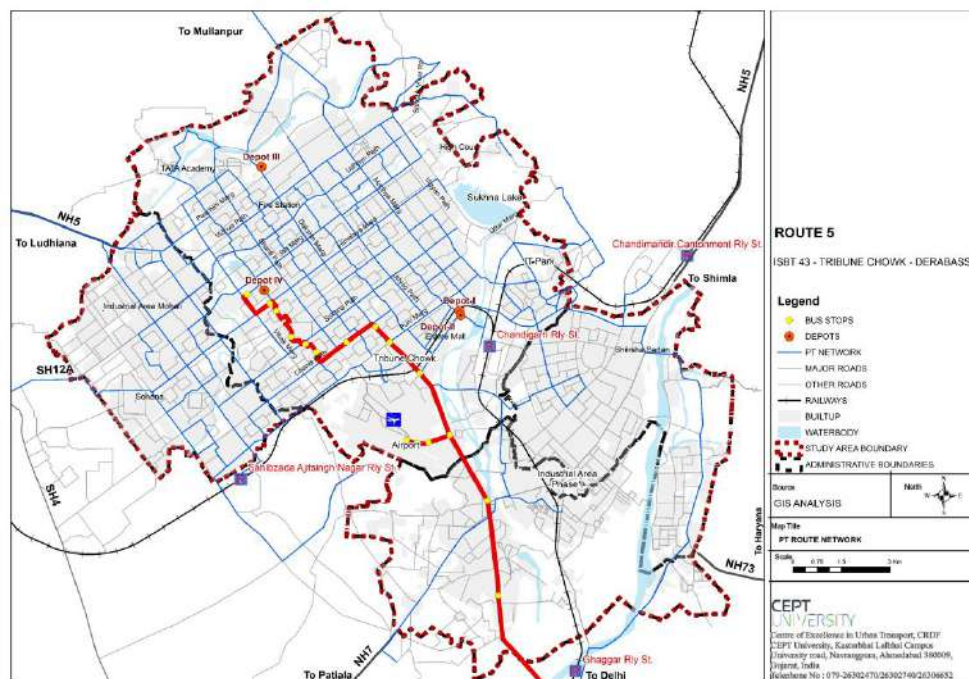


Figure 123: Proposed Route 5 (ISBT 43 - Tribune Chowk - Derabassi)

Route 6: ISBT 17 - PGI - Khizrabad via Kubaheri, Jhande Majra, Khera, Sialba, Majri, Bhootgarh, Majra, Siswan Junction, Paroul, Ferozpur, Sarangpur Tax Barrier, Mullanpur Barrier, Sarangpur Barrier, Khudda Lahora, PGI, Sector 11/15, Sector 10/16 Hospital, Rose Garden, Matka Chowk, Sector 16/17

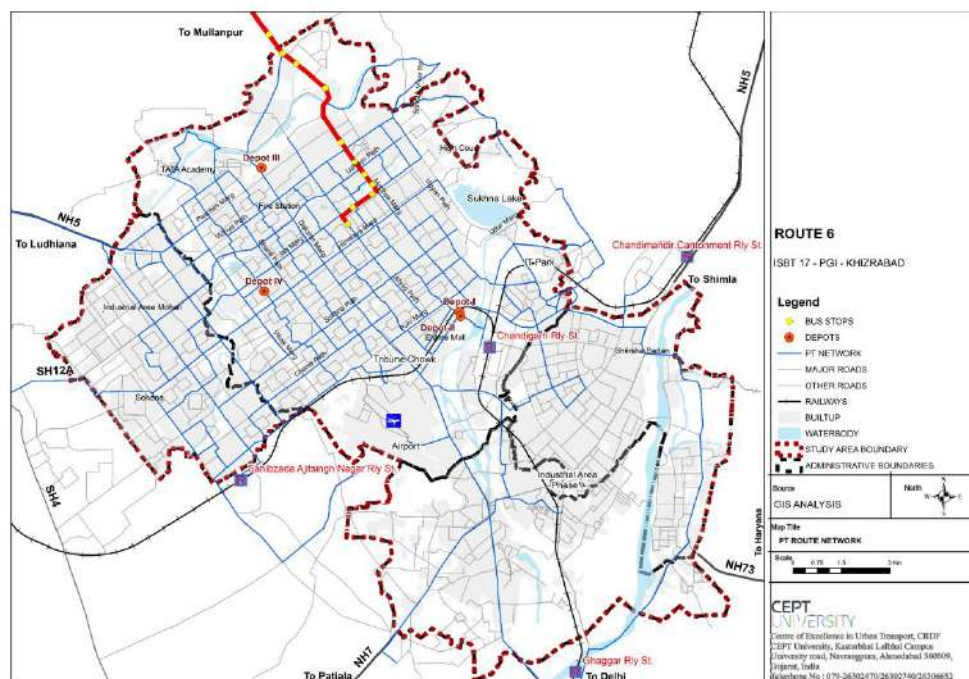


Figure 124: Proposed Route 6 (ISBT 17 - PGI - Khizrabad)

Route 7: ISBT 17 - Mani Majra via Sector 22/17, Sector 18/21, Sector 21/18, Sector 19 Market, Sector 26/27, Sector 26, Grain market 28/26, Transport Chowk, Railway Crossing, Shivalik Park, Indira Colony, Housing Board Mani Majra, Housing Board, Kalagram

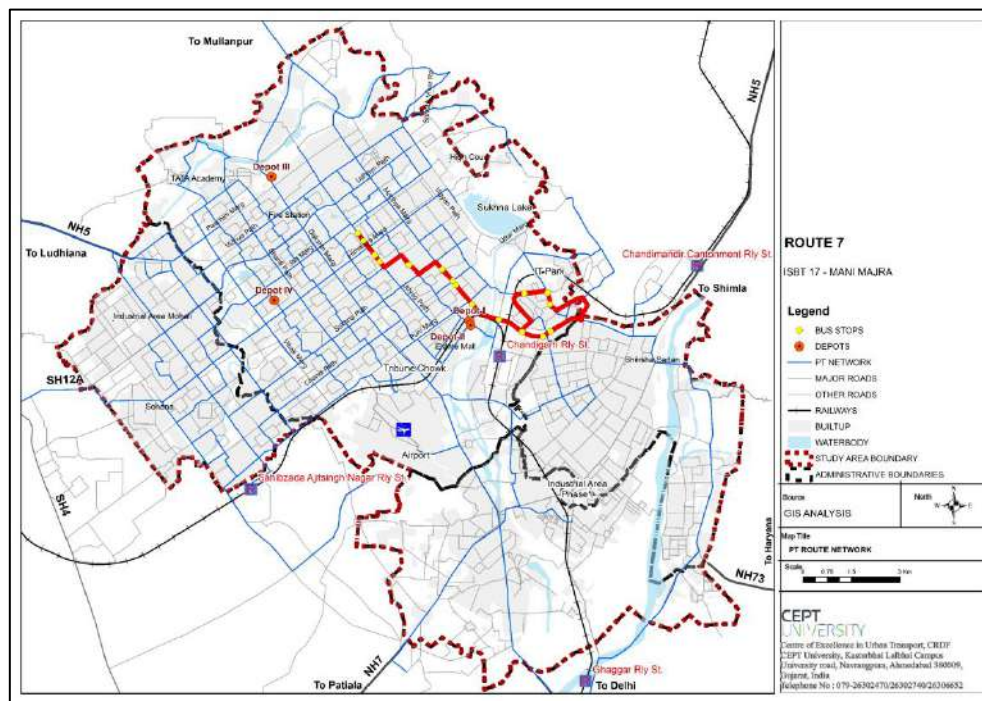


Figure 125: Proposed Route 7(ISBT 17 - Mani Majra)

Route 8: ISBT 17 - PGI - Mullanpur Barrier - ISBT 17 via ISBT 17, Sector 23/16, Sector 15/24, Sector 24 market, Sector 24/25, Sector 38/25, CTU Depot Sector 25, Daddu Majra Colony, Dhanas, Sarangpur Barrier, Mullanpur Barrier, Sarangpur Barrier, Khudda Lahora, PGI, Government College, New OPD, PEC College, Sector 2/11, Mini Secretariat

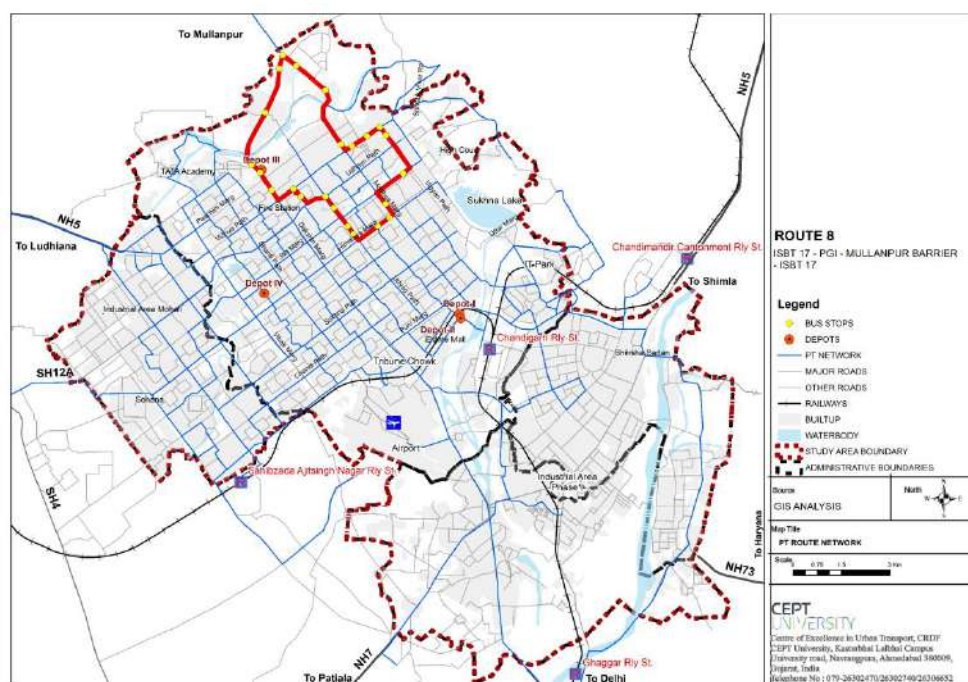


Figure 126: Proposed Route 8 (ISBT 17 - PGI - Mullanpur Barrier - ISBT 17)

Route 9: ISBT17 -Railway Station -Dhakoli via Sector 22/17, Sector 18/21, Sector 21/18, Sector 20/19, Sector 19/27 Market, Sector 27 Market, Sector 27/28, Sector 26, Transport Chowk, Railway Crossing, Railway Station, CTU Workshop, Hallo Majra, Vikas Nagar, Raipur Kalan, Parvathi Vihar, Tribune colony, Baltana village

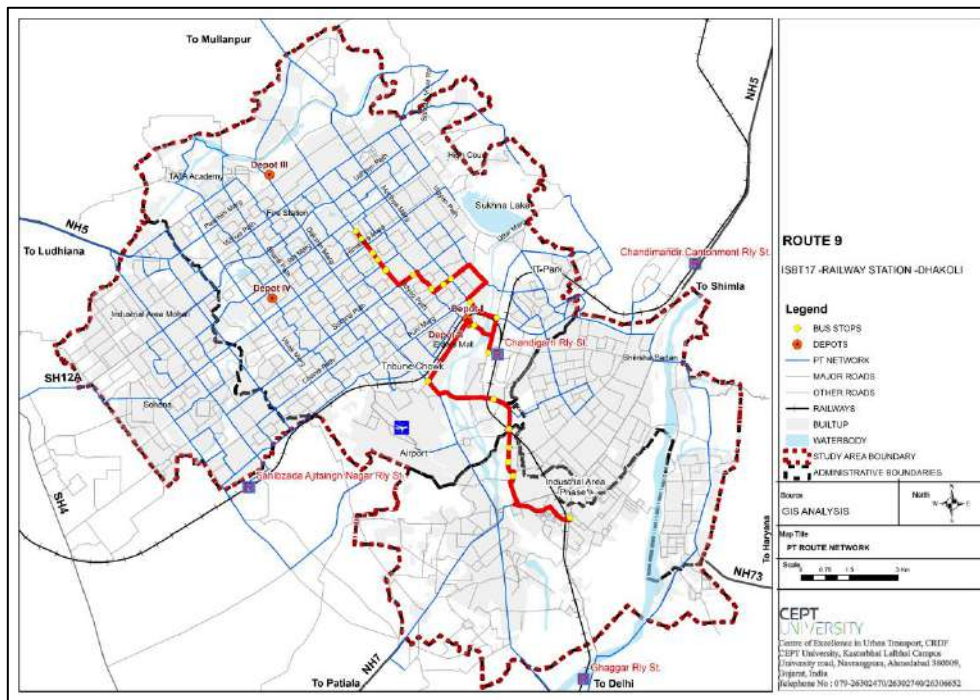


Figure 127: Proposed Route 9 (ISBT17 -Railway Station -Dhakoli)

Route 11: ISBT 17 - Mani Majra - Derabassi via Sector 22/17, Sector 21/18, Sector 20/19, Sector 26/27, Sector 26, Grain market 28/26, Railway Crossing, Kalagram, Housing Board, Housing Board Mani Majra, Mani Majra FR, Command Hospital, Tank Chowk,, Nadha Shaib, Moginand, Madanpur, Pkl, Ramgarh, Kakrali, Mubarikpur,

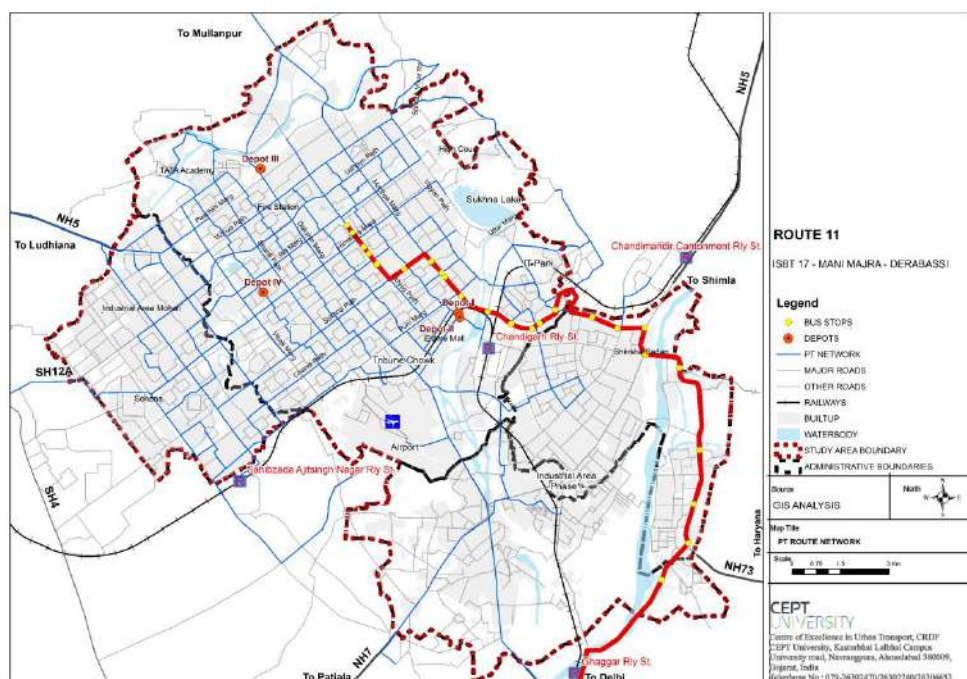


Figure 128: Proposed Route 11 (ISBT 17 - Mani Majra – Derabassi)

Route 12: PGI - ISBT 17 - Derabassi Via Government College, New OPD, PEC College, Sector 2/11, Sector 11/10, Sector 11, Sector 10/16 Hospital, Rose Garden, Matka Chowk, ISBT 17, Sector 22/17, Sector 18/21, Sector 21/20, Sector 33/20, Sector 32 Hospital, Sector 29/31, Tribune Chowk, Hallo Majra, Airport Chowk, Zirakpur bus stand

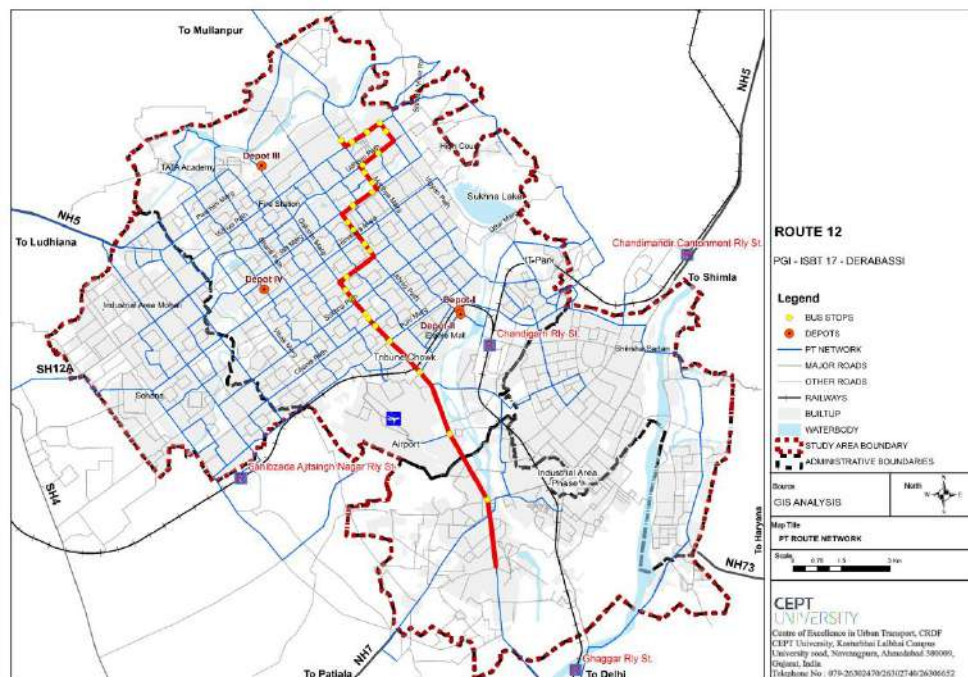


Figure 129: Proposed Route 12 (PGI - ISBT 17 – Derabassi)

Route 12E: PGI - ISBT 17 - Zirakpur - Derabassi

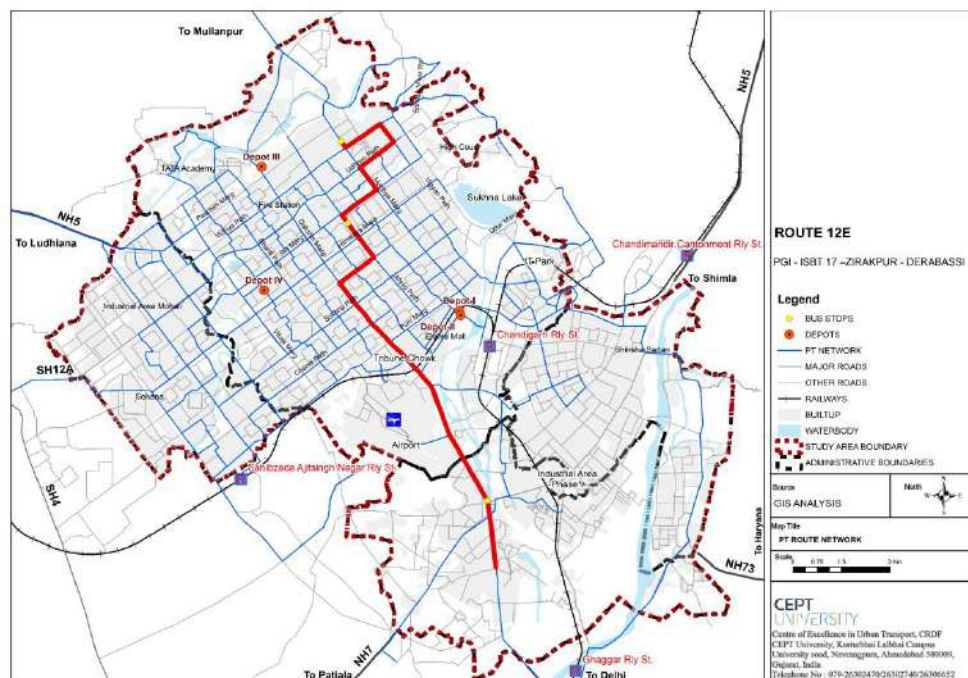


Figure 130: Proposed Route 12E (PGI - ISBT 17 - Zirakpur - Derabassi)

Route 13: Kharar - ISBT 43 - ISBT 17 – Airport via Airport Chowk, Hallo Majra, Elante mall, Industrial Area Phase 1, Sector 29 Market, Sector 21 Market, Aroma 22/21, Sector 22/17, ISBT 17, Sector 23/22, Kisan Bhawan, Sector 35 Market, Sector 44/43, ISBT 43, Sector 53/42, Furniture Market Chowk, Sector 55, Franco Hotel, Barrier, Phase 1, Phase 6, ISBT Mohali, Balungi, Desu Majra, Bhago Majra

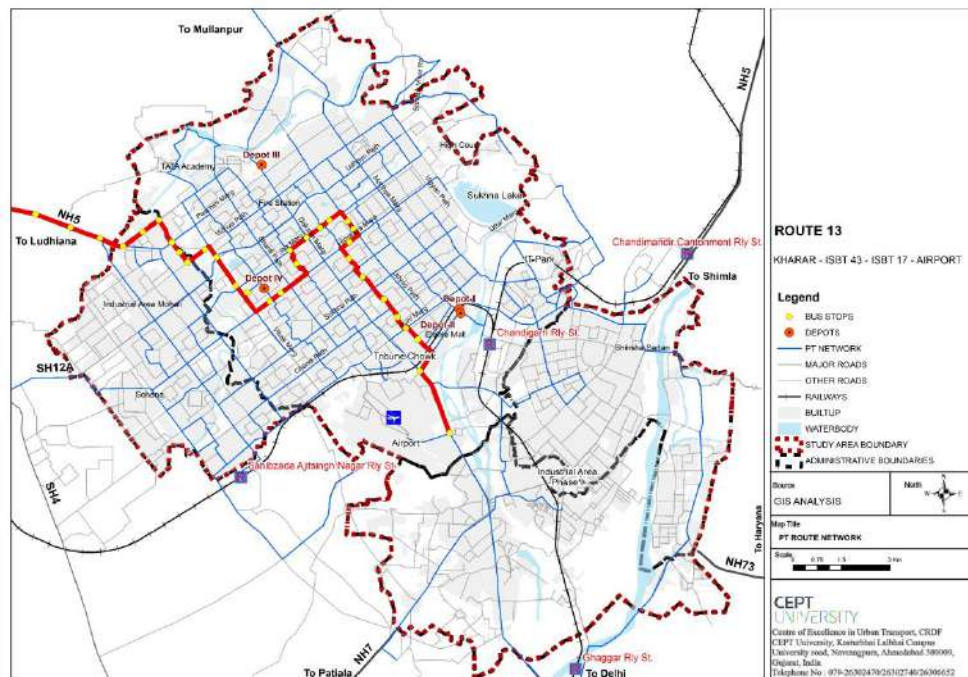


Figure 131: Proposed Route 13 (Kharar - ISBT 43 - ISBT 17 – Airport)

Route 14: ISBT 43 - Landran - Kharar Via Bhago Majra, Imperial Heights, Chappercheri Village, Landran, Lakhnaur, Gurudwara Singh Saheeda Sahib, Sector 68/69 Mohali, Phase 7, Sector 52/51

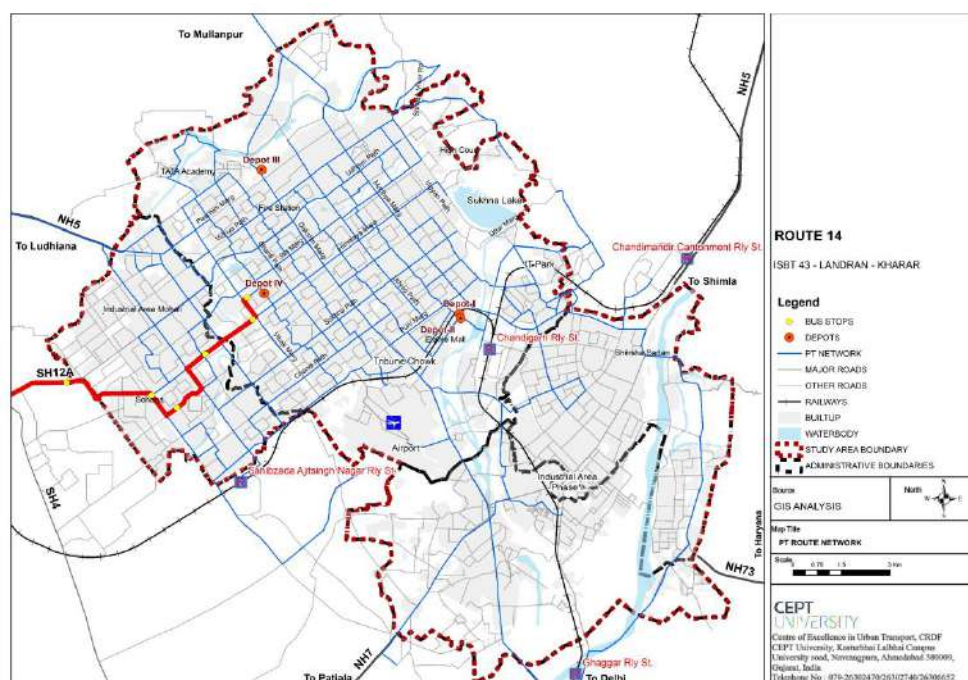


Figure 132: Proposed Route 14 (ISBT 43 - Landran – Kharar)

Route 15: ISBT 17 - PGI – Nayagaon - ISBT 17 Via Sector 23/16, Sector 15/24, Sector 14/15, PGI, Nayagaon, Sector 2, PEC College, Sector 2/11, Sector 10 market, Matka Chowk, Sector 16/17

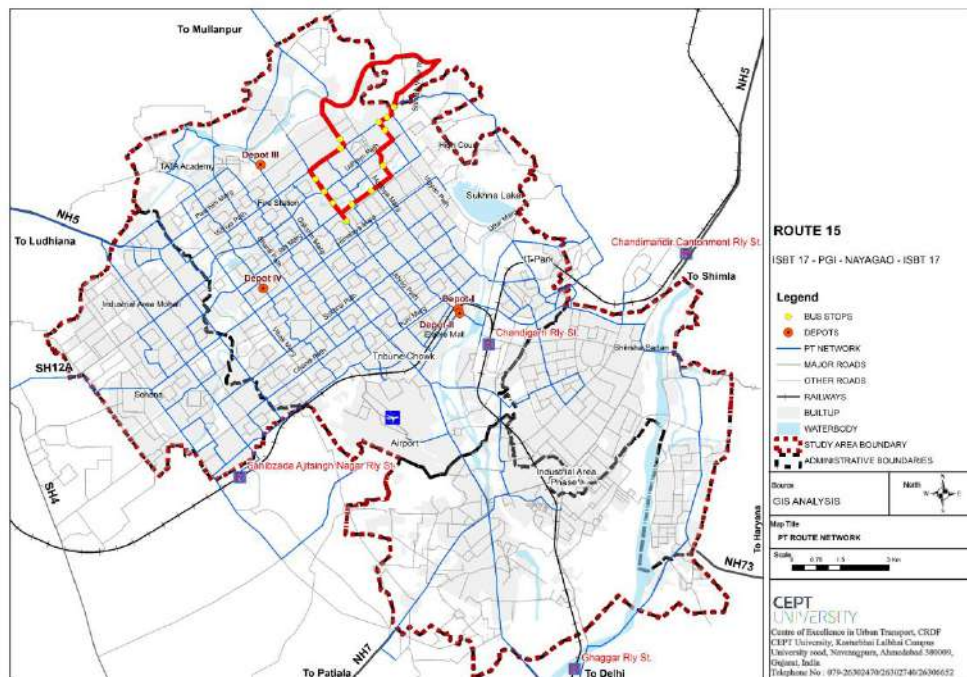


Figure 133: Proposed Route 15 (ISBT 17 - PGI – Nayagaon - ISBT 17)

Route 16: ISBT 43 - Maloya Colony via Sector 42 Market, Sector 41 Market, Sector 40/41, Sector 40 Market, Sector 39C, Sector 39/38 West, Maloya Bus Stand

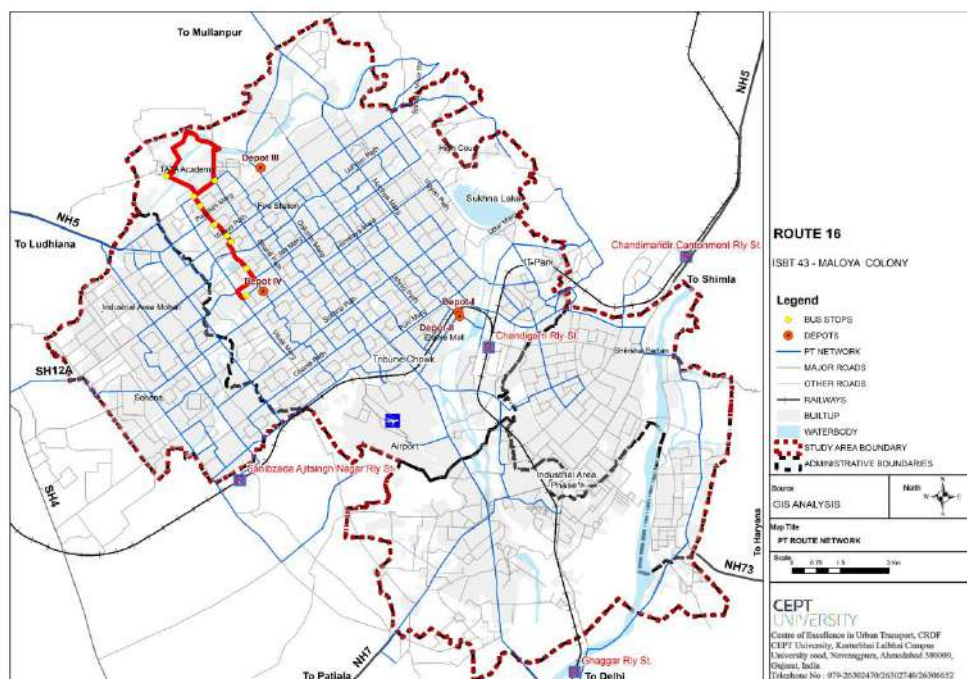


Figure 134: Proposed Route Route 16 (ISBT 43 - Maloya Colony)

Route 17: Sohna Gurudwara - ISBT Mohali - Kharar via Desu Majra, Balungi, ISBT Mohali, Phase 3/5, P.C.L. Chowk, IVY Hospital, Sector 69/70

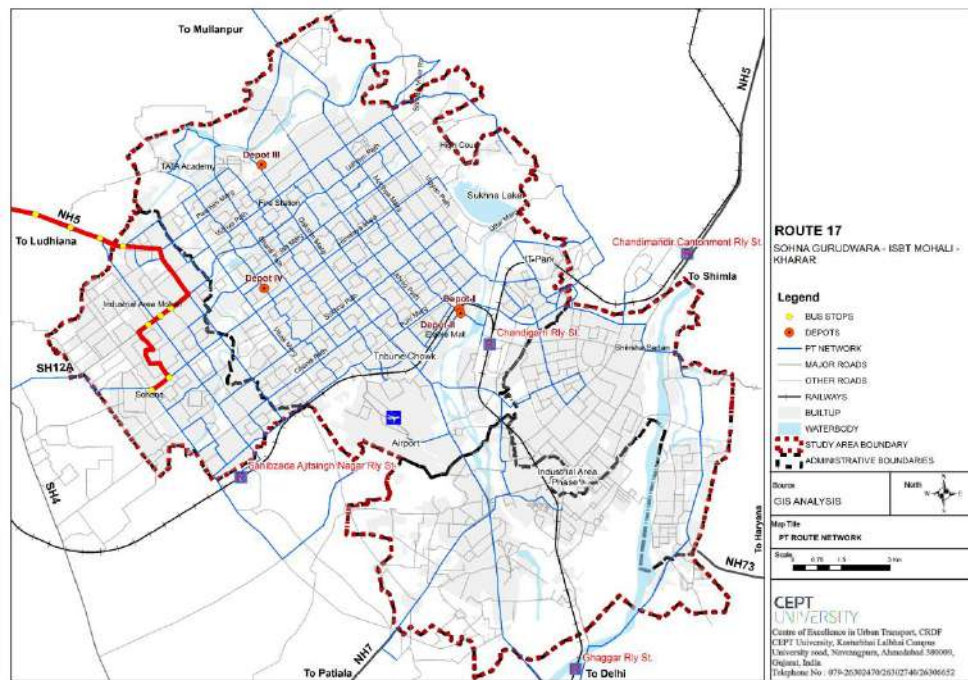


Figure 135: Proposed Route 17 (Sohna Gurudwara - ISBT Mohali - Kharar)

Route 18: PGI - ISBT 17 - Zirakpur Via Sector 11/15, Sector 10/16 Hospital, Sector 16 Market, Sector 16/17, ISBT 17, Sector 22/17, Sector 21/18, Sector 21/20, Sector 33/20, Sector 30/32 hospital, Sector 32 Hospital, Sector 29/31, Tribune Chowk, Hallo Majra, Airport Chowk

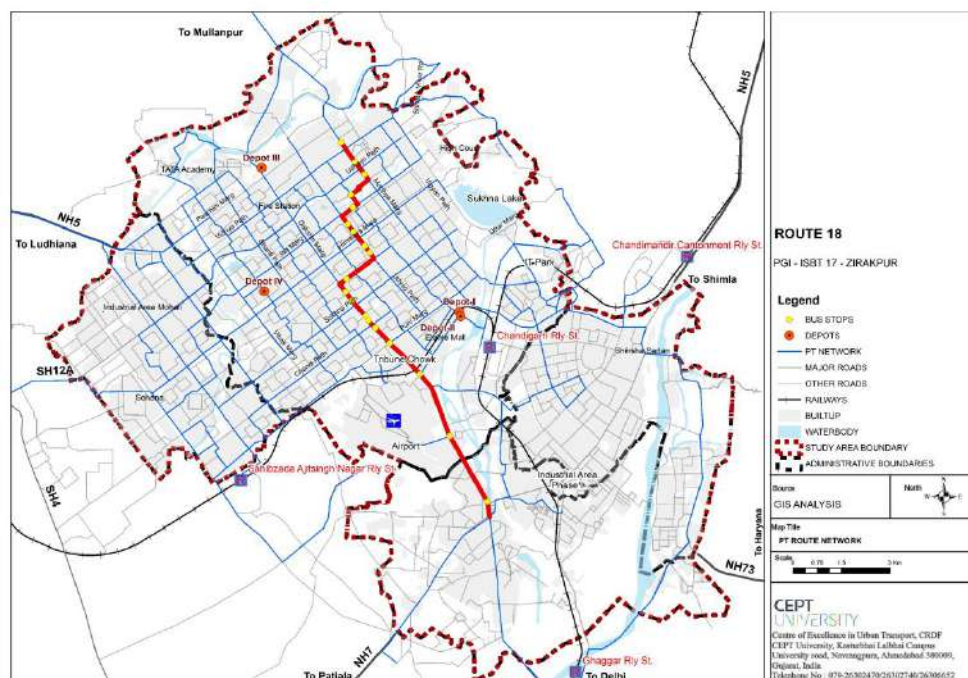


Figure 136: Proposed Route 18 (PGI - ISBT 17 - Zirakpur)

Route 19: ISBT 17 - High Court - Kaimbwala Via Sector 16/17, Matka Chowk, U.T Secretariat, Mini Secretariat, MLA Flats, Nayagoan, Shivalik Vihar, Khuda Alisher, Kansal

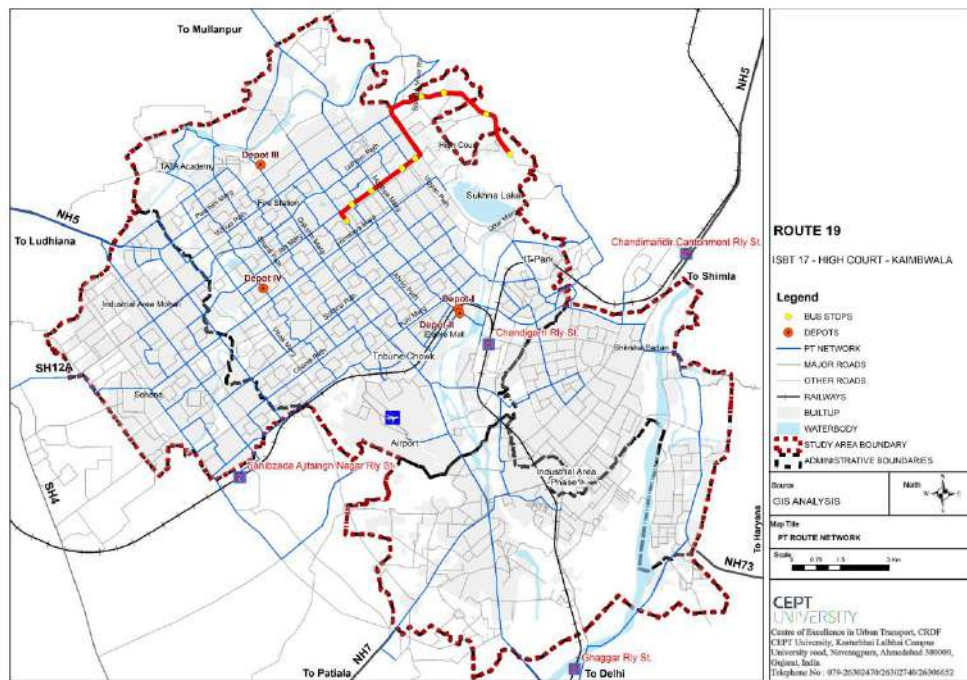


Figure 137: Proposed Route Route 19 (ISBT 17 - High Court - Kaimbwala)

Route 20: ISBT 17 - ISBT43 - New Airport via Sector 22/23, Kisan Bhawan, Sector 35/36, Attawa Chowk, Sector 42/43, Sector 44/51, Colony Number 5, Sector 49/50, Phase 10/11, Phase 11/Sector 65 Market, White House, Sector 82, New Airport Chowk

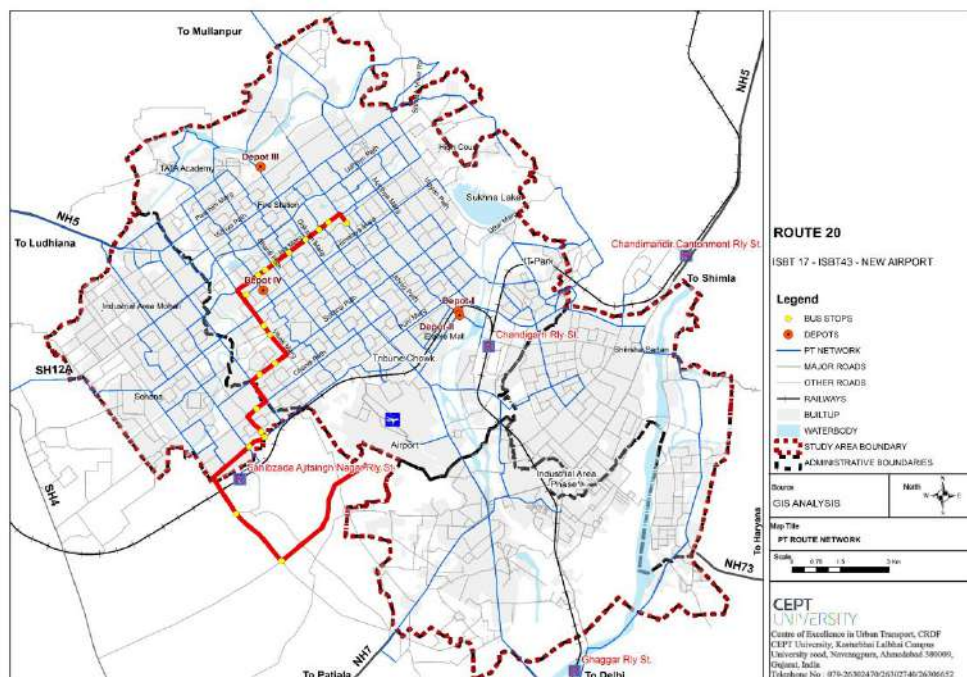


Figure 138: Proposed Route Route 20 (ISBT 17 - ISBT43 - New Airport)

Route 21: ISBT43-Landran via Sector 52/53, Phase 7/3, Sector 52, Phase 7, Phase 8 Mohali, Fortis Hospital, Sector 68/69 Mohali, Gurudwara Singh Saheeda Sahib, Lakhnaur

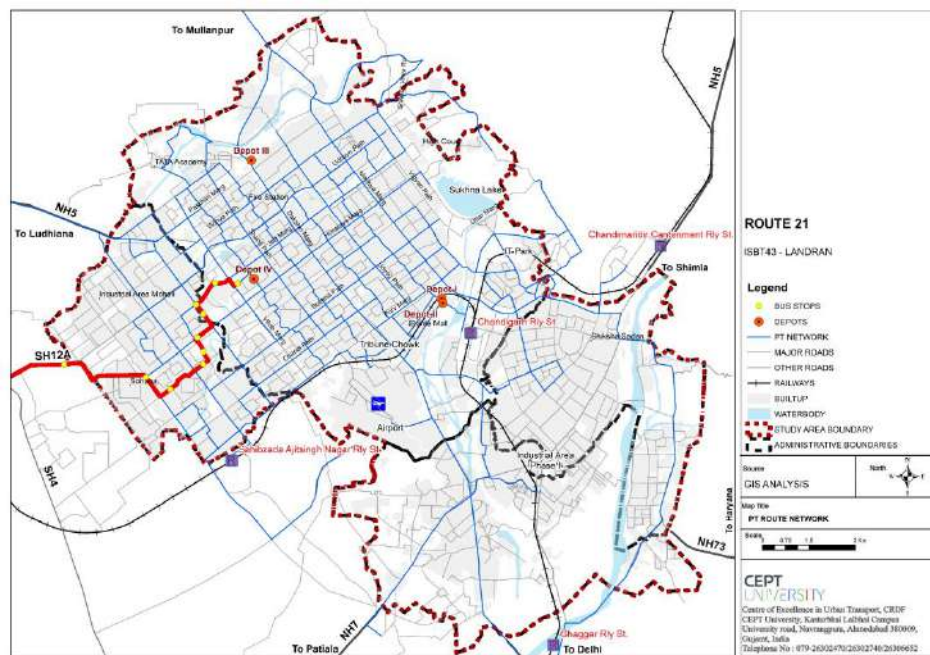


Figure 139: Proposed Route 21(ISBT43-Landran)

City Demand Routes

Route 22: Sohna Gurudwara - White House - Mata Mansa Devi Via Gurudwara Singh Saheeda Sahib, Sector 68/79, Sector 76, Sector 67 market, Sector 66/67, Sector 66, Phase 1 Ind area Mohali, White House, Phase 11, Sector 48/49, Sector 48, Sector 47, Industrial Area Phase 2, Hallo Majra, CTU Workshop, Railway Station, Railway Crossing, Kalagram, Housing Board, Housing Board Mani Majra, Bhainsa Tibba,

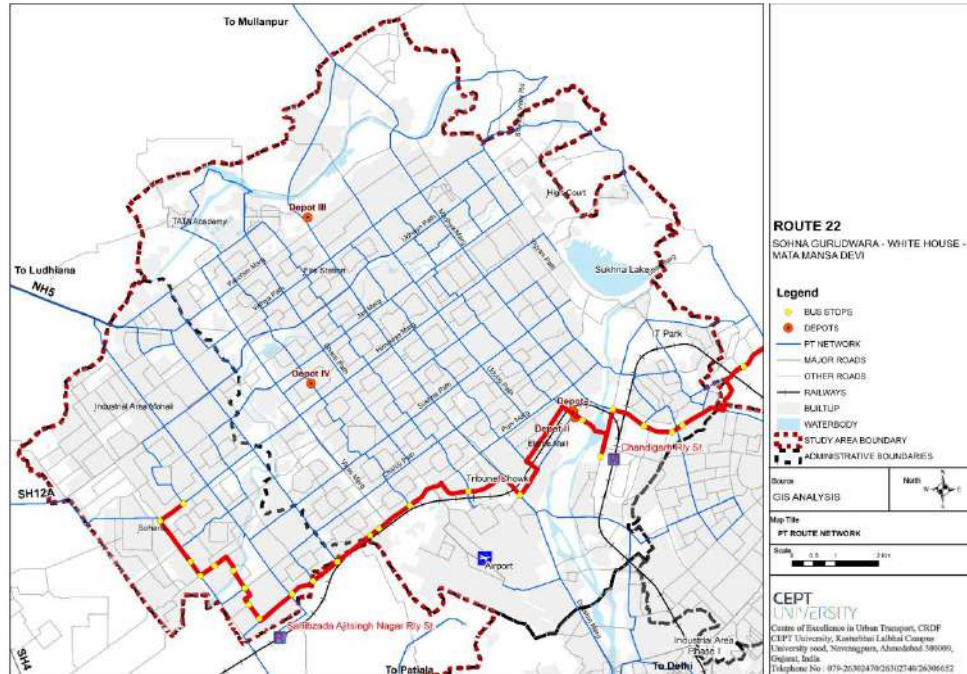


Figure 140: Proposed Route 22 (Sohna Gurudwara - White House - Mata Mansa Devi)

Route 23: ISBT17 – High Court – Kaimbwala Via Sector16/17, Matka Chowk, Mini Secretariat, MLA Flats, Nayagaon, Shivalik Vihar, Khuda Alisher, Kansal,

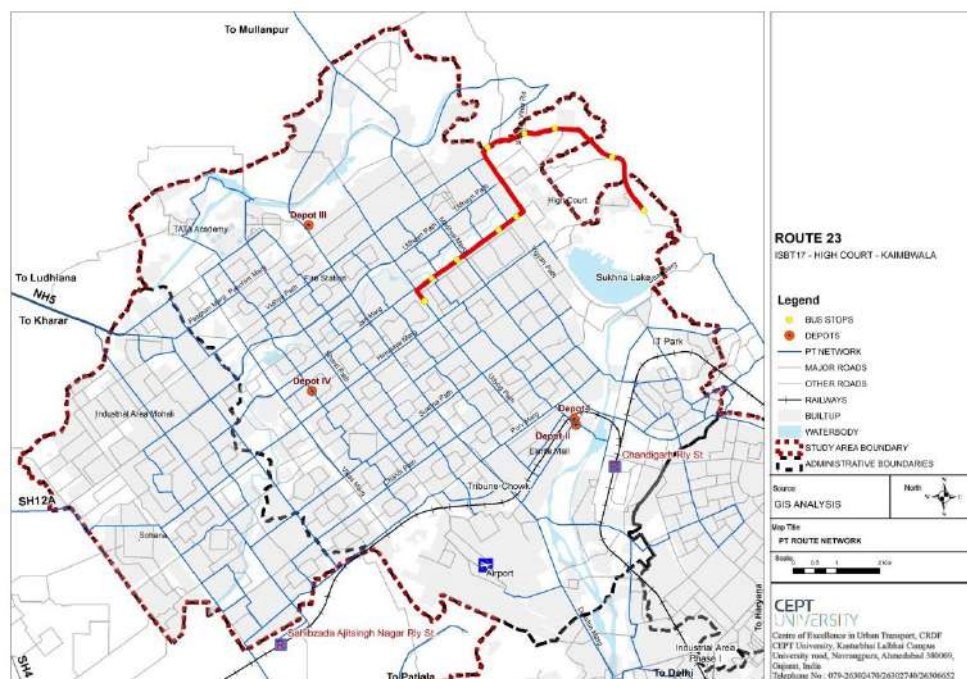


Figure 141 Proposed Route 23 (ISBT17 - High Court - Kaimbwala)

Route 24: Ramdarbar - ISBT43 -PGI - ISBT43 via Industrial Area Phase 2, Sector 47, Sector 47 Market, Sector 46, Sector 46/45, Sector 44, Sector 40 Market, Sector 40/39, Sector 39, Maloya, Sector 39, Sector 38/40, Sector 37/38, Sector 24/25, Sector 14/15, Sector 10/16 Hospital, Rose Garden, Sector 18/8, Sector 26 Guru Gobind School, Sector 26, Sector 28 Market, Industrial Area Phase 1, Tribune Chowk, Sector 29/31, Sector 32/31

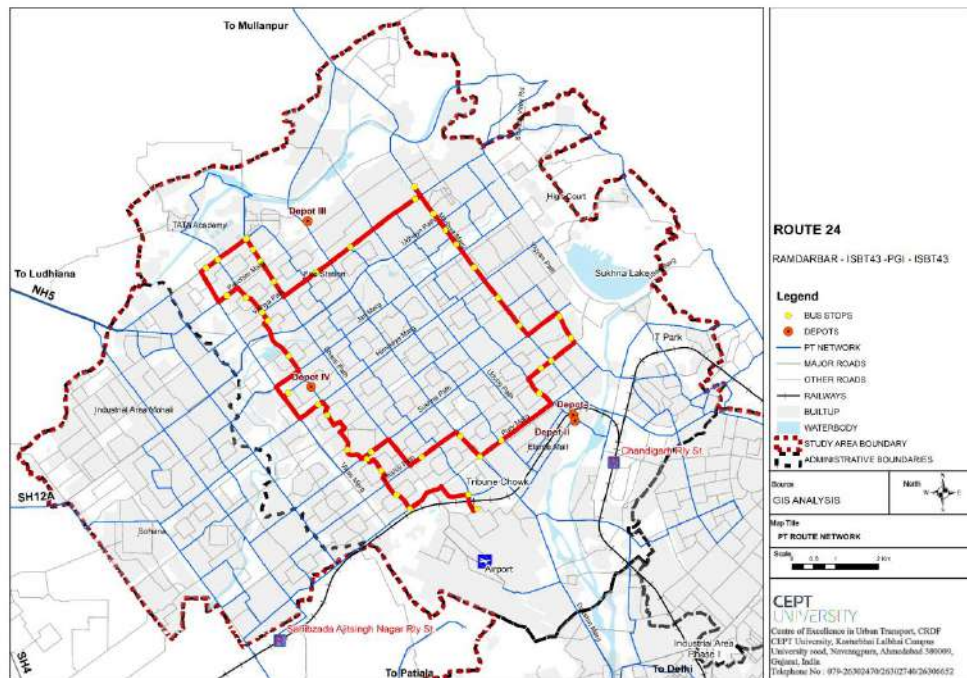


Figure 142: Proposed Route 24 (Ramdarbar - ISBT43 -PGI - ISBT43)

Route 25: ISBT 43 - CTU Workshop - Mata Mansa Devi via Bhainsa Tibba, Mani Majra FR, Housing Board, Kalagram, Railway Station, Railway Crossing, CTU Workshop, Sector 28 Petrol Pump, Grain market 28/26, Sector 27 Market, Sector 19/27 Market, Sector 33/32, Attawa Chowk, Sector 42/43

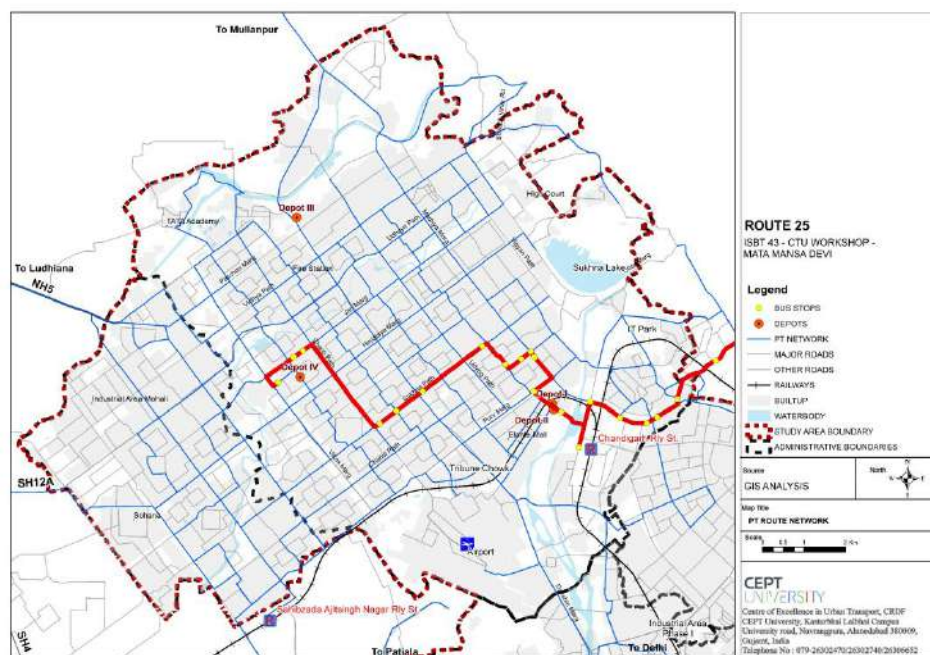


Figure 143: Proposed Route 25 (ISBT 43 - CTU Workshop - Mata Mansa Devi)

Route 26: Mata Mansa Devi - ISBT17 - Dhanas via Kalagram, Housing Board, Mani Majra FR, Bhainsa Tibba, It park, Kishangarh, Mani Majra town, Grain market 28/26, Sector 26/27, Sector 19/7, Sector 18 press, Sector 18/17, Sector 22/17, Sector 23/16, Sector 24/25, Sector 38/25, CTU Depot Sector 25, Daddu Majra Colony

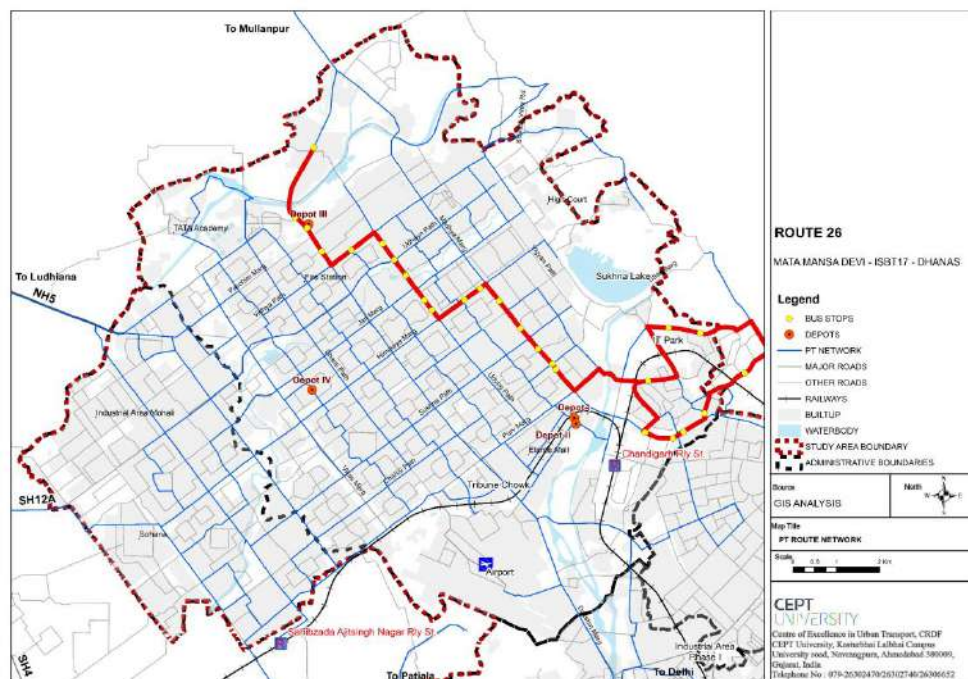


Figure 144: Proposed Route 26 (Mata Mansa Devi - ISBT17 - Dhanas)

Route 27: ISBT17 - SAS Railway Station Via Sector 22/17, Aroma 22/21, Sector 21 Market, Sector 20 Market, Sector 30 Market, Sector 32/31, Phase 10/11, Phase 11/Sector 65 Market, Phase 11, White House

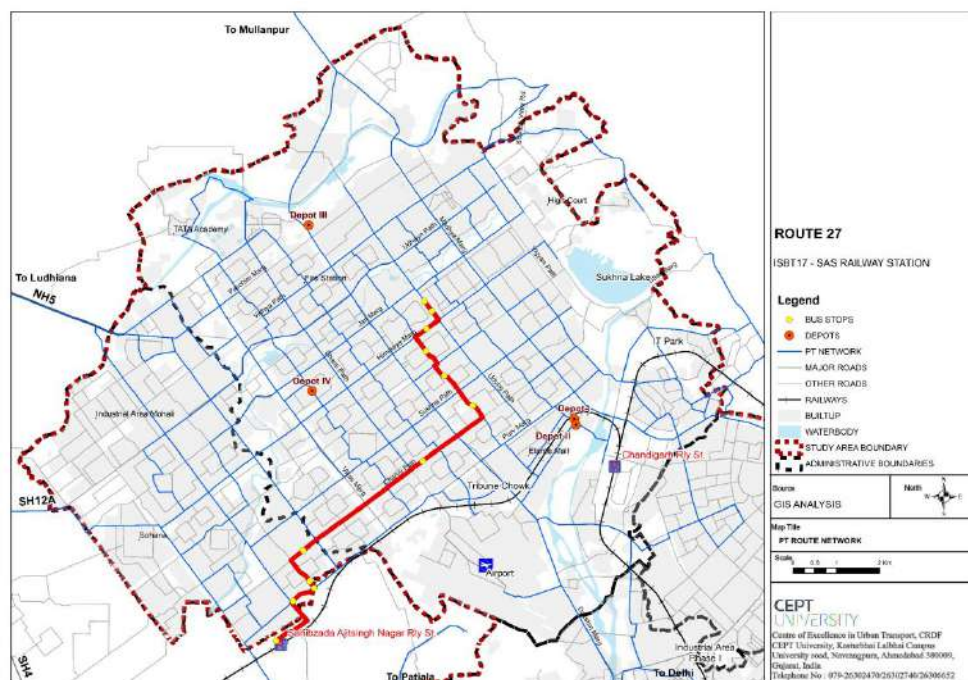


Figure 145: Proposed Route 27 (ISBT17 - SAS Railway Station)

Route 28: ISBT43- Mohali Municipal Corporation - Phase11 via Sector52/53, Phase 7/3,Chawla Chowk, Phase 7 Mohali, Phase 8 Mohali, Fortis Hospital, Sector 68/79,Sector 80/67,MC office Mohali, Sector 67 market, Sector 66,Phase 1 Ind area Mohali, White House

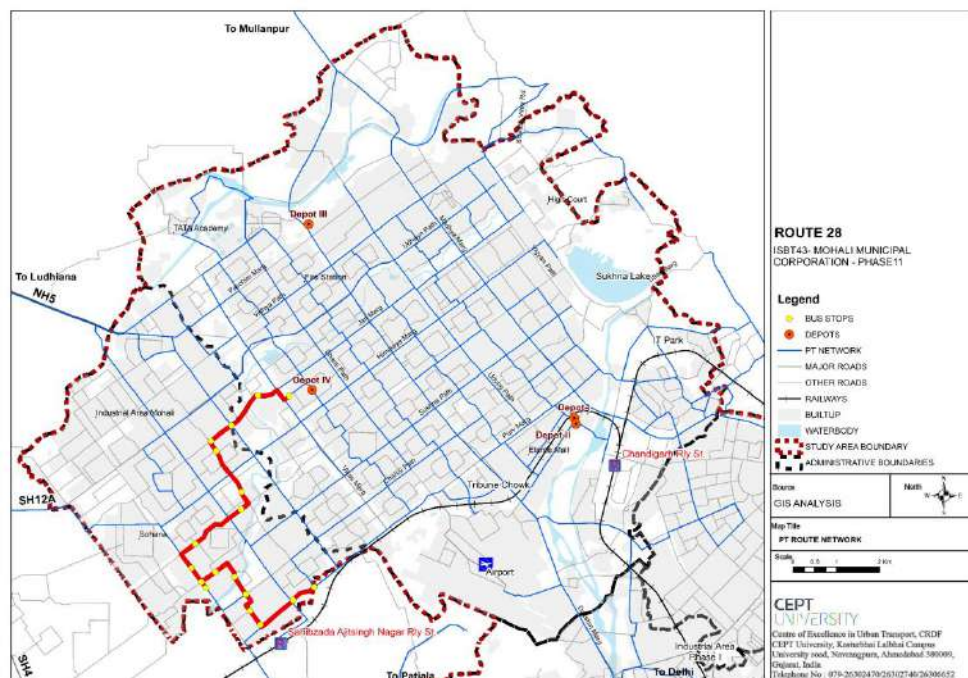


Figure 146: Proposed Route 28 (ISBT43- Mohali Municipal Corporation - Phase11)

Route 29E: ISBT43 – ISBT 17 - PGI

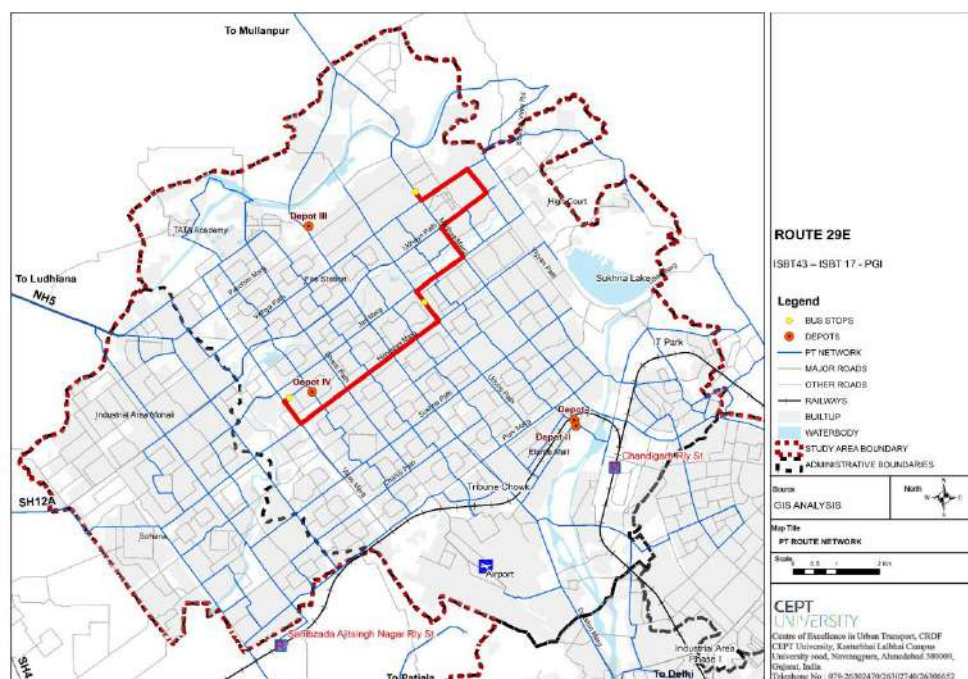


Figure 147: Proposed Route 29E (ISBT43- ISBT 17 - PGI)

Terminal Routes

Route 30: ISBT Mohali-ISBT 43 via Phase 1, Barrier, Franco Hotel, Sector 55, Furniture Market Chowk, Sector 53/42

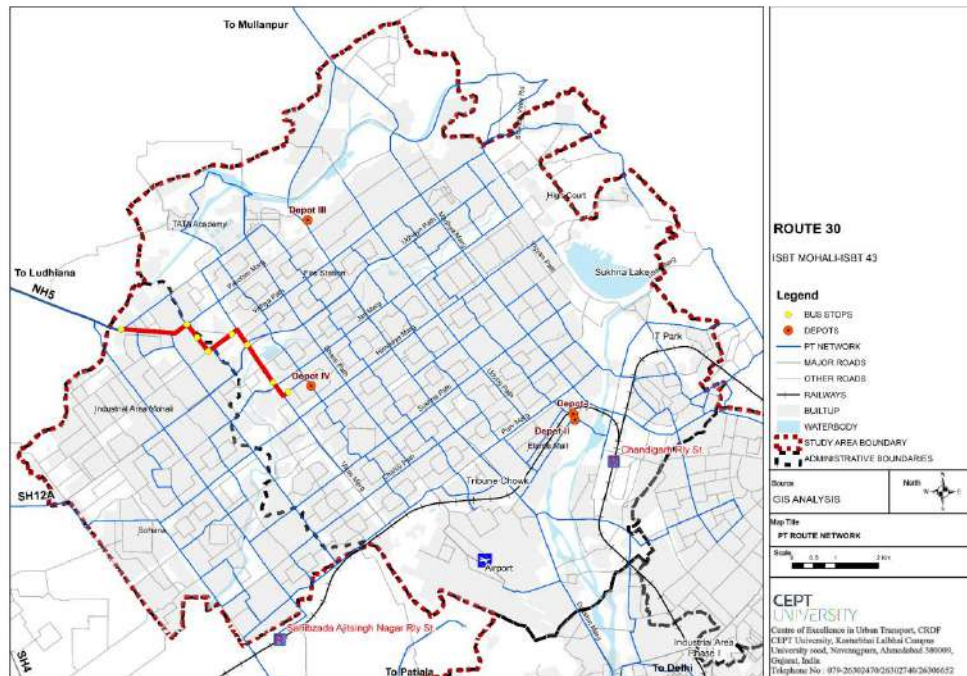


Figure 148: Proposed Route 30 (ISBT Mohali-ISBT 43)

Route 31: PGI-ISBT Mohali–Kharar Via Government College, New OPD, PEC College, Sec 2/11, Sec 11/10, Sect 11, Sec 10/16 Hospital, Sec 16 Market, Sec 16/17, Sec 23/16, Sec 15/24, Sec 36/37, Sec 43, Sec 42/41, Furniture Market Chowk, Sec 54/53, Phase 2, Franco Hotel, Barrier, Phase 1, ISBT Mohali, Balungi, Desu Majra

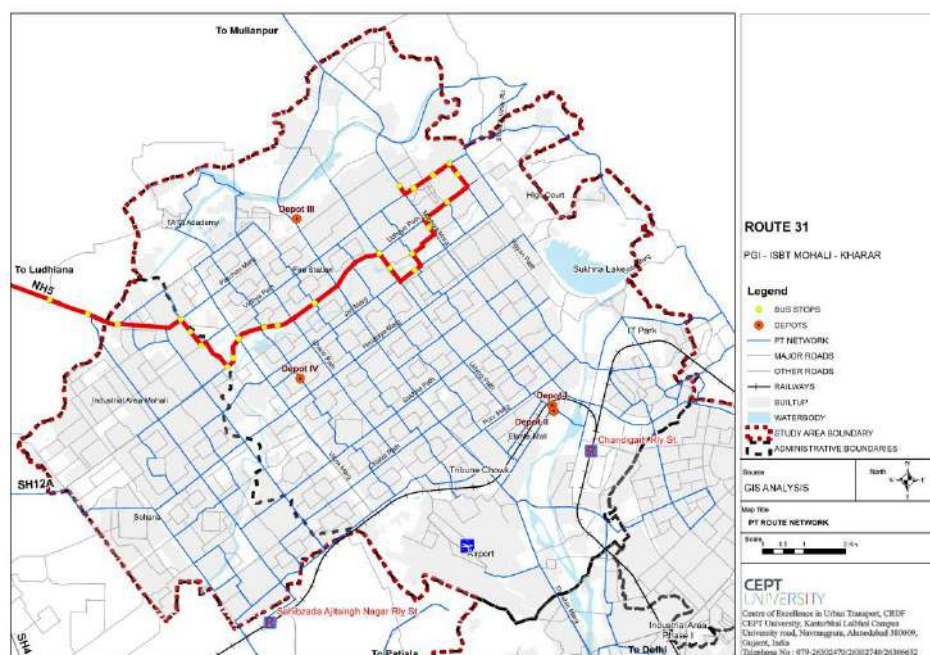


Figure 149: Proposed Route 31 (PGI - ISBT Mohali - Kharar)

Route 31E: PGI-ISBT Mohali–Kharar

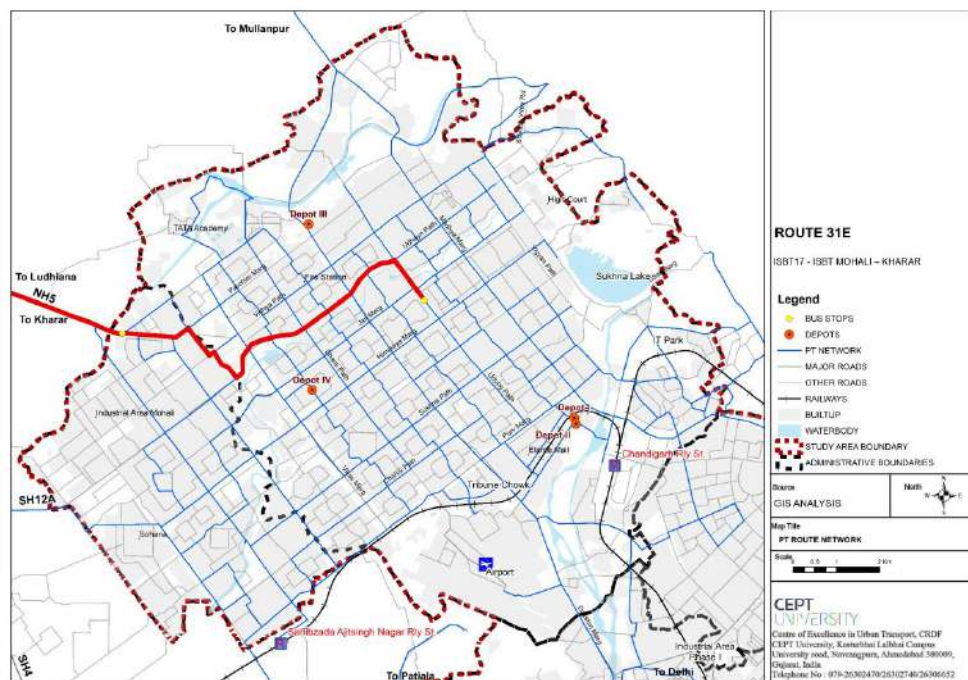


Figure 150: Proposed Route 31E (PGI - ISBT Mohali – Kharar)

Route 33: PGI - ISBT17 - ISBT 43 - Fortis Hospital via Government College, New OPD, PEC College, Sector 2/11, Sector 11/10, Sector 11, Sector 10/16 Hospital, Sector 16 Market, Sector 16/17, ISBT 17, Sector 22/17, Aroma 22/21, Sector 34/35 Piccadilly Chowk, Sector 43/44, ISBT 43, Sector 52/53, Phase 7/3, Chawla Chowk, Phase 7 Mohali, Phase 8 Mohali, Fortis Hospital

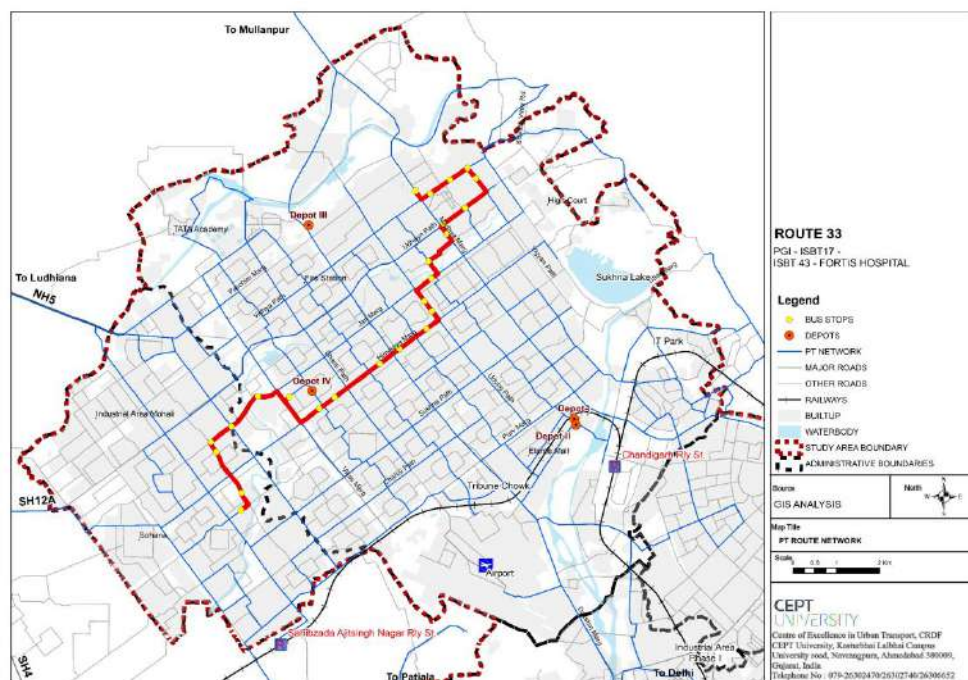


Figure 151: Proposed Route 33 (PGI - ISBT17 - ISBT 43 - Fortis Hospital)

Route 34E: PGI – ISBT 17- Grain Market-Chandigarh Rly St- ISBT Panchkula

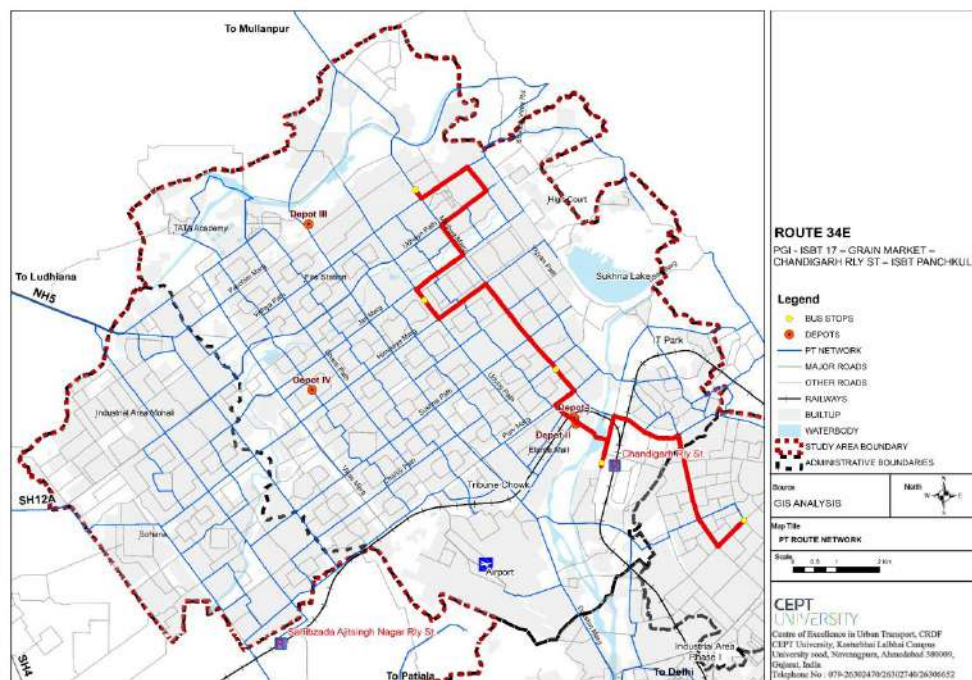


Figure 152: Proposed Route 34E (PGI - ISBT 17- Grain Mkt-Chandigarh Rly St- ISBT Panchkula)

Route 35: ISBT 43 -Tribune Chowk - Panchkula Via Sector 43/44, Sector 32/33, Sector 30/32 hospital, Sector 32 Hospital, Sector 29/31, Tribune Chowk, Halo Majra, Vikas Nagar, Sector 5 Bus Stand Panchkula

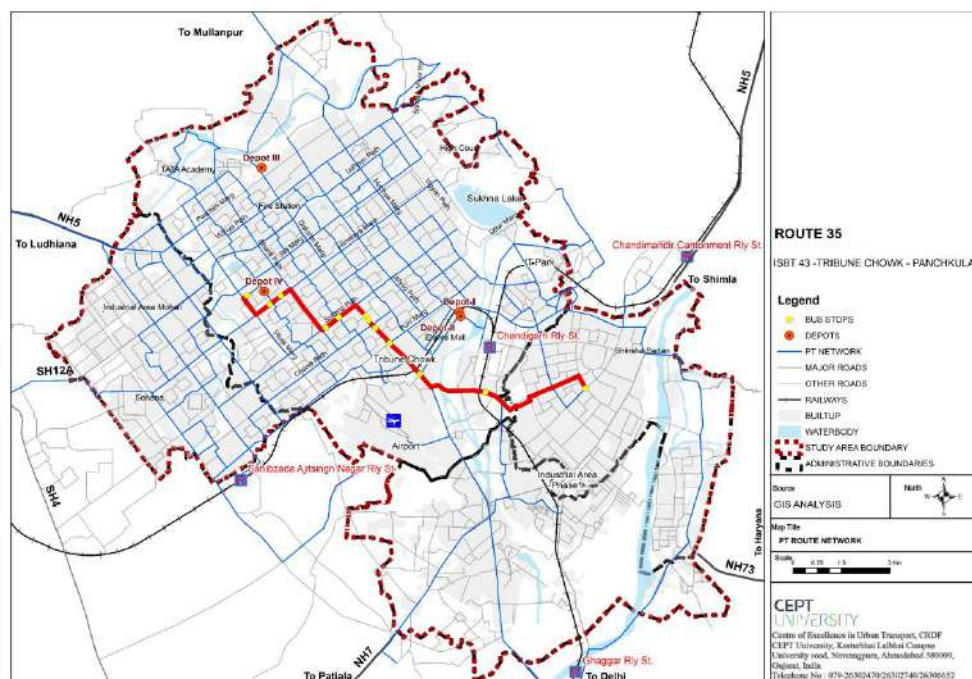


Figure 153: Proposed Route 35 (ISBT 43 -Tribune Chowk - Panchkula)

Route 36: ISBT 17 - Railway Station - Mata Mansa Devi Via It Park, Kishangarh, Shivalik Park, Indira Colony, Indira Colony, Mani Majra FR, Housing Board, Kalagram, Railway Crossing, CTU Workshop, Sector 28 Market, Sector 27 Market, Sector 19 Market, Sector 18 Market, Sector 22/17

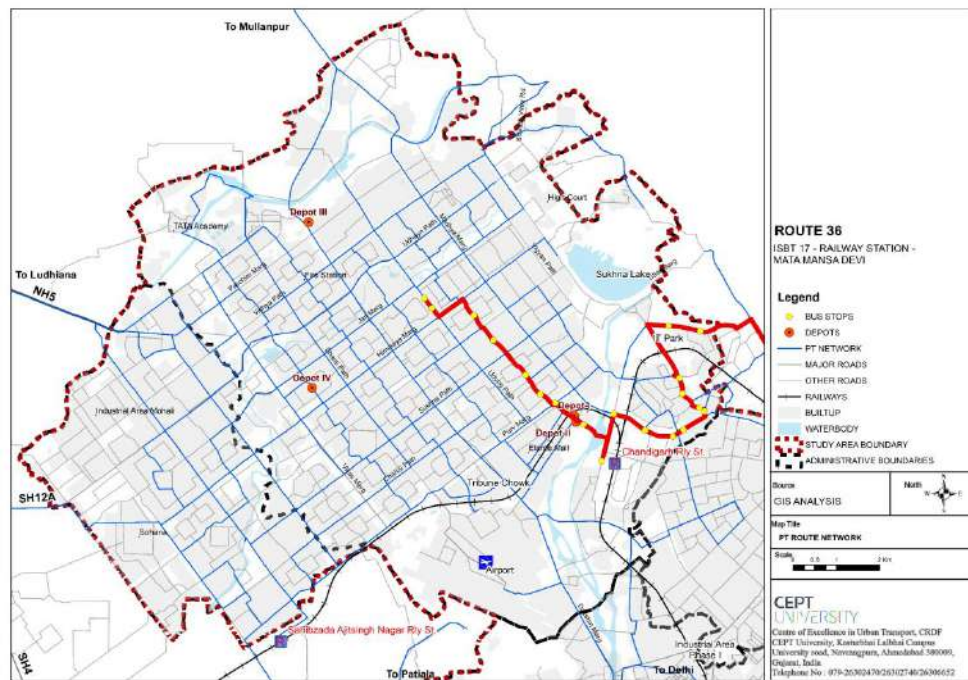


Figure 154: Proposed Route 36 (ISBT 17 - Railway Station - Mata Mansa Devi)

Loops Routes only for Scenario 2A

Route 37: Sohna Gurudwara - Sector 65 - Sohna Gurudwara via, Sector 69/70, Sector 61/70, Phase 8 Mohali, Fortis Hospital, Sector 68/67, phase 10 Mohali, Phase 1 Ind area Mohali, Mauli, sector 80/67, Sector 76, Sector 68/79

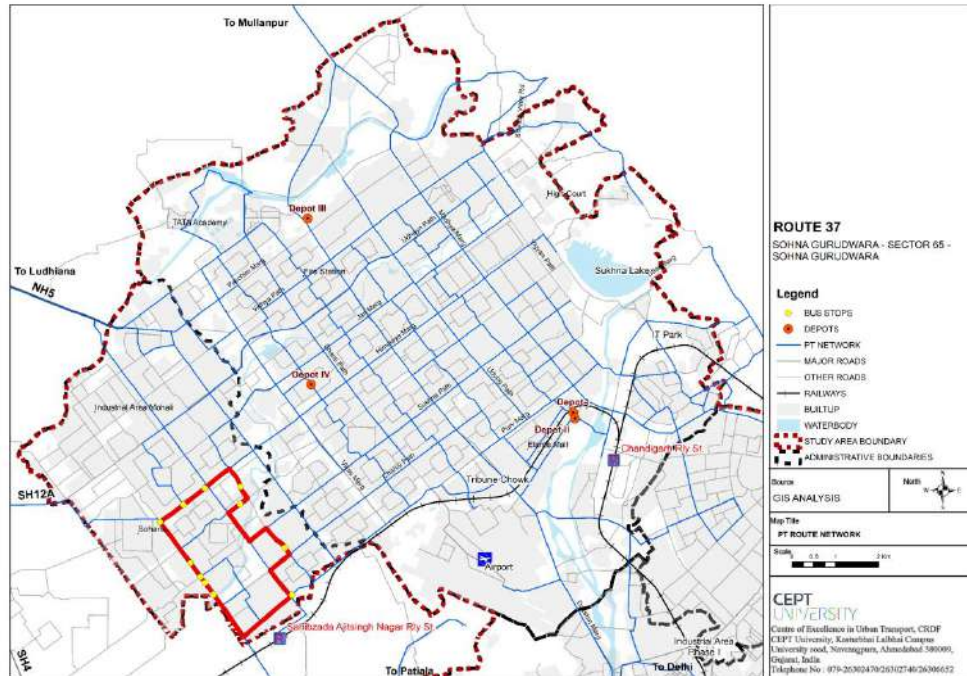


Figure 155: Proposed Route 37 (Sohna Gurudwara - Sector 65 - Sohna Gurudwara)

Route 38: ISBT 17 - Pgi - ISBT 17 Via ISBT 17, Sector 23/16, Sector 15/24, Sector 24/15, Sector 14/15, Pgi, Government College, New Opd, Pec College, Sector 2/11, Sector 2/3, Rock Garden, Sector 9/8, Sector 18 Press, Sector 18/17, Sector 22/17

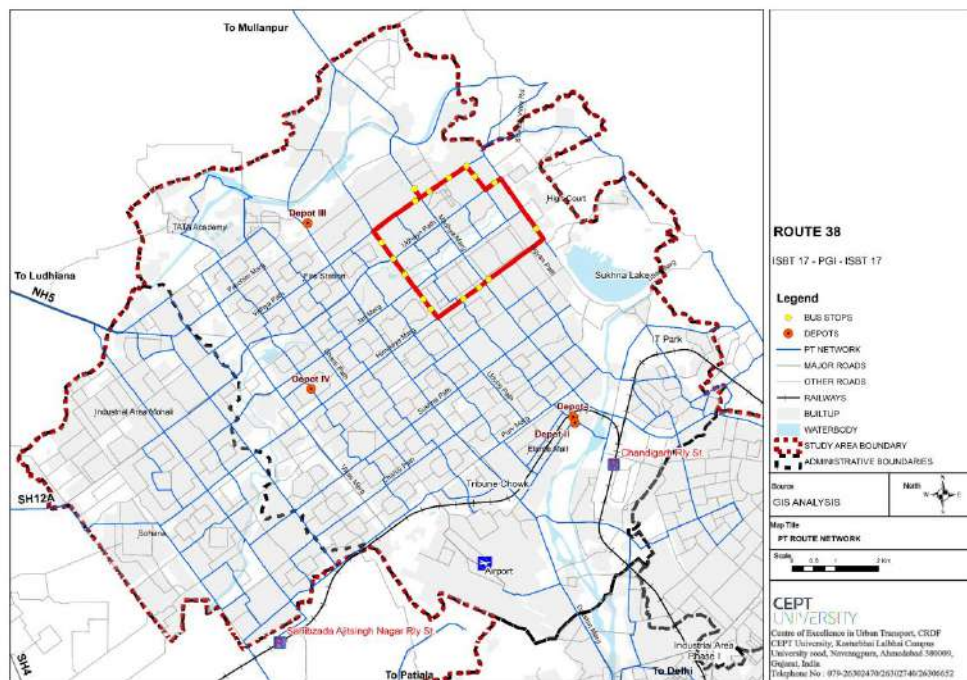


Figure 156: Proposed Route 38 (ISBT 17 - PGI - ISBT 17)

Route 39: ISBT17 - High Court - ISBT 17 Via ISBT 17, Sector 16/17, Matka Chowk, Ut Secretariat, Mini Secretariat, Mla Flats, High Court, Rock Garden, Sukhna Lake, Sector 9/8, Sector 18 Press, Sector 18/17, Sector 18 Market, Sector 21/18, Sector 18/21

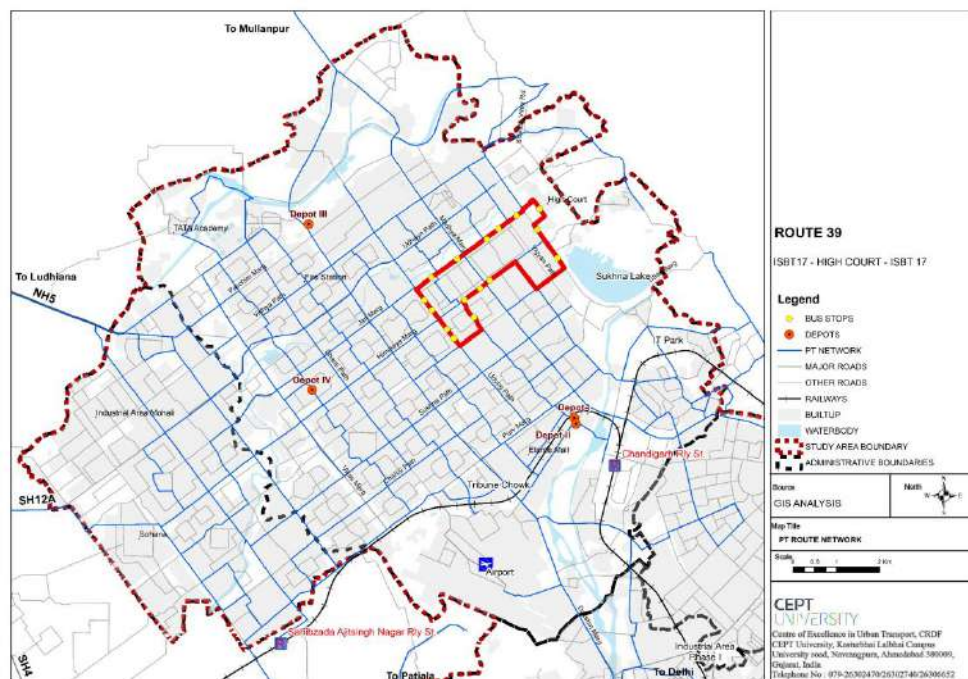


Figure 157: Proposed Route 39 (ISBT17 - High Court - ISBT 17)

Route 40: Labour Chowk - Sukhna Lake - Labour Chowk Via SECTOR 20/33, SECTOR 21/20, sector 7 market, sector 26/27, sector 26, sector 27/28, sector 29/30, sector 30, sector 20/30 gurudwara chowk, sector 33/20

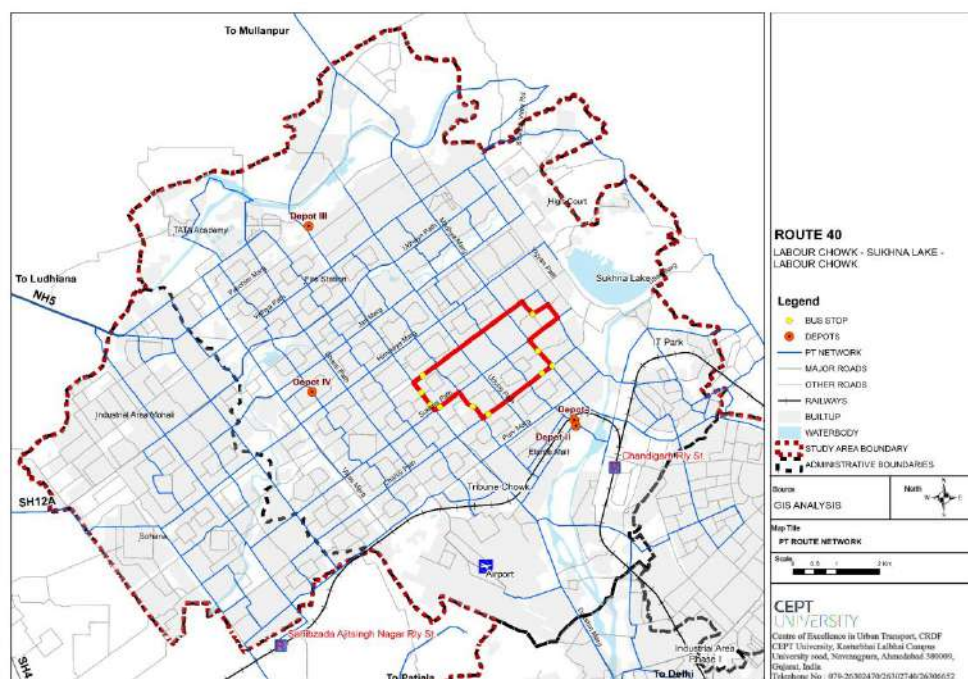


Figure 158: Proposed Route 40 (Labour Chowk - Sukhna Lake - Labour Chowk)

Route 41: Sector 21 - Purv Marg - Sector 21 via sector 21 market, aroma 22/21, sector 18/17, sector 18 market, sector 19 market, sector 27 market, sector 28 market, sector 28 petrol pump, industrial area phase 1, sector 29 market, sector 30 market, sector 20 market

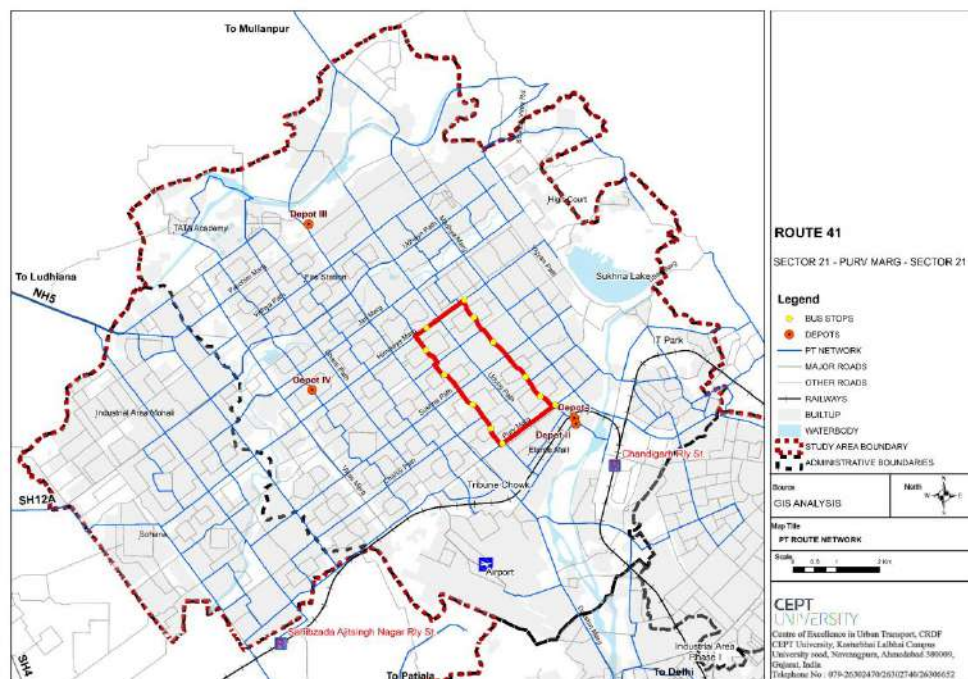


Figure 159: Proposed Route 41 (Sector 21 - Purv Marg - Sector 21)

Route 42: Sector 15 Market - Sector 21 Market - Sector 15 Market via sector 15, sector 16, sector 16/17, sector 17 market, sector 18/17, sector 18, sector 21, sector 22 market, sector 23/22, sector 23 market, sector 24 market, sector 24/25, sector 38/25

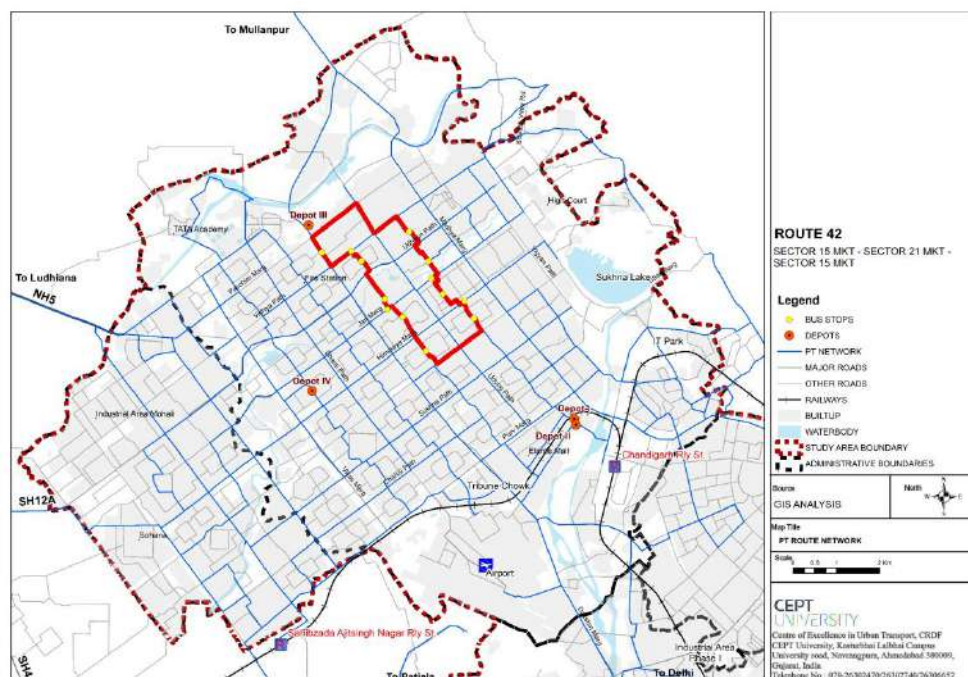


Figure 160: Proposed Route 42 (Sector 15 Market - Sector 21 Market - Sector 15 Market)

Route 43: ISBT 43 - Pgi - Sec 22 Market - ISBT 43 Via ISBT 43, Sec 52/51, Sec 52, Phase 7/3, Madanpura Chowk, Phase 2, Sec 55, Sec 40/41, Sec 37 Market, Sec 36/37, Sec 15/24, Sec 15 Markket, Sec 14/15, Pgi, Government Sollege, New Opd, Pec College, Sec 2/11, Mini Secretariat, Ut Secretariat, Matka Chowk, Sec 16/17, Sec 22/23, Sec 22 Market, Sec 22/21, Sec 34/35 Piccadily Chowk, Attawa Chowk, Sec 42/43

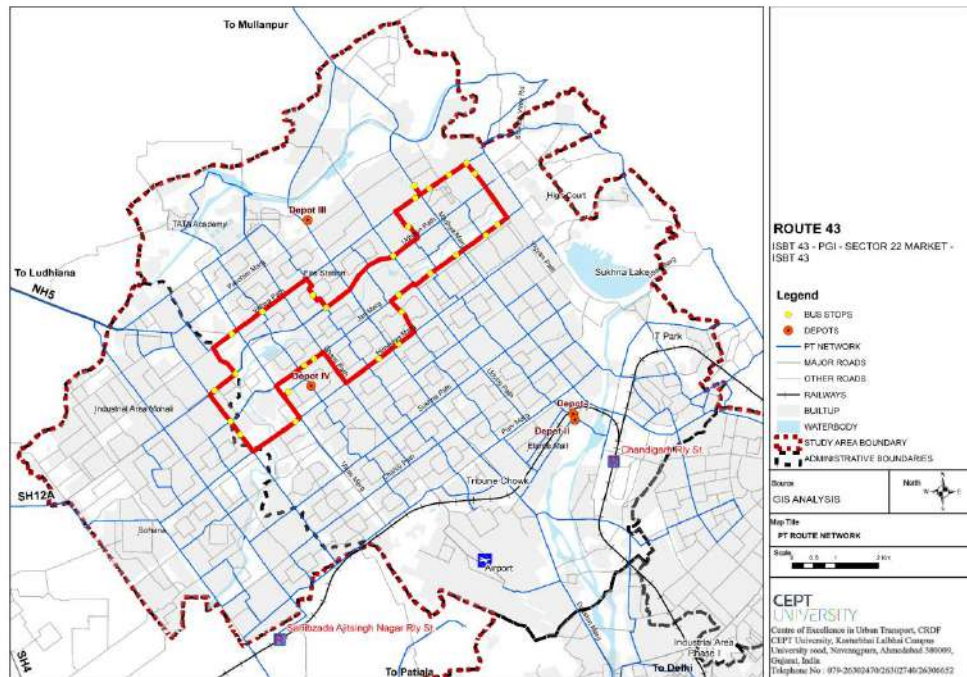


Figure 161: Proposed Route 43 (ISBT 43 - PGI - Sector 22 Market - ISBT 43)

Route 44: ISBT 43 - Industrial Phase 2 - Sector 21 - ISBT 43 Via ISBT 43, Sector 44/51, Colony Number 5, Sector 45/46, Sector 46, Sector 46/45, Sector 31/47, Industrial Area 2, Tribune Chowk, Sector 29/31, Sector 32 Hospital, Sector 30/32, Sector 33/20, Sector 20/33, Sector 21, Sector 34/21, Sector 35/22, Sector 22/35, Sector 35/36, Sector 36/36, Attawa Chowk, Sector 42/43

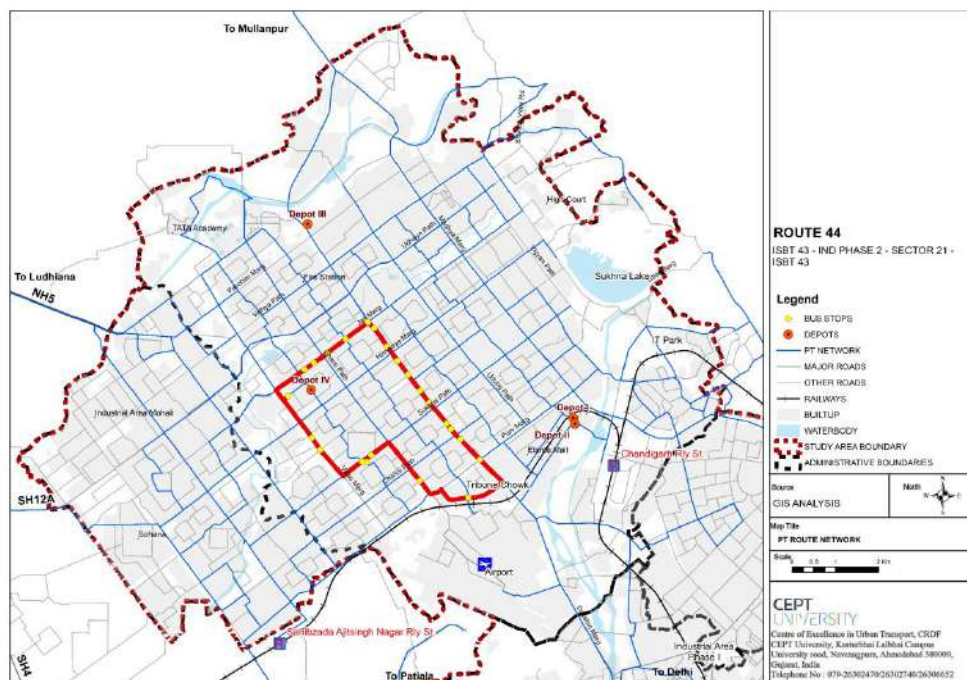


Figure 162: Proposed Route 44 (ISBT 43 - Industrial Phase 2 - Sector 21 - ISBT 43)

Route 45: ISBT Mohali - Sector 36/37 - ISBT Mohali Via Dera Studio Chowk, Franco Hotel, Sector 55, Sector 40/41, Sector 41 Market, Sector 42/41, Sector 43, Sector 36/42, Sector 36/37, Sector 37 Market, Sector 38 Main Market, Sector 38 West, Sector 39/38, Sector 39a, Sector 39/38, Maloya, Sector 39, Sector 56 West, Phase 6

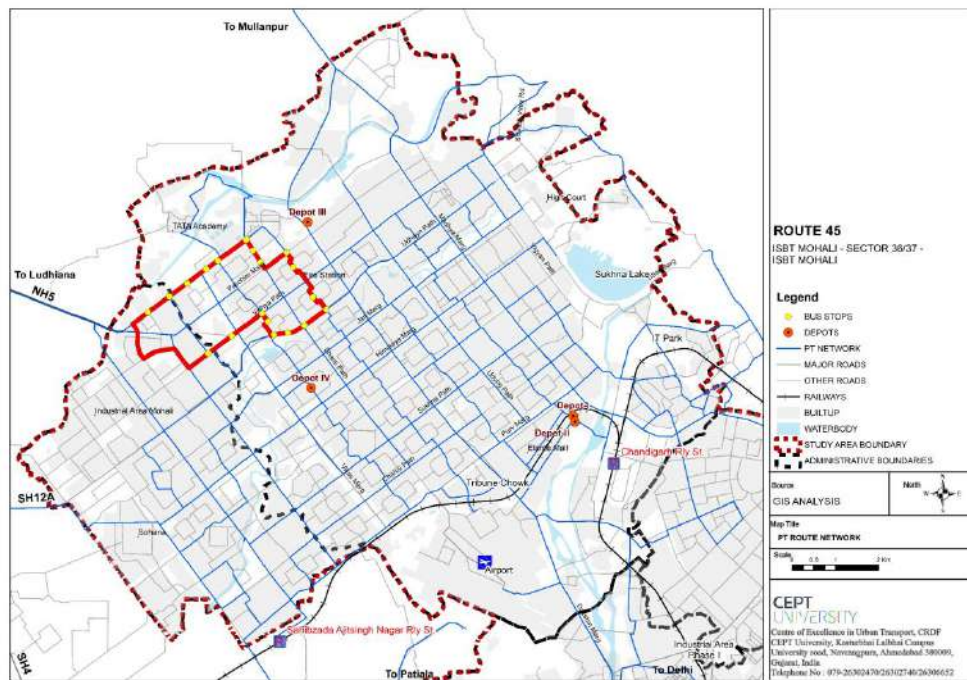


Figure 163: Proposed Route 45 (ISBT Mohali - Sector 36/37 - ISBT Mohali)

Route 46: ISBT 43 - Sector 46 - Sector 31 - ISBT 43 Via ISBT 43, Sector 52/51, Ut Barrier, Sector 48/49, Sector 48, Sector 48b, Sector 32/31, Sector 32 Market, Sector 33/32, Sector 32/33, Attawa Chowk, Sector 42/43

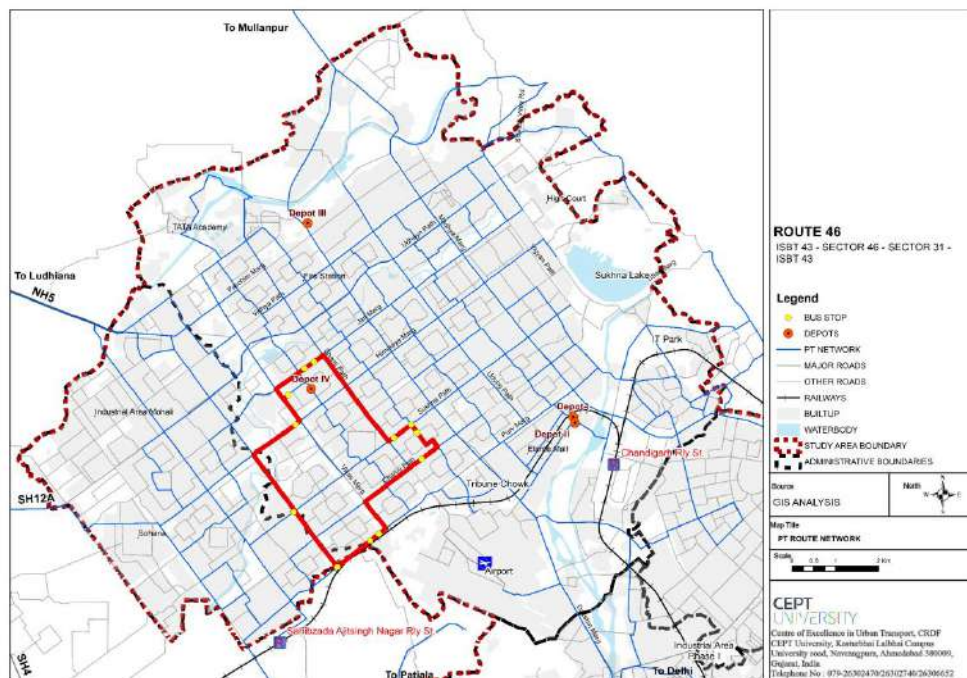


Figure 164: Proposed Route 46 (ISBT 43 - Sector 46 - Sector 31 – ISBT 43)

Route 47: Sector 65 - Sector 43 Market - Sector 65 Via Sector 65 Market, Phase 11, Sector 48/49, Sector 46 Market, Sector 46, Sector 45/46, Sector 45 Market, Sector 44 Market, Sector 43 Market, Sector 52/53, Phase 7/3, Chawla Chowk, Sector 61/70, Sector 68/67, Phase 10 Mohali

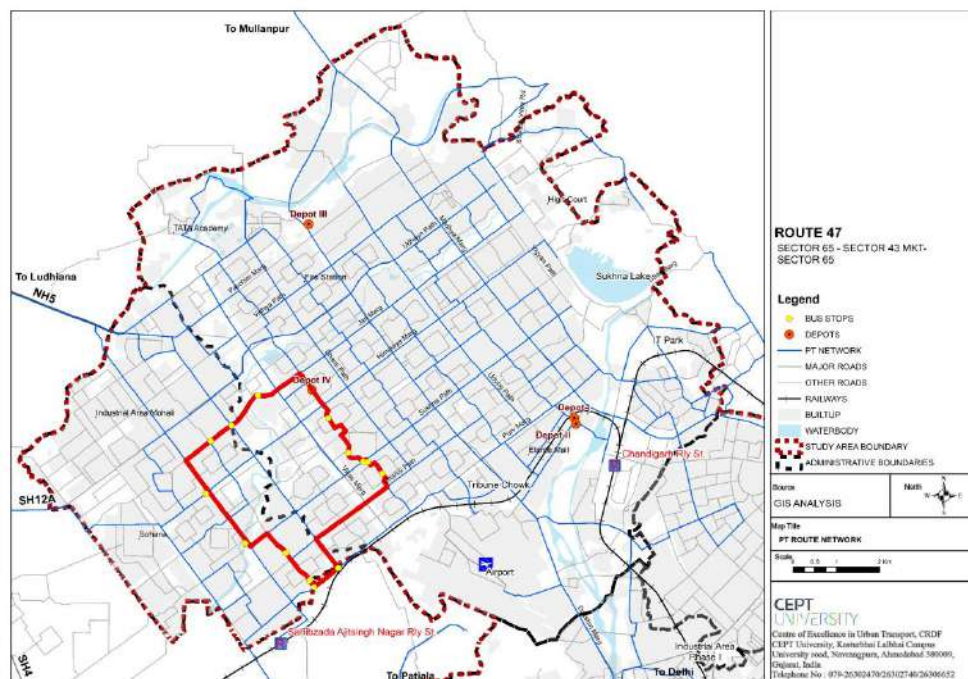


Figure 165: Proposed Route 47 (Sector 65 - Sector 43 Market - Sector 65)

Route 48: Sector 43 - Sohna Gurudwara - Ptl Chowk - ISBT 43 Via ISBT 43, Sector 52/51, Sector 52, Phase 7/3, Chawla Chowk, Sector 69/70, Gurudwara Singh Saheeda Sahib, Sector 71/72, Ptl Chowk, Madanpura Chowk, Phase 2 Sector 54/53, Sector 53

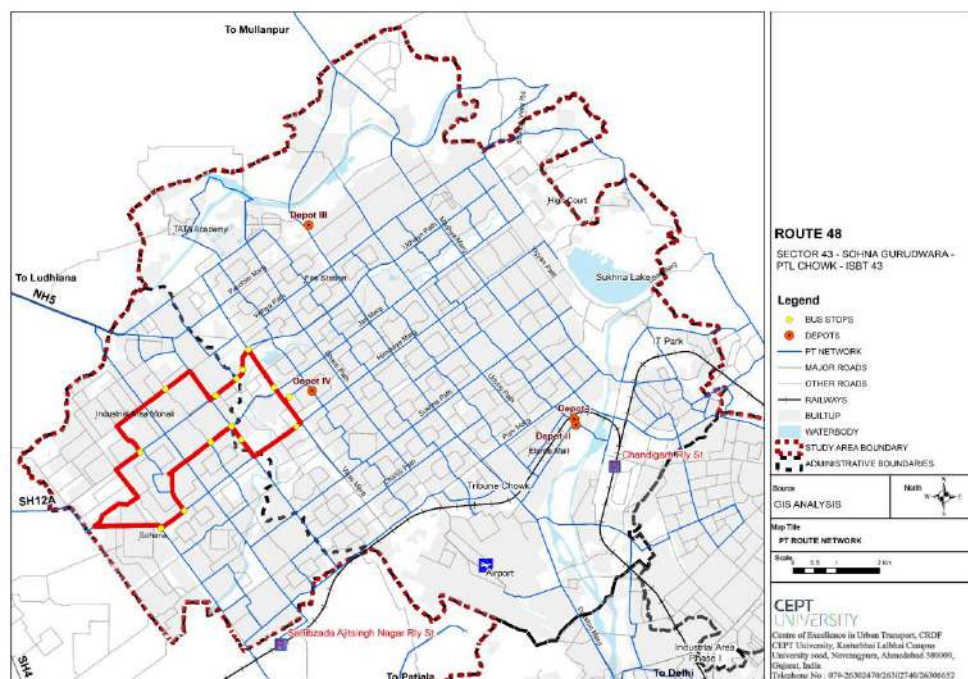


Figure 166: Proposed Route 48 (Sector 43 - Sohna Gurudwara - Ptl Chowk - ISBT 43)

Route 49: ISBT Mohali - Sector 43 - Fortis Hospital - ISBT Mohali Via ISBT Mohali, Phase 1, Sector 40/39, Sector 40 Market, Sector 40/41, Sector 41 Market, Sector 42/41, Sector 42 Market, Sector 43 Market, Sector 44 Market, Colony No. 5, Sector 62/63, Fortis Hospital, Sector 61/70, PTL CHOWK

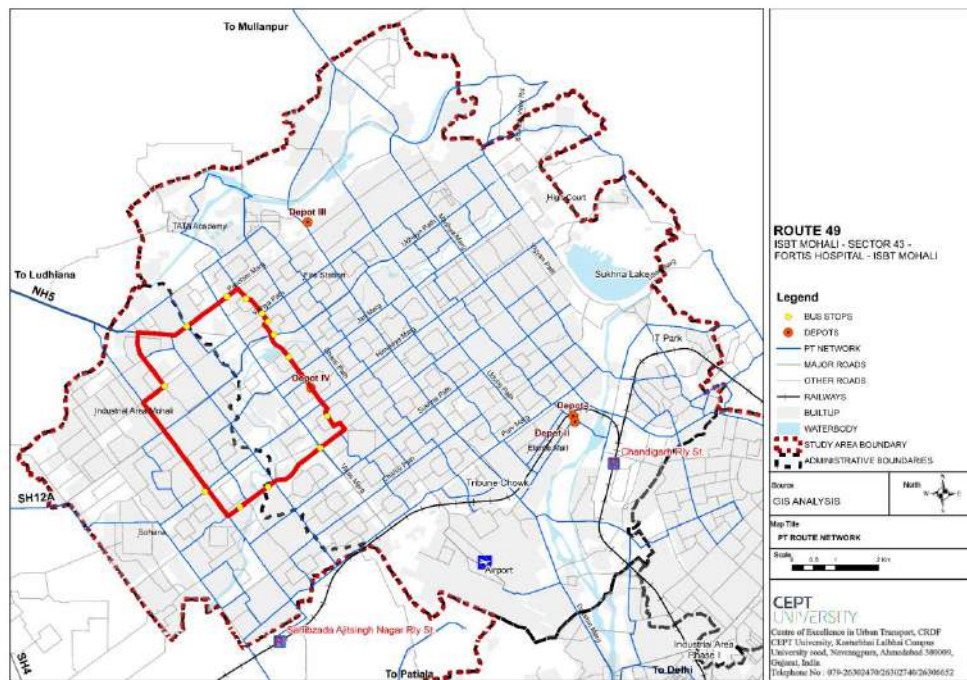


Figure 167: Proposed Route 49 (ISBT Mohali - Sector 43 - Fortis Hospital - ISBT Mohali)

Route 50: Daun Majra - ISBT 43 - Sector 65 - Daun Majra Via Daun Majra, Balongi, ISBT Mohali, Dera Studio Chowk, Phase 6, Sector 56 West, Furniture Market Chowk, Sector 53/42, Sector 52/53, Sector 51, Sector 50, Sector 49 Market, Sector 48 Market, Sector 48/49, Phase 11, White House, Phase 1 Ind Area Mohali, Sector 66, Sector 66/67, Sector 67 Market, Sector 68/67, Sector 61/70, Sector 69/70, Sector 71/72

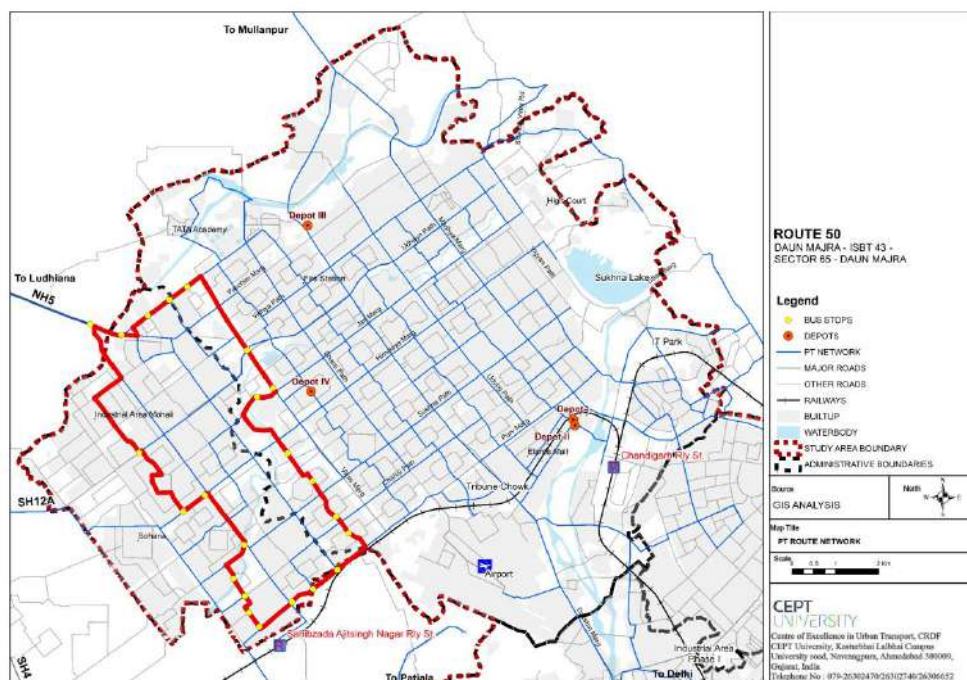


Figure 168: Proposed Route 50 (Daun Majra - ISBT 43 - Sector 65 - Daun Majra)

Routes on Major Corridors only for Scenario 2B

Route 53: Malloya - Sector 47 (Sector 40 Market Road) Via Sector 39C, Sector 40 Market, Sector 40/41, Sector 41 Market, Sector 42 Market, Sector 43 Market, Sector 44 Market, Sector 45 Market, Sector 46, Sector 46 Market, Sector 47 Market

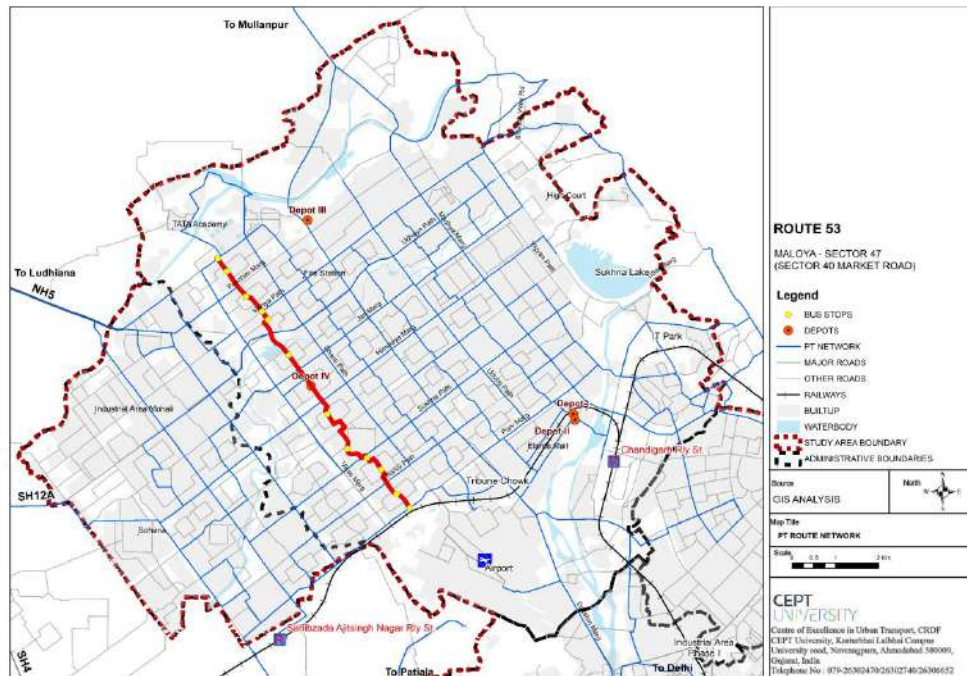


Figure 169: Proposed Route 53 (Malloya - Sector 47 (Sector 40 Market Road))

Route 54: Sector 38/39 -Industrial Area Phase 2 (Sector 36 Market Road) via Sector 38 West, Sector 38 Main Market, Sector 37 Market, Sector 36/37, Sector 36 market, Sector 35/36, Sector 35 Market, Sector 34 Gurudwara, Sector 34 Market, Sector 33 market, Sector 33/32, Sector 32 Market, Sector 31 Market

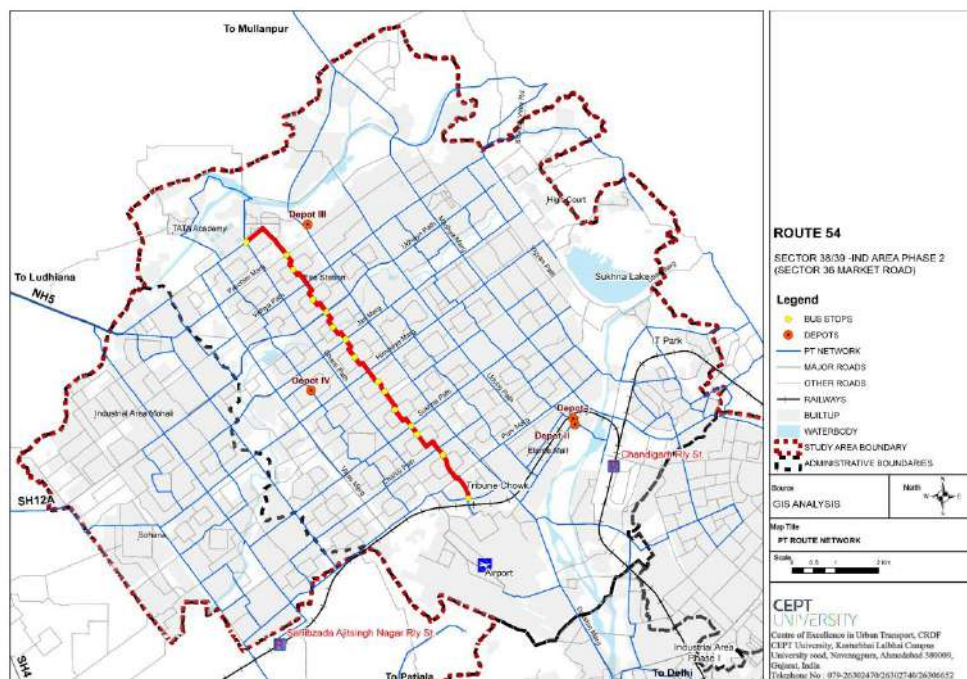


Figure 170: Proposed Route 54 (Sector 38/39 -Industrial Area Phase 2 (Sector 36 Market Road))

Route 55: Daddu Majra Colony - Hallo Majra (Dakshin Marg) Via CTU Depot Sector 25, Sector 38/25, Sector 37/24, Sector 36/23, Kisan Bhawan, Sector 22/35, Sector 35/22, Sector 34/21, Sector 21, Sector 20/33, Sector 33/20, Sector 30/32 hospital, Sector 32 Hospital, Sector 29/31, Tribune Chowk

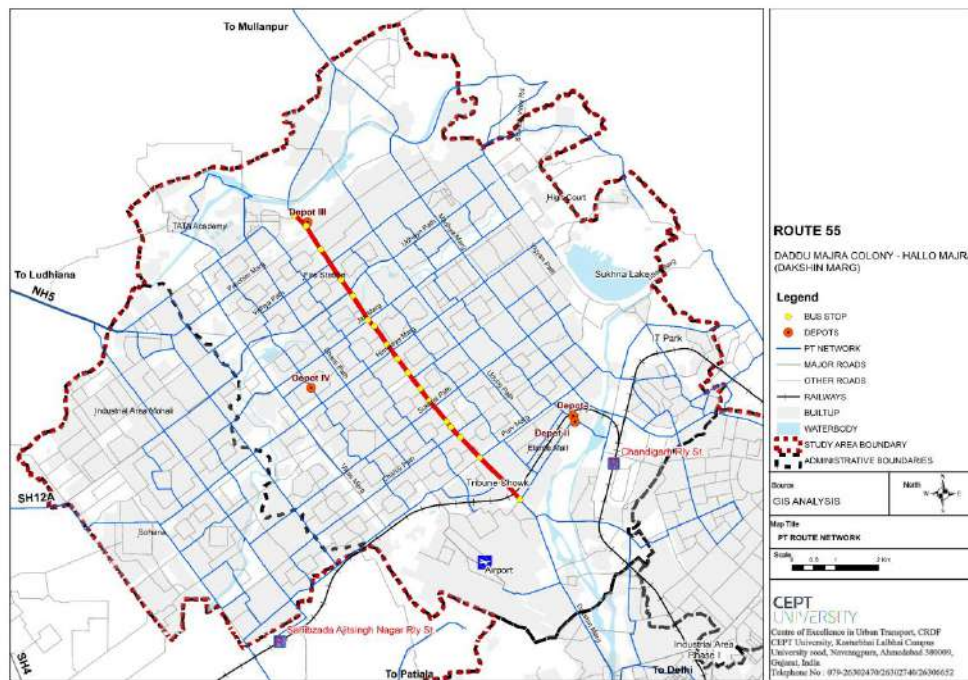


Figure 171: Proposed Route 55 (Daddu Majra Colony - Hallo Majra (Dakshin Marg))

Route 57: ISBT-43 TO MATA MANSA DEVI via Sector 24/25, Sector 15 market, Sector 16 Market, Sector 17 Market, Aroma 22/21, Sector 22 Market, Sector 24 Market, Sector 23 Market

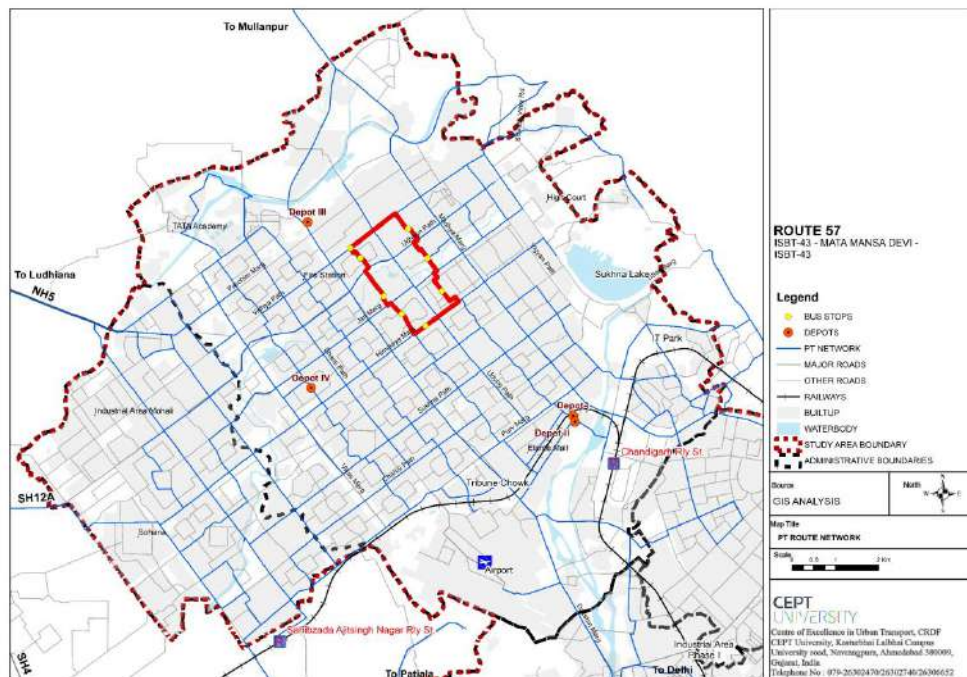


Figure 172: Proposed Route 57 (ISBT-43 TO MATA MANSA DEVI)

Route 58: IVY Hospital - Punjab Civil Secretariat (Udyan Path) via P.C.L. Chowk, Phase 3/5, Madanpura Chowk, Phase 2, Sector 54/53, Furniture Market Chowk, Sector 42/41, Sec 43, Sector 36/37, Sector 15/24, Sector 11, Sector 11/10, Sector 2/3

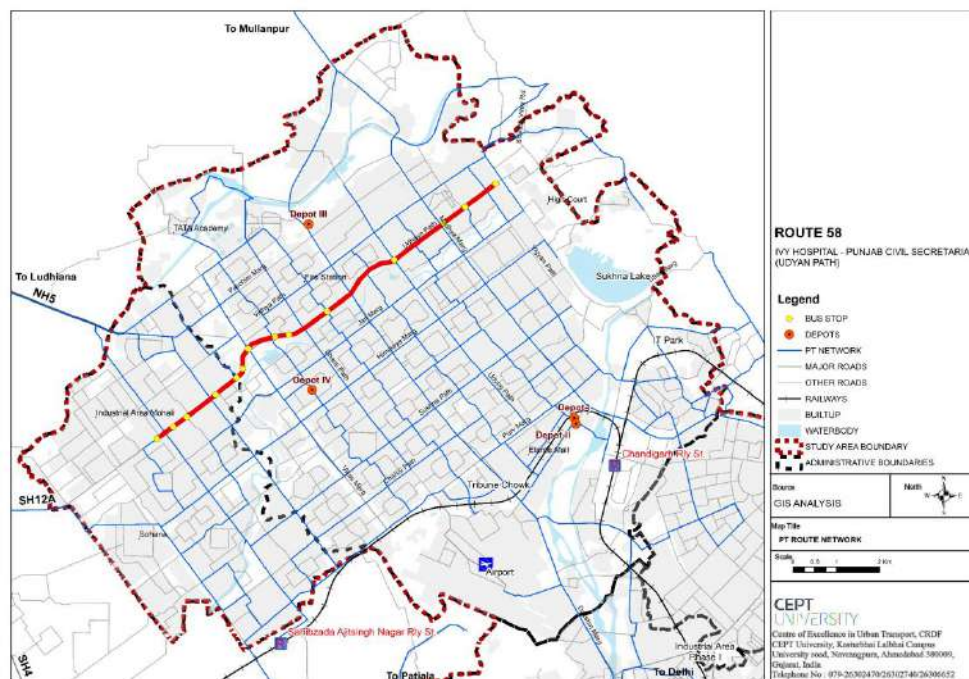


Figure 173: Proposed Route 58 (IVY Hospital - Punjab Civil Secretariat (Udyan Path))

Route 59: Sohna Gurudwara - Sukhna Lake (Himalaya Marg) via Sector 69/70, Sector 61/70, Phase 7, Sector 52/51, Sector 43/44, Sector 34/35 Piccadilly Chowk, Aroma 22/21, Sector 18/17, Sector 18 press, Sector 9/8

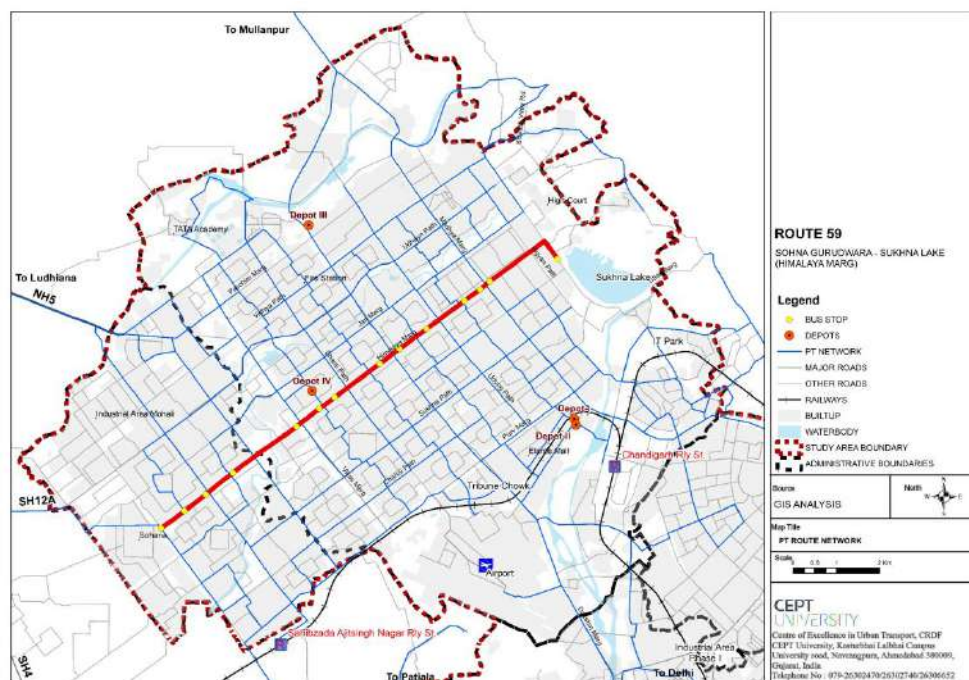


Figure 174: Proposed Route 59 (Sohna Gurudwara - Sukhna Lake (Himalaya Marg))

Route 60: Sector 68/67 - Sector 26 Guru Gobind School (Sukhna Path) via Sector 63/64, Sector 49/50, Sector 46/45, Sector 33/32, Sector 20/30 gurudwara chowk, Sector 19/27 Market

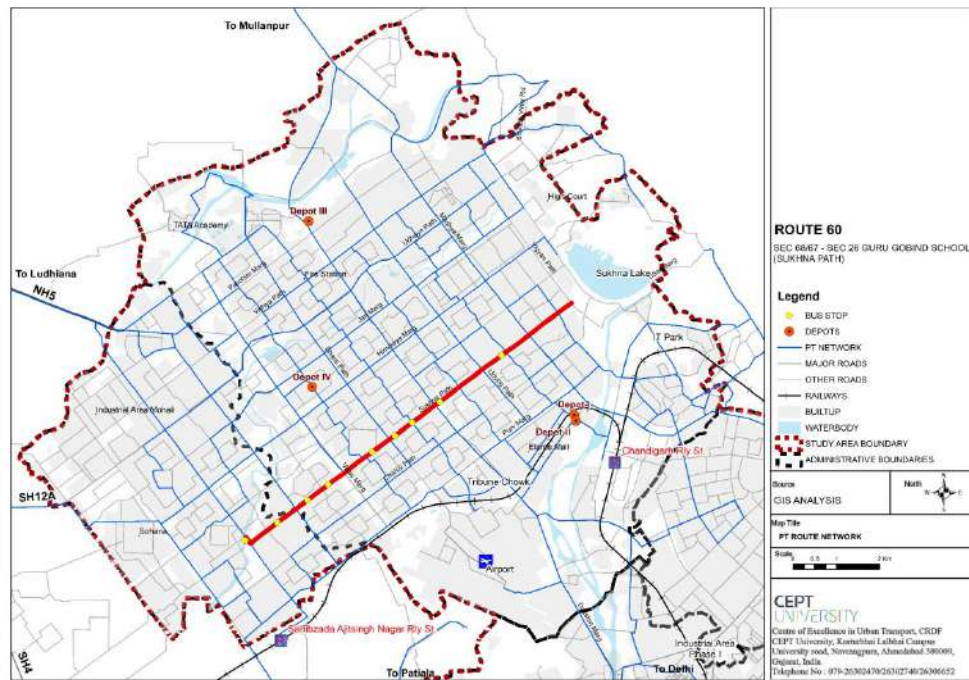


Figure 175: Proposed Route 60 (Sector 68/67 - Sector 26 Guru Gobind School (Sukhna Path))

Annexure 6: Service Plan – Scenario Evaluation

1.1 Evaluation Framework

Each scenario has been evaluated at three levels, i.e. supply indicators, model output and Service Level Benchmark (SLB) criteria by MoUHA. The Chandigarh and surrounding area consists of Chandigarh and other neighboring towns however Chandigarh and Mohali contribute to the major share of population and total demand. Also, in terms of the road connectivity, Chandigarh and Mohali are well connected with Mohali following same grid pattern as Chandigarh. Therefore, all the indicators are compiled for the entire Chandigarh and surrounding area as well as for Chandigarh and Mohali together separately.

1.1.1 Supply Indicators

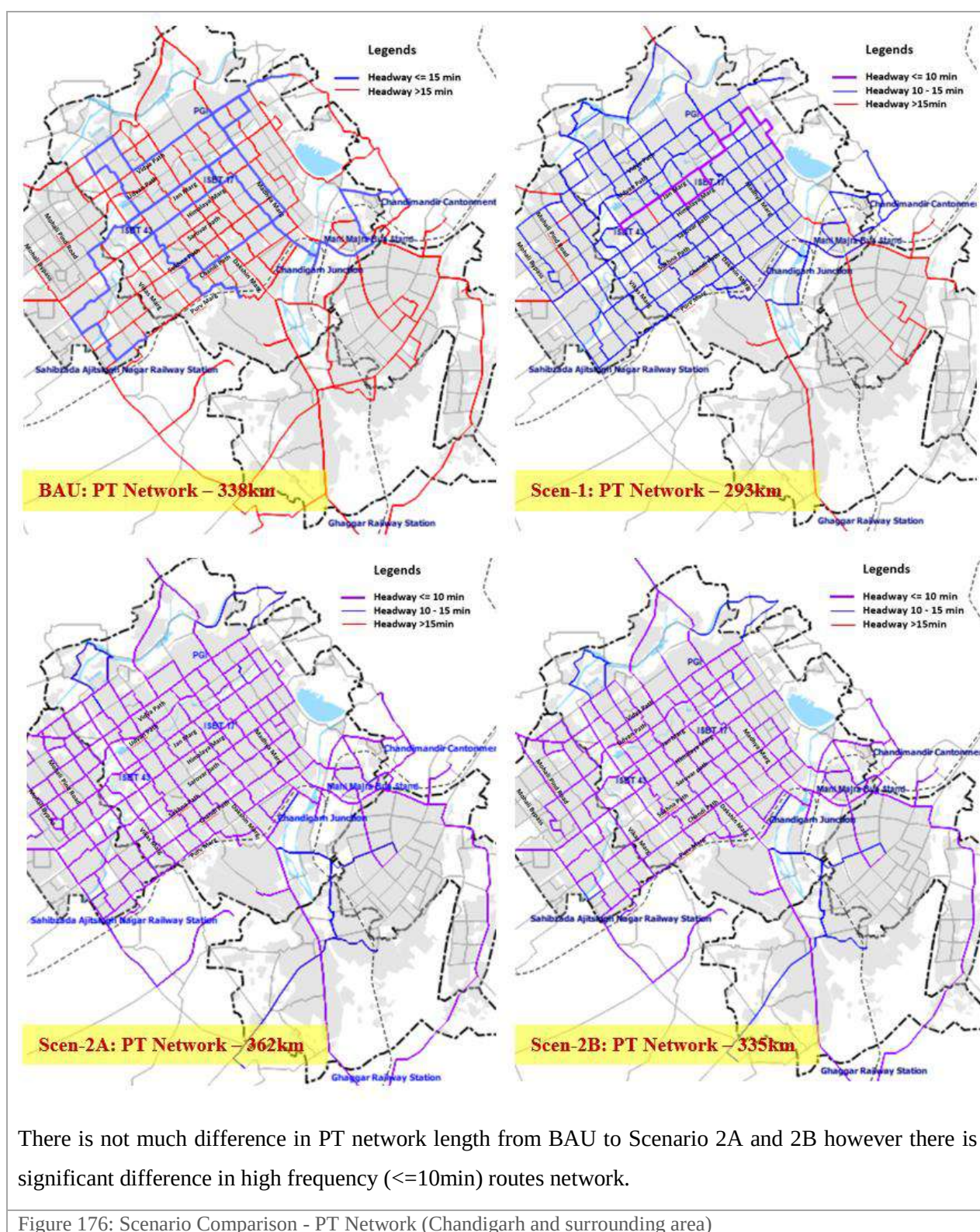
The following table shows the evaluation of the three scenarios with respect to supply characteristics - public transport network and area coverage, service headways, fleet size requirements, high frequency network coverage. The headways of proposed routes are assigned in line with demand pattern presented in the main report Chapter 2.

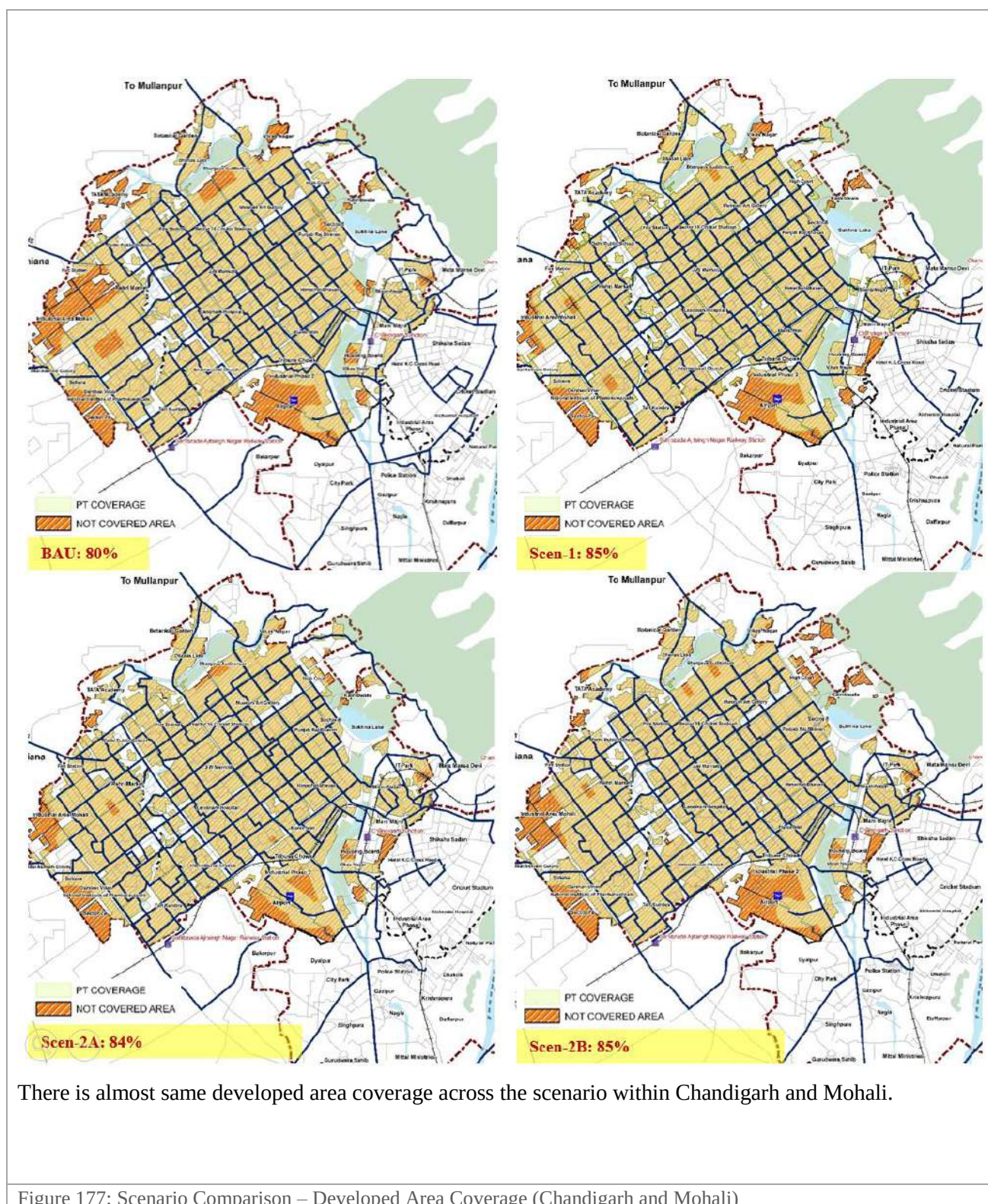
- More than 75% routes are with high frequency in proposed Scenario 2A and 2B against only 6% in Scenario 1.
- Similarly more than 90% of PT network is planned with high frequency (≤ 10 min) in Scenarios 2A and 2B compared to 6% in Scenario 1. As a result proposed scenarios serve about 92% population in Chandigarh Mohali and more than 75% in Chandigarh and surrounding area with high frequency service.
- Proposed scenario serves more than 95% stops with high frequency service in the Chandigarh connecting surrounding suburban areas.

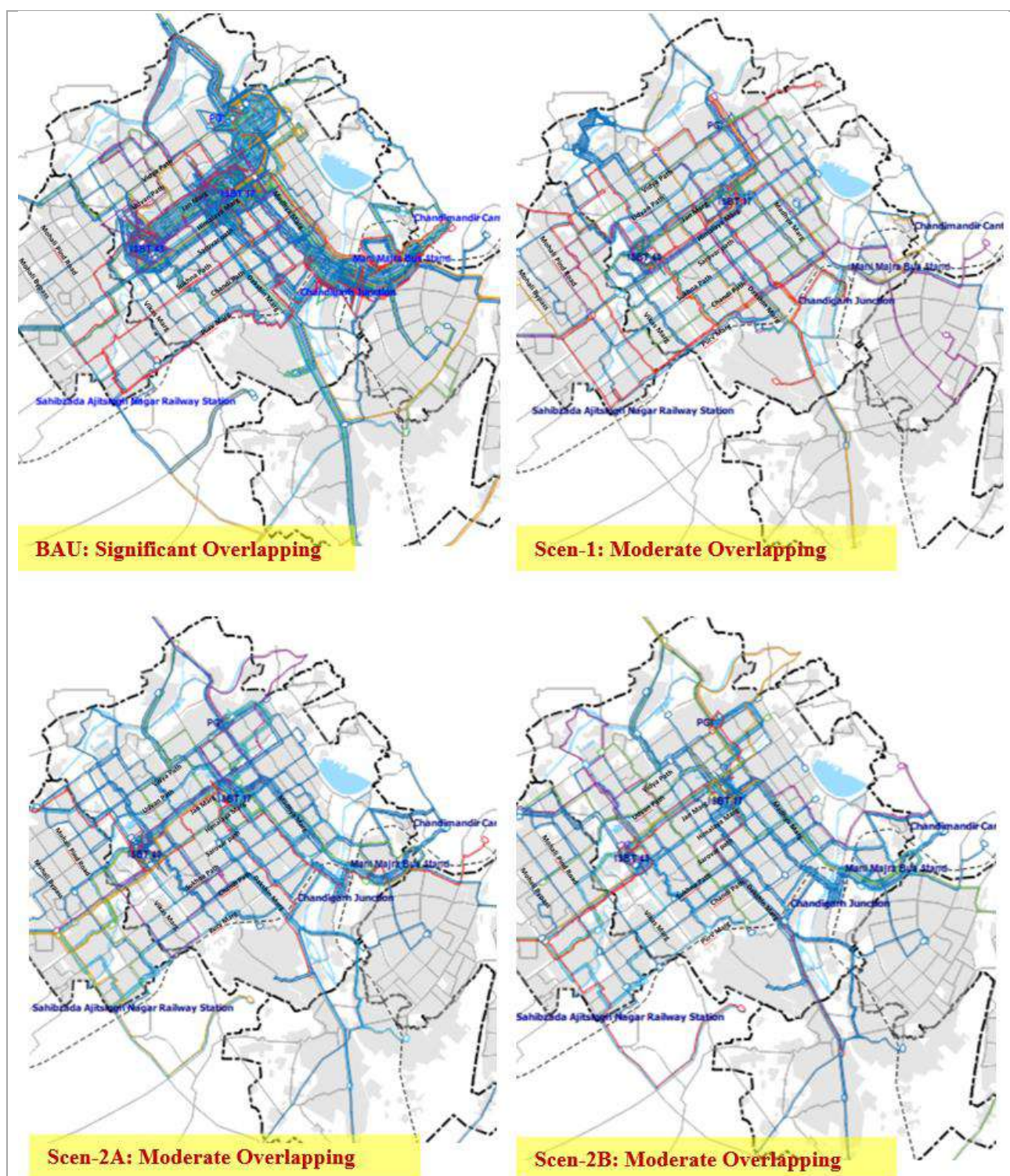
Table 4: Scenario evaluation based on supply analysis

Indicators	Study Area				Chandigarh and Mohali			
	BAU	Scenario 1	Scenario 2A	Scenario 2B	BAU	Scenario 1	Scenario 2A	Scenario 2B
No. of routes	68	35	50	42	68	35	50	42
No. of routes with headway <=10 min	0 (0%)	2 (6%)	40 (80%)	32 (76%)	0 (0%)	2 (6%)	40 (80%)	32 (76%)
No. of routes with headway <=15 min (& >10min)	5 (7%)	23 (66%)	10 (20%)	10 (24%)	5 (7%)	23 (66%)	10 (20%)	10 (24%)
No. of routes with headway <=20 min (& >15min)	13 (19%)	5 (14%)	0 (0%)	0 (0%)	13 (19%)	5 (14%)	0 (0%)	0 (0%)
No. of routes with headway >20 min	50 (74%)	5 (14%)	0 (0%)	0 (0%)	50 (74%)	5 (14%)	0 (0%)	0 (0%)
PT Network length - km	338	293	362	308	211	231	258	211
PT network length (km) <=10 min	0 (0%)	16 (6%)	316 (87%)	269 (87%)	0 (0%)	16 (7%)	235 (91%)	190 (90%)
PT network length (km) <=15 min	69 (20%)	209 (72%)	38 (10%)	39 (13%)	68 (32%)	201 (87%)	20 (8%)	21 (10%)
PT network length (km) <=20 min	102 (30%)	24 (8%)	0 (0%)	0 (0%)	76 (36%)	6 (3%)	0 (0%)	0 (0%)
PT network length (km) >20 min	167 (49%)	43 (15%)	0 (0%)	0 (0%)	67 (32%)	8 (3%)	0 (0%)	0 (0%)
Developed area coverage (Study area: 137sqkm, Chandigarh & Mohali: 95sqkm)	102 (74%)	95 (69%)	97 (71%)	94 (68%)	75 (80%)	81 (85%)	81 (85%)	77 (81%)
Accessibility to PT service								
Population served total PT network within 500mt.	16,52,152	15,50,760	17,05,099	16,71,905	13,49,726	13,84,084	13,81,370	14,54,378
Population served total PT network within 500mt. High frequency stops	9,66,214	10,64,565	13,48,019	12,65,211	9,06,219	10,50,483	11,48,631	12,09,338
% Population served total PT network within 500mt.	78%	73%	81%	79%	93%	95%	95%	95%
% population served with high frequency service (<=10min)	58%	69%	79%	76%	67%	76%	83%	83%
Total number of PT stops served	300	261	280	317	234	161	217	247
No of PT stops with effective <=10min bus route (Stops with avg. 5min waiting time)	183 (61%)	126 (48%)	266 (95%)	248 (78%)	157 (67%)	92 (57%)	213 (99%)	200 (81%)
Fleet size (On road)	330	373	733	622	330	373	733	622

The maps below shows the spatial network, developed area coverage and routes structure showing overlapping of routes.







There is significant overlapping in BAU compared to other three proposed scenarios. All three proposed scenario show moderate overlapping of route structure. However, Scenario 1 shows duplication of routes on a few corridors.

Figure 178: Scenario Comparison – Routes Structure (Chandigarh and Mohali)

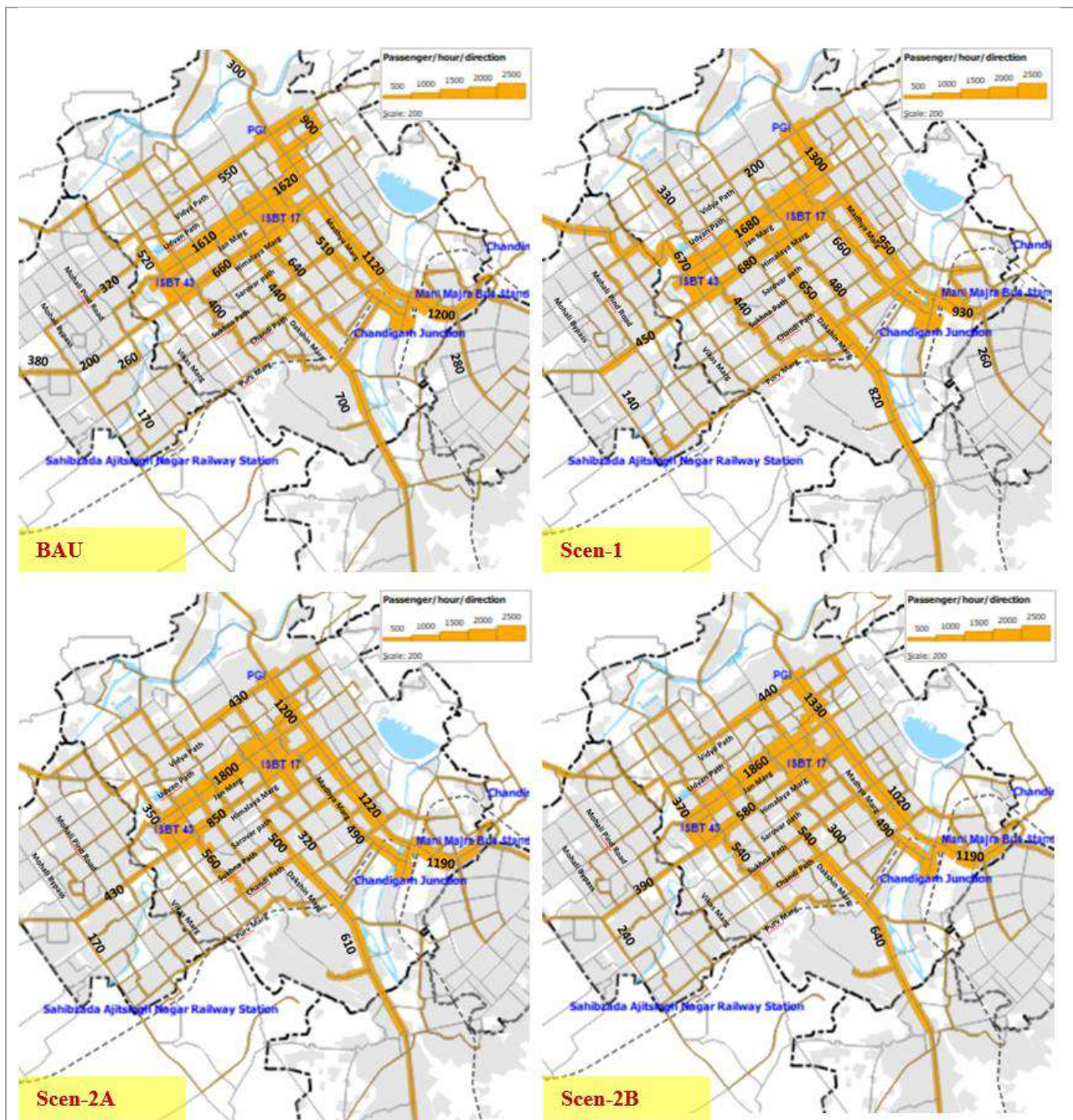
1.1.2 Model Outputs

The PT model outputs were also used to compare the four scenarios. The analysis has been undertaken for the year 2023 where about 35% shift from IPT is assumed. The purpose of this exercise is only to see how different scenarios perform with the same demand. However detail demand analysis has been conducted in Business Plan section.

It can be seen from the table below that average trip length is about 8-9 km. with such trip lengths and looking at city structure, one transfer is reasonable. BAU has less transfer because of numerous overlapping route structure whereas Scenario 2A and 2B scores well marginally in average trip length and reasonable transfer rate because of network based approach to route planning.

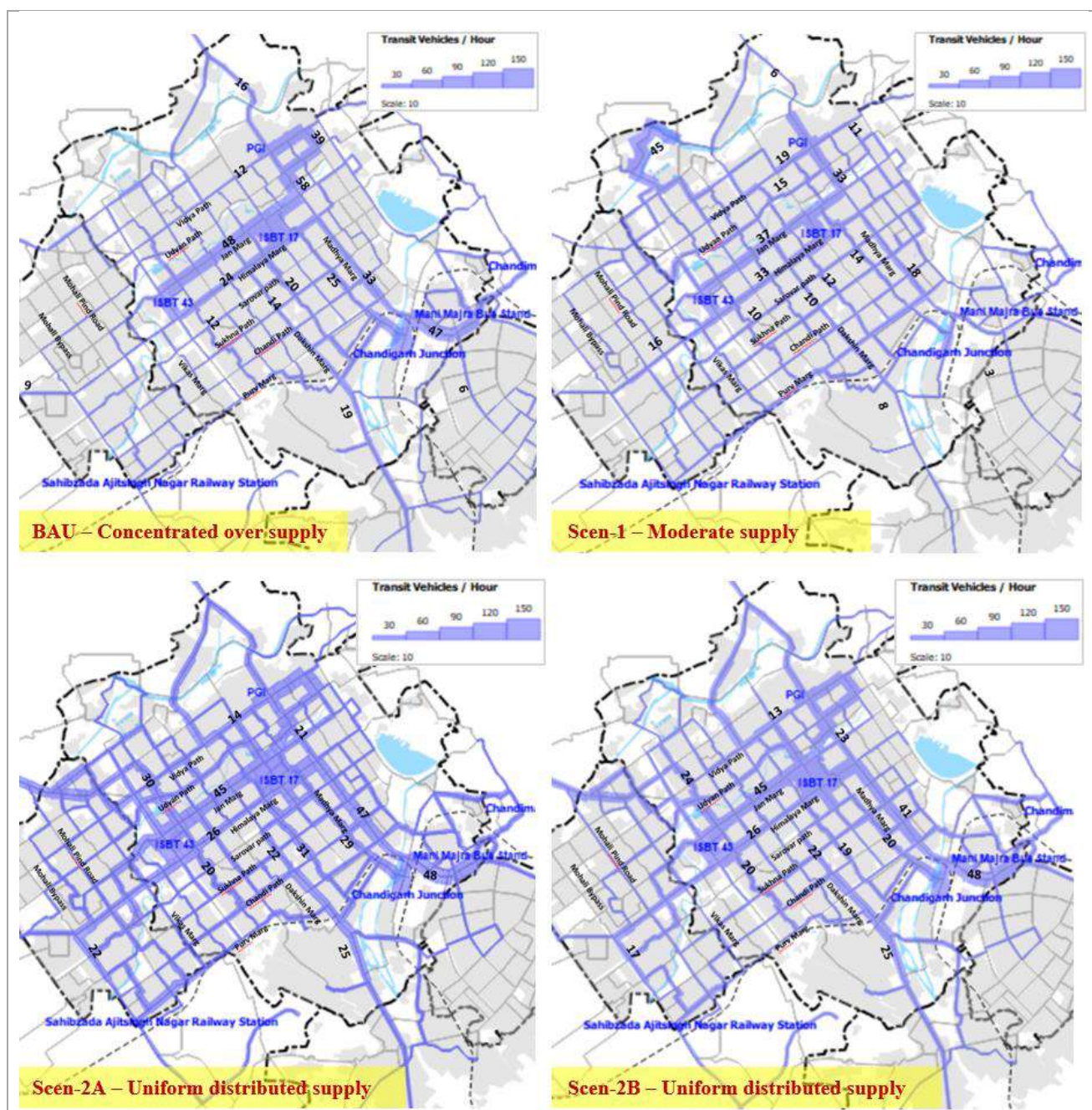
Table 5: Model Output - Scenario Comparison for Chandigarh and surrounding area

	BAU	Scenario 1	Scenario 2A	Scenario 2B
Demand (Full day – FY2023)	2,12,000	2,12,000	2,12,000	2,12,000
Total Boarding	2,47,382	3,08,814	2,82,337	2,85,816
Transfer Rate	1.17	1.46	1.33	1.35
Avg. trip length (km)	8.52	8.31	7.96	7.94



The total PT and IPT demand has been assigned on the proposed network scenarios. Demand (passenger flows) is very much concentrated on Jan Marg and Himalaya Marg in BAU and Scenario 1. Some demand from Himalaya Marg has shifted to other corridors in Scenario 2A and 2B compared to BAU and Scenario 1. This is because the proposed scenarios 2A and B have route designs based on the existing demand pattern reducing any detour travel. The same is reflected in trip length comparison across the scenario.

Figure 179: Scenario Comparison – Passenger Flow (Chandigarh and Mohali)



There is significant bus oversupply observed on Jan Marg and Himalaya Marg in BAU scenario because of overlapping route structure. In comparison, rest of the city does not have enough bus services. BAU and Scenario 1 also have inadequate level of services on Zirakpur road where demand seems to be higher. Similar observation can be made on the road leading to Mani Majra terminal. Overall Scenario 2A and 2B appears to have more distributed bus supply across Chandigarh and surrounding area, which would be important for attracting new passengers on board.

Figure 180: Scenario Comparison – Bus Supply (Buses/hr.) (Chandigarh and Mohali)

1.1.3 Service Level Benchmark

In addition to the above, MoHUA's Service Level Benchmark indicators for urban public transport have also been used to evaluate the scenarios. Scenario 2A and 2B perform better in extent of supply and average waiting time. As in case of supply indicators, SLB indicators 2, 3 and 4 have been computed for Chandigarh and surrounding area as well as for Chandigarh & Mohali cities separately. The outputs are shown in the table below.

Table 6: Service Level Benchmark Indicators

Service Level Benchmark		BAU	Scenario 1	Scenario 2A	Scenario 2B	Service Standards	LOS
1	Presence of Organized Public Transport System in Urban Area (%)						
	Total number of buses operating on road (a)	330	373	763	630	$\geq 60\%$	1
	Total number of buses under the ownership of STU/SPV or under concession agreement (b)	330	373	763	630	40-60%	2
	Presence of Organized Public Transport System in Urban Area($\% = (b/a) \times 100$)	100%	100%	100%	100%	20-40%	3
	SLB levels obtained	1	1	1	1	$< 20\%$	4
2	Extent of supply availability of public transport (Study Area)						
	No. of Buses available in a city on any day (a)	330	373	763	630	≥ 0.6	1
	Total Population of the study area (b) - FY 2023 Study Area Population	19,46,190	19,46,190	19,46,190	19,46,190	0.4-0.6	2
	Availability of Public transport/1000 population= $(a/b) \times 1000$	0.17	0.19	0.39	0.32	0.2-0.4	3
	SLB levels obtained	4	4	3	3	< 0.2	4
2	Extent of supply availability of public transport (Chandigarh + Mohali area)						
	No. of Buses available in a city on any day (a)	330	373	763	630	≥ 0.6	1
	Total Population of the study area (b) - FY 2023 Chandigarh + Mohali Population	15,32,260	15,32,260	15,32,260	15,32,260	0.4-0.6	2
	Availability of Public transport/1000 population= $(a/b) \times 1000$	0.22	0.24	0.5	0.41	0.2-0.4	3
	SLB levels obtained	3	3	2	2	< 0.2	4
3	Service Coverage of Public Transport in Study Area						
	Total length in road km of the corridor on which public transport system ply in the city (in km) (a) (Corridors along which the service	338	293	362	308	≥ 1	1

Service Level Benchmark		BAU	Scenario 1	Scenario 2A	Scenario 2B	Service Standards	LOS
	frequency is one hour or less are only consider compute the length)						
	Total study area (Sq.km) (b) --> Total road network length in the city (km)	783	783	783	783	0.7-1	2
	Service coverage =(a/b)	0.43	0.37	0.46	0.39	0.3-0.7	3
	SLB levels obtained	3	3	3	3	<0.3	4
3	Service Coverage of Public Transport (Chandigarh + Mohali area)						
	Total length in road km of the corridor on which public transport system ply in the city (in km) (a) (Corridors along which the service frequency is one hour or less are only consider compute the length)	211	231	258	211	>=1	1
	Total study area (Sq.km) (b) --> Total road network length in the city (km)	550	550	550	550	0.7-1	2
	Service coverage =(a/b)	0.38	0.42	0.47	0.38	0.3-0.7	3
	SLB levels obtained	3	3	3	3	<0.3	4
4	Average Waiting time for Public Transport Users (min) – Study Area					<=4	1
	Waiting time for headway	5.7	5.9	1.8	1.7	4-8	2
	Calculate the average waiting time of passenger for each route as half of the average headway for that particular route.					8-12	3
	SLB levels obtained	2	2	1	1	>12	4
4	Average Waiting time for Public Transport Users (min) – Chandigarh + Mohali					<=4	1
	Waiting time for headway	5.2	5.9	1.6	1.7	4-8	2
	Calculate the average waiting time of passenger for each route as half of the average headway for that particular route.					>12	4
	SLB levels obtained	2	2	1	1	>12	4

Annexure 7: List of proposed routes & details

Table 7: List of proposed routes

Route No	Route code	Route Description	Route type	Remarks	Service type	Length (km)	Total travel time (min)
1	SU13	ISBT Mohali - ISBT 17 - Sukteri	Suburban	Extended (C35)	Suburban	25.7	210
2	SU12	Sohna Gurudwara - ISBT 43 - ISBT 17 - PGI - Khizrabad	Suburban	Extended (C40)	Suburban	34.6	250
3	SU11	ISBT 17 - ISBT 43 - Airport	Suburban	Same Route (C38AAC)	Suburban	21.9	150
4	SU10	ISBT 17 - Railway Station - Cantonment	Suburban	Curtailed (C211AC)	Suburban	14.8	121
5	SU9	ISBT 43 - Tribune Chowk - Derabassi	Suburban	Curtailed (C213)	Suburban	24.4	188
6	SU8	ISBT 17 - PGI - Khizrabad	Suburban	Same Route (C25A)	Suburban	24.6	168
7	SU7	ISBT 17 - Mani Majra	Suburban	Curtailed (C2A/C2C)	Suburban	13.7	113
8	SU6	ISBT17 -Railway Station -Dhakoli	Suburban	Modified Alignment (C26AC)	Suburban	19.2	174
9	SU3	ISBT 17 - Mani Majra - Derabassi	Suburban	Curtailed (C32)	Suburban	29.4	205
10	SU2	PGI - ISBT 17 - Derabassi	Suburban	Curtailed and Extended (C216AC)	Suburban	26.2	227
11	SU15	Kharar - ISBT 43 - ISBT 17 - Airport	Suburban	Same Route (C35B, C17)	Suburban	29.8	290
12	SU16	ISBT 43 - Landran - Kharar	Suburban	Curtailed (C20A)	Suburban	20.9	199
13	SU17	ISBT 17 - PGI - Nayagao - ISBT 17	Suburban	New Route	Suburban	17.7	157
14	SU19	Sohna Gurudwara - ISBT Mohali - Kharar	Suburban	New Route	Suburban	13.4	128
15	SU20	PGI - ISBT 17 - Zirakpur	Suburban	Modified Alignment (C26AC)	Suburban	15.2	149
16	C38AC	ISBT 17 - ISBT43 - New Airport	Suburban	Modified Alignment (C38AC)	Suburban	21.1	173
17	C213	ISBT43 - Landran	Suburban	Curtailed (C213)	Suburban	12.1	103
18	SU21	ISBT 17 - High Court - Kaimbwala	City	Curtailed (C8)	City	12.5	107
19	C9C	Sohna Gurudwara - White House - Mata Mansa Devi	City	Curtailed (C9A/C9C)	City	22.8	201
20	C5A	Ramdarbar - ISBT43 -PGI - ISBT43	City	No change (C5A/C5C)	City	34.4	286
21	C241	ISBT 43 - CTU Workshop - Mata Mansa Devi	City	No change (C241)	City	23	175
22	C2D	Mata Mansa Devi - ISBT17 - Dhanas	City	Modified Alignment (C2D)	City	14.6	117

Route No	Route code	Route Description	Route type	Remarks	Service type	Length (km)	Total travel time (min)
23	C8	ISBT17 - SAS Railway Station	City	Curtailed (C8)	City	11.8	87
24	C12	ISBT43- Mohali Municipal Corporation - Phase11	City	Curtailed (C12)	City	16.7	139
25	SU14	ISBT 17 - PGI - Mullanpur Barrier - ISBT 17	City	Curtailed (C2A/C2C)	City	18.9	145
26	SU18	ISBT 43 - Maloya Colony	City	New Route	City	11.5	99
27	I4	ISBT Mohali-ISBT 43	Suburban - Terminal	Curtailed (C35B)	Suburban	5.2	50
28	I1	PGI - ISBT Mohali - Kharar	Suburban - Terminal	Modified Alignment (C35)	Suburban	20.8	191
29	SU4	ISBT 17 - CTU Workshop - Panchkula	Suburban - Terminal	Curtailed (C2F)	Suburban	14.2	114
30	I2	PGI - ISBT17 - ISBT 43 - Fortis Hospital	City - Terminal	Curtailed and Extended (C28C)	City	15.7	141
31	C22	ISBT 17 - Railway Station - Mata Mansa Devi	City - Terminal	Curtailed (C22)	City	16.4	149
32	SU2E	PGI - ISBT 17 – Zirakpur - Derabassi	Suburban - Express	Curtailed and Extended (C216AC)	Suburban	26.2	205
33	SU4E	PGI - ISBT 17 – Grain Market – Chandigarh Railway Station – ISBT Panchkula	Suburban - Terminal - Express	Curtailed (C2F)	Suburban	13.1	73
34	I1E	ISBT17 - ISBT Mohali – Kharar	Suburban - Terminal - Express	Modified Alignment (C35)	Suburban	15.5	137
35	C551E	ISBT43 – ISBT 17 - PGI	City - Express	Curtailed (C242)	City	10.5	65
36	V3	V3_Sector40 Market Rd	Grids	New Route	City	8.2	74
37	V4	V4_Daddu Majra Rd	Grids	New Route	Suburban	9.9	87
38	V5	V5_Dakshini Marg	Grids	New Route	City	8.6	77
39	V7	V7_ISBT17 - Sector25	Grids	New Route	City	9.4	71
40	H1	H1_Udyan Path	Grids	New Route	City	11.4	86
41	H2	H2_Himalaya Path	Grids	New Route	City	11.2	84
42	H3	H3_Sukhna Path	Grids	New Route	City	10.9	84

Annexure 8: Routes frequency (peak hours) and fleet requirement for fleet mix scenarios (2032)

Table 8: Route frequency and fleet requirement for fleet mix scenarios

EMME code	Scenario - 1			Scenario - 2		Scenario - 3		Scenario - 4			Scenario - 5			Scenario - 5A			Scenario - 6		
	10min HDW - Std+Midi Buses			10min HDW - All Midi Buses		LF 60% - All Midi, HDW cap at 15min		LF 60% - Stand+Midi, HDW cap at 15min			LF 60% - Midi+Mini, HDW cap at 15min			LF 60% - Midi+Mini, HDW cap at 10min			LF 60% - Stand+Midi+Mini, HDW cap at 15min		
	Bus/ Type	Fleet	Hdwy (min)	Fleet	Hdwy (min)	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)
SU13	Midi	24	10	24	10	16	15	Midi	16	15	Mini	16	15	Mini	24	10	Mini	19	13
SU12	Std	28	10	28	10	50	6	Std	50	6	Midi	50	6	Midi	50	6	Std	30	9
SU11	Midi	17	10	17	10	11	15	Midi	11	15	Mini	11	15	Mini	17	10	Mini	13	13
SU10	Std	14	10	14	10	11	13	Midi	11	13	Mini	11	13	Mini	21	7	Midi	11	13
SU9	Std	21	10	21	10	32	7	Midi	32	7	Midi	32	7	Midi	32	7	Midi	32	7
SU8	Midi	19	10	19	10	13	15	Midi	13	15	Mini	13	15	Mini	20	9	Mini	20	9
SU7	Midi	13	10	13	10	9	15	Midi	9	15	Mini	9	15	Mini	13	10	Mini	9	14
SU6	Midi	20	10	20	10	13	15	Midi	13	15	Mini	13	15	Mini	20	10	Mini	13	15
SU3	Std	23	10	23	10	34	7	Midi	34	7	Midi	34	7	Midi	34	7	Midi	34	7
SU2	Midi	26	10	26	10	17	15	Midi	17	15	Mini	17	15	Mini	26	10	Mini	24	11
SU15	Std	33	10	33	10	31	11	Midi	31	11	Mini	31	11	Mini	59	6	Midi	31	11
SU16	Midi	23	10	23	10	15	15	Midi	15	15	Mini	15	15	Mini	26	9	Mini	26	9
SU17	Midi	18	10	18	10	12	15	Midi	12	15	Mini	12	15	Mini	18	10	Mini	12	15
SU19	Midi	15	10	15	10	10	15	Midi	10	15	Mini	10	15	Mini	15	10	Mini	11	13
SU20	Midi	17	10	17	10	11	15	Midi	11	15	Mini	11	15	Mini	21	8	Mini	21	8
C38AC	Std	20	10	20	10	14	14	Midi	14	14	Mini	14	14	Mini	27	7	Midi	14	14
C213	Midi	12	10	12	10	8	15	Midi	8	15	Mini	8	15	Mini	12	10	Mini	10	12
SU21	Std	12	10	12	10	9	13	Midi	9	13	Mini	9	13	Mini	17	7	Midi	9	13
C9C	Midi	23	10	23	10	15	15	Midi	15	15	Mini	15	15	Mini	26	9	Mini	26	9
C5A	Std	33	10	33	10	49	7	Midi	49	7	Midi	49	7	Midi	49	7	Midi	49	7
C241	Std	20	10	20	10	26	8	Midi	26	8	Mini	26	8	Midi	26	8	Midi	26	8
C2D	Std	13	10	13	10	37	4	Std	37	4	Midi	37	4	Midi	37	4	Std	22	6
C8	Midi	10	10	10	10	7	15	Midi	7	15	Mini	7	15	Mini	12	8	Mini	12	8
C12	Midi	16	10	16	10	10	15	Midi	10	15	Mini	10	15	Mini	16	10	Mini	10	15
SU14	Midi	16	10	16	10	11	15	Midi	11	15	Mini	11	15	Mini	16	10	Mini	11	15
SU18	Midi	11	10	11	10	7	15	Midi	7	15	Mini	7	15	Mini	11	10	Mini	7	15
I4	Midi	6	10	6	10	4	15	Midi	4	15	Mini	4	15	Mini	6	10	Mini	4	15
I1	Midi	22	10	22	10	14	15	Midi	14	15	Mini	14	15	Mini	22	10	Mini	15	14
SU4	Std	13	10	13	10	11	12	Midi	11	12	Mini	11	12	Mini	21	6	Midi	11	12

EMME code	Scenario - 1			Scenario - 2		Scenario - 3		Scenario - 4			Scenario - 5			Scenario - 5A			Scenario - 6		
	10min HDW - Std+Midi Buses			10min HDW - All Midi Buses		LF 60% - All Midi, HDW cap at 15min		LF 60% - Stand+Midi, HDW cap at 15min			LF 60% - Midi+Mini, HDW cap at 15min			LF 60% - Midi+Mini, HDW cap at 10min			LF 60% - Stand+Midi+Mini, HDW cap at 15min		
	Bus/ Type	Fleet	Hdwy (min)	Fleet	Hdwy (min)	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)	Bus/ Type	Fleet	Hdwy (min)
I2	Std	16	10	16	10	21	8	Midi	21	8	Mini	21	8	Midi	21	8	Midi	21	8
C22	Midi	17	10	17	10	11	15	Midi	11	15	Mini	11	15	Mini	20	8	Mini	20	8
SU2E	Midi	22	10	22	10	15	15	Midi	15	15	Mini	15	15	Mini	22	10	Mini	17	13
SU4E	Midi	8	10	8	10	5	15	Midi	5	15	Mini	5	15	Mini	9	9	Mini	9	9
I1E	Std	15	10	15	10	11	14	Midi	11	14	Mini	11	14	Mini	22	7	Midi	11	14
C551E	Std	7	10	7	10	16	4	Std	16	4	Midi	16	4	Midi	16	4	Std	10	7
V3	Midi	8	10	8	10	5	15	Midi	5	15	Mini	5	15	Mini	8	10	Mini	5	15
V4	Midi	9	10	9	10	6	15	Midi	6	15	Mini	6	15	Mini	9	10	Mini	6	15
V5	Midi	8	10	8	10	5	15	Midi	5	15	Mini	5	15	Mini	8	10	Mini	5	15
V7	Midi	7	10	7	10	5	15	Midi	5	15	Mini	5	15	Mini	7	10	Mini	6	12
H1	Midi	9	10	9	10	6	15	Midi	6	15	Mini	6	15	Mini	9	10	Mini	6	15
H2	Midi	9	10	9	10	6	15	Midi	6	15	Mini	6	15	Mini	9	10	Mini	6	15
H3	Midi	9	10	9	10	6	15	Midi	6	15	Mini	6	15	Mini	9	10	Mini	6	14

Annexure 9: Operational cost, revenue estimation, operating ratio and load factor

Built Up	Cost (by component) per Km									Total annual Vehicle Km	Total No of Buses	Total Operational Cost (inc. Depreciation & Cost of fund)	Total Operational Cost (w/o Depreciation & Cost of fund)	Total Revenue (with Concession & Discount)	Operating Ratio (Revenue/ Operation Cost w/o Depreciation & Cost of fund)	Load Factors
	Fuel Cost	R & M Cost	Insurance + MVT + MACT + RR Tax	Staff Cost	ITMS Operating Cost	Cost of Depreciation - Total	Cost of Fund - (Expected returns and Interests)	Miscellaneous Cost	CPKM (with depreciation and Interest)							
Scenario 1: Standard + Midi buses - 10min headways																
FY2023	10.56	14.89	1.52	18.47	1.07	7.11	10.63	3.25	67.51	2,43,82,000	361	165	121	87	0.72	0.35
FY2024	10.91	15.94	1.52	19.21	1.10	7.11	19.67	3.87	79.34	2,43,82,000	361	193	128	96	0.75	0.40
FY2025	4.40	12.74	3.80	19.43	1.00	15.22	14.45	3.71	74.74	2,78,13,000	400	208	125	129	1.03	0.37
FY2026	4.51	13.08	3.98	21.17	1.03	15.90	11.77	3.74	75.20	2,75,21,000	415	207	131	151	1.16	0.44
FY2027	4.67	13.52	4.06	22.02	0.72	15.57	5.88	3.46	69.89	4,19,02,000	631	293	203	192	0.94	0.32
FY2028	4.78	13.86	4.34	22.90	0.74	16.63	4.02	3.52	70.79	4,19,02,000	631	297	210	209	0.99	0.34
FY2029	4.92	14.26	4.34	23.82	0.68	16.48	2.74	3.51	70.74	4,73,04,000	713	335	244	250	1.03	0.32
FY2030	5.05	14.62	4.84	24.77	0.69	18.33	2.74	3.75	74.78	4,73,04,000	713	354	254	258	1.02	0.33
FY2031	5.15	14.93	4.79	25.76	0.68	18.11	2.60	3.82	75.84	4,97,86,000	750	378	274	297	1.08	0.32
FY2032	5.28	15.31	5.20	26.79	0.70	19.62	2.60	4.04	79.54	4,97,86,000	750	396	285	303	1.06	0.33
Totals/ Average									73.92	38,20,82,000		2824	1976	1971	1.00	0.35
Scenario 2: All Midi buses - 10min headways																
FY2023	10.49	14.69	1.39	18.47	1.07	6.44	10.63	3.19	66.39	2,43,82,000	361	162	120	87	0.72	0.37
FY2024	10.84	15.74	1.39	19.21	1.10	6.44	19.67	3.81	78.20	2,43,82,000	361	191	127	96	0.75	0.42
FY2025	4.08	11.85	3.22	19.43	1.00	13.02	14.45	3.49	70.53	2,78,13,000	400	196	120	129	1.07	0.46
FY2026	4.18	12.15	3.38	21.17	1.03	13.58	11.77	3.51	70.78	2,75,21,000	415	195	125	151	1.21	0.54
FY2027	4.29	12.46	3.38	22.02	0.73	13.08	6.02	3.21	65.19	4,08,80,000	616	267	188	192	1.02	0.41
FY2028	4.40	12.77	3.64	22.90	0.75	14.07	4.12	3.25	65.91	4,08,80,000	616	269	195	209	1.07	0.45
FY2029	4.51	13.10	3.62	23.82	0.70	13.87	2.86	3.24	65.71	4,53,33,000	683	298	222	250	1.13	0.43
FY2030	4.62	13.43	4.04	24.77	0.72	15.45	2.86	3.45	69.34	4,53,33,000	683	314	231	258	1.11	0.44
FY2031	4.74	13.77	4.01	25.76	0.68	15.23	2.60	3.50	70.28	4,97,86,000	750	350	261	297	1.14	0.41
FY2032	4.86	14.12	4.31	26.79	0.70	16.36	2.60	3.68	73.42	4,97,86,000	750	366	271	303	1.12	0.42
Totals/ Average									69.32	37,60,96,000		2607	1861	1971	1.06	0.43
Scenario 3: All Midi buses at 60% LF - 15min headways																
FY2023	9.55	13.71	1.63	18.41	0.94	7.10	9.25	3.06	63.66	2,80,32,000	413	178	133	85	0.64	0.31
FY2024	9.87	14.61	1.63	19.14	0.97	7.10	17.11	3.60	74.03	2,80,32,000	413	208	140	91	0.65	0.35
FY2025	4.08	11.85	3.22	19.43	0.93	12.92	13.49	3.43	69.35	2,97,84,000	428	207	128	117	0.91	0.39
FY2026	4.18	12.15	3.38	21.17	0.97	13.48	10.99	3.45	69.77	2,94,92,000	444	206	134	133	0.99	0.44
FY2027	4.29	12.46	3.38	22.02	0.79	13.17	6.55	3.25	65.91	3,75,95,000	567	248	174	165	0.95	0.38
FY2028	4.40	12.77	3.77	22.90	0.81	14.63	4.48	3.32	67.09	3,75,95,000	567	252	180	177	0.98	0.41
FY2029	4.51	13.10	3.72	23.82	0.73	14.28	2.97	3.28	66.39	4,36,54,000	658	290	215	210	0.98	0.37
FY2030	4.62	13.43	4.12	24.77	0.75	15.76	2.97	3.48	69.90	4,36,54,000	658	305	223	216	0.97	0.38
FY2031	4.74	13.77	4.11	25.76	0.73	15.70	2.84	3.55	71.20	4,56,98,000	689	325	241	248	1.03	0.37
FY2032	4.86	14.12	4.31	26.79	0.75	16.44	2.84	3.71	73.82	4,56,98,000	689	337	249	253	1.01	0.38
Totals/ Average									69.22	36,92,34,000		2556	1816	1695	0.93	0.38
Scenario 4: Standard + Midi buses at 60% LF - 15min headways																
FY2023	10.16	14.59	1.55	18.44	1.02	6.72	10.06	3.16	65.70	2,57,69,000	381	169	126	84	0.67	0.33
FY2024	10.50	15.60	1.55	19.18	1.05	6.72	18.61	3.75	76.95	2,57,69,000	381	198	133	88	0.66	0.36
FY2025	4.17	12.11	3.39	19.43	1.05	13.50	15.25	3.59	72.49	2,63,53,000	379	191	115	109	0.94	0.38
FY2026	4.28	12.42	3.55	21.17	1.09	14.10	12.43	3.61	72.65	2,60,61,000	393	189	120	120	1.00	0.43
FY2027	4.40	12.76	3.57	22.02	0.88	13.84	7.29	3.37	68.12	3,37,99,000	509	230	159	147	0.93	0.35
FY2028	4.51	13.09	3.95	22.90	0.90	15.25	4.98	3.42	69.00	3,37,99,000	509	233	165	156	0.95	0.38

Built Up	Cost (by component) per Km									Total annual Vehicle Km	Total No of Buses	Total Operational Cost (inc. Depreciation & Cost of fund)	Total Operational Cost (w/o Depreciation & Cost of fund)	Total Revenue (with Concession & Discount)	Operating Ratio (Revenue/ Operation Cost w/o Depreciation & Cost of fund)	Load Factors
	Fuel Cost	R & M Cost	Insurance + MVT + MACT + RR Tax	Staff Cost	ITMS Operating Cost	Cost of Depreciation - Total	Cost of Fund - (Expected returns and Interests)	Miscellaneous Cost	CPKM (with depreciation and Interest)							
FY2029	4.62	13.42	3.90	23.82	0.80	14.89	3.29	3.37	68.11	3,94,20,000	594	268	197	184	0.93	0.34
FY2030	4.74	13.76	4.34	24.77	0.82	16.52	3.29	3.59	71.82	3,94,20,000	594	283	205	188	0.92	0.34
FY2031	4.85	14.08	4.29	25.76	0.78	16.27	3.03	3.64	72.70	4,27,05,000	644	310	228	216	0.95	0.32
FY2032	4.97	14.44	4.51	26.79	0.80	17.09	3.03	3.80	75.44	4,27,05,000	644	322	236	219	0.93	0.33
Totals/ Average									71.34	33,58,00,000		2396	1684	1512	0.90	0.36
Scenario 5: Midi + Mini buses at 60% LF - 15min headways																
FY2023	8.70	12.68	1.48	18.35	0.83	6.41	8.05	2.85	59.34	3,21,93,000	473	191	144	86	0.60	0.32
FY2024	8.98	13.51	1.48	19.08	0.85	6.41	14.90	3.33	68.53	3,21,93,000	473	221	152	94	0.62	0.36
FY2025	3.62	8.99	2.40	19.43	0.78	9.76	11.05	2.89	58.92	3,63,54,000	523	214	139	125	0.90	0.50
FY2026	3.71	9.19	2.51	21.17	0.80	10.15	8.98	2.91	59.44	3,60,62,000	543	214	145	145	1.00	0.59
FY2027	3.80	9.42	2.51	22.02	0.65	9.88	5.23	2.74	56.24	4,70,85,000	710	265	194	183	0.94	0.50
FY2028	3.90	9.66	2.84	22.90	0.66	11.10	3.58	2.80	57.43	4,70,85,000	710	270	201	198	0.98	0.54
FY2029	4.00	9.91	2.79	23.82	0.59	10.79	2.33	2.77	56.99	5,55,53,000	837	317	244	237	0.97	0.49
FY2030	4.10	10.16	3.04	24.77	0.60	11.72	2.33	2.91	59.64	5,55,53,000	837	331	253	244	0.96	0.50
FY2031	4.22	10.50	3.07	25.76	0.60	11.80	2.28	3.00	61.22	5,69,40,000	858	349	268	281	1.05	0.49
FY2032	4.32	10.77	3.23	26.79	0.62	12.41	2.28	3.12	63.54	5,69,40,000	858	362	278	286	1.03	0.50
Totals/ Average									59.95	45,59,58,000		2734	2019	1879	0.93	0.48
Scenario 5A: Midi _ Mini buses at 60% LF - 10min headways																
FY2023	8.45	12.43	1.50	18.33	0.79	6.59	7.69	2.81	58.59	3,37,26,000	495	198	149	88	0.59	0.32
FY2024	8.73	13.23	1.50	19.06	0.81	6.59	14.22	3.27	67.42	3,37,26,000	495	227	157	99	0.63	0.37
FY2025	3.60	8.85	2.36	19.43	0.71	9.63	9.97	2.81	57.36	4,02,96,000	580	231	152	137	0.90	0.50
FY2026	3.69	9.05	2.47	21.17	0.73	10.02	8.10	2.84	58.07	4,00,04,000	603	232	160	164	1.02	0.61
FY2027	3.78	9.28	2.46	22.02	0.59	9.74	4.71	2.69	55.27	5,22,68,000	788	289	213	210	0.98	0.53
FY2028	3.88	9.51	2.78	22.90	0.60	10.91	3.22	2.75	56.56	5,22,68,000	788	296	222	230	1.04	0.58
FY2029	3.98	9.76	2.74	23.82	0.53	10.61	2.10	2.73	56.26	6,16,85,000	930	347	269	276	1.03	0.52
FY2030	4.08	10.01	2.99	24.77	0.55	11.55	2.10	2.87	58.91	6,16,85,000	930	363	279	286	1.02	0.54
FY2031	4.19	10.34	3.02	25.76	0.55	11.63	2.05	2.95	60.50	6,30,72,000	950	382	295	329	1.12	0.53
FY2032	4.30	10.60	3.18	26.79	0.57	12.23	2.05	3.08	62.80	6,30,72,000	950	396	306	337	1.10	0.55
Totals/ Average									59.01	50,18,02,000		2961	2203	2155	0.98	0.51
Scenario 6: Standard + Midi + Mini buses at 60% LF - 15min headways																
FY2023	9.19	13.29	1.51	18.38	0.89	6.43	8.72	2.95	61.57	2,97,11,000	437	182	137	86	0.63	0.33
FY2024	9.50	14.17	1.51	19.12	0.91	6.43	16.14	3.46	71.54	2,97,11,000	437	212	145	93	0.64	0.36
FY2025	3.79	9.77	2.70	19.43	0.91	10.89	13.13	3.14	64.37	3,05,87,000	440	195	122	121	0.99	0.49
FY2026	3.88	9.99	2.83	21.17	0.95	11.34	10.69	3.16	64.55	3,02,95,000	457	194	127	139	1.09	0.57
FY2027	3.99	10.28	2.84	22.02	0.75	11.11	6.19	2.95	60.55	3,97,85,000	600	239	170	174	1.02	0.47
FY2028	4.09	10.54	3.21	22.90	0.77	12.46	4.23	3.01	61.60	3,97,85,000	600	244	177	188	1.06	0.51
FY2029	4.20	10.81	3.15	23.82	0.68	12.12	2.77	2.96	60.83	4,67,93,000	705	283	213	224	1.05	0.46
FY2030	4.30	11.08	3.44	24.77	0.70	13.18	2.77	3.12	63.69	4,67,93,000	705	297	222	230	1.04	0.47
FY2031	4.42	11.42	3.46	25.76	0.70	13.23	2.69	3.20	65.21	4,81,07,000	725	312	236	264	1.12	0.46
FY2032	4.53	11.71	3.65	26.79	0.72	13.94	2.69	3.35	67.71	4,81,07,000	725	324	244	269	1.10	0.47
Totals/ Average									63.69	38,96,74,000		2482	1793	1787	1.00	0.46

Annexure 10: Fleet requirement and procurement plan, capital and operating cost estimates for 25% PT share (shortlisted) scenarios

Table 9: Fleet procurement plan for short listed scenario – 25% PT share

Scenario -->	Scen-1: 10min HDW - Std+Midi Buses		Scen-2: 10min HDW - All Midi Buses	Scen-5A: LF 60% - Midi+Mini, hdw cap at 10min	
Bus Type -->	Standard	Midi	Midi	Midi	Mini
2023	36	126	161	119	179
2024	0	0	0	0	0
2025	142	185	327	51	286
2026	8	10	19	4	22
2027	130	145	256	53	150
2028	0	0	0	0	0
2029	56	48	86	42	112
2030	0	0	0	0	0
2031	0	48	86	21	0
2032	0	0	0	0	0
Total Standard/ Midi/ Mini Fleet	372	563	935	289	748
Total Fleet	935		935	1037	

Table 10: Total Capital Cost (in Cr.) Investment (2023 to 2032) – 25% PT share

Item	Scenario 1- Standard & Midi	Scenario 2 – All midi	Scenario 5A – Midi & mini
New Bus Stops + Existing Bus Stops reconstruction (Within Chandigarh)	31.28	31.28	31.28
Terminals improvement + Driver conductor facilities at few locations	11.82	12.26	12.26
Depots cost (with land)	212	173	164
ETVM	6.80	6.82	7.46
Utilities & Recovery Vans – Depots	1.25	1.25	1.25
Capital infrastructure cost	263	225	216
Fleet cost	1385	1150	920
Total Investment	1648	1375	1136

Table 11: Total Operating Cost (in Cr.) by operational cost components (2023 to 2032) – 25% PT share

Cost details	Cost in Cr		
	Scenario 1- Standard & Midi	Scenario 2 – All midi	Scenario 5A – Midi & mini
Fuel Cost	255.60	234.83	245.77
Lubricant Cost	0.65	0.65	0.65
R & M Cost	665.86	605.38	546.10
Insurance + MVT + MACT + RR Tax	194.26	159.76	141.29
Staff Cost	1078.73	1061.09	1256.18
ITMS Operating Costs	30.91	30.86	31.42
Cost of Depreciation - Total	745.56	618.42	554.38
Cost of Fund - (Expected returns & Interests)	239.76	239.76	239.76
Miscellaneous Cost	168.16	153.19	154.17
Total Operational Cost	3379	3104	3170